

GENERAL THETA FOUNDATIONS I: SADDLE GEOMETRY AND THE MEMORY OF CAUSAL EXPERIMENTS

QIAN QI

ABSTRACT. We study the memory required to attain a statistical saddle in a causal experiment. For finite interfaces, admissible registers give normalized positive realizations of the experiment's continuation channels. An unrestricted minimax saddle determines an exposed Bayes face; optimal finite-memory realizations must also satisfy contact and stationarity at the supporting parameters. These additional conditions can fix responses on Bayes ties and thereby force memory that a Bayes best response alone would not require. For a continuous family of marked product experiments we obtain exactly five states at the decision cut of an optimal two-preparation audit. For serial validation with the candidate read first, the minimum whole-schedule peak is exactly twelve. Both converses include stochastic encoders and continuations. They are proved by positive face packing, not by minimizing a chosen deterministic machine. We also identify the all-preparation limiting value and derive the closest-product geometry directly from the experimental probabilities. Every least-favorable prior localizes near two intrinsic contact points; symmetric priors converge to their equal mixture with an explicit quadratic-transport bound. Prepared predictive quotients, causal simulation, informative finite-memory control, exact one- and two-preparation saddles, and positive-noise physical realization are developed within the same resource framework.

CONTENTS

1. Introduction	2
1.1. Saddle geometry and its realizations	2
1.2. Exact memory in a continuous marked experiment	3
1.3. A law for every preparation number	4
1.4. Prepared foundations and retained consequences	4
1.5. Relation to earlier work	4
1.6. Organization	5
2. Causal experiments and their predictive quotients	5
2.1. The experimental object	5
2.2. Prepared predictive equivalence	6
3. Attainable resolution and resource-aware simulation	7
3.1. Checkpoint and online prediction	7
3.2. Causal morphisms and simultaneous error budgets	8
4. Experiments, architectures and attainable resources	10
5. A statistical dual that is uniform over architectures	11
6. Controlled experiments without a dominating action	13
6.1. An informative action choice	14
7. Controlled experiments at a prescribed memory profile	15

Date: September 24, 2026. Revision 27.

2020 Mathematics Subject Classification. Primary 62B05, 62C20; Secondary 60J10, 68Q45, 93E20.

Key words and phrases. Statistical experiments, causal realization, predictive distributions, finite memory, minimax duality, physical certification.

7.1. A parameterwise occupation state	15
7.2. An exact informative-control bottleneck	17
8. Saddle faces and causal realization	18
8.1. A normal form over a finite causal interface	19
8.2. The realization theorem	20
8.3. Faces as memory witnesses	21
9. Exact minimax laws for a marked product family	22
9.1. Complementary slackness and the event alphabet	24
10. A self-contained collision interface	24
10.1. Preparation and elastic scattering	25
10.2. Positive noise and the actual mark	26
11. The physical experiment and its sharp preparation boundary	27
11.1. An exact rational certificate against one preparation	28
11.2. A marked five-event selector	28
11.3. Causal compilation and the full register profile	29
12. The exact two-preparation marked saddle	31
13. A continuous two-preparation marked family	33
14. Exact memory from a marked saddle	35
14.1. The sharp serial-validation peak	37
15. Preparation limits and localization of least-favorable priors	38
16. An exact preparation–decision-memory law	41
17. Integer memory and the failure of a naive partition formula	43
18. Quantitative certificates for a finite compatible architecture	44
19. Autonomous noisy testing with a precision and time ledger	46
20. Physical consumers and the first-principles connection	47
21. A sharpened physical consumer and its simulation budget	48
22. Dependencies, retained results and remaining distinctions	49
References	50

1. INTRODUCTION

A statistical saddle determines the best expected decision, but need not determine a machine that can make that decision. A Bayes response can lose information that is immaterial under its prior yet necessary for uniform optimality over the parameter family. In a causal experiment, this distinction has a resource cost: discarded information cannot be recovered by an uncharged random selector or a terminal decoder. The central question of this paper is how to read that cost from the saddle itself.

The underlying experiment consists of a preparation, positive normalized kernels, a causal action interface, executable future tests, and charged register cuts. The unknown parameter is fixed throughout an execution. The preparation determines path laws and posterior continuations; no geometric reference measure is supplied independently of the experiment. The predictive quotient identifies histories that admit identical legal future tests. Its stochastic realizations, rather than ordinary linear rank alone, govern finite memory.

1.1. Saddle geometry and its realizations. For a finite causal interface, all controller wirings within a given profile are represented by common normalized positive matrix products. The generators at successive cuts must be compatible under every one-step continuation. Independently factoring each cut does not suffice. Let $\mathcal{C}_{\mathbf{K}}$ be this compact, generally nonconvex response set,

and let U be the unrestricted minimax value. A least-favorable prior exposes the Bayes face F_* . The profiles attaining U are precisely those admitting compatible positive realizations of

$$\mathcal{S}_* = F_* \cap \{C : g_\theta(C) \geq U \text{ for all } \theta\}.$$

Theorem 8.2 gives the exact loss decomposition, attainment criterion, supporting-parameter contact equations, and the factorizations imposed at a retained-state cut. This is not a minimax interchange over nonconvex machines. Its operative distinction is between a Bayes exposed face and its uniformly optimal part.

This distinction is decisive when some Bayes coefficients vanish. Contact and stationarity at interior supporting parameters may determine these coordinates even though the Bayes objective does not. Once they are determined, a convex-face obstruction applies to every stochastic encoder and decoder. Lemma 8.4 gives a simple reusable form: obligatory vertices and additional rows in disjoint faces require separate positive generators. Exact causal recodings preserve the attainable profiles; approximate simulations transport the value with explicit state products and error allowances.

1.2. Exact memory in a continuous marked experiment. Our main application is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad (p, q) \in [0, 1]^2,$$

where the last coordinate is the actual preparation mark. The target Q_γ has masses $((1 + \gamma)/4, (1 - \gamma)/4, (1 - \gamma)/4, (1 + \gamma)/4)$ on 000, 001, 110, 111, and zero mass elsewhere. After candidate training, the auditor freezes a row and event and applies them to independent candidate and target validations. The score is target indicator minus candidate indicator. Write $U_N(\gamma)$ for its unrestricted minimax value with N training preparations.

The exact two-preparation saddle of Theorem 13.1 has a two-point least-favorable prior for $6/25 \leq \gamma \leq 13/50$. Its two extreme-count response coordinates are Bayes ties. Theorem 14.1 proves that every minimax response, not only the displayed symmetric optimizer, must average these coordinates to the same explicitly determined number $t_\gamma \in (0, 1)$. Three other count rows are forced cube vertices. Two disjoint exposed faces then require two additional generators. Consequently

minimum decision width at the exact two-preparation value = 5.

The result applies to the original adaptive row–event interface: training, gates, stopping, retained randomness, and independent calibration are included. A common-garbling subfamily supplies the converse; the construction works against the original private candidate class.

Applying the same argument to future *output channels* at a validation cut, rather than only to event responses, yields a stronger scheduled result. If the candidate’s complete validation is read before any target validation report, eight deterministic continuation vertices and four additional disjoint faces force twelve generators. Theorem 14.3 therefore proves

minimum peak for exact forward serial validation = 12.

The matching implementation is the existing five-event residual machine. The lower bound does not assume that the event name remains stored after its residual is sufficient. This serial-order assertion is distinguished from the order-independent decision-width theorem. Reverse-order and interleaved validations are not assigned a peak optimum by this result.

For the collision bias $\gamma = 1/4$, the exact statistical values are

$$(1) \quad U_1 = \frac{85 - 7\sqrt{73}}{64}, \quad U_2 = \frac{-r^3 + 6r^2 - 3r + 14}{32}, \quad 2r^3 + 25r^2 - 3 = 0,$$

where $1/3 < r < 7/20$. Their full-parameter lower certificates and matching Bayes upper bounds are proved in the article. The new memory converses extract consequences from the entire optimal face of these calculations. They do not replace the exact values by a relaxation or a finite grid. Compactness also gives a strict four-state loss uniformly over the bias interval, and a strictly positive eleven-state loss for the forward serial class. The four-state loss survives sufficiently small positive sensor noise through an explicit vanishing Gaussian-tail modulus.

1.3. A law for every preparation number. The finite- N saddles also determine a natural limiting question. Theorem 15.1 proves that unrestricted values increase to the distance from the target to the nearest candidate experiment. More significantly, every least-favorable prior concentrates on that closest-experiment set. This follows from the same common empirical response used as a competitor in the saddle game; the prior is not replaced by a parameter-dependent algorithm.

For the marked family the closest candidates and a uniform quadratic growth inequality are derived directly from its four diagonal masses. With $s_\gamma = \sqrt{(1 + \gamma)}/2$, the exact limiting value is

$$d_\gamma = 2s_\gamma - 1 - \gamma/2,$$

and its contact set consists exactly of (s_γ, s_γ) and $(1 - s_\gamma, 1 - s_\gamma)$. For every $N \geq 1$,

$$d_\gamma - \sqrt{3/N} \leq U_N(\gamma) \leq d_\gamma.$$

Every symmetric least-favorable prior converges to the equal mixture of those points, with quadratic-transport distance at most $\sqrt{2300/7}(3/N)^{1/4}$. The intrinsic growth constant is derived, not assumed as a dimension or small-ball hypothesis. These are uniform bounds and a limiting classification, not a claim that a two-point prior is exact at every finite preparation number or that the rates are sharp.

1.4. Prepared foundations and retained consequences. The general foundations layer proves path construction, predictive quotient closure, checkpoint resolution, and acquired causal transport. The latter jointly charges calibration, preparation, quantization, numerical and model error, and the memory of a downstream controller. The informative routing family exhibits the exact values $1/(2\lceil m/K \rceil)$ with a real routing cut and $K/(2m)$ with a free persistent controller selector. The separate nonlinear occupation recursion retains all common-row constraints. These results distinguish an unrestricted saddle from its legal positive realizations.

The positive-noise collision construction realizes the marked experiment on the original report and actual-mark interface. The exact ideal score for $N = 2$, $W \geq 12$, the finite-precision implementation, and the transported confidence theorem remain part of the paper. So do the integer revelation laws, compatible subdivision certificates, and the explicitly attributed autonomous finite-memory comparison. For reference, the deterministic precursor constants are

$$(2) \quad u_1 = \frac{4035777}{10240000}, \quad m_2 = \frac{12 - 3\sqrt{3}}{16}, \quad \beta_0 = \frac{1}{300}.$$

Their earlier constructions and proofs are retained along with their later exact improvements.

1.5. Relation to earlier work. Positive stochastic realization and invariant cone methods have classical antecedents in Heller [17]; Vidyasagar [18] surveys the complete hidden-Markov realization problem. Nonnegative rank differs from ordinary linear rank, and its geometric interpretation is developed by Gillis and Glineur [19]. Our matrix normalization adds the specified causal action order and charged cuts. We do not claim the positive-realization mechanism itself as new. The additional argument links its generators to the uniformly optimal portion of a statistical saddle, including contact equations on Bayes-indifferent coordinates.

Blackwell comparison and convex minimax provide the statistical mechanisms [2, 3]. Belief-state dynamic programming is classical [1]. Controlled information acquisition and adaptive sensing have an established history [13, 14, 15]; neither causal action selection nor an adaptive advantage by itself is a priority claim here. Finite-memory testing and randomization are likewise classical [6, 7, 8, 9]. The autonomous stationary formula below is attributed to Hellman and Cover. Recent work on memory-constrained adversarial testing [10] and imperfect-recall optimization [16] concerns adjacent constraints with different adversarial or objective conventions. Polynomial policy optimization and positivity methods also have substantial antecedents [4, 5]. We retain the explicit distinctions in the literature crosswalk accompanying the source.

The empirical estimation bound used in the preparation limit is the same elementary finite-alphabet estimate appearing in the retained decision-spectrum theorem. The new use is localization of every least-favorable prior, followed by an explicit intrinsic contact calculation. Norberg's filtered-experiment comparison [11] remains an adjacent general theory; no proof-level non-overlap claim is made without a complete comparison of the original work.

1.6. Organization. The prepared foundations and finite interface are followed by statistical and controlled duality. Section 8 gives the central realization criterion. The exact marked saddles and physical bridge supply its hypotheses, and Section 14 extracts the decision and serial-peak converses. Section 15 gives the all-preparation law. The remaining resource and physical consumers follow. Complete predecessor volumes retain the development of the predictive, geometric, and operator interfaces. The independent historical A2 chain and the B4/C2 aggregate obligations are not turned into consequences of a finite-state theorem by a change of terminology.

2. CAUSAL EXPERIMENTS AND THEIR PREDICTIVE QUOTIENTS

We begin with the object and morphism structure. Its elementary probability statements fix the meaning of later resource theorems; classical kernel construction, conditional expectation and quantization are not assigned a new priority. The substantive comparisons below retain their preparation, memory and approximation coordinates simultaneously.

2.1. The experimental object. An experiment has a fixed unknown parameter $\theta \in \Theta$, hidden states X_t , visible actions A_t , reports Y_{t+1} , and preparation laws μ_θ . We impose the following six conventions.

A0 (types and visibility). These spaces are standard Borel. Visible histories include actions, reports and publicly recorded resource increments. A legal-action set and its remaining budget are part of the visible protocol type. Two histories with different legal continuations are not identified.

A1 (positive kernels). There are jointly measurable normalized nonnegative kernels $K_t^\theta(dx', dy, dc \mid x, a)$, with c a nonnegative resource increment. An absorbing symbol represents termination or failure. Dirac kernels are allowed. Positivity here does not require every outcome to have strictly positive probability. A hidden state can contain the necessary physical past; this notation does not assume a finite-dimensional physical closure.

A2 (causality). A policy is a kernel $\alpha_t(da \mid h_t)$ depending only on visible past data and allowed independent randomization. It cannot read the unknown parameter. Stopping is with respect to the visible filtration.

A3 (executable tests). The specified bounded future tests include constants, required report events and their legal one-step continuations. For discrete reports the one-step instrument is

$$(\mathbb{T}_{t,a,y}f)(\theta, x) = \int f(\theta, x') K_t^\theta(dx', \{y\}, dc \mid x, a).$$

The test spaces are closed under the required instruments, finite linear combinations and the bounded monotone limits needed to recover the stated event laws. For general reports one uses event instruments and regular conditional distributions, not division by the probability of a continuous singleton.

A4 (actual preparation). Average risks and acquired state laws refer to the specified preparation and exploration protocol. Failed trials, calibration, resets and conditioning events retain their probabilities and implementation costs. A convenient measure on a reachable set does not replace its acquired law.

A5 (executable resources). Persistent labels, temporary buffers, program constants, calibration precision, preparations and clocks are distinguished. A retained selector is a state. A parameter-dependent table is not a known program constant. A simulator with R_t labels composed with a controller with K_t labels has the direct product bound $R_t K_t$, before any justified quotient is taken. Atomic stochastic rows and fair-bit implementations are separately specified.

Proposition 2.1 (Prepared path laws). *For every fixed parameter and causal policy, the preparation and kernels uniquely determine their finite-dimensional path laws. On a finite horizon they are the iterated kernel product*

$$\mu_\theta(dx_0) \prod_{t < T} \alpha_t(da_t \mid h_t) K_t^\theta(dx_{t+1}, dy_{t+1}, dc_t \mid x_t, a_t).$$

These laws are consistent under marginalization. For a countable horizon they determine the usual kernel extension. Deterministic dynamics give their original trajectory laws when represented by Dirac kernels.

Proof. Successive integration constructs each finite law. Normalization of the last two kernels gives its preceding marginal, and induction proves consistency and uniqueness on cylinder events. The countable kernel extension gives the full law; equivalently one applies the standard extension theorem to these consistent iterated kernels. Integration against a Dirac transition is evaluation along that transition, which proves the last assertion. $\square$

2.2. Prepared predictive equivalence. For Bayesian predictions additionally fix a prior on the parameter, and regard (θ, X_t) as the hidden variable. A prepared posterior π at a given protocol type evaluates the future tests. Define $\pi \sim_t \tilde{\pi}$ if their evaluations agree on every specified future test. This definition concerns one prepared experiment; it does not identify a prior-dependent posterior with a statistic sufficient simultaneously for all fixed parameters.

Theorem 2.2 (Predictive quotient and causal closure). *For finite report alphabets, predictive equivalence is preserved by every legal positive-probability report update. Its class determines the laws of all specified future experiments. Every deterministic history statistic that determines those laws factors onto this quotient. For a measurable realization it is sufficient that a countable separating test family gives a measurable evaluation map with standard Borel image, and that the chosen statistic admits a measurable factor. In finite history spaces the latter conditions are automatic.*

Proof. For a legal action and report, Bayes' rule gives

$$\pi'(f) = \frac{\pi(\mathbb{T}_{t,a,y} f)}{\pi(\mathbb{T}_{t,a,y} 1)}.$$

Both numerator and denominator depend only on the old class by A3. Equivalent posteriors assign the same probability to the report, so updates agree on the next test space whenever that probability is positive. Null reports do not affect the prepared law and may be assigned a fixed

admissible version. Induction determines every finite future report word; the monotone-class extension gives the required future events. Conversely those event laws determine the tests. If two histories have the same sufficient statistic, they have the same future evaluations; the quotient is therefore constant on its fibers. This gives the factor set-theoretically. The stated measurable factor assumption supplies its measurable version; no claim that every abstract quotient of a standard Borel space is standard Borel is used. $\square$

Proposition 2.3 (Barycenters, entropy and normalized response). *In a fixed preparation, suppose a hidden target Z , observation Y and postprocessing W form a Markov chain $Z \rightarrow Y \rightarrow W$. For any bounded vector of tests f ,*

$$p_W = \mathbb{E}[p_Y | W], \quad p_Y = \mathbb{E}[f(Z) | Y], \quad \mathbb{E}\varphi(p_W) \leq \mathbb{E}\varphi(p_Y)$$

for every convex φ for which these expectations exist. For a parameter-independent kernel G , $P \ll Q$, and finite $D(P\|Q)$, regular conditional laws give

$$\begin{aligned} D(P\|Q) &= D(PG\|QG) \\ &\quad + \int D(P(dx | w)\|Q(dx | w)) (PG)(dw). \end{aligned}$$

Finally, if f, g are bounded and $\pi_u(dx) = e^{ug(x)}\pi(dx)/\pi(e^{ug})$, then

$$\left. \frac{d}{du} \pi_u(f) \right|_{u=0} = \text{Cov}_\pi(f, g).$$

Proof. The Markov property and the tower rule give the conditional barycenter; conditional Jensen gives convex order. The joint laws $P(dx)G(dw | x)$ and $Q(dx)G(dw | x)$ have relative entropy $D(P\|Q)$. Disintegrating by w and applying the entropy chain rule gives the identity, and hence data processing. The last equality follows by quotient differentiation; boundedness justifies differentiation under the integral. $\square$

The barycenter statement concerns a law of predictions, not a pointwise map between reachable sets. Likewise the covariance in the last formula is a normalized response, not an independently postulated Fisher matrix or a global sufficiency theorem.

3. ATTAINABLE RESOLUTION AND RESOURCE-AWARE SIMULATION

3.1. Checkpoint and online prediction. Fix an actual exploration protocol and its history law. Let $p(h)$ be a bounded d -dimensional conditional prediction vector, with law ν . For instance, a future test is sampled independently after encoding, and the coordinates include its sampling weights. Squared excess prediction loss is then $\mathbb{E}\|p(h) - \hat{p}\|^2$.

Proposition 3.1 (Checkpoint resolution). *For at most M retained labels the least checkpoint excess loss is*

$$(3) \quad \inf_{c_1, \dots, c_M} \int \min_{1 \leq j \leq M} \|p - c_j\|^2 \nu(dp).$$

Independent randomized encoders or decoders cannot improve it. This formula concerns the stated preparation and does not provide a common codebook by separately optimizing at each unknown true parameter.

Proof. Conditional squared loss decomposes into its Bayes loss and squared distance from the conditional mean. For each retained label replace a random decoder by its mean. Random assignment to the resulting centers cannot do better than a nearest center. Conversely a nearest-center assignment attains the infimum over codebooks arbitrarily closely. Conditioning on an independent shared random seed, if such a seed is allowed, gives a codebook for each seed,

each with risk at least (3). Averaging cannot improve the bound. The formula does not charge the computation of a checkpoint assignment as an online algorithm; online compatibility is a separate condition. $\square$

Theorem 3.2 (A matched online resolution law). *Suppose a fixed prepared experiment has a predictive state s_t in a compact metric space S , a known initial state, and updates $s_{t+1} = F_{t,a,y}(s_t)$. Assume every permitted update is ρ -Lipschitz for one $0 \leq \rho < 1$, and the prediction map $p : S \rightarrow \mathbb{R}^d$ is L -Lipschitz. Suppose S has a δ -net of at most $C\delta^{-d}$ points for all sufficiently small δ . At a checkpoint T , assume the actual law ν of $p(s_T)$ satisfies the small-ball bound $\nu(B(z,a)) \leq Aa^d$ for every center z and all sufficiently small $a > 0$. Then for all sufficiently large M , the minimum online excess prediction loss with M retained predictive labels satisfies*

$$(4) \quad c_0 M^{-2/d} \leq R_M^{\text{on}} \leq C_0 M^{-2/d}.$$

Here one may take

$$c_0 = \frac{1}{2}(2A)^{-2/d}, \quad C_0 = \frac{L^2 C^{2/d}}{(1-\rho)^2},$$

Any separate exploration controller, report buffer or clock is charged in addition to these M labels. The exploration protocol and the preparation are fixed before optimizing the encoder.

Proof. Any online M -label state is also a checkpoint encoder, so Proposition 3.1 applies. For any M centers, the union of balls of radius $a = (2AM)^{-1/d}$ has ν -mass at most one half. On its complement squared distance is at least a^2 . Integration gives $c_0 M^{-2/d}$. This argument also permits randomized online encoders.

For the upper bound take a δ -net with $\delta = (C/M)^{1/d}$, and a fixed measurable nearest-net assignment (with a fixed tie rule). Quantize the known initial state and, after each legal update, quantize again. The state error obeys $e_{t+1} \leq \rho e_t + \delta$, $e_0 \leq \delta$, and hence $e_t \leq \delta/(1-\rho)$. The prediction error is at most $L e_t$. Squaring proves the upper bound, uniformly in time. Each update uses only the current net label and the current permitted input, so the construction is online. It does not grant the encoder free access to the original discarded history. If the exploration policy needs another register, that register is combined by a product construction. $\square$

For example, a Lebesgue density bounded above by B gives the small-ball hypothesis with $A = Bv_d$, where v_d is unit-ball volume. The lower hypothesis is a condition on the *acquired prediction law*. A reachable set of dimension d without a mass condition does not imply the lower bound. Similarly, the contraction assumption is additional dynamics, not a consequence of kernel positivity.

3.2. Causal morphisms and simultaneous error budgets. A causal simulator $\Gamma : \mathfrak{E} \rightarrow \mathfrak{F}$ is a parameter-independent executable transducer that uses source preparations to supply the target interface under every allowed target policy. Its defect ε is a uniform total-variation bound on the resulting path laws, including the fields required by the task. Its resource map ρ_Γ bounds the source resources of the transducer composed with a target controller.

Proposition 3.3 (Composition and risk transfer). *Causal simulators compose. Their uniform defects add, capped at one, and their resource maps compose in the order $\rho_{\Gamma_1} \circ \rho_{\Gamma_2}$. For a terminal loss in $[0, L]$ and a common target task,*

$$\mathcal{R}_{\mathfrak{E}}(\rho_\Gamma(B)) \leq \mathcal{R}_{\mathfrak{F}}(B) + L\varepsilon.$$

For reward in $[-1, 1]$, the corresponding maximum worst-case rewards satisfy the reverse inequality with an error 2ε .

Proof. Interleave the two transducers; their pair of internal states is a causal implementation. Total variation contracts under a kernel and obeys the triangle inequality, proving the defect sum. If their state counts are R_t and M_t , the pair has at most $R_t M_t$ states. Other declared resource bounds compose by execution of the two protocols. A bounded loss changes expectation by at most its range times total variation. Apply this bound to every fixed parameter and target controller and take the appropriate infimum or supremum. Uniformity in the controller is essential to this step. $\square$

Theorem 3.4 (Acquisition, quantization and model-error transport). *Suppose an exact source transducer has predictive state $s_t \in S_t$ and additional charged buffers of size R_t , supplies T target operations, and has uniform model defect at most ε_{mod} . Its initialization and update depend on a calibration quantity ξ . From N_{cal} source preparations acquire an estimate $\hat{\xi}$ with at most C_{cal} retained labels, such that*

$$\sup_{\theta} \mathbb{P}_{\theta}(d(\hat{\xi}, \xi) > a) \leq \eta.$$

Subsequent preparation noises are independent of calibration conditional on the fixed parameter. Let finite nets of sizes M_t have radii δ_t . Assume, on matching input/output histories and on the good calibration event, that the quantized transducer's state error satisfies

$$(5) \quad e_0 \leq d_0 + a_0 a + \delta_0, \quad e_{t+1} \leq \rho_t e_t + d_{t+1} + a_{t+1} a + \delta_{t+1}.$$

Assume its next output kernel and the exact transducer's output kernel differ in total variation by at most $\epsilon_t + L_t e_t + b_t a$, uniformly in the fixed parameter and legal policy. All constants and nets are declared independently of that unknown parameter. Then the quantized acquired transducer has defect at most

$$(6) \quad \Delta = \min \left\{ 1, \varepsilon_{\text{mod}} + \eta + \sum_{t < T} (\epsilon_t + L_t e_t + b_t a) \right\}.$$

Composing it with any target controller of cut widths K_t requires at most $C_{\text{cal}} R_t M_t K_t$ labels at aligned execution cuts and transfers its bounded reward with loss at most 2Δ . Its preparations are N_{cal} plus all source preparations specified by the transducer; no successful-trial conditioning removes the failures.

Proof. On the good calibration event, iterate (5). More explicitly, with $D_0 = d_0 + a_0 a + \delta_0$ and $D_j = d_j + a_j a + \delta_j$,

$$e_t \leq \sum_{j=0}^t D_j \prod_{k=j}^{t-1} \rho_k.$$

Couple each pair of output kernels maximally while their external histories agree. On agreement the target controller sees the same inputs, so it uses the same legal action and may use the same independent randomization. Its discontinuous action choices require no Lipschitz assumption: before a disagreement their arguments are identical. The chance of a first discrepancy is bounded by the sum in (6). Add the probability of bad calibration and the exact transducer's model defect. This proves the path bound, including stopping after padding by absorbing states.

The executable approximate transducer stores its calibration label, additional buffer, and net index. Combine these with the actual target register, not its full history or an uncharged belief. The resulting product gives the state count. If an operation contains further buffer cuts, apply the same product at each such cut; R_t was assumed to include them. Preparation costs add by execution. Finally apply Proposition 3.3. $\square$

Corollary 3.5 (A finite acquisition register). *Suppose r_0 calibration rows have at most d outcomes, and take n independent preparations from each row on a fixed schedule. Their empirical rows satisfy, with probability at least $1 - \eta$ uniformly in the row probabilities,*

$$\max \text{TV}(\hat{P}, P) \leq \frac{dz}{2}, \quad \eta = \min\{1, 2r_0d \exp(-2nz^2)\}.$$

Their counts can be retained in at most $(n+1)^{r_0(d-1)}$ labels. Thus Theorem 3.4 can use $N_{\text{cal}} = r_0n$, $a = dz/2$ and this finite calibration register when its calibration metric is the maximum row total variation.

Proof. For each outcome the empirical frequency is the mean of independent Bernoulli indicators. Hoeffding's bound and a union bound over at most r_0d coordinates give the event stated. The total-variation estimate is half the sum of coordinate errors. Retain $d - 1$ counts per row; the last is reconstructed from the known number of draws at that stage of the clocked schedule. Each retained count takes at most $n + 1$ values. An autonomous implementation must additionally charge its schedule. $\square$

These results separate exact predictive equivalence, average checkpoint resolution, online realizability, and uniform simulation risk. In particular, neither an optimal checkpoint codebook nor the proof-state bundle of Section 7 is automatically an executable finite register.

4. EXPERIMENTS, ARCHITECTURES AND ATTAINABLE RESOURCES

Throughout, total variation is $\text{TV}(P, Q) = \sup_{0 \leq h \leq 1} (P - Q)h = \frac{1}{2} \sum_x |P(x) - Q(x)|$. An experimental parameter is unknown and fixed throughout an execution. Controller randomization is independent of that parameter. A reset preparation is a fresh draw, independent of earlier preparations given the parameter. A controller may not recover discarded information from an uncharged tape, clock, selector or calibration table.

Definition 4.1 (Finite architecture). A finite architecture A specifies a finite causal execution graph, its legal experimental operations, the observable inputs to each controller row, and a register alphabet at every cut. It also specifies which row occurrences share a stochastic table. A clocked architecture may use different rows at different scheduled times. An autonomous architecture may use time only through a charged state. For a parameter family Θ and terminal reward in $[-1, 1]$, let

$$V(A) = \sup_{q \in \prod_r \Delta_{s_r}} \inf_{\theta \in \Theta} g_\theta(A, q).$$

If $\mathcal{A}(R)$ is the class of architectures obeying a global resource budget R , its outer value is

$$V^{\text{out}}(R) = \sup_{A \in \mathcal{A}(R)} V(A).$$

Both the architecture and the simulator parameter are fixed before the controller's random choices. In particular, a mixture of complete controllers is not automatically an admissible controller at the original register profile.

For a finite hidden execution graph, compatibility can be stated without division. Let $x_{\theta ir}$ be the incoming mass of hidden history i at observable row r , and let $f_{\theta irj}$ be the outgoing mass through choice j . Include the usual fixed-kernel flow equations, as well as

$$(7) \quad f_{\theta irj} \geq 0, \quad \sum_j f_{\theta irj} = x_{\theta ir}, \quad f_{\theta irj} x_{\theta' i' r} = f_{\theta' i' rj} x_{\theta ir}.$$

For shared rows, the indices i, i' include all their occurrences.

Proposition 4.2 (Compatible realization). *For a specified finite architecture, nonnegative flow arrays satisfying (7) are exactly those realized by a common stochastic row at every declared observable input, including at zero incoming mass.*

Proof. Necessity follows from $f_{\theta irj} = x_{\theta ir} q_r(j)$. For sufficiency, if some incoming mass at row r is positive, define $q_r(j)$ by dividing the corresponding outgoing mass by that incoming mass. Conservation makes q_r stochastic, and the cross-products force the same ratio at every other positive incoming mass. If an incoming mass vanishes, nonnegativity and conservation force all its outgoing masses to vanish. If the entire row is unreachable, choose any legal stochastic row. Induction through the finite execution graph reconstructs the prescribed flows. The same ratio argument applies to identified occurrences of an autonomous row. $\square$

This formulation is inherited from the complete companion and is included to fix the controller class. Compact semialgebraic feasibility and an Archimedean positivity theorem provide a complete semialgebraic certificate hierarchy for a *fixed finite architecture*. They do not interchange the two optimizations in the definition. Section 18 gives a quantitative alternative for the inner problem. The next section supplies a different certificate that survives the outer optimization.

5. A STATISTICAL DUAL THAT IS UNIFORM OVER ARCHITECTURES

Let Θ be compact, let Ω be finite, and let $\theta \mapsto P_\theta \in \mathcal{P}(\Omega)$ be continuous. Fix $Q \in \mathcal{P}(\Omega)$. An N -preparation reset selector observes $Z = (Z_1, \dots, Z_N) \sim P_\theta^{\otimes N}$ and chooses a response $h_Z \in [0, 1]^\Omega$. Its expected validation score is

$$(8) \quad G_\theta(h) = \sum_{z \in \Omega^N} P_\theta^{\otimes N}(z) (Q - P_\theta) h_z.$$

The response is fixed before two independent validation draws. Operationally it can be realized by choosing a random event with those coordinate marginals and retaining that event for both validations. No memory bound is imposed in the definition of $U_N(P, Q) = \max_h \min_\theta G_\theta(h)$.

Theorem 5.1 (Posterior-predictive dual). *Both extrema in the following identity are attained:*

$$(9) \quad U_N(P, Q) = \min_{\mu \in \mathcal{P}(\Theta)} \sum_{z, \omega} \left[\int P_\theta^{\otimes N}(z) \{Q(\omega) - P_\theta(\omega)\} d\mu(\theta) \right]_+$$

$$(10) \quad = \min_{\mu \in \mathcal{P}(\Theta)} \sum_z m_\mu(z) \text{TV}(Q, \bar{P}_\mu(\cdot | z)).$$

Here $m_\mu(z) = \int P_\theta^{\otimes N}(z) d\mu(\theta)$ and, when $m_\mu(z) > 0$,

$$\bar{P}_\mu(\omega | z) = \frac{\int P_\theta^{\otimes N}(z) P_\theta(\omega) d\mu(\theta)}{m_\mu(z)}.$$

A zero-mass term in (10) is defined to be zero.

Proof. The response space is a finite-dimensional compact convex box. Probability measures on the compact space Θ form a weakly compact convex set. The integral of (8) is continuous and affine in each argument. The minimum over parameters equals the minimum over probability measures. The minimax theorem [3] therefore interchanges the maximum and minimum. For fixed μ , each response coordinate is independent in the maximization and its contribution is the positive part of its coefficient, proving (9). The coefficient vector at a fixed z has zero sum. For positive $m_\mu(z)$ it equals $m_\mu(z)(Q - \bar{P}_\mu(\cdot | z))$, proving (10). If $m_\mu(z) = 0$, every nonnegative likelihood integrand vanishes μ -almost surely and all coefficients are zero. Continuity on the two compact strategy spaces gives attainment. $\square$

Theorem 5.2 (Dominating-row converse). *Suppose every permitted training row is a parameter-independent causal garbling of one fresh draw from P_θ . Suppose every permitted validation row is the same garbling of the target and candidate laws, and every selected row–event pair is fixed before validation. Then $U_N(P, Q)$ bounds the worst-case expected score of every auditor using at most N candidate training preparations, regardless of its register profile, feedback schedule, independent randomization, or stopping rule. Independent target-only calibration, whose law does not depend on θ , does not invalidate the bound. If the all-on row and all its events are permitted, the bound is attained in the class with no memory cap and N training preparations.*

Proof. Generate N independent all-on observations in advance. At the t th requested preparation, apply the declared garbling to observation t , using only previously simulated reports and independent randomization. This simulates the adaptive transcript with its exact law. A stopped execution ignores the unused observations; it does not reveal them to the auditor. Pull back its final row–event response to the all-on alphabet. Conditional on the entire generated training word and the independent random seed, this is a function in $[0, 1]^\Omega$. Averaging the seed, including any target-only calibration, produces one response h_z for each full word z . The score is consequently (8). This proves the upper bound for the union of all architectures. Conversely, retaining the entire word and then a selected event implements any such finite response kernel when memory is unrestricted. Fresh validation gives exactly (8). $\square$

Remark 5.3 (What the dual does not convexify). The measure μ is an adversarial prior used to bound the minimum over *fixed* parameters. Its likelihood is $\int P_\theta^{\otimes N} d\mu(\theta)$, not $(\int P_\theta d\mu(\theta))^{\otimes N}$. It neither enlarges the private candidate class to mixtures nor gives the auditor a free persistent selector. Attainment of U_N without a memory cap does not assert attainment at any specified finite profile.

The state in (10) is generated by the actual preparation law, not by a geometric measure chosen independently of the experiment. For sequential continuation one retains the posterior μ_z or its future-test equivalence class; a single predictive marginal need not be closed under further observations. This is the distinction between a one-step response quotient and a future-predictive quotient.

Corollary 5.4 (Finite moment support for the product family). *For two independent Bernoulli report bits with parameters (p, q) and an independent fair mark, an optimal prior in (9) can be chosen with at most $(N + 2)^2$ atoms. The two report counts (k, ℓ) are a causally updatable sufficient statistic for computing the dual.*

Proof. Conditional on the two counts, all arrangements of report bits and all mark words have parameter-independent probabilities. Summing their coefficients reduces (9) to the $(N + 1)^2$ count pairs. Their likelihoods are

$$\binom{N}{k} p^k (1 - p)^{N-k} \binom{N}{\ell} q^\ell (1 - q)^{N-\ell}.$$

After multiplication by $Q(\omega) - P_{p,q}(\omega)$, every integrand has degree at most $N + 1$ in each variable. Thus the objective depends only on the $(N + 2)^2$ rectangular moments. Their constant coordinate is one. Carathéodory’s theorem in the remaining affine space replaces an optimal moment vector by a convex combination of at most $(N + 2)^2$ point evaluations. Counts update by addition of the current bits. No claim that this count register is optimal for the decision is made. $\square$

6. CONTROLLED EXPERIMENTS WITHOUT A DOMINATING ACTION

The reduction to one dominating row is not needed for a statistical dual, provided causal action choice is retained in the dual recursion. We give the finite-action theorem with a compact parameter space and an explicit protocol state. Its equality is an unrestricted-memory statement; its upper certificates apply to every smaller-memory class.

Let Θ be compact metric and E a finite set of protocol states. At $e \in E$ let $\mathcal{A}_e$ be a finite set of training actions. Action a produces x in a finite alphabet X_a with law P_θ^a and changes the protocol state to $T(e, a, x)$. The functions $\theta \mapsto P_\theta^a(x)$ are continuous. An unknown θ is fixed throughout the experiment. Conditional on it, each requested preparation uses a fresh independent draw from the selected row. Legal validation designs form a nonempty finite set $\mathcal{B}_e$. Design b has a finite outcome alphabet Y_b , target law Q^b and continuous candidate law V_θ^b . A policy may stop after at most N training actions. It then freezes $b \in \mathcal{B}_e$ and an event on Y_b , before one candidate and one target validation. Randomization is independent of θ . No training action is assumed to dominate another.

For a finite nonnegative measure ν on Θ write $\nu_{a,x}(d\theta) = P_\theta^a(x)\nu(d\theta)$ and set

$$(11) \quad \Phi_e(\nu) = \max_{b \in \mathcal{B}_e} \sum_{y \in Y_b} \left[Q^b(y)\nu(\Theta) - \int V_\theta^b(y) d\nu \right]_+,$$

$$F_0(e, \nu) = \Phi_e(\nu),$$

$$(12) \quad F_n(e, \nu) = \max \left\{ \Phi_e(\nu), \max_{a \in \mathcal{A}_e} \sum_{x \in X_a} F_{n-1}(T(e, a, x), \nu_{a,x}) \right\}.$$

If no training action is legal, only the stopping term is retained. All the functions vanish at the zero measure, so no posterior division is required on null histories. For positive mass, (11) is the mass times the largest posterior-predictive total variation.

Theorem 6.1 (Controlled posterior-predictive dual). *Let $\mathcal{U}_N(e_0)$ be the maximum worst-case expected validation score over all randomized causal policies in the declared protocol, with at most N training preparations and no memory cap. Then*

$$(13) \quad \mathcal{U}_N(e_0) = \min_{\mu \in \mathcal{P}(\Theta)} F_N(e_0, \mu).$$

Both extrema are attained. Each $F_n(e, \cdot)$ is the maximum of finitely many continuous linear functionals on the cone of measures; in particular it is convex, continuous and positively homogeneous. For every prior μ , $F_N(e_0, \mu)$ bounds the worst-case score of every finite-memory architecture implementing this protocol.

Proof. There are finitely many histories of length at most N and finitely many event masks at each stopping history. Consequently there are finitely many deterministic policy trees, denoted by $\Pi_N(e_0)$. For $\pi \in \Pi_N(e_0)$ its payoff $g_\pi(\theta)$ is a finite sum of finite products of continuous primitive probabilities and is continuous in θ .

A randomized causal policy induces the same outcome laws as a probability distribution on these trees: sample its finite random decision table in advance and condition on it. Conversely, with no memory cap, a sampled tree can be retained and followed. Thus the unrestricted payoff vectors are exactly the convex hull of the g_π . This step is not an assertion that mixing machines preserves a specified register width.

The simplex on $\Pi_N(e_0)$ and $\mathcal{P}(\Theta)$ are compact and convex. The expected payoff is continuous and bilinear. Minimax therefore gives

$$\max_{\lambda \in \Delta(\Pi_N)} \min_{\theta} \sum_{\pi} \lambda_{\pi} g_{\pi}(\theta) = \min_{\mu} \max_{\pi \in \Pi_N} \int g_{\pi} d\mu.$$

For a fixed measure, the terminal maximum is (11), because an event includes precisely the positive coordinates of its signed predictive measure. At a training node one selects an action and, after its observed outcome, selects a continuation tree separately on that branch. The unnormalized branch measure is exactly $\nu_{a,x}$. Backward induction proves (12) and identifies the right side with (13). It also proves the finite-linear-functional representation. Compactness gives an optimizing prior and an optimizing mixed tree. Finally, the minimum over fixed parameters of any policy is at most its average under μ , which is at most $F_N(e_0, \mu)$. This remains true after restricting its memory or architecture. $\square$

The minimization in (13) occurs once, outside the recursion. It cannot be replaced by fresh adversarial choices after observations. The posterior carries dependence on one fixed unknown parameter. Also, a Bayes-optimal tree for a minimizing prior need not by itself be a minimax policy; the saddle-point mixture is supplied by the finite game in the proof.

Corollary 6.2 (Bellman upper certificates). *Suppose functions $\Psi_n(e, \nu)$ satisfy $\Psi_0 \geq \Phi$ and, for $n \geq 1$, satisfy*

$$\Psi_n(e, \nu) \geq \Phi_e(\nu), \quad \Psi_n(e, \nu) \geq \sum_x \Psi_{n-1}(T(e, a, x), \nu_{a,x}) \quad (a \in \mathcal{A}_e).$$

Then $\Psi_N(e_0, \mu)$ is an upper certificate for every admissible architecture and every prior μ . The family F_n is the least such family. An optimal prior can be chosen with at most $d + 1$ support points, where d is the affine dimension of $\{(g_{\pi}(\theta))_{\pi \in \Pi_N} : \theta \in \Theta\}$.

Proof. Induction in (12) gives $F_n \leq \Psi_n$. The equality case is the recursion itself. The objective depends on a prior only through its vector of integrals of the finitely many g_{π} . An optimal such vector belongs to the convex hull of their compact evaluation image. Carathéodory's theorem gives the stated support bound while preserving every pure-tree payoff. $\square$

Proposition 6.3 (Causal stability). *Consider two models with the same protocol, alphabets and policy class. Suppose their training rows differ uniformly in total variation by at most ϵ , their candidate validation rows by at most η , and their target validation rows by at most β . Their unrestricted values, or their values in any common specified memory class, differ by at most $2N\epsilon + \eta + \beta$.*

Proof. Use the same policy randomization and couple corresponding training rows while the histories agree. The probability of a first discrepancy is at most $N\epsilon$, even with feedback and stopping. Conditional expected validation rewards lie in $[-1, 1]$, so differing training histories contribute at most $2N\epsilon$. On agreeing histories the same frozen design and event are used, and their conditional mean rewards differ by at most $\eta + \beta$. The bound is uniform in the fixed parameter and in the policy. Taking minima and suprema preserves it, with the reverse inequality obtained by interchanging the models. $\square$

6.1. An informative action choice. The following calculation distinguishes the controlled theorem from a common-row reduction. Let $\Theta = \{(i, j) : i, j \in \{0, 1\}\}$. Action R reports i . Action L reports j when $i = 0$ and otherwise reports an independent fair bit. Action H reports j when $i = 1$ and otherwise reports an independent fair bit. The candidate validation is $\delta_{(i,j)}$ and the target is uniform on Θ .

Proposition 6.4 (Strict adaptive advantage). *With two training preparations, the unrestricted adaptive value is $3/4$ and the best nonadaptive value is $5/8$. Nonadaptive policies here include mixtures of schedules, with their random schedule retained. No available single training action dominates all the others.*

Proof. Take R first, and then L or H according to the observed i . The parameter is recovered exactly. The complement of its validation atom has score $3/4$. No policy can exceed $\text{TV}(Q, \delta_\theta) = 3/4$ even with full information.

For the converse for fixed schedules, use the uniform prior. For a pair of actions a, b its Bayes value is explicitly

$$(14) \quad \sum_{x,y,\theta} \left[\frac{1}{16} \sum_{\vartheta} P_{\vartheta}^a(x) P_{\vartheta}^b(y) - \frac{1}{4} P_{\theta}^a(x) P_{\theta}^b(y) \right]_+.$$

Substitution of the three displayed kernels gives

	R	L	H
R	$1/2$	$5/8$	$5/8$
L	$5/8$	$7/16$	$1/2$
H	$5/8$	$1/2$	$7/16$

This is an exact finite sum, not a parameter-grid estimate. It bounds any mixture of fixed schedules by $5/8$; an early stop can be padded with an ignored observation. To attain the bound uniformly in θ , choose with equal probabilities the schedules (R, L) and (R, H) . When the chosen branch-specific action matches i , the parameter is known and the score is $3/4$. Otherwise only i is known, and the complement of its two-atom fiber gives score $1/2$. Each fixed parameter therefore has mean score $5/8$.

Action R cannot distinguish parameters differing only in j . Action L cannot distinguish $(1, 0)$ and $(1, 1)$, and H cannot distinguish $(0, 0)$ and $(0, 1)$. An experiment which fails to distinguish such a pair cannot garble to one that distinguishes it. Hence no action is a common dominating row. $\square$

If every binary report is additionally passed through an independent binary symmetric channel of crossover ϵ , the primitive training rows change by at most ϵ and have full support for $0 < \epsilon < 1/2$. Proposition 6.3, applied to both policy classes, leaves an adaptive advantage at least $1/8 - 8\epsilon$, which is positive for $0 < \epsilon < 1/64$. Thus the distinction is not confined to disjoint observation supports. The finite-horizon belief recursion and its piecewise-linear structure are classical [1]. The contribution here is their explicit integration with the frozen validation score, one fixed adversarial parameter, outer certificates, and the resource conventions; no priority for Bellman recursion itself is claimed.

7. CONTROLLED EXPERIMENTS AT A PRESCRIBED MEMORY PROFILE

An unrestricted posterior is a useful information state for a proof, but it is not a free runtime state. We first give a resource-exact dynamic formulation and then solve a controlled family in which replacing that formulation by a policy-tree mixture changes the value strictly.

7.1. A parameterwise occupation state. Fix a finite horizon T , a compact metric parameter space Θ , finite hidden execution spaces H_t , and register alphabets S_t of sizes at most K_t . The visible inputs to each stochastic row and all legal actions are specified. Any input that must survive a cut belongs to S_t : this includes a chosen action, an unread report buffer, a stop flag, a selected validation event and a validation accumulator. One may subdivide an operation into several cuts to expose these costs. The scheduled time is available only in the clocked convention.

For now rows at different scheduled times are independently prescribed. Let $\mathcal{Q}_t$ be the finite product of legal row simplices at stage t . A prescription $q \in \mathcal{Q}_t$ is a table, chosen at design time, whose runtime arguments are exactly the declared visible inputs. After summing the fixed environmental kernels its one-stage transition is a stochastic matrix $M_{t,\theta}^q$ on the hidden-state/register pairs. We assume this matrix is jointly continuous in (q, θ) . Both action selection and state updating occur in this matrix, in their specified causal order; it is not an arbitrary matrix chosen separately for each hidden state or each parameter.

For a continuous bundle of probability vectors $x = (x_\theta(h, s))_{\theta \in \Theta}$ define

$$(\mathcal{T}_{t,q}x)_\theta(h', s') = \sum_{h,s} x_\theta(h, s) M_{t,\theta}^q((h, s), (h', s')).$$

The initial bundle x_0 is fixed by preparation. Terminal rewards $r_\theta(h, s) \in [-1, 1]$ are continuous in θ . Absorbing stopped states and frozen validation phases are included in this execution, not added afterwards without charge. Put

$$(15) \quad B_T(x) = \min_{\theta \in \Theta} \sum_{h,s} x_\theta(h, s) r_\theta(h, s),$$

$$(16) \quad B_t(x) = \max_{q \in \mathcal{Q}_t} B_{t+1}(\mathcal{T}_{t,q}x), \quad 0 \leq t < T.$$

The argument x is an *offline occupation bundle*, not the realized history, not an observed parameter, and not an additional register made available to the machine.

Theorem 7.1 (Resource-exact dynamic value and supersolutions). *For the specified clocked profile and observable-row interface, $B_0(x_0)$ is the exact maximum worst-case reward over all stochastic controllers. The maximum is attained by a sequence of legal common prescriptions. With the norm $\|x - x'\| = \sup_\theta \sum_{h,s} |x_\theta(h, s) - x'_\theta(h, s)|$, each B_t is 1-Lipschitz. Moreover,*

$$(17) \quad B_0(x_0) = \inf_H H_0(x_0),$$

where the infimum ranges over families of functions on occupation bundles satisfying

$$H_T \geq B_T, \quad H_t(x) \geq H_{t+1}(\mathcal{T}_{t,q}x) \quad (q \in \mathcal{Q}_t).$$

The infimum is attained by $H = B$. No convexification over retained controllers and no new adversarial choice after a report occurs in these identities.

Proof. For a fixed sequence $q_0, \dots, q_{T-1}$, iteration of the displayed transition gives its actual terminal law for every fixed θ . The terminal minimum is therefore exactly its worst-case reward. A runtime randomized controller in this interface is precisely such a sequence of common conditional rows; independence of fresh randomization and the declared register specify all its conditional choices. Persistent random correlations are allowed only through that register.

The occupation bundle at the next stage is determined by the current bundle and the prescription, without drawing a realized history in the design problem. Hence optimizing over the remaining prescription sequence gives (16) by ordinary backward induction. This does not select a different continuation for each unrecorded history. Each maximizing q is still a single legal table common to all hidden histories and parameters. At x_0 one follows the maximizing sequence of design prescriptions, computing their occupation bundles offline; the runtime controller only reads its allowed inputs.

Stochastic matrices contract the ℓ^1 norm. The terminal minimum is 1-Lipschitz in the displayed norm. Induction, using the same q when comparing two bundles and then taking the two maxima, proves the Lipschitz assertion. Joint continuity and compactness of $\mathcal{Q}_t$ give attainment at each stage, and compactness of Θ gives the terminal minimum. Any family H

in (17) bounds the terminal reward along every prescription sequence. Backward induction therefore gives $H_t \geq B_t$. Equality is admissible, proving (17) and its attainment. $\square$

This is a nonlinear supersolution dual, not a least-favorable-prior saddle theorem at fixed width. Its optimization can be large and nonconvex. The principle of dynamic programming is classical; the point of this formulation is that the common-row and memory constraints remain in the maximizations. If rows are shared across scheduled times, as for an autonomous controller, choose the shared rows once and hold them fixed in all subsequent transitions; equivalently augment the *design* problem by that immutable prescription. Maximizing a shared row anew at each time would solve a different problem. A finite union of allowed interfaces can be handled by taking their maximum, or by a fully wired interface with the prescribed profile and its legal observable inputs. Neither operation enlarges a runtime register invisibly.

7.2. An exact informative-control bottleneck. Fix integers $m \geq 2$ and $1 \leq K \leq m$. The fixed parameter is $\theta = (i, j) \in [m] \times \{0, 1\}$. The first training experiment R reports i . Across the next cut the controller retains at most K labels, with an arbitrary stochastic encoder $E(s \mid i)$. It then chooses one of the m sensing actions with probabilities $D(a \mid s)$. The selected action is kept in a charged m -state action buffer and the old label is discarded. Action a reports j when $a = i$ and reports $\perp$ otherwise. The first report is no longer available.

The candidate validation law on $\{0, 1\}$ is δ_j ; the target is uniform. An event is frozen before one fresh candidate and one fresh target validation. Stopping before either training action is permitted. No independent persistent selector is given to the controller. The observation symbols are atomic; a bitwise input implementation would have additional cuts. Denote the maximum worst-case score by $V_{m,K}$.

Theorem 7.2 (Exact finite-memory control and its convexification gap). *In this protocol,*

$$(18) \quad V_{m,K} = \frac{1}{2 \lceil m/K \rceil}.$$

The value obtained by allowing an uncharged ex ante mixture of complete K -label controllers is instead

$$(19) \quad \bar{V}_{m,K} = \frac{K}{2m}.$$

The inequality is strict whenever K does not divide m . The matching controller for (18) has charged cut profile bounded by $(1, K, m, 3, 6, 3)$. The K -label cut controls both retention of the routing observation and the subsequent informative action choice.

Proof. For a fixed i , let

$$p_i = \sum_s E(s \mid i) D(i \mid s).$$

This is the probability of taking the informative action. On its complement, all retained information is independent of j . Averaging over the two values of j makes the expected validation score zero there, for any frozen event or randomized response. On the informative part the score is at most $\text{TV}(\text{Ber}(1/2), \delta_j) = 1/2$. Thus the minimum over j at a given i is at most $p_i/2$. A stop before sensing is also uninformative about j and obeys the same bound; it can be padded by an ignored sensing operation.

Set $L = \lceil m/K \rceil$. Since $p_i \leq \max_s D(i \mid s)$, a value $\min_i p_i > 1/L$ would require every i to have some row s with $D(i \mid s) > 1/L$. Each stochastic row has at most $L - 1$ such indices. But $K(L - 1) < m$. This contradiction proves the upper bound. Partition $[m]$ into at most K fibers of size at most L , retain the fiber label after R , and choose uniformly inside that fiber. On a successful report freeze $\{1\}$ when $j = 0$ and $\{0\}$ when $j = 1$; on $\perp$ freeze the empty event. The

score for every j is $1/(2|\text{fiber}(i)|)$, proving equality. The event alphabet has three labels. Storing the event and its candidate indicator requires at most six labels, and the final difference has three. The chosen sensing action was separately charged, proving the displayed profile.

For the convexified upper bound use the uniform prior on (i, j) . For any one controller,

$$\frac{1}{m} \sum_i p_i = \frac{1}{m} \sum_{i,s} E(s | i) D(i | s) \leq \frac{1}{m} \sum_{i,s} D(i | s) = \frac{K}{m}.$$

The same upper bound holds for mixtures. For the matching mixture, choose uniformly one of the m cyclic intervals of K indices. For that interval hardcode a deterministic K -label encoder which routes each included index to its own sensing action, and assigns other indices arbitrarily. Each i belongs to exactly K intervals. A successful action is followed by the complementary singleton event and failure by the empty event, as above. Every fixed (i, j) has mean score $K/(2m)$. This proves (19). $\square$

For example, $V_{3,2} = 1/4$ whereas $\bar{V}_{3,2} = 1/3$. The mixture just constructed can be implemented with a retained interval selector and a local label, using at most mK labels at the bottleneck. It cannot be declared a K -label realization: when the displayed gap is positive, (18) proves that no such realization exists. The action buffer of size m is not part of the K coordinate and has not been omitted from the profile. The theorem is a multicut resource law, not an assertion that its uniform peak equals K .

The actions are statistically incomparable. R cannot distinguish the two values of j at any fixed i , while action a does distinguish them at $i = a$. For $a \neq b$, action a does not distinguish the pair with $i = b$, while b does. A parameter-independent garbling cannot create a distinction absent from its input experiment.

Corollary 7.3 (A strict gap with positive report probabilities). *Replace each of the two training kernels by a mixture with weight $\epsilon > 0$ of the uniform law on its report alphabet. Let $\Delta_{m,K} = K/(2m) - 1/(2\lceil m/K \rceil) > 0$. For $\epsilon < \Delta_{m,K}/8$ the resulting full-support model still has a strict gap between the genuine profile-constrained value and its free-selector convexification.*

Proof. Each training row changes by at most ϵ in total variation. Couple corresponding rows until the first disagreement. For any fixed controller or mixed controller the probability of such a disagreement is at most 2ϵ , and its conditional reward lies in $[-1, 1]$. Each of the two optimal values changes by at most 4ϵ . Their difference is therefore at least $\Delta_{m,K} - 8\epsilon > 0$. All row arguments and register alphabets are unchanged. $\square$

The distinction between mixed and behavioral strategies without perfect recall is classical; it is not first asserted here. Contemporary polynomial formulations over behavioral simplices include [16]. Theorem 7.2 evaluates the discrepancy in this specific frozen-validation experiment for every m, K , with a matching actual register and a noisy persistence statement. In particular, a single prior followed by an unrestricted policy-tree best response cannot be substituted for (16).

8. SADDLE FACES AND CAUSAL REALIZATION

The statistical saddle and its implementation are different objects. The former is convex; the latter need not be. The purpose of this section is to retain that distinction while extracting memory obstructions from an unrestricted saddle. In particular, a zero Bayes coefficient must not be discarded before the contact conditions at the worst parameters have been imposed.

8.1. A normal form over a finite causal interface. Fix a finite horizon and finite action and report alphabets. An action a_t is chosen before report y_t . A terminal output is included as one last action, followed by a dummy report. Illegal actions can be removed from each protocol type; equivalently, use a fixed padding of the finite protocol tree and impose its zero constraints. Define a *causal response array* $C(\mathbf{a} \mid \mathbf{y})$ by nonnegativity, normalization in $\mathbf{a}$ for every $\mathbf{y}$, and the requirement that its marginal on $a_1, \dots, a_t$ depends only on $y_1, \dots, y_{t-1}$. The resulting set $\mathcal{C}$ is a finite polytope. Conditioning successive marginals realizes every such array by a full-history randomized controller; arbitrary versions on zero marginals do not change the array.

For a fixed parameter θ , the environment has a history-dependent likelihood $L_\theta(\mathbf{a}, \mathbf{y})$, obtained by marginalizing its hidden states. The bounded reward has the linear form

$$(20) \quad g_\theta(C) = \sum_{\mathbf{a}, \mathbf{y}} L_\theta(\mathbf{a}, \mathbf{y}) r_\theta(\mathbf{a}, \mathbf{y}) C(\mathbf{a} \mid \mathbf{y}).$$

Assume its coefficient vector is continuous on a compact metric parameter space. The parameter is fixed once, not selected after a report. Protocol-specific equalities, including a frozen validation choice, are represented by the corresponding finite protocol tree. A discarded protocol choice is not supplied again as a free input.

Here is an explicit positive realization of a response array. At cut t use k_t states, an initial distribution α , and nonnegative matrices $M_t^{a,y}$ of size k_{t-1} by k_t such that

$$(21) \quad M_t^{a,y} \mathbf{1} = p_t(a), \quad \sum_a p_t(a) = \mathbf{1}, \quad p_t(a) \text{ is independent of } y.$$

Then

$$(22) \quad C(\mathbf{a} \mid \mathbf{y}) = \alpha M_1^{a_1, y_1} \dots M_T^{a_T, y_T} \mathbf{1}.$$

The independence in (21) is essential: otherwise an action could use its not-yet-observed report. If an action or an input must survive an intermediate cut, split this formula at that cut and include its buffer in the state alphabet. Likewise, impose any declared shared-row constraints before optimizing. An externally fixed clock permits time-dependent matrices; an autonomous implementation has to store its phase or use the prescribed shared rows.

Proposition 8.1 (Outer normal form and compatible continuations). *For the finite interface and declared cuts, the response arrays of all controllers with profile at most $\mathbf{K}$ are exactly the arrays (22) with $k_t \leq K_t$, the normalization (21), and the declared interface constraints. This set, denoted $\mathcal{C}_{\mathbf{K}}$, is compact. The assertion is over all wirings of the allowed register, not one chosen transition graph.*

Equivalently, there are at most K_t normalized continuation arrays $B_{t,s}$ at each cut, with terminal continuation 1, and common nonnegative coefficients satisfying

$$(23) \quad B_{t-1,s}(a\mathbf{a} \mid y\mathbf{y}) = \sum_{s'} M_t^{a,y}(s, s') B_{t,s'}(\mathbf{a} \mid \mathbf{y}), \quad C = \sum_s \alpha(s) B_{0,s}.$$

Separate positive factorizations at the cuts are not sufficient without these common shift identities.

Proof. In a controller, let $p_t(a \mid s)$ be its action row and $q_t(s' \mid s, a, y)$ its update row. Their product defines $M_t^{a,y}$ and gives (21). Multiplication over a complete execution gives (22). Every finite wiring embeds into the fully connected register by assigning zero to unused transitions; smaller alphabets are padded by unused states. Thus no further maximization over graph presentations is necessary.

Conversely choose action a with probability $p_t(a \mid s)$, and, after observing y , use the row $M_t^{a,y}(s, s')/p_t(a \mid s)$ when the denominator is positive. At a zero denominator the entire corresponding nonnegative row vanishes; choose an arbitrary legal update there. This gives

exactly the required runtime arguments. It does not expose a posterior, the parameter, or an offline occupation array to the machine. Apply the same construction at each declared buffer cut. A product of finitely many stochastic row simplices is compact; closed legality and sharing constraints preserve compactness. Its continuous image is $\mathcal{C}_{\mathbf{K}}$.

The future execution starting at a state s defines $B_{t,s}$ and satisfies (23). Conversely, induction from the terminal continuation and these identities gives the product formula. Thus the continuation description has exactly the same realizations. In particular, independently selected factorizations at different cuts cannot be spliced unless the shift equations hold. $\square$

This is a finite controlled positive-realization statement. The role of positivity and invariant generating cones in stochastic realization is classical [17, 18, 19]. Ordinary linear rank need not be a state count. Here the additional statistical question is which positive realizations meet an optimal saddle face. Proposition 8.1 provides a normal form, not an efficient factorization algorithm. It also does not compactify an unbounded number of observable operations.

8.2. The realization theorem. Let

$$U = \max_{C \in \mathcal{C}} \min_{\theta} g_{\theta}(C), \quad V_{\mathbf{K}} = \max_{C \in \mathcal{C}_{\mathbf{K}}} \min_{\theta} g_{\theta}(C).$$

By compact convex minimax there is a least-favorable prior μ_* with $U = \max_{C \in \mathcal{C}} \int g_{\theta}(C) d\mu_*$. Put

$$\begin{aligned} \ell_*(C) &= \int g_{\theta}(C) d\mu_*, \\ F_* &= \{C \in \mathcal{C} : \ell_*(C) = U\}, \\ \mathcal{S}_* &= F_* \cap \bigcap_{\theta} \{C : g_{\theta}(C) \geq U\}. \end{aligned} \tag{24}$$

The set F_* is an exposed face. The second intersection is important: a Bayes best response to a least-favorable prior need not be minimax.

Here $\mathcal{C}$ is the full-history convex reference for the finite protocol. A nonconvex restriction, such as reuse of one stochastic table at several visits, is imposed on $\mathcal{C}_{\mathbf{K}}$, not inserted into $\mathcal{C}$ before invoking convex minimax. In that case U is a reference upper value unless exactness for the unrestricted physical class is separately known. In the reset applications below that exactness is supplied by Theorems 5.1 and 5.2.

Theorem 8.2 (Saddle-face realization and exact resource loss). *Under the preceding assumptions, the constrained outer value satisfies*

$$U - V_{\mathbf{K}} = \min_{C \in \mathcal{C}_{\mathbf{K}}} \left\{ U - \ell_*(C) + \ell_*(C) - \min_{\theta} g_{\theta}(C) \right\}. \tag{25}$$

Both summands are nonnegative. Consequently $V_{\mathbf{K}} = U$ if and only if some array in $\mathcal{S}_$ has compatible positive continuation generators of profile at most $\mathbf{K}$. If no such generators exist, $V_{\mathbf{K}} < U$ strictly.*

For $C \in \mathcal{S}_$ one has $g_{\theta}(C) = U$ on $\text{supp } \mu_*$. At an interior supporting parameter at which the payoff is differentiable, $\nabla_{\theta} g_{\theta}(C) = 0$. These equalities and stationarity conditions hold also in directions whose Bayes coefficient vanishes.*

At any reset continuation cut, let R_h denote the conditional channel of future outputs under each externally specified future input word. It factors as

$$R_h = \sum_{s=1}^{K_t} E(s \mid h) B_s, \quad E(\cdot \mid h) \in \Delta_{K_t}, \tag{26}$$

with one common family of continuation channels B_s . Any obstruction to this positive factorization for every member of $\mathcal{S}_*$ is therefore a memory obstruction for every admissible architecture, regardless of its training wiring.

Proof. For every C , its prior average is at most U and at least its minimum over fixed parameters. The expression in braces in (25) is identically $U - \min_{\theta} g_{\theta}(C)$. Taking its minimum over the compact set $\mathcal{C}_{\mathbf{K}}$ proves the identity and attainment. Its two nonnegative summands vanish exactly when C belongs to $\mathcal{S}_*$. Proposition 8.1 then proves the realization criterion. Compactness makes nonintersection a strictly positive value loss, not merely failure of one optimizer.

For an optimal array, $g_{\theta}(C) - U$ is continuous, nonnegative, and has zero μ_* integral. It is zero on the support of that measure. Every interior zero is a local minimum, so its gradient vanishes. No strict-sign hypothesis on individual coordinates is needed. Finally condition on the actual retained state at a reset cut. Given that state, the future program receives the same specified inputs and has the same output law, irrespective of the discarded past. Averaging gives (26). All persistent random variables are part of that state. The assertion can equivalently be checked on controller input tapes, using the generators in (23); it does not assume that a hidden physical state resets at a cut where the interface has not reset it. $\square$

The identity (25) is an exact loss decomposition, not a new interchange of a minimum and maximum over nonconvex machines. Its use is that an unrestricted certificate identifies a finite face, while contact at its supporting parameters can determine the remaining coordinates. The resulting channels are then tested for causal realizability. This is stronger than choosing an arbitrary Bayes tie rule and counting the states of that one rule.

Corollary 8.3 (Explicit coefficient form for frozen validation). *In the reset experiment of (8), write*

$$C_{z\omega}^* = \int P_{\theta}^{\otimes N}(z)(Q(\omega) - P_{\theta}(\omega)) d\mu_*(\theta).$$

The first summand of (25) is exactly

$$(27) \quad \sum_{C_{z\omega}^* > 0} C_{z\omega}^* (1 - h_z(\omega)) + \sum_{C_{z\omega}^* < 0} |C_{z\omega}^*| h_z(\omega).$$

Thus optimality fixes every strict-sign coordinate, while the support contact and stationarity equations must still be imposed at the zeros. For K labels at the post-training decision cut, an arbitrary upstream encoder gives the compact relaxation

$$(28) \quad h_z(\omega) = \sum_{s=1}^K E(s | z) D_s(\omega), \quad E \in (\Delta_K)^{\Omega^N}, \quad D_s \in [0, 1]^{\Omega}.$$

If this relaxation misses the saddle set, its strictly positive gap bounds every smaller architectural class.

Proof. Maximizing each coefficient separately gives the sum of its positive part. Subtraction gives (27). Conditioning on the decision state gives (28); the post-cut randomization is independent of the discarded word. Conversely an arbitrary encoder and a response list realize this factorization when other memory is unrestricted. The parameter spaces in (28) are compact, so the final assertion follows from Theorem 8.2. $\square$

8.3. Faces as memory witnesses. For a finite input set $\mathcal{Y}$ and output alphabet $\mathcal{D}$, continuation channels form the polytope $\mathcal{P} = \prod_{y \in \mathcal{Y}} \Delta(\mathcal{D})$. Its vertices are deterministic output functions. The analogous polytope for scalar test responses is $[0, 1]^{\mathcal{Y}}$.

Lemma 8.4 (Vertex and face packing). *Suppose channels to be factored through one common state alphabet include s distinct vertices $v_1, \dots, v_s$ of a polytope $\mathcal{P}$ and rows $r_1, \dots, r_j$. Suppose $r_i \in F_i$, the faces F_i are pairwise disjoint, and*

$$r_i \notin \text{conv}\{v_a : v_a \in F_i\}.$$

Every factorization of these rows as stochastic mixtures of generators in $\mathcal{P}$ needs at least $s + j$ generators.

Proof. A convex decomposition of a vertex uses that same vertex at every positive coefficient. Hence all s vertices must occur among the generators. If a convex combination belongs to a face, all generators with positive coefficients belong to that face. The stated exclusion therefore forces an additional generator in each F_i . Pairwise disjointness prevents any such generator from serving two faces. $\square$

The lemma does not assert that a stochastic state must itself be a history residual. Its generator can be any channel in the ambient polytope. This is why the argument also excludes randomized encoders and randomized continuations. It is a positive-factorization argument, not ordinary matrix rank or a restriction to a deterministic compiler.

Proposition 8.5 (Resource equivalence and stable obstructions). *A causal, cut-preserving recoding with a causal inverse, both using only one transducer state and preserving the payoff, preserves every $V_{\mathbf{K}}$, U , and the set of profiles attaining U . More generally suppose simulations in both directions have defect at most ϵ and require at most R_t states at cut t . For rewards in $[-1, 1]$,*

$$V_{\epsilon}(\mathbf{RK}) \geq V_{\mathfrak{F}}(\mathbf{K}) - 2\epsilon, \quad V_{\mathfrak{F}}(\mathbf{RK}) \geq V_{\epsilon}(\mathbf{K}) - 2\epsilon.$$

For a common controller class and training-validation kernels whose values differ by at most η , a gap $U - V_{\mathbf{K}} = \delta$ remains at least $\delta - 2\eta$ after the perturbation.

Proof. Transport a controller through each simulator and retain the product of its state and the simulator state. Apply the uniform bounded-reward variation bound and optimize. With an exact one-state inverse the two inequalities are equalities at the same profile. In the last assertion both the unrestricted and the constrained optimum change by at most η ; subtract their two bounds. The state products must be applied at the actual buffer cuts, not only at the terminal decision. $\square$

These statements place the saddle face, the future-test quotient, and resource-aware causal morphisms in one theorem chain. The channel factorizations are presentation-independent under the stated causal recodings. The memory obstruction in the next section is derived from the experimental coefficients themselves, not from a metric dimension or a covering exponent supplied as an additional hypothesis.

9. EXACT MINIMAX LAWS FOR A MARKED PRODUCT FAMILY

The posterior-predictive dual can be solved explicitly for a continuous family containing the collision target. Throughout this section, the training and candidate validation law is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad (p, q) \in [0, 1]^2.$$

In the order 000, 001, 110, 111, let Q_{γ} have masses

$$((1 + \gamma)/4, (1 - \gamma)/4, (1 - \gamma)/4, (1 + \gamma)/4),$$

and zero mass elsewhere. Write $U_1(\gamma) = U_1(b, Q_\gamma)$, with exactly the training-validation convention of (8). Set

$$(29) \quad \gamma_* = 1 - 1/\sqrt{2}, \quad s_\gamma = \sqrt{\gamma^2 - 2\gamma + 5}, \quad u_\gamma = \gamma - 2 + s_\gamma, \quad \tau_\gamma = \frac{2u_\gamma}{u_\gamma + 2 - \gamma}.$$

Theorem 9.1 (Exact marked minimax family). *For every $0 \leq \gamma \leq \gamma_*$,*

$$(30) \quad U_1(\gamma) = \frac{3 + u_\gamma^2}{8}.$$

An optimal response depends on the training word only through the report count $K = X_1 + X_2$. On the four target atoms its rows are

$$(31) \quad h_0 = (\tau_\gamma, 0, 1, 1), \quad h_1 = (1, 1, 1, 1), \quad h_2 = (1, 1, 0, \tau_\gamma),$$

and all responses are zero off these atoms. A least-favorable prior is

$$(32) \quad \mu_\gamma = \frac{1}{2}\delta_{(t_\gamma, t_\gamma)} + \frac{1}{2}\delta_{(1-t_\gamma, 1-t_\gamma)}, \quad t_\gamma = \frac{1 - \sqrt{u_\gamma}}{2}.$$

The assertion optimizes over all stochastic responses to the full marked training word, not merely over count-based or symmetric responses.

Proof. First, u_γ is the positive root of

$$(33) \quad u^2 + (4 - 2\gamma)u - (1 + 2\gamma) = 0.$$

On the asserted interval, $0 < u_\gamma < 1$ and $0 < \tau_\gamma < 1$. Indeed the polynomial is negative at zero and positive at one; also $u_\gamma < 2 - \gamma$ because $\gamma < 1/2$. Substitution of $1 - 2\gamma$ into the polynomial gives $8(\gamma^2 - 2\gamma + 1/2)$, which is nonnegative precisely on the present interval. Monotonicity on $[0, \infty)$ therefore gives

$$(34) \quad u_\gamma \leq 1 - 2\gamma.$$

Put $a = (1 - p)(1 - q)$, $c = pq$, and $z = 1 - a - c$. The probabilities of the three training counts are a, z, c . The independent training mark may be discarded by the displayed construction. Let $G_\gamma(p, q)$ denote its score. With $u = (p + q - 1)^2$ and $v = (p - q)^2$, direct expansion gives

$$(35) \quad G_\gamma(p, q) - \frac{3 + u_\gamma^2}{8} = \frac{2 - \tau_\gamma}{16} \{(u - v - u_\gamma)^2 + 4v\}.$$

For clarity, this identity follows by substituting the three rows of (31) into

$$a(Q_\gamma - b)h_0 + z(Q_\gamma - b)h_1 + c(Q_\gamma - b)h_2,$$

using $a + c = (1 + u - v)/2$, $a - c = -(p + q - 1)$, and then eliminating u_γ^2 by (33). The coefficient of the resulting square is $(2 - \tau_\gamma)/16$ and that of v is four times this coefficient. Both are nonnegative. Equality holds at both points of (32), since there $v = 0$ and $u = u_\gamma$. This proves the lower bound and identifies its equality set.

It remains to bound every response, including nonsymmetric ones. Use (32) in Theorem 5.1. At $t = t_\gamma$ put $a = (1 - t)^2$, $c = t^2$, and $T = a + c = (1 + u_\gamma)/2$. For the training report pair 00, the four dual coefficients, apart from a common positive factor, are

$$(36) \quad \left(Q_H T - \frac{a^2 + c^2}{2}, \quad Q_L T - \frac{a^2 + c^2}{2}, \quad Q_L T - ac, \quad Q_H T - ac \right),$$

where $Q_H = (1 + \gamma)/4$ and $Q_L = (1 - \gamma)/4$. The first coordinate is zero by (33); the second is $-\gamma T/2 \leq 0$. The last two are positive: $T \geq 1/2$, $ac \leq 1/16$, and $\gamma < 1/2$ imply $Q_L T > 1/16 \geq ac$. Thus $(\tau_\gamma, 0, 1, 1)$ maximizes this linear functional. For the training pair 11, reversing all report bits and the mark reverses the four coordinates and gives the row h_2 .

For either training pair 01 or 10, the two prior support points have the same likelihood $t(1-t)$. The posterior is still equally weighted, and the candidate predictive mass at each target atom is $T/4$. Every one of the four coefficients is therefore nonnegative by $Q_L - T/4 = (1 - 2\gamma - u_\gamma)/8 \geq 0$. The response h_1 is optimal. At off-diagonal validation atoms the target mass vanishes and the coefficient is nonpositive. Finally, conditioning on either training mark multiplies all these coefficients by $1/2$; it creates no further optimizing choice. We have therefore optimized every coordinate of the full marked-word response space.

The prior average of the displayed response equals its value at either support point, namely $(3 + u_\gamma^2)/8$, by (35). Every other response has prior average at most this number and hence has minimum over fixed (p, q) at most this number. Together with the lower bound this proves (30), optimality of the response, and least favorability of the prior. $\square$

Corollary 9.2 (The exact one-preparation collision value). *For $Q_* = Q_{1/4}$,*

$$(37) \quad U_1(b, Q_*) = u_{\text{ex}} := \frac{85 - 7\sqrt{73}}{64}, \quad u_{1/4} = \frac{\sqrt{73} - 7}{4}, \quad \tau_{1/4} = 2 - \frac{14}{\sqrt{73}}.$$

If only the target changes to Q , then $|U_1(b, Q) - u_{\text{ex}}| \leq \text{TV}(Q, Q_)$.*

Proof. Substitute $\gamma = 1/4$ in Theorem 9.1. For a fixed trained response the target perturbation is at most total variation, uniformly in (p, q) . Taking first the minimum and then the maximum preserves this bound in both directions. $\square$

9.1. Complementary slackness and the event alphabet. The tie in the first coordinate of (36) explains the randomization. Any value of that response coordinate is Bayes-optimal for the two-point prior. The minimax choice makes the uniform payoff stationary at its two support points; solving this condition gives τ_γ and the square identity (35). This derives the prior and response together, rather than postulating that a numerically effective prior is optimal.

Operationally, for $K = 0$ one draws once between the events with indicator vectors $(0, 0, 1, 1)$ and $(1, 0, 1, 1)$, with probabilities $1 - \tau_\gamma$ and τ_γ . For $K = 2$ one draws between $(1, 1, 0, 0)$ and $(1, 1, 0, 1)$. For $K = 1$ the event is $(1, 1, 1, 1)$. The resulting event identity is retained and used for both fresh validations. These are exactly the five events in the two-preparation construction below. The one-preparation theorem supplies their complementary-slackness interpretation; it does not prove that the two-preparation selector is optimal. The four-point rational prior retained below is a useful rational certificate, not a least-favorable prior.

The exact response uses an atomic stochastic row. It does not receive free persistent randomness. Replacing τ_γ by a dyadic number within 2^{-b} changes each conditional response, and hence the score, by at most 2^{-b} . A sequential fair-bit implementation must retain its bit prefix and event identity. None of these implementation costs is needed by the deterministic two-preparation physical audit.

Proposition 9.3 (The zero-preparation endpoint). *For $0 \leq \gamma \leq 1$, $U_0(b, Q_\gamma) = \gamma/2$.*

Proof. The fixed event $\{000, 111\}$ has score $(1 + \gamma)/2 - (a + c)/2 \geq \gamma/2$ because $a + c \leq 1$. Conversely, the equally weighted prior on $b(0, 0)$ and $b(1, 1)$ has predictive validation law uniform on the four target atoms. Its total variation from Q_γ is $\gamma/2$. Theorem 5.1 with $N = 0$ gives the matching upper bound. $\square$

10. A SELF-CONTAINED COLLISION INTERFACE

We give the preparation, mechanical estimate, observation law and private simulator class needed for the physical specialization. The following argument does not appeal to an unpublished

physical lemma. It retains the same positive-volume preparation and positive sensor noise as the earlier collision construction.

Definition 10.1 (The reset row–event architecture). A preparation supplies two sequential report slots and, afterwards, an actual mark. The source reports in the two slots are constant. A private width-one simulator has no retained information between those slots and no persistent hidden seed. It may use fresh independent coins. Its first emission is $X_1 \sim \text{Ber}(p)$; after receiving the current gate $A \in \{0, 1\}$, its second emission has a fixed Bernoulli kernel with parameter q_A . The parameters (p, q_0, q_1) are fixed throughout the audit. The simulator does not observe the actual mark. All its coins are independent of that mark. Different preparations reset all unretained variables and are independent conditional on the parameters.

The auditor observes the first report, chooses a gate using its retained state and that report, observes the second report, and then observes the mark. During training it may choose rows and stop adaptively, using at most the declared number of candidate preparations. It next freezes one gate policy and one event, with any retained random choice included in its register. It uses that same row–event continuation on one fresh candidate preparation and one independent target preparation and returns the difference of their event indicators, target minus candidate. It may not reselect the event after either validation. Independent target-only calibration is allowed but all retained calibration and randomness are charged. Peak width counts every bit, mark, event and validation-accumulator cut. Clocked schedules are declared externally; an autonomous implementation must retain its phase. An unretained transcript, seed or stopping-time value is not available to a continuation.

In the all-on row the complete private law is $b(p, q_1)$, with an independent fair actual mark. For lower bounds it suffices to restrict the adversary to $q_0 = 0$, $q_1 = q$. In this subfamily every deterministic gate policy is the pushforward

$$T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w)$$

of $b(p, q)$. The gate, when included in the transcript, is another function of these same coordinates. Randomized gate policies are covered by averaging their independent random tables. This proves both the all-on product characterization for the upper construction and the common-garbling subfamily used for the outer converse. The width-one constraint is a simulator constraint; it is not an uncharged restriction on the auditor’s register.

10.1. Preparation and elastic scattering. Consider two labeled equal-mass hard spheres in $\mathbb{R}^3$, with no walls and contact distance d . Write $r = q_1 - q_2$, $g = v_1 - v_2$, $V = (v_1 + v_2)/2$, and let the center of mass position range over any fixed rational box of positive volume. Prepare $(B, W) \in \{0, 1\}^2$ with probability $5/16$ at $B = W$ and $3/16$ at $B \neq W$. Conditional on (B, W) , use the uniform product preparation on the following boxes:

$$(38) \quad \begin{aligned} r &= (-a_0, (2B - 1)b_0, c_0), & a_0 &\in [29/10, 31/10], & b_0 &\in [9/20, 11/20], \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & & & |V_z - (2W - 1)| &\leq 1/100. \end{aligned}$$

Thus the actual mark is $W = \mathbf{1}_{\{V_z > 0\}}$, preserved by momentum conservation. Set

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector, identically zero on the incoming transverse-velocity range.

Lemma 10.2 (Uniform collision signal). *For every preparation in (38) and every $9/10 \leq d \leq 11/10$, exactly one collision occurs and its time lies in $(4/5, 3/2)$. Afterwards*

$$(2B - 1)v_{1,y} > 1/2, \quad (2B - 1)h(v_{1,y}) > 1/4.$$

For $0 < d \leq 2/5$ no collision occurs on $[0, 3]$ and the detector is zero throughout that interval.

Proof. Before contact, $r(t) = r + tg$. At $t \leq 4/5$ its x coordinate is at most $-29/10 + (4/5)(21/10) = -61/50$, so its norm exceeds $11/10$. At $t = 3/2$ its absolute coordinate bounds are $1/4$, $113/200$ and $1/40$, whose squared sum is less than $(9/10)^2$. The first contact therefore occurs in the claimed interval. At the free zero of the x coordinate, whose time is at most $31/19$, the transverse norm is less than $9/10$. Hence the entry contact has $n_x < 0$, where $n = r(t_c)/d$. Its remaining coordinates give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B - 1)n_y \geq 87/220.$$

It follows that $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$. The elastic rule gives

$$v_1^+ = V + g/2 - (g \cdot n)n, \quad g^+ = g - 2(g \cdot n)n.$$

Consequently

$$(2B - 1)v_{1,y}^+ > -3/200 + (1443/1000)(87/220) > 1/2.$$

The definition of h implies the signal bound. Since $r(t_c) \cdot g^+ > 0$, squared separation increases strictly afterwards; there is no second collision. Finally, for $d \leq 2/5$ and $0 \leq t \leq 3$, $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$. There is no contact, and the incoming $|v_{1,y}| \leq 3/200$ makes h zero. $\square$

10.2. Positive noise and the actual mark. Let $\mathcal{B}_t$ be Brownian motion independent of the preparation and set

$$Y_t = \int_0^t h(v_{1,y}(s)) ds + \sigma \mathcal{B}_t, \quad 0 < \sigma \leq 1/24.$$

The two potential all-on reports are

$$Z_1 = \mathbf{1}_{\{2(Y_2 - Y_{3/2}) > 0\}}, \quad Z_2 = \mathbf{1}_{\{2(Y_3 - Y_{5/2}) > 0\}}.$$

The actual second report is AZ_2 . The gate changes acquisition, not the mechanical path. The continuous path is implemented by the target sensor; it is not made available to the private simulator or supplied as an uncharged auditor buffer. Denote the law of (Z_1, Z_2, W) by $Q_{d,\sigma}$.

Theorem 10.3 (Uniform physical presentation). *For $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$,*

$$(39) \quad \text{TV}(Q_{d,\sigma}, Q_*) < 1/300.$$

The same estimate holds after every declared feedback gate, including its observable gate value. The all-on private product family and the common-garbling subfamily in Definition 10.1 apply to this very experiment.

Proof. Every collision occurs before $3/2$. Conditional on the preparation, the two normalized sensor windows are independent Gaussians with mean $h(v_{1,y}^+)$ and standard deviation $\sqrt{2}\sigma \leq 1/12$. Lemma 10.2 makes their means have sign $2B - 1$ and magnitude greater than $1/4$. Each potential report therefore differs from B with probability less than $\Phi(-3)$. The Gaussian tail estimate

$$\Phi(-x) \leq \frac{e^{-x^2/2}}{x\sqrt{2\pi}}$$

gives $\Phi(-3) < 1/600$: $e^{9/2} > 80$ and $\sqrt{2\pi} > 5/2$ suffice. Couple the ideal and noisy reports using the same preparation, actual mark and gate random table. Except on a set of probability less than $2\Phi(-3) < 1/300$, both potential reports equal B and the entire gated transcript agrees. The ideal marked law is precisely Q_* , by the prepared (B, W) table. This proves (39) and the feedback-uniform version. Finally, the source is blind, its actual mark is unobserved by the simulator, and the gate is its only current input at the second emission. The product and subfamily assertions follow from the two independent private-emission kernels specified above. $\square$

For $d \leq 2/5$, the two windows instead contain only independent centered Gaussian noise. Their signs are independent fair bits, independent of W , and a private width-one simulator reproduces every gated row exactly. Thus the informative regime and its no-collision control use the same detector, preparation convention and actual mark.

11. THE PHYSICAL EXPERIMENT AND ITS SHARP PREPARATION BOUNDARY

Take $\Omega = \{0, 1\}^3$, with coordinates (x_1, x_2, w) . The all-on private candidate is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad 0 \leq p, q \leq 1.$$

The mark w is the actual mark, not an invented simulator bit. Put $a = (1-p)(1-q)$ and $c = pq$. On the ordered atoms

$$(40) \quad 000, \quad 001, \quad 110, \quad 111,$$

let Q_* have masses $5/16, 3/16, 3/16, 5/16$, and set it to zero elsewhere. The candidate masses on these atoms are $a/2, a/2, c/2, c/2$. A target Q is allowed whenever $\text{TV}(Q, Q_*) \leq \beta$.

The original feedback operation has the form

$$(41) \quad T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w), \quad g : \{0, 1\} \longrightarrow \{0, 1\}.$$

The selected gate is observable. It may depend on earlier preparations and on the first current report. The target rows are pushforwards by (41). For the converse, restrict the adversary to the admissible private subfamily whose second emission is AX_2 , where A is the current gate and $X_2 \sim \text{Ber}(q)$ is independent of the first emission and of the fair mark. Every row is then a causal garbling of $b(p, q)$. For the construction only the all-on row is used, and every private candidate in the original width-one class has the all-on product law just specified. These two facts, proved for the microscopic interface in Definition 10.1 and Theorem 10.3, explain why using a subfamily for the lower bound does not narrow the upper construction.

The admissible audit first trains, then selects one row–event pair, then uses one fresh candidate and one fresh target validation. It reports $D = \mathbf{1}_A(Y) - \mathbf{1}_A(X)$. The event and row cannot be reselected after seeing a validation draw. All bit, mark, decision and validation cuts are charged. Define $N_c^{\text{phys}}(W, Q)$ to be the minimum number of candidate training preparations among these original-interface auditors, with peak register width at most W , whose score is uniformly strictly greater than c . Randomization, feedback and stopping are allowed. If no such audit exists, the value is $+\infty$.

Theorem 11.1 (Two-preparation physical segment). *Let u_1, m_2 be as in (2). For every $\beta \geq 0$, $\text{TV}(Q, Q_*) \leq \beta$, and*

$$W \geq 12, \quad u_1 + \beta \leq c < m_2 - \beta,$$

one has $N_c^{\text{phys}}(W, Q) = 2$. The two-preparation audit is deterministic, uses no acquired calibration and no random bits, and has the full profile

$$(42) \quad (1, 2, 3, 3, 4, 5, 5, 10, 12, 10, 10, 7, 3)$$

from the initial cut through its twelve bit/mark updates. In particular, its decision alphabet has five labels and its peak has twelve. The lower bound remains valid with no memory cap.

The proof occupies the next three subsections. The interval in the statement can be empty for large β ; no assertion is then made. For $\beta = 1/300$, it contains $c = 2/5$ strictly between its endpoints.

11.1. An exact rational certificate against one preparation.

Lemma 11.2. *Every original-interface auditor using at most one candidate training preparation has worst-case score at most $u_1 + \beta$, regardless of memory.*

Proof. By Theorem 5.2, it suffices to bound $U_1(b, Q_*)$ and then perturb the target. In (9) use the four fixed parameters

$$(p, q) = (1/8, 1/8), (1/5, 1/5), (4/5, 4/5), (7/8, 7/8)$$

with weights $1/16, 7/16, 7/16, 1/16$, respectively. This measure is not asserted to be least favorable. Since $p = q$ on its support, the training count $K = x_1 + x_2$ is sufficient. The independent training mark and the order of unequal bits contain no further information about this prior.

For $k = 0, 1, 2$ and the four validation atoms in (40), the signed coefficients in (9), after grouping by K , are the entries of

$$(43) \quad \frac{1}{40960000} \begin{pmatrix} -1977 & -1775177 & 2170423 & 3943623 \\ 1765554 & 191954 & 191954 & 1765554 \\ 3943623 & 2170423 & -1775177 & -1977 \end{pmatrix}.$$

For example, the coefficient in column j is

$$\sum_{i=1}^4 \mu_i \binom{2}{k} p_i^k (1-p_i)^{2-k} \left(Q_{*,j} - \frac{(1-p_i)^2 \mathbf{1}_{j \leq 2} + p_i^2 \mathbf{1}_{j \geq 3}}{2} \right).$$

This identity proves every entry of (43) by rational arithmetic. All coefficients at off-diagonal validation atoms are nonpositive. The sum of the positive entries in (43) is $4035777/10240000 = u_1$. Each response coordinate is in $[0, 1]$, so this sum bounds its prior-averaged score and hence its minimum over the four fixed candidates.

Replacing Q_* by Q changes the score of any trained response by at most β . The training law is a candidate law and is unchanged. Consequently the same prior bounds every score by $u_1 + \beta$. A procedure with no training, or one that stops before taking its permitted sample, is covered by ignoring the generated all-on observation. $\square$

11.2. A marked five-event selector. Take two independent all-on training preparations. Let S be the total number of ones among their four report bits; discard their marks. Choose the following events, represented by their indicators on (40) and with indicator zero at every off-diagonal atom:

$$(44) \quad \begin{array}{c|cccc} S & 000 & 001 & 110 & 111 \\ \hline 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & 0 \end{array}$$

Denote these indicators by $h_0, \dots, h_4$ and report $D = h_S(Y) - h_S(X)$ on independent validation draws.

Lemma 11.3 (Exact uniform score of the selector). *Against the ideal target, the selector (44) has uniform minimum score $m_2 = (12 - 3\sqrt{3})/16$. Against Q , its score is at least $m_2 - \beta$.*

Proof. The generating polynomial of S is

$$((1-p) + pz)^2((1-q) + qz)^2.$$

Writing $g_s = (Q_* - b(p, q))h_s$, its expected score is $G(p, q) = \sum_{s=0}^4 [z^s]((1-p) + pz)^2((1-q) + qz)^2 g_s$. Substitution of the five rows in (44) and collection of terms with $x = p + q - 1$, $y = p - q$, $u = x^2$ and $v = y^2$ gives

$$(45) \quad G(p, q) = H(u, v) = -\frac{u^3}{32} + \frac{3u^2v}{32} + \frac{3u^2}{16} - \frac{3uv^2}{32} - \frac{uv}{4} - \frac{3u}{32} \\ + \frac{v^3}{32} + \frac{v^2}{16} + \frac{15v}{32} + \frac{7}{16}.$$

This is a polynomial identity, not an interpolation over a grid. Since $0 \leq u \leq 1$ and $v \geq 0$ on the parameter square,

$$\partial_v H(u, v) = \frac{3(u-v)^2 - 8u + 4v + 15}{32} \geq \frac{7}{32}.$$

Thus $H(u, v) \geq H(u, 0) = (-u^3 + 6u^2 - 3u + 14)/32$. The derivative of the last expression is $(-3u^2 + 12u - 3)/32$. On $[0, 1]$ it changes sign from negative to positive at $u = 2 - \sqrt{3}$. Substitution gives $(12 - 3\sqrt{3})/16$. The minimizing values are feasible: take $p = q = (1 \pm \sqrt{2 - \sqrt{3}})/2$. This proves exactness of the minimum for this specified selector, not optimality of its score over all selectors with two preparations. Finally, for each s , the target expectation of $h_s \in [0, 1]^\Omega$ changes by at most β . Averaging over training proves the claimed robustness. $\square$

11.3. Causal compilation and the full register profile.

Lemma 11.4. *The selector and its two validations admit the deterministic profile (42). No selector, first-bit buffer or validation accumulator is outside the charged register. An autonomous phase-tagged realization uses at most 75 states.*

Proof. After each training report bit retain the count accumulated so far. The two marks are read and discarded. The first six updates therefore have widths 2, 3, 3, 4, 5, 5 after the initial singleton. At the decision cut the five count values select the five events.

During candidate validation, after the first bit retain the event and that bit, using at most ten states. After the second bit retain the event and the function of the unread mark that gives its candidate-event indicator. For events 0, 1, 2, 3, 4, respectively, the possible functions are

$$\{0, 1\}, \quad \{0, 1 - w, 1\}, \quad \{0, 1\}, \quad \{0, w, 1\}, \quad \{0, 1\}.$$

There are twelve event–function pairs. After the candidate mark, retain the event and its now known candidate indicator $c_0 \in \{0, 1\}$, using at most ten states.

During target validation, after the first bit discard the event name and retain c_0 and its residual function of the second bit and mark. These residual functions belong to

$$\{0, x_2, (1 - x_2)(1 - w), 1 - x_2, x_2w\}.$$

This uses at most ten states. After the second bit, the residual *difference* as a function of the unread mark belongs to

$$\{-1, 0, 1, w, 1 - w, w - 1, -w\},$$

so seven states suffice. The mark update gives the three possible outputs $-1, 0, 1$. Each transition depends only on its retained residual and the current report. Thus it is a legal common observable row, not a family of independently optimized hidden-history maps.

For completeness, equality of two residual functions implies equality of their restrictions after either next bit, so identifying equal residuals is closed under every subsequent update. The supplied transition tables are obtained by these restrictions and can be reconstructed directly from (44). Enumerating residuals is a convenience for reproducing the tables, not the proof of their sufficiency. Finally, disjointly tag the states at the thirteen cuts by their phase and use terminal self-loops. The sum of the displayed widths is 75. This implementation charges the clock; it does not assert that 75 is optimal. $\square$

Proof of Theorem 11.1. Lemma 11.2 excludes $N \leq 1$ for a strict score greater than $c \geq u_1 + \beta$, for every architecture and every memory budget. Lemmas 11.3 and 11.4 supply a two-preparation architecture of peak twelve with score at least $m_2 - \beta > c$. Larger width budgets can use the same machine. $\square$

Corollary 11.5 (The positive-noise collision family). *For the original marked collision targets with $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$, the local microscopic estimate $\text{TV}(Q_{d,\sigma}, Q_*) < 1/300$ implies*

$$N_{2/5}^{\text{phys}}(W, Q_{d,\sigma}) = 2 \quad (W \geq 12).$$

There are four preparations in the complete audit: two candidate training, one candidate validation, and one target validation.

Proof. Apply Theorem 11.1. The lower-end margin is the exact rational number

$$\frac{2}{5} - u_1 - \frac{1}{300} = \frac{78269}{30720000} > 0.$$

For the other endpoint, $\sqrt{3} < 7/4$ implies $m_2 > 27/64$ and $27/64 - 1/300 > 2/5$. The physical total-variation estimate, including the actual mark and positive sensor noise, is proved in Theorem 10.3; the current deduction does not replace those hypotheses by a zero-noise limit. $\square$

Remark 11.6 (The part of the frontier determined here). The deterministic theorem evaluates the minimum preparation coordinate for $W \geq 12$. Theorem 12.1 and Corollary 12.2 below additionally determine the ideal maximum score with two preparations on this width range. The least peak width over all validation orders and the full boundary for $3 \leq W \leq 11$ remain separate questions. Theorem 14.3 below determines the peak for forward serial validation, and Theorem 14.1 determines the order-independent decision cut. The earlier exact decision-width result $\kappa_{2/5}^{\text{rows}} = 3$ is unchanged: it allows longer training, whereas this shortest audit uses five decision states. The older points (399, 40602), (320000, 1604) and $(12800 \cdot 2^{400}, 12)$ remain valid constructions with decision width three. They are dominated by (2, 12) in the projection onto training preparations and peak width, but not necessarily in a resource order that also constrains the decision alphabet separately.

Corollary 11.7 (The enlarged physical preparation segment). *With the full architecture of Definition 10.1,*

$$N_c^{\text{phys}}(W, Q) = 2 \quad \text{if} \quad W \geq 12, \quad \text{TV}(Q, Q_*) \leq \beta, \quad u_{\text{ex}} + \beta \leq c < m_2 - \beta.$$

Here $u_{\text{ex}} = (85 - 7\sqrt{73})/64$ is the exact one-preparation all-on value, rather than the older rational upper bound.

Proof. Restrict the adversary to the common-garbling subfamily in Definition 10.1. Theorem 5.2 and Corollary 9.2 bound every at-most-one-preparation auditor, even with unlimited memory, by $u_{\text{ex}} + \beta$. The same deterministic two-preparation machine in Lemmas 11.3 and 11.4 has score at least $m_2 - \beta$ against the full private candidate family. These inequalities prove the assertion. The four-preparation total, peak twelve and positive-noise specialization are unchanged. $\square$

12. THE EXACT TWO-PREPARATION MARKED SADDLE

The one-preparation saddle has an exact successor. A new zero coefficient of the least-favorable predictive measure makes it necessary to randomize at the two extreme training counts. This randomization changes the value, but does not enlarge the alphabet of frozen validation events.

Let $b(p, q)$ and Q_* be as in Section 11. Let r be the unique root in $(1/3, 7/20)$ of

$$(46) \quad 2r^3 + 25r^2 - 3 = 0,$$

and define

$$(47) \quad \tau = \frac{6(-r^2 + 4r - 1)}{r(3r + 25)}, \quad v_2 = \frac{-r^3 + 6r^2 - 3r + 14}{32}, \quad p_{\pm} = \frac{1 \pm \sqrt{r}}{2}.$$

The polynomial in (46) is strictly increasing on the positive half-line and has opposite signs at the indicated endpoints. These observations prove the asserted existence, uniqueness and isolating interval. That interval also gives $0 < \tau < 1$.

Theorem 12.1 (Exact two-preparation value). *For the full marked-word response class of (8),*

$$U_2(b, Q_*) = v_2.$$

A least-favorable prior gives mass $1/2$ to each of (p_-, p_-) and (p_+, p_+) . An optimal response discards the two training marks, retains the total number S of ones in the four training report bits, and uses

$$(48) \quad \begin{array}{c|cccc} S & 000 & 001 & 110 & 111 \\ \hline 0 & \tau & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & \tau \end{array}$$

with zero response at off-diagonal validation atoms. Numerically, $r = 0.3417694155\dots$, $\tau = 0.1688236447\dots$ and $v_2 = 0.4261127787\dots$. The exact value is determined by (46)–(47), not by these decimals.

Proof. Write $a = (1 - p)(1 - q)$, $c = pq$, $u = (p + q - 1)^2$ and $v = (p - q)^2$. Both u and v belong to $[0, 1]$. The score of the deterministic five-event selector is the polynomial $H(u, v)$ in (45). Only counts zero and four are changed by (48). Their probabilities are a^2 and c^2 . Consequently the new score is

$$G(u, v) = H(u, v) + \tau J(u, v), \quad J = a^2(5/16 - a/2) + c^2(5/16 - c/2).$$

Expansion gives the polynomial identity

$$(49) \quad \begin{aligned} J(u, v) = & -\frac{u^3}{64} + \frac{3u^2v}{64} - \frac{25u^2}{128} - \frac{3uv^2}{64} + \frac{13uv}{64} \\ & + \frac{v^3}{64} - \frac{v^2}{128} - \frac{v}{32} + \frac{3}{128}. \end{aligned}$$

Thus

$$H(u, 0) = \frac{-u^3 + 6u^2 - 3u + 14}{32}, \quad J(u, 0) = \frac{3 - 25u^2 - 2u^3}{128}.$$

Equation (46) says $J(r, 0) = 0$, and the choice of τ says $\partial_u G(r, 0) = 0$. Direct factorization, using these two identities, yields

$$(50) \quad G(u, 0) - v_2 = \frac{(u - r)^2 \{75 + 12r + 6u - 37r^2 - 74ru\}}{64r(3r + 25)}.$$

The expression in braces is strictly positive for $0 \leq u \leq 1$: it is at least $75 - 37(7/20)^2 - 74(7/20) > 0$. Moreover,

$$\partial_v H = \frac{3(u-v)^2 - 8u + 4v + 15}{32} \geq \frac{7}{32}, \quad \partial_v J = \frac{3(u-v)^2 + 13u - v - 2}{64} \geq -\frac{3}{64}.$$

Because $0 < \tau < 1$, $\partial_v G \geq 11/64$ on the unit square. It follows that $G(u, v) \geq G(u, 0) \geq v_2$, with equality precisely when $v = 0$ and $u = r$. Both equality points are feasible and are the two parameters in the asserted prior. This proves the global lower bound.

It remains to prove an upper bound against every response, not only those depending on S . Under the asserted prior the four training report bits are independent Bernoulli variables with common parameter p_- or p_+ . Conditional on $S = s$, their arrangements and the two independent training marks have parameter-independent distributions. The signed predictive coefficients can therefore be grouped by S without restricting any response coordinate. For the four displayed validation atoms put

$$C_{s,j} = \frac{1}{2} \sum_{p \in \{p_-, p_+\}} \binom{4}{s} p^s (1-p)^{4-s} \{Q_{*,j} - b_j(p, p)\}.$$

The first three rows of $256C$ are

$$\begin{aligned} s = 0 : & \quad (-(2r^3 + 25r^2 - 3), -(2r^3 + 27r^2 + 12r - 1), \\ & \quad -(2r^3 - 5r^2 - 20r - 1), -(2r^3 - 7r^2 - 32r - 3)), \\ s = 1 : & \quad 4(r-1)(2r^2 + 7r - 3, 2r^2 + 9r - 1, \\ & \quad 2r^2 - 7r - 1, 2r^2 - 9r - 3), \\ s = 2 : & \quad -6(r-1)^2(2r-3, 2r-1, 2r-1, 2r-3). \end{aligned}$$

Rows three and four are the reversals of rows one and zero, respectively. Their signs, using $1/3 < r < 7/20$, are

$$(0, -, +, +), \quad (+, -, +, +), \quad (+, +, +, +), \quad (+, +, -, +), \quad (+, +, -, 0).$$

Every off-diagonal validation coefficient is nonpositive because its target mass is zero. Hence (48) maximizes every signed coefficient: it is one at positive coefficients, zero at negative ones, and arbitrary at the two zeros. The lower-bound equality at both prior points shows that this Bayes maximum is exactly v_2 . The posterior-predictive dual, Theorem 5.1, now gives the matching upper bound for the full response class. $\square$

Corollary 12.2 (Exact physical score on the peak-twelve segment). *Use the atomic stochastic-row convention and the original marked interface of Definition 10.1. At two candidate training preparations, every peak budget $W \geq 12$ has exact ideal worst-case score v_2 . For a target Q with $\text{TV}(Q, Q_*) \leq \beta$, the corresponding value lies in $[v_2 - \beta, v_2 + \beta]$. In particular, with $u_{\text{ex}} = (85 - 7\sqrt{73})/64$,*

$$(51) \quad N_c^{\text{phys}}(W, Q) = 2 \quad \text{if} \quad W \geq 12, \quad u_{\text{ex}} + \beta \leq c < v_2 - \beta.$$

Proof. Let $h_0, \dots, h_4$ be the five deterministic events in (44). At count zero select h_1 with probability τ and h_0 otherwise; at count four select h_3 with probability τ and h_4 otherwise. At the other counts retain the indicated event. Discard the count and retain the selected event for both validations. This realizes (48); the event alphabet remains five. The full charged profile, now including the separate event-selection update, is

$$(1, 2, 3, 3, 4, 5, 5, 5, 10, 12, 10, 10, 7, 3).$$

The common residual-function compiler in Lemma 11.4 applies unchanged. In particular, no persistent random selector is outside the register. The all-on law for every private candidate is $b(p, q)$, so Theorem 12.1 proves the constructive lower bound. For the converse restrict to the admissible common-garbling product subfamily and apply Theorem 5.2 followed by Theorem 12.1. This bounds all architectures, including those using feedback, stopping and more memory. Target perturbation changes each frozen-event expectation by at most β . Finally the exact one-preparation theorem excludes a strict score above c at $N \leq 1$, whereas the displayed machine achieves it at $N = 2$ under (51). $\square$

The root r is larger than $2 - \sqrt{3}$. The old diagonal polynomial $H(u, 0)$ is strictly increasing between these points, whence $v_2 > m_2$. The deterministic theorem, including its zero-random-bit implementation, remains valid; the new exact score uses an additional stochastic row.

Proposition 12.3 (Finite precision for the new row). *Replacing τ by a dyadic $\tau_b = k/2^b \in [0, 1]$ with $|\tau_b - \tau| \leq 2^{-b}$ loses at most $(5/16)2^{-b}$ in uniform score. A clocked fair-bit implementation uses b extra random bits and at most $\max\{12, 6(b+1) + 5\}$ persistent states, including their phase, in addition to the unchanged preparation count.*

Proof. At each affected training word the response changes at a single atom; $|5/16 - a/2|$ and $|5/16 - c/2|$ are at most $5/16$. The two affected word probabilities sum to at most one. This gives the score estimate. To implement the dyadic coin, compare b fresh fair bits lexicographically with the fixed b -bit integer k . Retain the origin (count zero or four), one of the three comparison flags, and a phase in $\{0, \dots, b\}$. Once the comparison ends retain its chosen one of five events. Unaffected counts can wait in their event states. These states total at most $6(b+1) + 5$; the other cuts have peak twelve. The cases $k = 0, 2^b$ are deterministic. Exact atomic randomization and a finite fair-bit implementation are therefore distinct resource claims. $\square$

13. A CONTINUOUS TWO-PREPARATION MARKED FAMILY

The preceding saddle persists on an explicit interval of target laws, with exact rather than perturbative values. Use the target Q_γ from Section 9 and write $U_2(\gamma) = U_2(b, Q_\gamma)$.

Theorem 13.1 (A two-preparation saddle family). *For $6/25 \leq \gamma \leq 13/50$, let r_γ be the unique root in $(1/3, 7/20)$ of*

$$(52) \quad f_\gamma(r) = r^3 + (13 - 2\gamma)r^2 + (3 - 12\gamma)r - (1 + 2\gamma) = 0.$$

Set

$$D_\gamma(r) = 3r^2 + (26 - 4\gamma)r + 3 - 12\gamma, \quad t_\gamma = -\frac{2\{3r_\gamma^2 + (8\gamma - 14)r_\gamma + 3\}}{D_\gamma(r_\gamma)}.$$

Then $0 < t_\gamma < 1$, and

$$(53) \quad U_2(\gamma) = \frac{-r_\gamma^3 + (7 - 4\gamma)r_\gamma^2 - 3r_\gamma + 13 + 4\gamma}{32}.$$

The response (48), with $\tau = t_\gamma$, is optimal. A least-favorable prior gives equal masses to the two diagonal parameters $p = q = (1 \pm \sqrt{r_\gamma})/2$. The response has the same five-event, peak-twelve atomic realization. At $\gamma = 1/4$ these formulas reduce to Theorem 12.1.

Proof. For brevity write $r = r_\gamma$, $D = D_\gamma(r)$ and $t = t_\gamma$. The derivative D_γ is positive on the rectangle $\gamma \in [6/25, 13/50]$, $r \in [1/3, 7/20]$. For example its lower bound is $1/3 + (26 - 52/50)/3 - 3/25 > 8$. The polynomial f_γ decreases with γ at a fixed positive argument. Direct rational evaluations give $f_{6/25}(1/3) < 0$ and $f_{13/50}(7/20) > 0$. Monotonicity therefore gives the root and its uniqueness in this interval. The numerator of t is $-16\gamma r - 6r^2 + 28r - 6$. It is bounded

below by $28/3 - 6 - 16(13/50)(7/20) - 6(7/20)^2 > 0$, and above by $28(7/20) - 6 < 4 < D$. Thus $0 < t < 1$.

Let H, J be the polynomials from (45) and (49). Changing the target masses linearly in γ gives

$$\begin{aligned} H_\gamma(u, v) &= H(u, v) + (\gamma - 1/4) \frac{1 - (u - v)^2}{8}, \\ J_\gamma(u, v) &= J(u, v) + (\gamma - 1/4) \frac{(u - v)^2 + 6u - 2v + 1}{32}. \end{aligned}$$

The response's full-parameter score is $G_\gamma = H_\gamma + tJ_\gamma$. One has $J_\gamma(r, 0) = -f_\gamma(r)/64 = 0$ and $\partial_u G_\gamma(r, 0) = 0$. Algebraic division gives

$$(54) \quad G_\gamma(u, 0) - H_\gamma(r, 0) = \frac{(u - r)^2 A_\gamma(r, u)}{16D},$$

where

$$\begin{aligned} A_\gamma(r, u) &= 24\gamma^2 + (3\gamma - 10)r^2 + (6\gamma - 20)ru \\ &\quad + 12\gamma r + 6\gamma u - 51\gamma + 30. \end{aligned}$$

For $0 \leq u \leq 1$ this is at least $30 - 10(7/20)^2 - 20(7/20) - 51(13/50) > 0$. The transverse derivatives satisfy

$$\begin{aligned} 32\partial_v H_\gamma &= 3(u - v)^2 + (8\gamma - 10)u + (6 - 8\gamma)v + 15 \geq 5, \\ 64\partial_v J_\gamma &= 3(u - v)^2 + (14 - 4\gamma)u + (4\gamma - 2)v - 4\gamma - 1 \geq -3 \end{aligned}$$

on $0 \leq u, v \leq 1$. Consequently $\partial_v G_\gamma \geq 7/64$. Equation (54) proves a lower bound $H_\gamma(r, 0)$, with equality precisely at the proposed prior points. This quantity is the right side of (53).

For the upper bound group the full training words by S , as in the preceding theorem. Put $L(r) = r^2 + 6r + 1$. Using $f_\gamma(r) = 0$, the first three rows of $256C$ are

$$\begin{aligned} 0: & \quad (0, -8\gamma L(r), 32r(r+1) - 8\gamma L(r), 32r(r+1)), \\ 1: & \quad 8(r-1)(r^2 + 4r - 1 - 2\gamma(r+1), \\ & \quad \quad r^2 + 4r - 1 + 2\gamma(r+1), \\ & \quad \quad r^2 - 4r - 1 + 2\gamma(r+1), r^2 - 4r - 1 - 2\gamma(r+1)), \\ 2: & \quad -12(r-1)^2(r-1-2\gamma, r-1+2\gamma, r-1+2\gamma, r-1-2\gamma). \end{aligned}$$

The other rows are their reversals. The sign pattern is again

$$(0, -, +, +), \quad (+, -, +, +), \quad (+, +, +, +), \quad (+, +, -, +), \quad (+, +, -, 0).$$

Here are explicit interval checks for the potentially small margins. In row zero the third coordinate is bounded below by $32(1/3)(4/3) - 8(13/50)((7/20)^2 + 6(7/20) + 1) > 0$. The first parenthesis of row one is at most $(7/20)^2 + 4(7/20) - 1 - 2(6/25)(4/3) < 0$; the third is at most $(7/20)^2 - 4/3 - 1 + 2(13/50)(27/20) < 0$. The second parenthesis is positive already from $r > 1/3$ and $\gamma > 0$, and the fourth is negative. Finally $r - 1 + 2\gamma \leq -13/100 < 0$. All off-diagonal validation coefficients are nonpositive. Thus the stated response is a Bayes best response for this prior, including its two zero coefficients. Its score at both supporting parameters equals the lower bound. The full posterior-predictive dual proves (53). $\square$

This interval is a uniform certificate region for the displayed saddle, not a claim that the same masks remain optimal for every mark bias. The theorem supplies a continuous family at genuinely growing training history $N = 2$; it does not infer a recurrence for every N from the one-preparation formulas.

14. EXACT MEMORY FROM A MARKED SADDLE

The realization criterion is now applied to a saddle with genuine Bayes indifference. The zero coefficients in the two-preparation problem are not harmless: their values are fixed by contact with the parameter family. This supplies a converse for all optimal audits, rather than for a preselected response table.

Put $I = [6/25, 13/50]$. Throughout this section $r = r_\gamma$, $t = t_\gamma$, and $v = U_2(\gamma)$ are those of Theorem 13.1. Denote the five deterministic events in (44) by $e_0, \dots, e_4$. Responses displayed below are ordered on 000, 001, 110, 111 and vanish at the other four atoms.

Theorem 14.1 (Contact rigidity and exact decision memory). *For every $\gamma \in I$, every minimax optimal full-word response with two training preparations has, after averaging the independent training marks at the two extreme report counts, the five rows*

$$(55) \quad (t, 0, 1, 1), \quad (1, 0, 1, 1), \quad (1, 1, 1, 1), \quad (1, 1, 0, 1), \quad (1, 1, 0, t).$$

At the three interior counts the displayed response is forced at each individual training word, not merely after averaging.

In the original reset row–event interface, the least number of states at the post-training cut that permits the exact ideal score v is five. This converse allows arbitrary upstream finite memory, randomized encoding, adaptive training gates, stopping, independent target-only calibration, and all legal post-cut randomizations. Other cuts are unrestricted in this decision-memory assertion.

Proof. First consider a response to the full two-preparation word. Under the least-favorable prior of Theorem 13.1, every training word has positive probability. The strict coefficient signs force e_1, e_2, e_3 at report counts 1, 2, 3. At count zero only the 000 coordinate is free; at count four only the 111 coordinate is free. All off-diagonal validation coefficients are strictly negative: the prior parameters are interior, each word likelihood is positive, and the target assigns zero mass to those atoms. Thus those coordinates are zero for every optimal response.

Let a and b be the averages of the two free coordinates over the four equiprobable training mark words at their respective extreme counts. On the diagonal parameter (p, p) the score is

$$(56) \quad H_\gamma((2p-1)^2, 0) + aA_\gamma(p) + bB_\gamma(p),$$

where

$$A_\gamma(p) = (1-p)^4 \left(\frac{1+\gamma}{4} - \frac{(1-p)^2}{2} \right), \quad B_\gamma(p) = p^4 \left(\frac{1+\gamma}{4} - \frac{p^2}{2} \right).$$

Here H_γ is the deterministic five-event score in the proof of Theorem 13.1. At $p_+ = (1 + \sqrt{r})/2$, $A_\gamma(p_+) + B_\gamma(p_+) = J_\gamma(r, 0) = 0$ and $A_\gamma(p_+) > 0$. The latter follows from $p_+ > 1/2$ and $(1 + \gamma)/4 > 1/4$. Reflection exchanges A and B at $p_- = 1 - p_+$. Also $H_\gamma(r, 0) = v$. Optimality forces contact at both prior points, by Theorem 8.2. Subtracting v from (56) therefore gives $(a - b)A_\gamma(p_+) = 0$, hence $a = b$.

With this common value the diagonal score is $H_\gamma(u, 0) + aJ_\gamma(u, 0)$, where

$$H_\gamma(u, 0) = \frac{-u^3 + (7 - 4\gamma)u^2 - 3u + 13 + 4\gamma}{32}, \quad J_\gamma(u, 0) = -\frac{f_\gamma(u)}{64}.$$

The supporting value r lies in $(0, 1)$, so the derivative at r vanishes. Since $f'_\gamma(r) = D_\gamma(r) > 8$, the derivative equation has the unique solution

$$a = b = \frac{64 \partial_u H_\gamma(r, 0)}{D_\gamma(r)} = t_\gamma.$$

This proves (55). Individual extreme mark words may still use different tie values; no unnecessary uniqueness is asserted.

Now take an arbitrary audit with K states at the decision cut. Restrict the candidate to the common-garbling subfamily in Definition 10.1. Generate two all-on words and simulate the adaptive training and its stopping with independent coins. The retained state is a parameter-independent stochastic function of those two words. Conditional on that state, the frozen row-event continuation pulls back to a response $D_s \in [0, 1]^8$ on the all-on alphabet. Any calibration is part of the same encoding and retained state. Thus (28) holds without increasing K . If the audit attains v against the full physical class, its pullback attains v against this subfamily, since the unrestricted upper bound is v . Averaging selected rows preserves their factorization.

The three interior rows in (55) are distinct vertices of $[0, 1]^8$, so the generators must include e_1, e_2, e_3 themselves. The first row lies in the face

$$F_0 = \{d : d(001) = 0, d(110) = d(111) = 1, d = 0 \text{ off the diagonal}\}.$$

Of those three obligatory vertices only e_1 lies in F_0 ; since $t < 1$, this row is not e_1 . It needs another generator in F_0 . Similarly the last row needs another generator in

$$F_4 = \{d : d(000) = d(001) = 1, d(110) = 0, d = 0 \text{ off the diagonal}\}.$$

Only e_3 among the obligatory vertices lies there. The two faces are disjoint. Lemma 8.4 gives $K \geq 3 + 2 = 5$. Conversely the optimal response mixes e_0, e_1 at count zero and e_4, e_3 at count four and otherwise selects the indicated event. Only its five-valued event label survives the cut. The inherited compiler realizes its subsequent validation with finite memory and attains v . This proves exactness of the decision count. $\square$

The contact argument is essential. Setting both free coordinates equal to one gives a Bayes best response with only the three events e_1, e_2, e_3 , but its derivative at r is nonzero because $t < 1$. It is not minimax. Counting the states of a Bayes best response alone would give an incorrect resource conclusion.

Corollary 14.2 (A strict gap and its physical persistence). *There is $\delta_* > 0$, uniform for $\gamma \in I$, such that every two-preparation audit with at most four states at the decision cut has ideal worst-case score at most $U_2(\gamma) - \delta_*$. In particular, this bound holds for every original-interface audit of whole-schedule peak at most four. If only the target changes by total variation at most $\beta < \delta_*/2$, the five-state decision construction still strictly outperforms this class.*

Proof. Let $W_4(\gamma)$ be the maximum worst-case score over the relaxed factorizations (28) with $K = 4$ and two full training words. Their encoder and decoder spaces are compact and fixed, independently of γ . The score is jointly continuous in these variables, the target bias, and (p, q) . Consequently W_4 is continuous. By Theorem 14.1 it is strictly less than $U_2(\gamma)$ for each $\gamma \in I$. The minimum of this positive continuous gap on compact I is positive. This defines δ_* without pretending to have evaluated it numerically. The domination argument in that theorem makes it an upper bound for every original-interface audit in question. Changing only the target changes any fixed audit's mean score by at most β . The difference between the two classes therefore decreases by at most 2β . $\square$

For the microscopic family at $\gamma = 1/4$, the signal is larger than $1/4$ and the standard deviation of each normalized sensor noise is $\sqrt{2}\sigma$. The Gaussian tail bound used in Theorem 10.3 gives the sharper, vanishing estimate

$$(57) \quad \text{TV}(Q_{d,\sigma}, Q_*) \leq \frac{8\sigma}{\sqrt{\pi}} \exp\{-1/(64\sigma^2)\}, \quad 9/10 \leq d \leq 11/10.$$

Thus Corollary 14.2 holds uniformly in d for all sufficiently small positive σ . This is an existence statement with the displayed physical modulus. The present proof does not certify that every $\sigma \leq 1/24$ lies below the unknown gap threshold.

14.1. The sharp serial-validation peak. A decision cut and the whole execution have different resource costs. We now determine the latter in a declared validation schedule. In the *forward serial class*, both training preparations finish before validation; the candidate validation's two report slots and mark are read before any target validation report is read. The same frozen row–event continuation is used on both preparations, as in Definition 10.1. Training gates, stochastic state updates and frozen gate policies are unrestricted within that interface. No intermediate output can be reread as external memory.

Theorem 14.3 (Sharp forward serial peak). *For every $\gamma \in I$, the least whole-schedule peak width attaining $U_2(\gamma)$ in the forward serial class is twelve. In particular, a peak of at most eleven in this class incurs a strictly positive loss, which can be chosen uniformly for $\gamma \in I$.*

Proof. Use the common-garbling subfamily and the two raw all-on training words as in Theorem 14.1. Any optimal audit must finish both training preparations almost surely under its least-favorable prior. Indeed a stopped branch after a first preparation would have a response independent of the unused second word. For each first report count 0, 1, 2, two choices of the unused word force two different interior vertices from e_1, e_2, e_3 . Each has positive conditional probability. A positive stopped component cannot be a convex component of two distinct vertices. The same argument excludes stopping before the first draw. Null branches may be completed arbitrarily.

Consider the cut after the candidate's second report slot and before its mark. On a canonical raw input tape its continuation is a channel from $(w, y) \in \{0, 1\} \times \{000, 001, 110, 111\}$ to $\{-1, 0, 1\}$, where w is the unread candidate mark and y the fresh target word. All these target words have positive probability. Each retained state supplies one such continuation channel, common to every past leading to it. Hence the rows indexed by selected pasts factor through at most the peak number of generators.

For the strict interior training counts, the following eight rows are forced deterministic channels:

$$(58) \quad e_1(y) - f(w), \quad f \in \{0, 1 - w, 1\}; \quad e_2(y) - f(w), \quad f \in \{0, 1\}; \quad e_3(y) - f(w), \quad f \in \{0, w, 1\}.$$

For example e_1 has candidate residual 0 after an off-diagonal report pair, $1 - w$ after 00, and 1 after 11. The other cases follow by restriction of their masks. Each listed past has positive probability under both supporting candidate parameters. The eight functions are distinct: subtracting two with the same mask compares their distinct w functions; for different masks their difference depends nontrivially on y and cannot equal a function of w alone. They are eight distinct vertices of the product of output simplices.

There are four additional required rows. Average the extreme training marks as in Theorem 14.1, and take the candidate raw report pair to be, respectively, off diagonal and 11 at count zero, and off diagonal and 00 at count four. Independence of a fresh raw candidate preparation from the frozen continuation preserves the averaged tie probability t . On the four target words these rows have means

$$(59) \quad (t, 0, 1, 1), \quad (t - 1, -1, 0, 0), \quad (1, 1, 0, t), \quad (0, 0, -1, t - 1),$$

for both values of w . More precisely, each integer coordinate here is a point mass at that integer, and the remaining coordinate is the Bernoulli mixture between the two neighboring integers. This stronger channel statement, not a statement about a zero mean alone, follows because the

candidate indicator is fixed at 0 or 1 on the selected raw pair, while the target indicator is a bit. Fresh gate randomness is already part of this channel; it supplies no additional past-dependent argument to a retained state.

The four rows lie in four pairwise disjoint faces of the product of output simplices. These faces fix, for both w , respectively,

	000	001	110	111
$F_{0,+}$	*	0	1	1
$F_{0,-}$	*	-1	0	0
$F_{4,+}$	1	1	0	*
$F_{4,-}$	0	0	-1	*

where a number specifies a point mass and $*$ is unrestricted. Among the eight vertices (58), each face contains exactly one: e_1 , $e_1 - 1$, e_3 , and $e_3 - 1$, respectively. Since $t < 1$, none of the four required rows is that vertex. Face packing forces four more generators, for a total of twelve. This argument uses conditional future-output laws and not the continued storage of an event name after a residual has replaced it.

The optimal five-event selection followed by the inherited forward compiler has peak twelve, including its stochastic selection cut. It proves the matching upper bound. To see strictness below twelve, use the finite-protocol normal form of Proposition 8.1 with all logical cuts bounded by eleven, including selection and frozen-continuation buffers. Unobserved internal random computations can be collapsed to common transition kernels on the same register; an unretained random duration cannot be supplied as a later input. The normalized row spaces and closed frozen-interface restrictions are compact. If the constrained value equalled U_2 , the preceding face obstruction would apply. Continuity and compactness in γ give a uniform strictly positive gap, just as in Corollary 14.2. $\square$

Theorem 14.3 is a minimum-width theorem for a specified serial order, not a claim that an arbitrary validation order has the same minimum. The ideal target has support restrictions that can change future residuals when the target is read first. Reverse-order and interleaved validations require their own continuation analysis. The decision-memory converse in Theorem 14.1 does not make this serial-order assumption. Neither statement replaces the original physical interface by an uncharged atomic validation.

15. PREPARATION LIMITS AND LOCALIZATION OF LEAST-FAVORABLE PRIORS

The contact structure has a second consequence, now for an arbitrary number of training preparations. We separate a general localization principle from the special geometry of the marked product family. An exact finite- N two-point prior is not assumed.

Theorem 15.1 (Nearest-experiment limit and prior localization). *Let P_θ be a continuous family on a finite alphabet of size d , with compact metric parameter space, and let Q be fixed. Put*

$$d_Q(\theta) = \text{TV}(Q, P_\theta), \quad d_* = \min_{\theta} d_Q(\theta), \quad C_* = \{\theta : d_Q(\theta) = d_*\}.$$

For $N \geq 1$, the unrestricted frozen-validation values satisfy

$$(60) \quad d_* - \sqrt{(d-1)/N} \leq U_N(P, Q) \leq d_*, \quad U_N \uparrow d_*.$$

Every least-favorable prior μ_N in Theorem 5.1 satisfies

$$(61) \quad \int (d_Q - d_*) d\mu_N \leq \sqrt{(d-1)/N}.$$

Consequently every weak limit of such priors is supported on C_* . For every $a > 0$ their mass on $\{d_Q - d_* \geq a\}$ is at most $\sqrt{(d-1)/N}/a$. More generally, an upper bound b_N on twice the expected total-variation error of a common empirical reconstruction replaces $\sqrt{(d-1)/N}$ in both conclusions.

Proof. Let $\hat{P}$ be the empirical law of the N reports, and choose the indicator $h_{\hat{P}}$ maximizing $(Q - \hat{P})h$. If h_θ maximizes $(Q - P_\theta)h$, the two event inequalities give

$$(Q - P_\theta)h_{\hat{P}} \geq d_Q(\theta) - 2 \text{TV}(\hat{P}, P_\theta).$$

This inequality holds for the same parameter-independent response rule at every parameter. For any law P on d atoms,

$$\mathbb{E} \text{TV}(\hat{P}, P)^2 \leq \frac{d}{4N} \left(1 - \sum_x P(x)^2\right) \leq \frac{d-1}{4N}.$$

Thus its expected score is at least $d_Q(\theta) - \sqrt{(d-1)/N}$. Every response has score at most $d_Q(\theta)$ at that parameter, which proves the upper bound at a minimizing parameter. Ignoring the last observation proves monotonicity and hence the limit.

Against a minimizing prior, the Bayes optimum is U_N . It is at least the prior average of the same empirical rule. Therefore

$$U_N \geq \int d_Q d\mu_N - \sqrt{(d-1)/N}.$$

Combine this with $U_N \leq d_*$ to get (61). Markov's inequality proves the mass estimate. The function $d_Q - d_*$ is nonnegative and continuous on the compact parameter space, so its integral under any weak limit is zero. Such a limit is supported on its zero set. The proof uses only the common reconstruction bound, which proves the last assertion. $\square$

The empirical estimate in (60) is the same finite-alphabet mechanism underlying the retained decision-spectrum classification. Its role here is different: it controls every least-favorable prior, not merely a constructive selector. In particular, finite-preparation saddle supports cannot remain far from the closest candidate experiments as the preparation number grows.

Theorem 15.2 (Intrinsic marked contact geometry). *For every $\gamma \in I$, set $s_\gamma = \sqrt{(1+\gamma)/2}$ and*

$$d_\gamma = 2s_\gamma - 1 - \gamma/2, \quad C_\gamma = \{(s_\gamma, s_\gamma), (1 - s_\gamma, 1 - s_\gamma)\}.$$

Then

$$(62) \quad \min_{p,q} \text{TV}(Q_\gamma, b(p, q)) = d_\gamma, \quad \operatorname{argmin}_{p,q} \text{TV}(Q_\gamma, b(p, q)) = C_\gamma.$$

Moreover, for Euclidean distance in $[0, 1]^2$,

$$(63) \quad \text{TV}(Q_\gamma, b(p, q)) - d_\gamma \geq \frac{7}{2300} \operatorname{dist}((p, q), C_\gamma)^2.$$

The constant is uniform on I and is derived from the experiment's atom probabilities, not from a metric-entropy or small-ball assumption.

Proof. Put $a = (1-p)(1-q)$, $c = pq$, $l = (1-\gamma)/2$ and $h = (1+\gamma)/2$. The common mass of target and candidate is $\psi(a) + \psi(c)$, where

$$\psi(z) = \begin{cases} z, & 0 \leq z \leq l, \\ z/2 + (1-\gamma)/4, & l \leq z \leq h, \\ 1/2, & h \leq z \leq 1. \end{cases}$$

This follows by summing the smaller mass on each of the two marked atoms corresponding to the report pair. Reflection $(p, q) \mapsto (1-p, 1-q)$ exchanges a, c ; assume $c \geq a$. If $c < 1/4$, both terms are in the first segment of ψ , and their sum is at most $1/2$.

For $c \geq 1/4$, put $x = \sqrt{c} \geq 1/2$. By the arithmetic-geometric mean inequality,

$$a \leq (1-x)^2 \leq 1/4 < l.$$

Consequently the common mass is at most $F(x) = (1-x)^2 + \psi(x^2)$, with equality only when $p = q = x$. On the three consecutive intervals with endpoints $1/2, \sqrt{l}, \sqrt{h}, 1$, this function is respectively

$$(1-x)^2 + x^2, \quad (1-x)^2 + x^2/2 + (1-\gamma)/4, \quad (1-x)^2 + 1/2.$$

The first two are strictly convex and the last strictly decreases. The first increases away from $1/2$. Thus a maximum must occur at $\sqrt{l}$ or $\sqrt{h}$. Their difference is

$$(64) \quad F(\sqrt{h}) - F(\sqrt{l}) = \gamma \left(\frac{3}{2} - \frac{2}{\sqrt{h} + \sqrt{l}} \right) > \frac{7}{575}.$$

Indeed $\sqrt{h} > 39/50$, $\sqrt{l} > 3/5$, and $\gamma \geq 6/25$, so the right side is greater than $(6/25)(3/2 - 100/69) = 7/575$. The unique maximizing diagonal point on this side is $x = \sqrt{h}$. Its common mass is $(1 - \sqrt{h})^2 + 1/2 > 1/2$, also excluding $c < 1/4$. Subtracting it from one gives d_γ . Reflection proves (62) and the assertion about all minimizers.

We record a uniform quantitative form. Write $s = \sqrt{h}$ and $\Delta = 7/575$. For $1/2 \leq x \leq \sqrt{l}$, the common mass is at most $F(\sqrt{l})$, so its deficit from $F(s)$ is at least Δ . For $\sqrt{l} \leq x \leq s$, convexity puts F below the chord between its endpoint values. Since this interval has length at most one, its deficit is at least $\Delta(s-x)$. For $s \leq x \leq 1$ the deficit is

$$(1-s)^2 - (1-x)^2 = (x-s)(2-s-x) \geq (1-s)(x-s).$$

Since $s < 4/5$, this too is at least $\Delta|x-s|$. Together the three intervals give $F(s) - F(x) \geq \Delta|x-s|$.

Let $z = (p-q)^2$. At fixed $c = x^2$ the additional loss from the nondiagonal candidate is exactly

$$(1-x)^2 - a = p + q - 2x = \frac{z}{p+q+2x} \geq z/4.$$

The geometric mean x lies between p and q , and hence

$$(p-s)^2 + (q-s)^2 \leq 4z + 4(x-s)^2 \leq 4z + 4|x-s|.$$

It follows that the total deficit is at least $(\Delta/4)((p-s)^2 + (q-s)^2)$, proving (63) in this case. If $c < 1/4$, the total variation is at least $1/2$, whereas $d_\gamma < 12/25$ from $s < 4/5$ and $\gamma \geq 6/25$. The excess is therefore greater than $1/50$; Euclidean squared distance in the square is at most two, and $1/50 > 2(7/2300)$. This proves the remaining case. Reflection completes the proof. $\square$

Corollary 15.3 (An all-preparation law and a limiting two-point prior). *For all $N \geq 1$ and $\gamma \in I$,*

$$(65) \quad d_\gamma - \sqrt{3/N} \leq U_N(b, Q_\gamma) \leq d_\gamma, \quad U_N(b, Q_\gamma) \uparrow d_\gamma.$$

Every least-favorable prior $\mu_{N,\gamma}$ satisfies

$$(66) \quad \int \text{dist}(\theta, C_\gamma)^2 d\mu_{N,\gamma}(\theta) \leq \frac{2300}{7} \sqrt{3/N}.$$

A least-favorable prior can be chosen invariant under simultaneous complementation of the two parameters. Every such symmetric choice satisfies, for the Euclidean quadratic transport distance,

$$(67) \quad W_2 \left(\mu_{N,\gamma}, \frac{1}{2} \delta_{(s_\gamma, s_\gamma)} + \frac{1}{2} \delta_{(1-s_\gamma, 1-s_\gamma)} \right) \leq \sqrt{2300/7} (3/N)^{1/4}.$$

In particular, the limiting symmetric least-favorable law is unique.

Proof. Estimate the four-atom report-pair law from training and tensor it with the known fair mark. Tensoring with a common probability preserves total variation. The proof of Theorem 15.1 therefore uses $d = 4$ rather than eight, proving (65) and an excess bound $\sqrt{3/N}$. Combining the latter with (63) gives (66).

Simultaneously complement both report bits and the mark in every training and validation word. This leaves Q_γ invariant and sends $b(p, q)$ to $b(1 - p, 1 - q)$. The Bayes objective in (9) is invariant under this transformation and convex in the prior. Averaging a minimizing prior with its reflected image is therefore again minimizing and is symmetric.

Map each parameter to a closest point of C_γ , choosing the two points with equal probabilities at a tie. This is a Borel kernel, equivariant under reflection. A symmetric prior is pushed forward to the equal two-point law in (67); the induced coupling has squared transport cost exactly the left side of (66). Taking square roots proves the assertion. $\square$

These conclusions identify the endpoint and limiting contact geometry for every preparation number. They do not assert that the exact prior has two atoms at every finite N , or that the displayed convergence rates and constants are sharp. The finite- $N = 1, 2$ saddles are preserved as exact results. The new theorem explains their relationship to an intrinsically determined limiting contact set without extrapolating their particular event tables to unproved values of N .

16. AN EXACT PREPARATION–DECISION-MEMORY LAW

A second solvable family separates a genuine joint resource law from an implementation bound. Here the memory coordinate is the number of states at the training–validation cut. It is not identified with the bit-level peak width of the collision audit.

Fix $m = 2^d$, $0 < \alpha < 1$, and an unknown parameter $\theta \in \{1, \dots, m\}$. A fresh training preparation reports θ with probability α and an erasure $\perp$ otherwise. Conditional on θ , the preparations are independent. Candidate validation has law δ_θ and target validation is uniform on $\{1, \dots, m\}$. At the decision cut the auditor retains at most $K = 2^k$ states, $0 \leq k \leq d$. Its subsequent response may depend only on that state, current validation observations and fresh independent randomness. In particular, a calibration table, remembered seed or stopping time outside the retained state is not available. Upstream and downstream memory are otherwise unrestricted for the converse. Let $V_N^{\text{er}}(K)$ be the best worst-case expected signed validation score with at most N training preparations.

Theorem 16.1 (Exact joint law). *For every $N \geq 0$ and $K = 2^k \leq m$,*

$$(68) \quad V_N^{\text{er}}(K) = (1 - (1 - \alpha)^N) \left(1 - \frac{1}{K}\right).$$

The converse covers stochastic encoders and arbitrary legal stopping. The value is attained by a clocked machine with K training states. With atomic preparation and validation outcomes, the complete audit has peak at most $\max\{2K, 3\}$ and uses at most $N + 2$ preparations.

Proof. Write $q = (1 - \alpha)^N$ and pre-generate N potential training reports. A stopped auditor ignores the unused reports; this does not disclose extra information to it. On the event that all N reports are erasures, its retained-state law $r = (r_j)_{j=1}^K$ is independent of θ . After padding any smaller register with unused states, its channel to the decision register has the form

$$(69) \quad C(j \mid \theta) = qr_j + (1 - q)S(j \mid \theta)$$

for a stochastic channel S . This identity remains valid with randomized stopping and randomized state updates.

Use the uniform prior on parameters as an upper certificate. The part qr_j has zero expected score for every response: averaging δ_θ under this prior gives the target itself. For the other part, optimizing over all responses gives a convex function of S , a sum of positive parts of linear functions. Every stochastic channel is a convex combination of deterministic channels $j = g(\theta)$; therefore its value is at most the largest deterministic-channel value. This convexification is used only to bound a channel. It does not presuppose that a mixture of width- K machines is realizable at width K .

If the fibers of g have sizes $n_1, \dots, n_K$, a nonempty fiber has posterior uniform on that fiber, so its total variation from the uniform target is $1 - n_j/m$. The prior-averaged optimal score is

$$\sum_j \frac{n_j}{m} \left(1 - \frac{n_j}{m}\right) = 1 - \sum_j (n_j/m)^2 \leq 1 - 1/K.$$

Empty fibers contribute zero. Cauchy–Schwarz proves the inequality. This is an upper bound on the average score and hence on its minimum over fixed parameters. Multiplying by $1 - q$ proves the upper bound in (68).

For attainment, partition the parameter alphabet into K equal groups and let $g(\theta)$ denote its group. Initialize the register uniformly on the K group labels. On an erasure leave it unchanged; on a revealed parameter overwrite it by $g(\theta)$. No extra “revelation occurred” flag is needed. At the decision cut its law is $q \text{Unif}[K] + (1 - q)\delta_{g(\theta)}$. For register j , freeze the event which is the complement of group j . On a revelation its score is $1 - 1/K$. Under a uniformly random initial label, the mean score is zero for each fixed θ : one of the K labels has score $1 - 1/K$ and the others have score $-1/K$. This proves the exact worst-case value.

The k initialization bits can be accumulated in the K -state register; all earlier prefix counts are at most K . Thus no persistent seed lies outside it. Candidate validation retains the group label and its event indicator, using at most $2K$ states. Target validation produces one of the three values $-1, 0, 1$. This proves the atomic-outcome peak bound. The controller uses a declared external schedule; an autonomous stopping implementation must additionally retain its phase. A phase-tagged implementation is finite, but its cost is not included in K . $\square$

Corollary 16.2 (Thresholds and matching resource scales). *For $K \geq 2$ and $0 \leq c < 1 - 1/K$, the least integer number of training preparations giving a score strictly greater than c is*

$$(70) \quad N_c^{\text{er}}(K) = \left\lfloor \frac{\log(1 - c/(1 - 1/K))}{\log(1 - \alpha)} \right\rfloor + 1.$$

If $c \geq 1 - 1/K$, no finite N suffices. For $0 < \varepsilon < 1$, achieving $V_N^{\text{er}}(K) \geq 1 - \varepsilon$ requires

$$K \geq \varepsilon^{-1}, \quad N \geq \frac{\log(1/\varepsilon)}{-\log(1 - \alpha)}.$$

Conversely, whenever an allowed $K \leq m$ satisfies $K \geq 2/\varepsilon$, the choice $N \geq \log(2/\varepsilon)/[-\log(1 - \alpha)]$ suffices.

Proof. The strict threshold is equivalent to $(1 - \alpha)^N < 1 - c/(1 - 1/K)$. Since the denominator logarithm is negative, the least integer satisfying this strict inequality is (70), including equality cases of the real quotient. Also

$$1 - V_N^{\text{er}}(K) = q + (1 - q)/K.$$

This is at least q and at least $1/K$, and is at most $q + 1/K$. These three inequalities prove the necessary and sufficient bounds. $\square$

The exact law concerns an entire (N, K, α, m) family, not a single finite diagnostic. Its mechanism is a parameter-independent erasure component followed by the convex geometry of

the decision channel. It demonstrates how a preparation obstruction and a memory obstruction can be matched in one theorem. It does not identify the least physical peak width at two collision preparations. Encoding the m -symbol reports bit by bit also introduces cuts not present in this atomic model; the physical bit-level accounting is given separately.

17. INTEGER MEMORY AND THE FAILURE OF A NAIVE PARTITION FORMULA

The divisibility assumptions in Theorem 16.1 can be removed, but a balanced-partition Bayes bound must not be mistaken for a minimax equality when a persistent random partition is not free. This distinction leads to another exact finite-resource transition.

Keep the revelation interface, but allow any integers $m \geq 2$ and $1 \leq K \leq m$. Put $\rho = 1 - (1 - \alpha)^N$, and denote the genuine K -label minimax value by $V_{N,m}^{\text{er}}(K)$. Write $m = Kq + s$, $0 \leq s < K$, and set

$$(71) \quad B_{m,K} = 1 - \frac{(K-s)q^2 + s(q+1)^2}{m^2}.$$

Here and below all-erasure initialization randomness is a state initialization, not an unrecorded seed used again at validation.

Proposition 17.1 (Arbitrary integer alphabet bounds and low-reveal equality). *The optimal uniform-prior Bayes score is $\rho B_{m,K}$. The minimax value satisfies*

$$(72) \quad V_{N,m}^{\text{er}}(K) \leq \min \left\{ \rho B_{m,K}, 1 - \frac{\lceil m/K \rceil}{m} \right\}.$$

If $\rho \leq m/(m+s)$, equality with the first term holds. At perfect revelation $\rho = 1$ the exact value is the second term. In particular, when $s > 0$ the uniform-prior Bayes optimum at perfect revelation is strictly larger than the minimax value.

Proof. The all-erasure decomposition (69) holds without any divisibility condition. Its parameter-independent part has zero uniform-prior score. Optimizing the other part over responses gives a convex function of its stochastic channel. A deterministic channel with fiber sizes n_j has Bayes score $1 - \sum_j (n_j/m)^2$. If two nonempty fiber sizes differ by at least two, moving one index from the larger to the smaller strictly decreases the sum of squares. Allowing empty fibers and repeating gives sizes q and $q+1$, proving the Bayes bound. It is attained by this partition and its complementary events, with any parameter-independent initialization. This proves the first upper bound as well as the asserted Bayes optimum.

For perfect revelation, fix any K response vectors. For each known parameter choose a response maximizing its score; this deterministic assignment cannot decrease any parameter's payoff, even when the original encoder was randomized. Some assigned fiber has size at least $\lceil m/K \rceil$. Averaging its parameters, the score of their common response is at most the total variation between the uniform target and the uniform law on that fiber, namely $1 - |\text{fiber}|/m$. The minimum is no larger. Balanced fibers with their complementary events attain the resulting upper bound. A perfectly informed encoder can simulate any less informative encoder, so this is an upper bound at every ρ .

For the low-reveal equality let balanced fibers have weights $w_j = n_j/m$. When $\rho < 1$, initialize their label register with probabilities

$$(73) \quad r_j = \frac{\rho w_j + (1 - 2\rho)/K}{1 - \rho}.$$

These sum to one, and are nonnegative precisely throughout the stated range: the smallest numerator is $\rho q/m + (1 - 2\rho)/K$. On a revealed parameter overwrite the label by its fiber; on

erasure do nothing. No revelation flag is retained. For a parameter in fiber i , the complementary-event score is

$$\begin{aligned} g_i &= \rho(1 - w_i) + (1 - \rho) \left(r_i - \sum_j r_j w_j \right) \\ &= \rho \left(1 - \sum_j w_j^2 \right) = \rho B_{m,K}. \end{aligned}$$

This matches the upper bound uniformly in the fixed parameter. If $\rho = 1$ and $s = 0$, the deterministic revealed label already attains that value, so the singular denominator in (73) is not used. Finally, with unequal balanced fibers the weighted average of their sizes is strictly smaller than the largest size; equivalently $B_{m,K} > 1 - \lceil m/K \rceil / m$. This proves the last assertion. $\square$

Theorem 17.2 (The complete two-label revelation law). *For every $m \geq 2$, every $N \geq 0$, and $0 < \alpha < 1$, set $w = \lfloor m/2 \rfloor / m$ and $b = \lceil m/2 \rceil / m$. Then*

$$(74) \quad V_{N,m}^{\text{er}}(2) = \min\{2wb\rho, w\}, \quad \rho = 1 - (1 - \alpha)^N.$$

The same formula holds at the limiting endpoints $\rho = 0, 1$. For odd m it changes regime at $\rho = m/(m+1)$; for even m it reduces to $\rho/2$. The attaining machine uses two training labels, a frozen two-event alphabet, and atomic peak at most four.

Proof. Proposition 17.1 gives the upper bound and its attainment for $\rho \leq 1/(2b)$. Above this threshold, partition into fibers of weights w, b and initialize deterministically in the larger fiber's label. Continue to overwrite on a reveal and leave the state unchanged on an erasure. A parameter in the larger fiber has score w ; a parameter in the smaller fiber has score $(2\rho - 1)b$. The latter is at least w exactly when $\rho \geq 1/(2b)$. This attains the second upper bound. For even m the threshold is one, so the low-reveal construction suffices. The endpoint $\rho = 0$ may instead freeze a constant event. Candidate validation retains the selected label and its indicator, with at most four states; target validation produces the difference. Initialization is atomic randomization when its probabilities are not dyadic. $\square$

For example, $m = 3, K = 2$ gives $V_{N,3}^{\text{er}}(2) = \min\{4\rho/9, 1/3\}$, rather than $4\rho/9$ for all ρ . The balanced-integer expression is therefore not the unqualified replacement of the divisible minimax formula. For $K \geq 3$ outside the low-reveal range, (72) remains a bound rather than an asserted equality. The compatible dynamic formulation of Section 7 retains the exact optimization in that regime. These results strengthen the revelation example without identifying its decision-cut coordinate with the collision model's bit-level peak.

18. QUANTITATIVE CERTIFICATES FOR A FINITE COMPATIBLE ARCHITECTURE

We return to an arbitrary finite architecture, now with a finite parameter set. It can have hidden states, controlled experiments, more than two hypotheses and registers of any size. Enumerate its distinct stochastic rows by $r = 1, \dots, R$ and write $q_r \in \Delta_{s_r}$. A reused autonomous row appears only once in this product. Assume each execution uses at most H controller rows, and uses row r at most d_r times. Delete unused rows and put $S = \max_r s_r$. Fixed environmental kernels are not controller rows.

For an integer $m \geq 1$, intersect Δ_{s_r} with the coordinate boxes of side $1/m$. The resulting nonempty closed polytopes cover the simplex, with overlaps permitted at their boundaries. Write C for a product of such row cells and choose a vertex q_r^C in each factor. A row-cell vertex has at most one coordinate not fixed at a box boundary; consequently it has at most $v_r = s_r 2^{s_r - 1}$ vertices, all on the $1/m$ grid. There are at most $m^{\sum_r s_r}$ product cells.

Write the vertices of row cell C_r as $v_{r,j}$ and parametrize $q_r = \sum_j \lambda_{r,j} v_{r,j}$ with λ_r in a simplex. A reward polynomial has degree at most d_r in block λ_r . Use the homogeneous simplex Bernstein basis of degree d_r in that block, obtaining

$$(75) \quad g_\theta(q(\lambda)) = \sum_\alpha b_{\theta,C,\alpha} B_\alpha(\lambda), \quad B_\alpha \geq 0, \quad \sum_\alpha B_\alpha = 1.$$

Degree elevation includes the factors $\sum_j \lambda_{r,j} = 1$. No independence between different visits of a shared row is imposed on its unknown row vector.

Lemma 18.1 (Causal interpretation of the coefficients). *Every coefficient in (75) satisfies*

$$|b_{\theta,C,\alpha} - g_\theta(q^C)| \leq HS/m.$$

Proof. For each row r , independently prepare an ordered list of d_r vertex labels, sampled independently with law λ_r . At its k th visit, use the stochastic row given by the k th vertex label in that list. Before conditioning, the next label is independent of the previous execution, so this experiment has exactly the row q_r at every visit and expected reward $g_\theta(q(\lambda))$. Unused labels may be sampled after termination without altering the reward.

Condition on the counts of every vertex label in every list. The probability of those counts is $B_\alpha(\lambda)$. Conditional on the counts, each list is a uniformly ordered multiset. Its expected reward is independent of λ and is therefore the corresponding Bernstein coefficient. This also proves (75) directly.

Each row used by this conditioned experiment is a vertex of C_r or a mixture of its remaining vertices. Every two vectors in C_r have coordinate differences at most $1/m$, hence total variation at most $s_r/(2m) \leq S/(2m)$. Couple the conditioned experiment to the one using q^C , using the same environmental randomness while their executions agree. At each actual controller update the probability of a first row-level disagreement is at most $S/(2m)$. There are at most H such updates on the agreeing path. The terminal laws consequently differ in total variation by at most $HS/(2m)$. A reward in $[-1, 1]$ has expectation difference at most twice that quantity. This proves the estimate. The conditioned urn is used to analyze a coefficient; it is not declared to be an admissible free-memory controller. $\square$

Theorem 18.2 (Explicit subdivision certificate). *Define*

$$\ell_C = \min_\theta g_\theta(q^C), \quad u_C = \min_\theta \max_\alpha b_{\theta,C,\alpha}, \quad L_m = \max_C \ell_C, \quad U_m = \max_C u_C.$$

For the exact all-stochastic value of the specified architecture,

$$(76) \quad L_m \leq V(A) \leq U_m \leq L_m + HS/m.$$

On each cell, a witnessing parameter θ_C supplies the nonnegative coefficient certificate

$$u_C - g_{\theta_C}(q(\lambda)) = \sum_\alpha (u_C - b_{\theta_C,C,\alpha}) B_\alpha(\lambda).$$

Its degree in row block r is d_r , and its total degree is at most $\sum_r d_r$. At most $\prod_r \binom{d_r + v_r - 1}{v_r - 1}$ coefficients per parameter and cell are needed. If the primitive data are rational, all coefficients and bounds can be checked by rational arithmetic. Taking $m \geq HS/\varepsilon$ gives a certified interval of length at most ε .

Proof. The point q^C is feasible, so ℓ_C is a lower bound on the value restricted to C . For any $q \in C$, choose a convex representation of its rows. Equation (75) implies

$$\min_\theta g_\theta(q) \leq \min_\theta \max_\alpha b_{\theta,C,\alpha} = u_C.$$

Choose a parameter attaining the minimum defining ℓ_C and apply Lemma 18.1; this gives $u_C \leq \ell_C + HS/m$. Taking the maximum over the finite cover proves (76). Choose θ_C attaining

the definition of u_C ; all displayed coefficients of its certificate are nonnegative. The degree and count statements follow from the product Bernstein basis. Cell vertices are rational grid points, and a finite execution sums finite products, proving the rational statement. $\square$

Remark 18.3. The number of cells and coefficients can be enormous. Neither polynomial running time nor a uniform degree bound for the original sum-of-squares hierarchy follows. Refinement changes the domain cover, not merely a single global SOS degree. For a finite collection of architectures obeying a specified budget, maxima of these intervals give outer intervals; an unbounded architecture class needs a separate uniform argument such as Theorem 5.2. For compact parameter families, an effective net and a verified modulus for the primitive kernels are additional inputs, not consequences of compactness alone.

The resource estimate also makes row precision explicit. With $m = 2^b$, the lower witnesses have dyadic rows and lose at most $HS2^{-b}$ in value. Thus increasing row precision provides a quantified approximation for a fixed architecture, while its random-bit implementation still has to be charged. The physical theorem avoids this cost altogether because its rows are deterministic.

19. AUTONOMOUS NOISY TESTING WITH A PRECISION AND TIME LEDGER

This section supplies a substantive autonomous comparison rather than merely identifying shared rows. The stationary optimum is classical [6]; the proof and explicit finite-time implementation are recorded to make the resource conventions unambiguous. They are not used as novelty evidence for the physical theorem.

Let reports be independent, with law $\text{Ber}(p)$ under hypothesis $-$ and $\text{Ber}(q)$ under hypothesis $+$, where $0 < p < 1/2$, $q = 1 - p$, and $r = q/p$. A W -state autonomous machine has two time-independent stochastic transition matrices T_0, T_1 . Assume their induced chains $P_- = qT_0 + pT_1$ and $P_+ = pT_0 + qT_1$ are irreducible. A final decision uses only the state, with optional fresh terminal randomization.

Lemma 19.1 (Likelihood-range bound). *If equivalent probability laws ν_+, ν_- on a finite set satisfy*

$$\frac{\max_i \nu_+(i)/\nu_-(i)}{\min_i \nu_+(i)/\nu_-(i)} \leq R,$$

then $\text{TV}(\nu_+, \nu_-) \leq (\sqrt{R} - 1)/(\sqrt{R} + 1)$.

Proof. Write $L = d\nu_+/d\nu_-$, $a = \min L$ and $b = \max L$. Since $\mathbb{E}_{\nu_-} L = 1$, one has $a \leq 1 \leq b$. The chord bound for $(L - 1)_+$ gives

$$\text{TV}(\nu_+, \nu_-) = \mathbb{E}_{\nu_-} (L - 1)_+ \leq \frac{(1 - a)(b - 1)}{b - a}.$$

For fixed b/a , maximizing the last expression gives $a = 1/\sqrt{b/a}$; substitution yields $(\sqrt{b/a} - 1)/(\sqrt{b/a} + 1)$. It is increasing in b/a , proving the assertion. The identical-law case follows directly. $\square$

Theorem 19.2 (Autonomous stationary bound and an explicit approximation). *Put $D = W - 1$, $W \geq 2$. In the irreducible autonomous class the infimum stationary minimax error is*

$$(77) \quad e_W = \frac{1}{1 + r^D}.$$

For every $\epsilon = 2^{-b}$, $b \geq 0$, there is a machine whose rows have denominator dividing 2^{b+1} , and whose minimax error at report time n from any initial state is at most

$$(78) \quad e_W + \epsilon(W - 2) + \exp \left\{ - \left(\frac{p\epsilon}{2} \right)^D \left\lfloor \frac{n}{D} \right\rfloor \right\}.$$

The stationary lower bound is not a finite-horizon lower bound.

Proof. Every positive induced transition has its likelihood ratio in $[r^{-1}, r]$, and the supports of the two induced matrices are identical. The Markov chain tree formula writes each stationary weight as a sum of products over directed spanning trees rooted at that state. Each product has D transitions; its ratio lies in $[r^{-D}, r^D]$. The ratio of the two unnormalized stationary weights lies in the same interval. Normalizing multiplies all state likelihood ratios by a common number, so their maximum-to-minimum ratio is at most r^{2D} . Lemma 19.1 therefore bounds stationary total variation by $(r^D - 1)/(r^D + 1)$. The equal-prior Bayes error is at least $1/(1 + r^D)$, and a minimax error cannot be smaller than that Bayes error. This is the equal-prior binary specialization of the Hellman–Cover bound.

For the upper approximation, take states $i = 0, \dots, D$, weights $c_0 = c_D = 1$, and $c_i = \epsilon$ at interior states. On report 1, move from i to $i + 1$ with probability $a_i = \frac{1}{2} \min(1, c_{i+1}/c_i)$, if that neighbor exists; otherwise stay. On report 0, move to $i - 1$ with probability $b_i = \frac{1}{2} \min(1, c_{i-1}/c_i)$, if it exists; otherwise stay. All other probability is assigned to staying. Detailed balance gives

$$\pi_+(i) = \frac{c_i r^i}{\sum_j c_j r^j}, \quad \pi_-(i) = \frac{c_i r^{-i}}{\sum_j c_j r^{-j}}.$$

Reflection $i \mapsto D - i$ exchanges these laws. Decide $+$ for $i > D/2$, $-$ for $i < D/2$, and use a fair terminal decision at $i = D/2$ when this state exists. The two errors agree. Conditional on being at an endpoint, the error is exactly e_W . The total interior stationary mass is at most $\epsilon(W - 2)$, since each $r^i \leq r^D$ and the denominator is at least r^D under $+$, with the reflected argument under $-$. This proves the first two terms of (78).

Every existing neighbor transition under either hypothesis has probability at least $p\epsilon/2$, and every self-loop has probability at least $1/2$. From any state, move down to zero and then stay until D steps have elapsed. This yields the common minorization $P_\pm^D(i, \{0\}) \geq \delta = (p\epsilon/2)^D$. Subtracting that common atom gives contraction in total variation by $1 - \delta$ every D steps. Comparison with the stationary law yields $\text{TV}(P_\pm^n(i, \cdot), \pi_\pm) \leq (1 - \delta)^{\lfloor n/D \rfloor} \leq e^{-\delta \lfloor n/D \rfloor}$. A bounded decision changes error by at most this amount, proving (78). Finally, letting n tend to infinity and then ϵ to zero proves that the stationary infimum equals (77). All transition probabilities belong to $\{0, 1, 1/2, \epsilon/2, 1 - 1/2, 1 - \epsilon/2\}$. $\square$

The W states in this theorem refer to an atomic stochastic update after each report. A sequential fair-bit implementation must also retain the report, the current state, a bit counter and a flag. Each nontrivial move probability is $1/2$ or $2^{-(b+1)}$; test for a string of zero bits of the required length, and pad to $b + 1$ bits if a uniform report cycle is required. At most $4W(b + 2)$ states suffice for this implementation, including its bit-phase counter and a ready phase. This conservative bound is an implementation claim, not an optimal precision–state tradeoff. The time index n in (78) is an observation time at which an external analyst reads the autonomous machine, not a free input clock used by its transitions. The adversarial model of [10] is not covered by the independent-report proof above.

20. PHYSICAL CONSUMERS AND THE FIRST-PRINCIPLES CONNECTION

The physical preparation theorem needs no estimate of the target law as a stored input: the five events are fixed and their uniform validity uses a proved neighborhood of Q_* . This removes calibration memory from this particular consumer without asserting that target acquisition is unnecessary in an arbitrary comparison problem. The general acquired calibration theorem, with its full counter register, remains in the complete companion.

Corollary 20.1 (A raw-interface confidence construction). *Let $m = m_2 - \beta > 0$ and $0 < c < m$. Repeat the full four-preparation audit independently R times and write $\bar{D}_R$ for the mean score. Under every private candidate,*

$$\mathbb{P}(\bar{D}_R \leq c) \leq \exp\{-R(m - c)^2/2\}.$$

Under the null in which the independent candidate and target validation laws agree, with the same event chosen before both validations,

$$\mathbb{P}(\bar{D}_R > c) \leq \exp\{-Rc^2/2\}.$$

A clocked implementation uses $4R$ preparations and at most $12(2R + 1)$ persistent states, with no calibration and no random bits.

Proof. The independent block scores lie in $[-1, 1]$ and have mean at least m under a private candidate. Hoeffding's inequality gives the first bound. Under the null, conditional on the selected event, the two independent validation indicators have the same mean. Each block therefore has mean zero, and the same inequality gives the second bound. Retain the integer sum in $\{-R, \dots, R\}$ along with the current twelve-state block. At a block boundary add its terminal score and reset its register. The deterministic external schedule is part of the clocked convention. $\square$

This is an upper construction on the raw marked interface. Its proof does not transfer a lower bound from a deliberately reduced Bernoulli interface, and no optimal confidence-memory frontier is claimed.

Proposition 20.2 (Target transport without a training multiplier). *For the two-preparation audit, changing only the target from Q to Q' changes its expected score by at most $\text{TV}(Q, Q')$. Its candidate training law, register profile and event selector remain unchanged.*

Proof. Condition on the candidate training word. The expectation difference is $(Q - Q')h_S$, whose absolute value is at most $\text{TV}(Q, Q')$. Average over that unchanged training law. $\square$

In particular, every common-preparation transport estimate verified for the original positive-noise collision family can be inserted here without multiplying it by a long training horizon. If candidate kernels also change, their training and validation defects must be added by a causal coupling; the target-only statement does not remove those terms.

The first-principles route is now concrete. The positive experimental kernels and actual reset preparations give the likelihoods in (9); these generate posterior continuation states. The validation-test quotient gives the positive parts in the statistical dual. A prior then supplies an architecture-uniform impossibility certificate. On the constructive side, the five-event decision is compiled by taking its future residual functions, which are closed under observation shifts. These are compatible causal states and determine the charged profile. Thus the preparation law, predictive continuation and finite-resource realization are used in the same upper/lower theorem. This is not a deduction of geometry or nonlinear operator compactness from the existence of finite stochastic rows.

21. A SHARPENED PHYSICAL CONSUMER AND ITS SIMULATION BUDGET

The exact two-preparation theorem has a downstream use with explicit hypotheses, rather than a merely renamed dependency arrow. It sharpens the mean margin of a raw-interface confidence test, and the causal morphism theorem transfers that test, with its full memory and calibration costs, to another executable interface.

Theorem 21.1 (Confidence and transported resource certificate). *Let Q be a physical marked target with $\text{TV}(Q, Q_*) \leq \beta$. Use the two-preparation event selector with a dyadic coin τ_b satisfying $|\tau_b - \tau| \leq 2^{-b}$, and put*

$$a_b = v_2 - \beta - \frac{5}{16}2^{-b}, \quad W_b = \max\{12, 6(b+1) + 5\}.$$

If $0 < c < a_b$, R independent four-preparation blocks, followed by the test $\overline{D}_R > c$, have error bounds

$$(79) \quad \sup_{\text{private}} \mathbb{P}(\overline{D}_R \leq c) \leq \exp\{-R(a_b - c)^2/2\},$$

$$(80) \quad \sup_{\text{equal}} \mathbb{P}(\overline{D}_R > c) \leq \exp\{-Rc^2/2\}.$$

They use $4R$ preparations, at most bR fresh random bits, and at most $W_b(2R+1)$ persistent states under a clocked schedule. With an exact atomic coin, replace a_b, W_b by $v_2 - \beta, 12$.

Suppose a source interface simulates this entire repeated audit under Theorem 3.4, uniformly over the private and equal-law families, with path defect Δ_R . Then the corresponding source error bounds are the right sides of (79)–(80) plus Δ_R , capped at one. At aligned cuts its state bound is $C_{\text{cal}}R_tM_tW_b(2R+1)$, and its preparation count is N_{cal} plus all source calls required to supply the $4R$ audit preparations. Calibration is acquired and retained, not supplied as a free real-valued table.

Proof. Corollary 12.2 and Proposition 12.3 give mean block score at least a_b under every fixed private parameter. Independent fresh preparations and independent coins give independent block scores in $[-1, 1]$. Hoeffding’s inequality proves (79). Under the equal-law null, conditional on training and the frozen event, the two validation indicators have the same mean. Each block therefore has mean zero, proving (80) by the same inequality. Retain the integer sum in $\{-R, \dots, R\}$ alongside the current block register. Updating the sum and resetting the block is a deterministic boundary operation. This proves the raw resource bounds.

Apply Theorem 3.4 to the whole repeated experiment, including its stopping and validation fields. Its defect bound applies to each of the two terminal error events, whose indicators have range one; each probability therefore changes by at most Δ_R , not $2\Delta_R$. The same theorem gives the state product and preparation count. The source blocks need not be independent after a shared calibration stage: independence was used only for the reference audit, and path-law comparison transfers its final event probability. $\square$

The first part uses the exact saddle, the original microscopic bridge and the compiled residual machine. The second additionally needs the uniform acquired causal simulation theorem. A source which does not satisfy that theorem’s stability and calibration hypotheses is not covered. Likewise this consumer supplies no reverse simulation and therefore does not transfer an optimal-width lower bound to an arbitrary source. It is an actual conditional theorem interface for subsequent models, not an assertion that the independent geometric A2 chain or the unbounded B4/C2 operator obligations have already been discharged.

22. DEPENDENCIES, RETAINED RESULTS AND REMAINING DISTINCTIONS

The exact three-report binary noisy frontier, arbitrary-horizon binary-register minimax recursion and compatible Bayesian interval recursion remain as proved in the complete companion. Its exact odd-horizon majority realization is also retained; it is not a new result of this revision. Section 18 applies to registers of any finite width, controlled rows and any finite parameter set, but provides certified intervals rather than a closed minimax recursion for all such classes. The

physical theorem supplies a matched preparation segment in a continuous two-parameter family with overlapping stochastic reports; its converse is not a finite parameter-grid check.

The stationary autonomous result in Section 19 has an explicit formula, but its stationary lower bound must not be applied to a finite-horizon clocked machine. Its finite-time upper bound is purposely conservative and exposes the slowing caused by finite transition precision. An arbitrary compact-family presentation still requires additional effective information to construct a kernel net. A subdivided positive-basis degree bound must not be called a degree bound for a global SOS hierarchy.

The physical result is a genuine consumer of the experimental and continuation root. It does not make the independent A2 geometric proofs dependent on this root. The original B4 normalized-resolvent, range, compactness and control-transfer obligations, and the broad C2 strict-dual, form, rigidity and optional-projection obligations, are not proved by our finite-state theorem. Their statements, derivations and current status remain in the retained historical pipeline. No aggregate closure is inferred from preserving their source files.

Finally, the rational certificate (43), polynomial identity (45), and residual transitions are independently reproducible. The program in the accompanying source checks their exact identities and includes deliberately invalid controls. These checks are not substitutes for the proofs above. Neither compilation, preservation of earlier pages nor a successful finite test establishes novelty or an editorial decision.

REFERENCES

- [1] R. D. Smallwood and E. J. Sondik, The optimal control of partially observable Markov processes over a finite horizon, *Operations Research* **21** (1973), 1071–1088.
- [2] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272.
- [3] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176.
- [4] J. Müller and G. Montúfar, The geometry of memoryless stochastic policy optimization in infinite-horizon POMDPs, arXiv:2110.07409 (2021; revised versions).
- [5] M. Putinar, Positive polynomials on compact semi-algebraic sets, *Indiana Univ. Math. J.* **42** (1993), 969–984.
- [6] M. E. Hellman and T. M. Cover, Learning with finite memory, *Ann. Math. Statist.* **41** (1970), 765–782.
- [7] M. E. Hellman and T. M. Cover, On memory saved by randomization, *Ann. Math. Statist.* **42** (1971), 1075–1078.
- [8] R. A. Flower and M. E. Hellman, Hypothesis testing with finite memory in finite time, *IEEE Trans. Inform. Theory* **18** (1972), 429–431.
- [9] T. M. Cover, M. A. Freedman and M. E. Hellman, Optimal finite memory learning algorithms for the finite sample problem, *Information and Control* **30** (1976), 49–85.
- [10] M. A. Managoli and V. M. Prabhakaran, Memory constrained adversarial hypothesis testing, arXiv:2605.12063v1 (May 12, 2026).
- [11] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28, doi:10.1524/strm.2002.20.14.1.
- [12] Q. Qi, General Theta Foundations I: Compatible statistical experiments and resource frontiers, revision 23, September 24, 2026, complete mathematical companion, source commit 8a6043aca644043bd805668185eb227468f21170; packaged commit e926b783b643b3e7ff275a18105528927b76cc97b. The 96-page article and 649-page development are reproduced without alteration after the present article in the two complete volumes.
- [13] H. Chernoff, Sequential design of experiments, *Ann. Math. Statist.* **30** (1959), 755–770.
- [14] M. Naghshvar and T. Javidi, Active sequential hypothesis testing, *Ann. Statist.* **41** (2013), 2703–2738, doi:10.1214/13-AOS1144.
- [15] S. Nitinawarat, G. K. Atia and V. V. Veeravalli, Controlled sensing for multihypothesis testing, *IEEE Trans. Automat. Control* **58** (2013), 2451–2464, arXiv:1205.0858v6.
- [16] R. Zheng, R. Sim and A. Varvitsiotis, Solving imperfect-recall games via sum-of-squares optimization, arXiv:2602.21722v1 (2026).
- [17] A. Heller, On stochastic processes derived from Markov chains, *Ann. Math. Statist.* **36** (1965), 1286–1291, doi:10.1214/aoms/1177700000.

- [18] M. Vidyasagar, The complete realization problem for hidden Markov models: a survey and some new results, *Math. Control Signals Systems* **23** (2011), 1–65, doi:10.1007/s00498-011-0066-7.
- [19] N. Gillis and F. Glineur, On the geometric interpretation of the nonnegative rank, *Linear Algebra Appl.* **437** (2012), 2685–2712, arXiv:1009.0880.

PRESERVED PREDECESSOR

General Theta Foundations I, revision 26

The following predecessor volume is reproduced without alteration.
The complete current article precedes this divider.

Frozen predecessor commit:

466bfcdcb8b7d594d3924dbfa82b4c6e29368858

GENERAL THETA FOUNDATIONS I: CAUSAL EXPERIMENTS, FINITE-MEMORY CONTROL, AND EXACT MINIMAX LAWS

QIAN QI

ABSTRACT. We study which statistical comparisons are attainable when both the choice of experiment and its retained continuation are resources. Prepared positive kernels generate a future-test quotient; causal simulators then transport prediction and decision guarantees with explicit acquisition, quantization and memory costs. At a prescribed finite register profile we give an occupation-state dynamic formulation and an exact controlled family whose value is $1/(2\lceil m/K \rceil)$, whereas a free mixture of complete controllers has value $K/(2m)$. This strict discrepancy persists under positive-noise perturbations. For a marked product experiment with two unknown Bernoulli parameters, we determine the exact two-preparation minimax value and its least-favorable two-point prior. The value is $(-r^3 + 6r^2 - 3r + 14)/32$, where $2r^3 + 25r^2 - 3 = 0$ and $1/3 < r < 7/20$. An explicit interval of mark biases has a corresponding exact two-preparation saddle family. A global polynomial certificate proves optimality over all responses to the full marked training word. The optimal response uses five frozen events and peak register width twelve, yielding an exact ideal physical score and an enlarged preparation boundary for positive-noise collision experiments. We also obtain arbitrary-integer revelation bounds, a complete two-label law, and a matched online prediction-resolution law under explicit dynamical and acquired-mass hypotheses.

CONTENTS

1. Introduction	2
1.1. Relation to comparison, control and finite memory	4
1.2. Organization	4
2. Causal experiments and their predictive quotients	4
2.1. The experimental object	5
2.2. Prepared predictive equivalence	6
3. Attainable resolution and resource-aware simulation	7
3.1. Checkpoint and online prediction	7
3.2. Causal morphisms and simultaneous error budgets	8
4. Experiments, architectures and attainable resources	9
5. A statistical dual that is uniform over architectures	10
6. Controlled experiments without a dominating action	12
6.1. An informative action choice	14
7. Controlled experiments at a prescribed memory profile	15
7.1. A parameterwise occupation state	15
7.2. An exact informative-control bottleneck	16
8. Exact minimax laws for a marked product family	18

Date: September 24, 2026. Revision 26.

2020 Mathematics Subject Classification. Primary 62B05, 62C20; Secondary 60J10, 68Q45, 93E20.

Key words and phrases. Statistical experiments, causal realization, predictive distributions, finite memory, minimax duality, physical certification.

8.1. Complementary slackness and the event alphabet	19
9. A self-contained collision interface	20
9.1. Preparation and elastic scattering	21
9.2. Positive noise and the actual mark	21
10. The physical experiment and its sharp preparation boundary	22
10.1. An exact rational certificate against one preparation	23
10.2. A marked five-event selector	24
10.3. Causal compilation and the full register profile	24
11. The exact two-preparation marked saddle	26
12. A continuous two-preparation marked family	28
13. An exact preparation–decision–memory law	30
14. Integer memory and the failure of a naive partition formula	32
15. Quantitative certificates for a finite compatible architecture	33
16. Autonomous noisy testing with a precision and time ledger	35
17. Physical consumers and the first-principles connection	37
18. A sharpened physical consumer and its simulation budget	38
19. Dependencies, retained results and remaining distinctions	39
References	39

1. INTRODUCTION

An experiment supplies observations, but a causal procedure must also retain enough information to choose its next experiment and interpret its eventual answer. These are different requirements. A posterior may summarize all relevant observations without fitting into the available register. Conversely, a small register sufficient for one terminal decision may not support a subsequent informative action. The purpose of this paper is to place these questions in one prepared causal model and to determine several of its exact resource boundaries.

The underlying object consists of a preparation, positive normalized experimental kernels, a causal action interface, executable future tests and a resource profile. No information metric or geometric measure is postulated independently of that object. The future-test quotient identifies prepared histories with identical legal continuations. Their actual probability law determines checkpoint resolution, whereas closure of the quotient under observations determines whether a quantization can be updated online. Sections 2 and 3 develop these distinctions. Under contractive updates and an acquired small-ball condition, online squared prediction loss is bounded above and below by constants times $M^{-2/d}$. A separate transport theorem simultaneously accounts for acquired calibration, state quantization, primitive model errors and the register used by a downstream controller.

The finite-memory question is not solved by unrestricted belief-state convexification. Section 7 retains the common observable rows in a parameterwise occupation-state recursion. The occupation state belongs to offline design, not to the executing controller. Its exact supersolution formulation contains neither an unobserved parameter oracle nor a free retained strategy tree. The distinction is quantitative. In a routing-and-sensing family with m incomparable branch-specific experiments and K labels at the routing cut, we prove

$$V_{m,K} = \frac{1}{2\lceil m/K \rceil}, \quad \bar{V}_{m,K} = \frac{K}{2m}.$$

The second value permits a persistent controller selector without charging it. The first does not. The values differ whenever K does not divide m , including $1/4$ versus $1/3$ at $(m, K) = (3, 2)$.

All other cuts, including the sensing-action buffer and the frozen validation event, are specified. A small full-support perturbation preserves the strict discrepancy.

Our principal explicit statistical calculation concerns

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad (p, q) \in [0, 1]^2.$$

The last bit is the mark of the actual preparation. On atoms 000, 001, 110, 111, the target Q_* has masses $5/16, 3/16, 3/16, 5/16$, and it vanishes elsewhere. Training is followed by a frozen event applied to independent candidate and target validations. If U_N is the unrestricted-memory worst-case mean difference using N candidate training preparations, the exact values at one and two are

$$(1) \quad U_1 = \frac{85 - 7\sqrt{73}}{64}, \quad U_2 = \frac{-r^3 + 6r^2 - 3r + 14}{32}, \quad 2r^3 + 25r^2 - 3 = 0,$$

where $r \in (1/3, 7/20)$. The first value belongs to the continuous marked-target family in Theorem 8.1. The second is proved in Theorem 11.1. Its optimal response depends on the total of four training bits, with a nontrivial randomization at the two extreme counts. A square factorization and strict transverse monotonicity prove its lower bound on the full parameter square. The upper bound uses a matching least-favorable prior and the signs of all posterior-predictive coefficients, including the real training marks. No restriction to symmetric responses or parameter grids is imposed on the converse. Theorem 12.1 extends this construction to every target mark bias $6/25 \leq \gamma \leq 13/50$, with a cubic-defined least-favorable prior and a matching exact $U_2(\gamma)$ formula.

The physical realization is part of the same argument. The five-event residual machine has peak twelve even after the new randomized selection step is charged. Its all-on law covers the original private candidate class; a common-garbling subfamily suffices for the converse. Thus the ideal two-preparation physical score is exactly U_2 at every $W \geq 12$. If $\text{TV}(Q, Q_*) \leq \beta$, then

$$N_c^{\text{phys}}(W, Q) = 2 \quad \text{whenever} \quad W \geq 12, \quad U_1 + \beta \leq c < U_2 - \beta.$$

One candidate and one target validation are charged in addition to the two training preparations. The local collision geometry and positive-noise estimate are proved in Section 9. The exact atomic stochastic row and its finite fair-bit approximations have distinct, explicit precision costs.

For comparison with the deterministic construction we retain the constants

$$(2) \quad u_1 = \frac{4035777}{10240000}, \quad m_2 = \frac{12 - 3\sqrt{3}}{16}, \quad \beta_0 = \frac{1}{300}.$$

The rational u_1 is a valid one-preparation certificate and m_2 is the exact minimum of the deterministic five-event selector. The exact values in (1) sharpen these two distinct bounds without discarding their proofs or implementations.

A revelation family illustrates another integer obstruction. For arbitrary alphabet size m , two decision labels have the complete law

$$V_{N,m}^{\text{er}}(2) = \min\{2wb[1 - (1 - \alpha)^N], w\}, \quad w = \lfloor m/2 \rfloor / m, \quad b = \lceil m/2 \rceil / m.$$

For odd m there is a saturation transition before perfect revelation. The corresponding balanced-partition Bayes formula is not the minimax value at all reveal probabilities. General integer bounds and an exact low-reveal range accompany this calculation. This example is a controlled comparison of decision-cut resources; it is not a surrogate for the collision model's bit-level width.

1.1. Relation to comparison, control and finite memory. Blackwell comparison and minimax theory provide the statistical mechanisms [2, 3]. The finite-horizon belief-state piecewise-linear representation is classical [1]. The unrestricted controlled dual in Section 6 specializes that machinery to a fixed adversarial parameter and a frozen validation score. It is not a new principle of dynamic programming. Its relation to the genuine constrained value is separated by Theorem 7.2.

Controlled information acquisition has a substantial sequential-design history, beginning with Chernoff [13]. Naghshvar and Javidi [14], Section 3 and equation (4), formulate a posterior-state stopping problem minimizing expected observation count plus a wrong-declaration penalty. Their continuation operator is the same Bayesian mechanism that underlies our unrestricted recursion. Our terminal functional instead compares fresh candidate and target draws, and our finite-profile recursion optimizes common row prescriptions without storing the full posterior. Their cost and asymptotic optimality results are not replaced by a finite-horizon validation formula.

Nitinawarat, Atia and Veeravalli [15], Theorem 2 and its controlled sensing example, already distinguish causal from open-loop experiments in a multihypothesis error-exponent problem. Thus informative adaptation or a strict adaptive advantage alone is not a novelty claim here. The exact ceiling law and its charged-selector discrepancy concern a different, fixed-horizon resource constraint; the marked saddle concerns a continuous two-parameter composite family and an exact finite- N validation value rather than an asymptotic error exponent.

Finite-memory testing and the role of randomization are treated in [6, 7, 8, 9]. Our autonomous stationary comparison explicitly retains the Hellman–Cover attribution. Managoli and Prabhakaran [10] study a time-invariant finite-state tester against adaptive sample-distribution choices. Here the unknown parameter is fixed throughout an execution. The lack of mixed/behavioral equivalence with imperfect recall is also established; Zheng, Sim and Varvitsiotis [16] develop sum-of-squares methods over behavioral strategies. We do not claim polynomial policy optimization, the mixed/behavioral distinction or a general positivity hierarchy as new [4, 5]. Our explicit finite-family values and full-parameter saddle certificates are the mathematical claims requiring comparison.

Norberg’s filtered comparison [11] remains relevant to the general object-level setting. No non-overlap theorem is asserted without a proof-level comparison of that original work. The foundational kernel, barycenter and quantization arguments here are proved to fix their resource and preparation meanings, not to establish priority for their classical ingredients.

1.2. Organization. The causal object, acquired resolution and simultaneous simulation budget come first. Compatible finite architectures and the posterior-predictive dual then lead to controlled finite memory, exact marked minimax laws and the physical realization. Integer revelation, quantitative finite-architecture certification and the autonomous comparison follow. A final consumer theorem transfers the sharpened physical audit to repeated confidence tests and causally simulated interfaces. The predecessor’s complete mathematical and development volumes are preserved separately. In particular, preservation of A2, B4 and C2 derivations is not a claim that their geometric or operator hypotheses follow from finite stochastic kernels.

2. CAUSAL EXPERIMENTS AND THEIR PREDICTIVE QUOTIENTS

We begin with the object and morphism structure. Its elementary probability statements fix the meaning of later resource theorems; classical kernel construction, conditional expectation and quantization are not assigned a new priority. The substantive comparisons below retain their preparation, memory and approximation coordinates simultaneously.

2.1. The experimental object. An experiment has a fixed unknown parameter $\theta \in \Theta$, hidden states X_t , visible actions A_t , reports Y_{t+1} , and preparation laws μ_θ . We impose the following six conventions.

A0 (types and visibility). These spaces are standard Borel. Visible histories include actions, reports and publicly recorded resource increments. A legal-action set and its remaining budget are part of the visible protocol type. Two histories with different legal continuations are not identified.

A1 (positive kernels). There are jointly measurable normalized nonnegative kernels $K_t^\theta(dx', dy, dc \mid x, a)$, with c a nonnegative resource increment. An absorbing symbol represents termination or failure. Dirac kernels are allowed. Positivity here does not require every outcome to have strictly positive probability. A hidden state can contain the necessary physical past; this notation does not assume a finite-dimensional physical closure.

A2 (causality). A policy is a kernel $\alpha_t(da \mid h_t)$ depending only on visible past data and allowed independent randomization. It cannot read the unknown parameter. Stopping is with respect to the visible filtration.

A3 (executable tests). The specified bounded future tests include constants, required report events and their legal one-step continuations. For discrete reports the one-step instrument is

$$(\mathbb{T}_{t,a,y}f)(\theta, x) = \int f(\theta, x') K_t^\theta(dx', \{y\}, dc \mid x, a).$$

The test spaces are closed under the required instruments, finite linear combinations and the bounded monotone limits needed to recover the stated event laws. For general reports one uses event instruments and regular conditional distributions, not division by the probability of a continuous singleton.

A4 (actual preparation). Average risks and acquired state laws refer to the specified preparation and exploration protocol. Failed trials, calibration, resets and conditioning events retain their probabilities and implementation costs. A convenient measure on a reachable set does not replace its acquired law.

A5 (executable resources). Persistent labels, temporary buffers, program constants, calibration precision, preparations and clocks are distinguished. A retained selector is a state. A parameter-dependent table is not a known program constant. A simulator with R_t labels composed with a controller with K_t labels has the direct product bound $R_t K_t$, before any justified quotient is taken. Atomic stochastic rows and fair-bit implementations are separately specified.

Proposition 2.1 (Prepared path laws). *For every fixed parameter and causal policy, the preparation and kernels uniquely determine their finite-dimensional path laws. On a finite horizon they are the iterated kernel product*

$$\mu_\theta(dx_0) \prod_{t < T} \alpha_t(da_t \mid h_t) K_t^\theta(dx_{t+1}, dy_{t+1}, dc_t \mid x_t, a_t).$$

These laws are consistent under marginalization. For a countable horizon they determine the usual kernel extension. Deterministic dynamics give their original trajectory laws when represented by Dirac kernels.

Proof. Successive integration constructs each finite law. Normalization of the last two kernels gives its preceding marginal, and induction proves consistency and uniqueness on cylinder events. The countable kernel extension gives the full law; equivalently one applies the standard extension theorem to these consistent iterated kernels. Integration against a Dirac transition is evaluation along that transition, which proves the last assertion. $\square$

2.2. Prepared predictive equivalence. For Bayesian predictions additionally fix a prior on the parameter, and regard (θ, X_t) as the hidden variable. A prepared posterior π at a given protocol type evaluates the future tests. Define $\pi \sim_t \tilde{\pi}$ if their evaluations agree on every specified future test. This definition concerns one prepared experiment; it does not identify a prior-dependent posterior with a statistic sufficient simultaneously for all fixed parameters.

Theorem 2.2 (Predictive quotient and causal closure). *For finite report alphabets, predictive equivalence is preserved by every legal positive-probability report update. Its class determines the laws of all specified future experiments. Every deterministic history statistic that determines those laws factors onto this quotient. For a measurable realization it is sufficient that a countable separating test family gives a measurable evaluation map with standard Borel image, and that the chosen statistic admits a measurable factor. In finite history spaces the latter conditions are automatic.*

Proof. For a legal action and report, Bayes' rule gives

$$\pi'(f) = \frac{\pi(\mathbb{T}_{t,a,y}f)}{\pi(\mathbb{T}_{t,a,y}1)}.$$

Both numerator and denominator depend only on the old class by A3. Equivalent posteriors assign the same probability to the report, so updates agree on the next test space whenever that probability is positive. Null reports do not affect the prepared law and may be assigned a fixed admissible version. Induction determines every finite future report word; the monotone-class extension gives the required future events. Conversely those event laws determine the tests. If two histories have the same sufficient statistic, they have the same future evaluations; the quotient is therefore constant on its fibers. This gives the factor set-theoretically. The stated measurable factor assumption supplies its measurable version; no claim that every abstract quotient of a standard Borel space is standard Borel is used. $\square$

Proposition 2.3 (Barycenters, entropy and normalized response). *In a fixed preparation, suppose a hidden target Z , observation Y and postprocessing W form a Markov chain $Z \rightarrow Y \rightarrow W$. For any bounded vector of tests f ,*

$$p_W = \mathbb{E}[p_Y | W], \quad p_Y = \mathbb{E}[f(Z) | Y], \quad \mathbb{E}\varphi(p_W) \leq \mathbb{E}\varphi(p_Y)$$

for every convex φ for which these expectations exist. For a parameter-independent kernel G , $P \ll Q$, and finite $D(P\|Q)$, regular conditional laws give

$$\begin{aligned} D(P\|Q) &= D(PG\|QG) \\ &\quad + \int D(P(dx | w)\|Q(dx | w)) (PG)(dw). \end{aligned}$$

Finally, if f, g are bounded and $\pi_u(dx) = e^{ug(x)}\pi(dx)/\pi(e^{ug})$, then

$$\left. \frac{d}{du} \pi_u(f) \right|_{u=0} = \text{Cov}_\pi(f, g).$$

Proof. The Markov property and the tower rule give the conditional barycenter; conditional Jensen gives convex order. The joint laws $P(dx)G(dw | x)$ and $Q(dx)G(dw | x)$ have relative entropy $D(P\|Q)$. Disintegrating by w and applying the entropy chain rule gives the identity, and hence data processing. The last equality follows by quotient differentiation; boundedness justifies differentiation under the integral. $\square$

The barycenter statement concerns a law of predictions, not a pointwise map between reachable sets. Likewise the covariance in the last formula is a normalized response, not an independently postulated Fisher matrix or a global sufficiency theorem.

3. ATTAINABLE RESOLUTION AND RESOURCE-AWARE SIMULATION

3.1. Checkpoint and online prediction. Fix an actual exploration protocol and its history law. Let $p(h)$ be a bounded d -dimensional conditional prediction vector, with law ν . For instance, a future test is sampled independently after encoding, and the coordinates include its sampling weights. Squared excess prediction loss is then $\mathbb{E}\|p(h) - \hat{p}\|^2$.

Proposition 3.1 (Checkpoint resolution). *For at most M retained labels the least checkpoint excess loss is*

$$(3) \quad \inf_{c_1, \dots, c_M} \int \min_{1 \leq j \leq M} \|p - c_j\|^2 \nu(dp).$$

Independent randomized encoders or decoders cannot improve it. This formula concerns the stated preparation and does not provide a common codebook by separately optimizing at each unknown true parameter.

Proof. Conditional squared loss decomposes into its Bayes loss and squared distance from the conditional mean. For each retained label replace a random decoder by its mean. Random assignment to the resulting centers cannot do better than a nearest center. Conversely a nearest-center assignment attains the infimum over codebooks arbitrarily closely. Conditioning on an independent shared random seed, if such a seed is allowed, gives a codebook for each seed, each with risk at least (3). Averaging cannot improve the bound. The formula does not charge the computation of a checkpoint assignment as an online algorithm; online compatibility is a separate condition. $\square$

Theorem 3.2 (A matched online resolution law). *Suppose a fixed prepared experiment has a predictive state s_t in a compact metric space S , a known initial state, and updates $s_{t+1} = F_{t,a,y}(s_t)$. Assume every permitted update is ρ -Lipschitz for one $0 \leq \rho < 1$, and the prediction map $p : S \rightarrow \mathbb{R}^d$ is L -Lipschitz. Suppose S has a δ -net of at most $C\delta^{-d}$ points for all sufficiently small δ . At a checkpoint T , assume the actual law ν of $p(s_T)$ satisfies the small-ball bound $\nu(B(z, a)) \leq Aa^d$ for every center z and all sufficiently small $a > 0$. Then for all sufficiently large M , the minimum online excess prediction loss with M retained predictive labels satisfies*

$$(4) \quad c_0 M^{-2/d} \leq R_M^{\text{on}} \leq C_0 M^{-2/d}.$$

Here one may take

$$c_0 = \frac{1}{2}(2A)^{-2/d}, \quad C_0 = \frac{L^2 C^{2/d}}{(1-\rho)^2},$$

Any separate exploration controller, report buffer or clock is charged in addition to these M labels. The exploration protocol and the preparation are fixed before optimizing the encoder.

Proof. Any online M -label state is also a checkpoint encoder, so Proposition 3.1 applies. For any M centers, the union of balls of radius $a = (2AM)^{-1/d}$ has ν -mass at most one half. On its complement squared distance is at least a^2 . Integration gives $c_0 M^{-2/d}$. This argument also permits randomized online encoders.

For the upper bound take a δ -net with $\delta = (C/M)^{1/d}$, and a fixed measurable nearest-net assignment (with a fixed tie rule). Quantize the known initial state and, after each legal update, quantize again. The state error obeys $e_{t+1} \leq \rho e_t + \delta$, $e_0 \leq \delta$, and hence $e_t \leq \delta/(1-\rho)$. The prediction error is at most Le_t . Squaring proves the upper bound, uniformly in time. Each update uses only the current net label and the current permitted input, so the construction is online. It does not grant the encoder free access to the original discarded history. If the exploration policy needs another register, that register is combined by a product construction. $\square$

For example, a Lebesgue density bounded above by B gives the small-ball hypothesis with $A = Bv_d$, where v_d is unit-ball volume. The lower hypothesis is a condition on the *acquired prediction law*. A reachable set of dimension d without a mass condition does not imply the lower bound. Similarly, the contraction assumption is additional dynamics, not a consequence of kernel positivity.

3.2. Causal morphisms and simultaneous error budgets. A causal simulator $\Gamma : \mathfrak{E} \rightarrow \mathfrak{F}$ is a parameter-independent executable transducer that uses source preparations to supply the target interface under every allowed target policy. Its defect ε is a uniform total-variation bound on the resulting path laws, including the fields required by the task. Its resource map ρ_Γ bounds the source resources of the transducer composed with a target controller.

Proposition 3.3 (Composition and risk transfer). *Causal simulators compose. Their uniform defects add, capped at one, and their resource maps compose in the order $\rho_{\Gamma_1} \circ \rho_{\Gamma_2}$. For a terminal loss in $[0, L]$ and a common target task,*

$$\mathcal{R}_{\mathfrak{E}}(\rho_\Gamma(B)) \leq \mathcal{R}_{\mathfrak{F}}(B) + L\varepsilon.$$

For reward in $[-1, 1]$, the corresponding maximum worst-case rewards satisfy the reverse inequality with an error 2ε .

Proof. Interleave the two transducers; their pair of internal states is a causal implementation. Total variation contracts under a kernel and obeys the triangle inequality, proving the defect sum. If their state counts are R_t and M_t , the pair has at most $R_t M_t$ states. Other declared resource bounds compose by execution of the two protocols. A bounded loss changes expectation by at most its range times total variation. Apply this bound to every fixed parameter and target controller and take the appropriate infimum or supremum. Uniformity in the controller is essential to this step. $\square$

Theorem 3.4 (Acquisition, quantization and model-error transport). *Suppose an exact source transducer has predictive state $s_t \in S_t$ and additional charged buffers of size R_t , supplies T target operations, and has uniform model defect at most ε_{mod} . Its initialization and update depend on a calibration quantity ξ . From N_{cal} source preparations acquire an estimate $\hat{\xi}$ with at most C_{cal} retained labels, such that*

$$\sup_{\theta} \mathbb{P}_{\theta}(d(\hat{\xi}, \xi) > a) \leq \eta.$$

Subsequent preparation noises are independent of calibration conditional on the fixed parameter. Let finite nets of sizes M_t have radii δ_t . Assume, on matching input/output histories and on the good calibration event, that the quantized transducer's state error satisfies

$$(5) \quad e_0 \leq d_0 + a_0 a + \delta_0, \quad e_{t+1} \leq \rho_t e_t + d_{t+1} + a_{t+1} a + \delta_{t+1}.$$

Assume its next output kernel and the exact transducer's output kernel differ in total variation by at most $\epsilon_t + L_t e_t + b_t a$, uniformly in the fixed parameter and legal policy. All constants and nets are declared independently of that unknown parameter. Then the quantized acquired transducer has defect at most

$$(6) \quad \Delta = \min \left\{ 1, \varepsilon_{\text{mod}} + \eta + \sum_{t < T} (\epsilon_t + L_t e_t + b_t a) \right\}.$$

Composing it with any target controller of cut widths K_t requires at most $C_{\text{cal}} R_t M_t K_t$ labels at aligned execution cuts and transfers its bounded reward with loss at most 2Δ . Its preparations are N_{cal} plus all source preparations specified by the transducer; no successful-trial conditioning removes the failures.

Proof. On the good calibration event, iterate (5). More explicitly, with $D_0 = d_0 + a_0a + \delta_0$ and $D_j = d_j + a_ja + \delta_j$,

$$e_t \leq \sum_{j=0}^t D_j \prod_{k=j}^{t-1} \rho_k.$$

Couple each pair of output kernels maximally while their external histories agree. On agreement the target controller sees the same inputs, so it uses the same legal action and may use the same independent randomization. Its discontinuous action choices require no Lipschitz assumption: before a disagreement their arguments are identical. The chance of a first discrepancy is bounded by the sum in (6). Add the probability of bad calibration and the exact transducer's model defect. This proves the path bound, including stopping after padding by absorbing states.

The executable approximate transducer stores its calibration label, additional buffer, and net index. Combine these with the actual target register, not its full history or an uncharged belief. The resulting product gives the state count. If an operation contains further buffer cuts, apply the same product at each such cut; R_t was assumed to include them. Preparation costs add by execution. Finally apply Proposition 3.3. $\square$

Corollary 3.5 (A finite acquisition register). *Suppose r_0 calibration rows have at most d outcomes, and take n independent preparations from each row on a fixed schedule. Their empirical rows satisfy, with probability at least $1 - \eta$ uniformly in the row probabilities,*

$$\max \text{TV}(\hat{P}, P) \leq \frac{dz}{2}, \quad \eta = \min\{1, 2r_0d \exp(-2nz^2)\}.$$

Their counts can be retained in at most $(n+1)^{r_0(d-1)}$ labels. Thus Theorem 3.4 can use $N_{\text{cal}} = r_0n$, $a = dz/2$ and this finite calibration register when its calibration metric is the maximum row total variation.

Proof. For each outcome the empirical frequency is the mean of independent Bernoulli indicators. Hoeffding's bound and a union bound over at most r_0d coordinates give the event stated. The total-variation estimate is half the sum of coordinate errors. Retain $d - 1$ counts per row; the last is reconstructed from the known number of draws at that stage of the clocked schedule. Each retained count takes at most $n + 1$ values. An autonomous implementation must additionally charge its schedule. $\square$

These results separate exact predictive equivalence, average checkpoint resolution, online realizability, and uniform simulation risk. In particular, neither an optimal checkpoint codebook nor the proof-state bundle of Section 7 is automatically an executable finite register.

4. EXPERIMENTS, ARCHITECTURES AND ATTAINABLE RESOURCES

Throughout, total variation is $\text{TV}(P, Q) = \sup_{0 \leq h \leq 1} (P - Q)h = \frac{1}{2} \sum_x |P(x) - Q(x)|$. An experimental parameter is unknown and fixed throughout an execution. Controller randomization is independent of that parameter. A reset preparation is a fresh draw, independent of earlier preparations given the parameter. A controller may not recover discarded information from an uncharged tape, clock, selector or calibration table.

Definition 4.1 (Finite architecture). A finite architecture A specifies a finite causal execution graph, its legal experimental operations, the observable inputs to each controller row, and a register alphabet at every cut. It also specifies which row occurrences share a stochastic table. A clocked architecture may use different rows at different scheduled times. An autonomous

architecture may use time only through a charged state. For a parameter family Θ and terminal reward in $[-1, 1]$, let

$$V(A) = \sup_{q \in \prod_r \Delta_{s_r}} \inf_{\theta \in \Theta} g_\theta(A, q).$$

If $\mathcal{A}(R)$ is the class of architectures obeying a global resource budget R , its outer value is

$$V^{\text{out}}(R) = \sup_{A \in \mathcal{A}(R)} V(A).$$

Both the architecture and the simulator parameter are fixed before the controller's random choices. In particular, a mixture of complete controllers is not automatically an admissible controller at the original register profile.

For a finite hidden execution graph, compatibility can be stated without division. Let $x_{\theta ir}$ be the incoming mass of hidden history i at observable row r , and let $f_{\theta irj}$ be the outgoing mass through choice j . Include the usual fixed-kernel flow equations, as well as

$$(7) \quad f_{\theta irj} \geq 0, \quad \sum_j f_{\theta irj} = x_{\theta ir}, \quad f_{\theta irj} x_{\theta' i' r} = f_{\theta' i' rj} x_{\theta ir}.$$

For shared rows, the indices i, i' include all their occurrences.

Proposition 4.2 (Compatible realization). *For a specified finite architecture, nonnegative flow arrays satisfying (7) are exactly those realized by a common stochastic row at every declared observable input, including at zero incoming mass.*

Proof. Necessity follows from $f_{\theta irj} = x_{\theta ir} q_r(j)$. For sufficiency, if some incoming mass at row r is positive, define $q_r(j)$ by dividing the corresponding outgoing mass by that incoming mass. Conservation makes q_r stochastic, and the cross-products force the same ratio at every other positive incoming mass. If an incoming mass vanishes, nonnegativity and conservation force all its outgoing masses to vanish. If the entire row is unreachable, choose any legal stochastic row. Induction through the finite execution graph reconstructs the prescribed flows. The same ratio argument applies to identified occurrences of an autonomous row. $\square$

This formulation is inherited from the complete companion and is included to fix the controller class. Compact semialgebraic feasibility and an Archimedean positivity theorem provide a complete semialgebraic certificate hierarchy for a *fixed finite architecture*. They do not interchange the two optimizations in the definition. Section 15 gives a quantitative alternative for the inner problem. The next section supplies a different certificate that survives the outer optimization.

5. A STATISTICAL DUAL THAT IS UNIFORM OVER ARCHITECTURES

Let Θ be compact, let Ω be finite, and let $\theta \mapsto P_\theta \in \mathcal{P}(\Omega)$ be continuous. Fix $Q \in \mathcal{P}(\Omega)$. An N -preparation reset selector observes $Z = (Z_1, \dots, Z_N) \sim P_\theta^{\otimes N}$ and chooses a response $h_Z \in [0, 1]^\Omega$. Its expected validation score is

$$(8) \quad G_\theta(h) = \sum_{z \in \Omega^N} P_\theta^{\otimes N}(z) (Q - P_\theta) h_z.$$

The response is fixed before two independent validation draws. Operationally it can be realized by choosing a random event with those coordinate marginals and retaining that event for both validations. No memory bound is imposed in the definition of $U_N(P, Q) = \max_h \min_\theta G_\theta(h)$.

Theorem 5.1 (Posterior-predictive dual). *Both extrema in the following identity are attained:*

$$(9) \quad U_N(P, Q) = \min_{\mu \in \mathcal{P}(\Theta)} \sum_{z, \omega} \left[\int P_{\theta}^{\otimes N}(z) \{Q(\omega) - P_{\theta}(\omega)\} d\mu(\theta) \right]_+$$

$$(10) \quad = \min_{\mu \in \mathcal{P}(\Theta)} \sum_z m_{\mu}(z) \text{TV}(Q, \bar{P}_{\mu}(\cdot | z)).$$

Here $m_{\mu}(z) = \int P_{\theta}^{\otimes N}(z) d\mu(\theta)$ and, when $m_{\mu}(z) > 0$,

$$\bar{P}_{\mu}(\omega | z) = \frac{\int P_{\theta}^{\otimes N}(z) P_{\theta}(\omega) d\mu(\theta)}{m_{\mu}(z)}.$$

A zero-mass term in (10) is defined to be zero.

Proof. The response space is a finite-dimensional compact convex box. Probability measures on the compact space Θ form a weakly compact convex set. The integral of (8) is continuous and affine in each argument. The minimum over parameters equals the minimum over probability measures. The minimax theorem [3] therefore interchanges the maximum and minimum. For fixed μ , each response coordinate is independent in the maximization and its contribution is the positive part of its coefficient, proving (9). The coefficient vector at a fixed z has zero sum. For positive $m_{\mu}(z)$ it equals $m_{\mu}(z)(Q - \bar{P}_{\mu}(\cdot | z))$, proving (10). If $m_{\mu}(z) = 0$, every nonnegative likelihood integrand vanishes μ -almost surely and all coefficients are zero. Continuity on the two compact strategy spaces gives attainment. $\square$

Theorem 5.2 (Dominating-row converse). *Suppose every permitted training row is a parameter-independent causal garbling of one fresh draw from P_{θ} . Suppose every permitted validation row is the same garbling of the target and candidate laws, and every selected row–event pair is fixed before validation. Then $U_N(P, Q)$ bounds the worst-case expected score of every auditor using at most N candidate training preparations, regardless of its register profile, feedback schedule, independent randomization, or stopping rule. Independent target-only calibration, whose law does not depend on θ , does not invalidate the bound. If the all-on row and all its events are permitted, the bound is attained in the class with no memory cap and N training preparations.*

Proof. Generate N independent all-on observations in advance. At the t th requested preparation, apply the declared garbling to observation t , using only previously simulated reports and independent randomization. This simulates the adaptive transcript with its exact law. A stopped execution ignores the unused observations; it does not reveal them to the auditor. Pull back its final row–event response to the all-on alphabet. Conditional on the entire generated training word and the independent random seed, this is a function in $[0, 1]^{\Omega}$. Averaging the seed, including any target-only calibration, produces one response h_z for each full word z . The score is consequently (8). This proves the upper bound for the union of all architectures. Conversely, retaining the entire word and then a selected event implements any such finite response kernel when memory is unrestricted. Fresh validation gives exactly (8). $\square$

Remark 5.3 (What the dual does not convexify). The measure μ is an adversarial prior used to bound the minimum over *fixed* parameters. Its likelihood is $\int P_{\theta}^{\otimes N} d\mu(\theta)$, not $(\int P_{\theta} d\mu(\theta))^{\otimes N}$. It neither enlarges the private candidate class to mixtures nor gives the auditor a free persistent selector. Attainment of U_N without a memory cap does not assert attainment at any specified finite profile.

The state in (10) is generated by the actual preparation law, not by a geometric measure chosen independently of the experiment. For sequential continuation one retains the posterior μ_z or its future-test equivalence class; a single predictive marginal need not be closed under further

observations. This is the distinction between a one-step response quotient and a future-predictive quotient.

Corollary 5.4 (Finite moment support for the product family). *For two independent Bernoulli report bits with parameters (p, q) and an independent fair mark, an optimal prior in (9) can be chosen with at most $(N + 2)^2$ atoms. The two report counts (k, ℓ) are a causally updatable sufficient statistic for computing the dual.*

Proof. Conditional on the two counts, all arrangements of report bits and all mark words have parameter-independent probabilities. Summing their coefficients reduces (9) to the $(N + 1)^2$ count pairs. Their likelihoods are

$$\binom{N}{k} p^k (1 - p)^{N-k} \binom{N}{\ell} q^\ell (1 - q)^{N-\ell}.$$

After multiplication by $Q(\omega) - P_{p,q}(\omega)$, every integrand has degree at most $N + 1$ in each variable. Thus the objective depends only on the $(N + 2)^2$ rectangular moments. Their constant coordinate is one. Carathéodory's theorem in the remaining affine space replaces an optimal moment vector by a convex combination of at most $(N + 2)^2$ point evaluations. Counts update by addition of the current bits. No claim that this count register is optimal for the decision is made. $\square$

6. CONTROLLED EXPERIMENTS WITHOUT A DOMINATING ACTION

The reduction to one dominating row is not needed for a statistical dual, provided causal action choice is retained in the dual recursion. We give the finite-action theorem with a compact parameter space and an explicit protocol state. Its equality is an unrestricted-memory statement; its upper certificates apply to every smaller-memory class.

Let Θ be compact metric and E a finite set of protocol states. At $e \in E$ let $\mathcal{A}_e$ be a finite set of training actions. Action a produces x in a finite alphabet X_a with law P_θ^a and changes the protocol state to $T(e, a, x)$. The functions $\theta \mapsto P_\theta^a(x)$ are continuous. An unknown θ is fixed throughout the experiment. Conditional on it, each requested preparation uses a fresh independent draw from the selected row. Legal validation designs form a nonempty finite set $\mathcal{B}_e$. Design b has a finite outcome alphabet Y_b , target law Q^b and continuous candidate law V_θ^b . A policy may stop after at most N training actions. It then freezes $b \in \mathcal{B}_e$ and an event on Y_b , before one candidate and one target validation. Randomization is independent of θ . No training action is assumed to dominate another.

For a finite nonnegative measure ν on Θ write $\nu_{a,x}(d\theta) = P_\theta^a(x)\nu(d\theta)$ and set

$$(11) \quad \Phi_e(\nu) = \max_{b \in \mathcal{B}_e} \sum_{y \in Y_b} \left[Q^b(y)\nu(\Theta) - \int V_\theta^b(y) d\nu \right]_+,$$

$$F_0(e, \nu) = \Phi_e(\nu),$$

$$(12) \quad F_n(e, \nu) = \max \left\{ \Phi_e(\nu), \max_{a \in \mathcal{A}_e} \sum_{x \in X_a} F_{n-1}(T(e, a, x), \nu_{a,x}) \right\}.$$

If no training action is legal, only the stopping term is retained. All the functions vanish at the zero measure, so no posterior division is required on null histories. For positive mass, (11) is the mass times the largest posterior-predictive total variation.

Theorem 6.1 (Controlled posterior-predictive dual). *Let $\mathcal{U}_N(e_0)$ be the maximum worst-case expected validation score over all randomized causal policies in the declared protocol, with at*

most N training preparations and no memory cap. Then

$$(13) \quad \mathcal{U}_N(e_0) = \min_{\mu \in \mathcal{P}(\Theta)} F_N(e_0, \mu).$$

Both extrema are attained. Each $F_n(e, \cdot)$ is the maximum of finitely many continuous linear functionals on the cone of measures; in particular it is convex, continuous and positively homogeneous. For every prior μ , $F_N(e_0, \mu)$ bounds the worst-case score of every finite-memory architecture implementing this protocol.

Proof. There are finitely many histories of length at most N and finitely many event masks at each stopping history. Consequently there are finitely many deterministic policy trees, denoted by $\Pi_N(e_0)$. For $\pi \in \Pi_N(e_0)$ its payoff $g_\pi(\theta)$ is a finite sum of finite products of continuous primitive probabilities and is continuous in θ .

A randomized causal policy induces the same outcome laws as a probability distribution on these trees: sample its finite random decision table in advance and condition on it. Conversely, with no memory cap, a sampled tree can be retained and followed. Thus the unrestricted payoff vectors are exactly the convex hull of the g_π . This step is not an assertion that mixing machines preserves a specified register width.

The simplex on $\Pi_N(e_0)$ and $\mathcal{P}(\Theta)$ are compact and convex. The expected payoff is continuous and bilinear. Minimax therefore gives

$$\max_{\lambda \in \Delta(\Pi_N)} \min_{\theta} \sum_{\pi} \lambda_{\pi} g_{\pi}(\theta) = \min_{\mu} \max_{\pi \in \Pi_N} \int g_{\pi} d\mu.$$

For a fixed measure, the terminal maximum is (11), because an event includes precisely the positive coordinates of its signed predictive measure. At a training node one selects an action and, after its observed outcome, selects a continuation tree separately on that branch. The unnormalized branch measure is exactly $\nu_{a,x}$. Backward induction proves (12) and identifies the right side with (13). It also proves the finite-linear-functional representation. Compactness gives an optimizing prior and an optimizing mixed tree. Finally, the minimum over fixed parameters of any policy is at most its average under μ , which is at most $F_N(e_0, \mu)$. This remains true after restricting its memory or architecture. $\square$

The minimization in (13) occurs once, outside the recursion. It cannot be replaced by fresh adversarial choices after observations. The posterior carries dependence on one fixed unknown parameter. Also, a Bayes-optimal tree for a minimizing prior need not by itself be a minimax policy; the saddle-point mixture is supplied by the finite game in the proof.

Corollary 6.2 (Bellman upper certificates). *Suppose functions $\Psi_n(e, \nu)$ satisfy $\Psi_0 \geq \Phi$ and, for $n \geq 1$, satisfy*

$$\Psi_n(e, \nu) \geq \Phi_e(\nu), \quad \Psi_n(e, \nu) \geq \sum_x \Psi_{n-1}(T(e, a, x), \nu_{a,x}) \quad (a \in \mathcal{A}_e).$$

Then $\Psi_N(e_0, \mu)$ is an upper certificate for every admissible architecture and every prior μ . The family F_n is the least such family. An optimal prior can be chosen with at most $d + 1$ support points, where d is the affine dimension of $\{(g_\pi(\theta))_{\pi \in \Pi_N} : \theta \in \Theta\}$.

Proof. Induction in (12) gives $F_n \leq \Psi_n$. The equality case is the recursion itself. The objective depends on a prior only through its vector of integrals of the finitely many g_π . An optimal such vector belongs to the convex hull of their compact evaluation image. Carathéodory's theorem gives the stated support bound while preserving every pure-tree payoff. $\square$

Proposition 6.3 (Causal stability). *Consider two models with the same protocol, alphabets and policy class. Suppose their training rows differ uniformly in total variation by at most ϵ , their*

candidate validation rows by at most η , and their target validation rows by at most β . Their unrestricted values, or their values in any common specified memory class, differ by at most $2N\epsilon + \eta + \beta$.

Proof. Use the same policy randomization and couple corresponding training rows while the histories agree. The probability of a first discrepancy is at most $N\epsilon$, even with feedback and stopping. Conditional expected validation rewards lie in $[-1, 1]$, so differing training histories contribute at most $2N\epsilon$. On agreeing histories the same frozen design and event are used, and their conditional mean rewards differ by at most $\eta + \beta$. The bound is uniform in the fixed parameter and in the policy. Taking minima and suprema preserves it, with the reverse inequality obtained by interchanging the models. $\square$

6.1. An informative action choice. The following calculation distinguishes the controlled theorem from a common-row reduction. Let $\Theta = \{(i, j) : i, j \in \{0, 1\}\}$. Action R reports i . Action L reports j when $i = 0$ and otherwise reports an independent fair bit. Action H reports j when $i = 1$ and otherwise reports an independent fair bit. The candidate validation is $\delta_{(i,j)}$ and the target is uniform on Θ .

Proposition 6.4 (Strict adaptive advantage). *With two training preparations, the unrestricted adaptive value is $3/4$ and the best nonadaptive value is $5/8$. Nonadaptive policies here include mixtures of schedules, with their random schedule retained. No available single training action dominates all the others.*

Proof. Take R first, and then L or H according to the observed i . The parameter is recovered exactly. The complement of its validation atom has score $3/4$. No policy can exceed $\text{TV}(Q, \delta_\theta) = 3/4$ even with full information.

For the converse for fixed schedules, use the uniform prior. For a pair of actions a, b its Bayes value is explicitly

$$(14) \quad \sum_{x,y,\theta} \left[\frac{1}{16} \sum_{\vartheta} P_{\vartheta}^a(x) P_{\vartheta}^b(y) - \frac{1}{4} P_{\theta}^a(x) P_{\theta}^b(y) \right]_+.$$

Substitution of the three displayed kernels gives

	R	L	H
R	$1/2$	$5/8$	$5/8$
L	$5/8$	$7/16$	$1/2$
H	$5/8$	$1/2$	$7/16$

This is an exact finite sum, not a parameter-grid estimate. It bounds any mixture of fixed schedules by $5/8$; an early stop can be padded with an ignored observation. To attain the bound uniformly in θ , choose with equal probabilities the schedules (R, L) and (R, H) . When the chosen branch-specific action matches i , the parameter is known and the score is $3/4$. Otherwise only i is known, and the complement of its two-atom fiber gives score $1/2$. Each fixed parameter therefore has mean score $5/8$.

Action R cannot distinguish parameters differing only in j . Action L cannot distinguish $(1, 0)$ and $(1, 1)$, and H cannot distinguish $(0, 0)$ and $(0, 1)$. An experiment which fails to distinguish such a pair cannot garble to one that distinguishes it. Hence no action is a common dominating row. $\square$

If every binary report is additionally passed through an independent binary symmetric channel of crossover ϵ , the primitive training rows change by at most ϵ and have full support for $0 < \epsilon < 1/2$. Proposition 6.3, applied to both policy classes, leaves an adaptive advantage at

least $1/8 - 8\epsilon$, which is positive for $0 < \epsilon < 1/64$. Thus the distinction is not confined to disjoint observation supports. The finite-horizon belief recursion and its piecewise-linear structure are classical [1]. The contribution here is their explicit integration with the frozen validation score, one fixed adversarial parameter, outer certificates, and the resource conventions; no priority for Bellman recursion itself is claimed.

7. CONTROLLED EXPERIMENTS AT A PRESCRIBED MEMORY PROFILE

An unrestricted posterior is a useful information state for a proof, but it is not a free runtime state. We first give a resource-exact dynamic formulation and then solve a controlled family in which replacing that formulation by a policy-tree mixture changes the value strictly.

7.1. A parameterwise occupation state. Fix a finite horizon T , a compact metric parameter space Θ , finite hidden execution spaces H_t , and register alphabets S_t of sizes at most K_t . The visible inputs to each stochastic row and all legal actions are specified. Any input that must survive a cut belongs to S_t : this includes a chosen action, an unread report buffer, a stop flag, a selected validation event and a validation accumulator. One may subdivide an operation into several cuts to expose these costs. The scheduled time is available only in the clocked convention.

For now rows at different scheduled times are independently prescribed. Let $\mathcal{Q}_t$ be the finite product of legal row simplices at stage t . A prescription $q \in \mathcal{Q}_t$ is a table, chosen at design time, whose runtime arguments are exactly the declared visible inputs. After summing the fixed environmental kernels its one-stage transition is a stochastic matrix $M_{t,\theta}^q$ on the hidden-state/register pairs. We assume this matrix is jointly continuous in (q, θ) . Both action selection and state updating occur in this matrix, in their specified causal order; it is not an arbitrary matrix chosen separately for each hidden state or each parameter.

For a continuous bundle of probability vectors $x = (x_\theta(h, s))_{\theta \in \Theta}$ define

$$(\mathcal{T}_{t,q}x)_\theta(h', s') = \sum_{h,s} x_\theta(h, s) M_{t,\theta}^q((h, s), (h', s')).$$

The initial bundle x_0 is fixed by preparation. Terminal rewards $r_\theta(h, s) \in [-1, 1]$ are continuous in θ . Absorbing stopped states and frozen validation phases are included in this execution, not added afterwards without charge. Put

$$(15) \quad B_T(x) = \min_{\theta \in \Theta} \sum_{h,s} x_\theta(h, s) r_\theta(h, s),$$

$$(16) \quad B_t(x) = \max_{q \in \mathcal{Q}_t} B_{t+1}(\mathcal{T}_{t,q}x), \quad 0 \leq t < T.$$

The argument x is an *offline occupation bundle*, not the realized history, not an observed parameter, and not an additional register made available to the machine.

Theorem 7.1 (Resource-exact dynamic value and supersolutions). *For the specified clocked profile and observable-row interface, $B_0(x_0)$ is the exact maximum worst-case reward over all stochastic controllers. The maximum is attained by a sequence of legal common prescriptions. With the norm $\|x - x'\| = \sup_\theta \sum_{h,s} |x_\theta(h, s) - x'_\theta(h, s)|$, each B_t is 1-Lipschitz. Moreover,*

$$(17) \quad B_0(x_0) = \inf_H H_0(x_0),$$

where the infimum ranges over families of functions on occupation bundles satisfying

$$H_T \geq B_T, \quad H_t(x) \geq H_{t+1}(\mathcal{T}_{t,q}x) \quad (q \in \mathcal{Q}_t).$$

The infimum is attained by $H = B$. No convexification over retained controllers and no new adversarial choice after a report occurs in these identities.

Proof. For a fixed sequence $q_0, \dots, q_{T-1}$, iteration of the displayed transition gives its actual terminal law for every fixed θ . The terminal minimum is therefore exactly its worst-case reward. A runtime randomized controller in this interface is precisely such a sequence of common conditional rows; independence of fresh randomization and the declared register specify all its conditional choices. Persistent random correlations are allowed only through that register.

The occupation bundle at the next stage is determined by the current bundle and the prescription, without drawing a realized history in the design problem. Hence optimizing over the remaining prescription sequence gives (16) by ordinary backward induction. This does not select a different continuation for each unrecorded history. Each maximizing q is still a single legal table common to all hidden histories and parameters. At x_0 one follows the maximizing sequence of design prescriptions, computing their occupation bundles offline; the runtime controller only reads its allowed inputs.

Stochastic matrices contract the ℓ^1 norm. The terminal minimum is 1-Lipschitz in the displayed norm. Induction, using the same q when comparing two bundles and then taking the two maxima, proves the Lipschitz assertion. Joint continuity and compactness of $\mathcal{Q}_t$ give attainment at each stage, and compactness of Θ gives the terminal minimum. Any family H in (17) bounds the terminal reward along every prescription sequence. Backward induction therefore gives $H_t \geq B_t$. Equality is admissible, proving (17) and its attainment. $\square$

This is a nonlinear supersolution dual, not a least-favorable-prior saddle theorem at fixed width. Its optimization can be large and nonconvex. The principle of dynamic programming is classical; the point of this formulation is that the common-row and memory constraints remain in the maximizations. If rows are shared across scheduled times, as for an autonomous controller, choose the shared rows once and hold them fixed in all subsequent transitions; equivalently augment the *design* problem by that immutable prescription. Maximizing a shared row anew at each time would solve a different problem. A finite union of allowed interfaces can be handled by taking their maximum, or by a fully wired interface with the prescribed profile and its legal observable inputs. Neither operation enlarges a runtime register invisibly.

7.2. An exact informative-control bottleneck. Fix integers $m \geq 2$ and $1 \leq K \leq m$. The fixed parameter is $\theta = (i, j) \in [m] \times \{0, 1\}$. The first training experiment R reports i . Across the next cut the controller retains at most K labels, with an arbitrary stochastic encoder $E(s | i)$. It then chooses one of the m sensing actions with probabilities $D(a | s)$. The selected action is kept in a charged m -state action buffer and the old label is discarded. Action a reports j when $a = i$ and reports $\perp$ otherwise. The first report is no longer available.

The candidate validation law on $\{0, 1\}$ is δ_j ; the target is uniform. An event is frozen before one fresh candidate and one fresh target validation. Stopping before either training action is permitted. No independent persistent selector is given to the controller. The observation symbols are atomic; a bitwise input implementation would have additional cuts. Denote the maximum worst-case score by $V_{m,K}$.

Theorem 7.2 (Exact finite-memory control and its convexification gap). *In this protocol,*

$$(18) \quad V_{m,K} = \frac{1}{2\lceil m/K \rceil}.$$

The value obtained by allowing an uncharged ex ante mixture of complete K -label controllers is instead

$$(19) \quad \bar{V}_{m,K} = \frac{K}{2m}.$$

The inequality is strict whenever K does not divide m . The matching controller for (18) has charged cut profile bounded by $(1, K, m, 3, 6, 3)$. The K -label cut controls both retention of the routing observation and the subsequent informative action choice.

Proof. For a fixed i , let

$$p_i = \sum_s E(s \mid i) D(i \mid s).$$

This is the probability of taking the informative action. On its complement, all retained information is independent of j . Averaging over the two values of j makes the expected validation score zero there, for any frozen event or randomized response. On the informative part the score is at most $\text{TV}(\text{Ber}(1/2), \delta_j) = 1/2$. Thus the minimum over j at a given i is at most $p_i/2$. A stop before sensing is also uninformative about j and obeys the same bound; it can be padded by an ignored sensing operation.

Set $L = \lceil m/K \rceil$. Since $p_i \leq \max_s D(i \mid s)$, a value $\min_i p_i > 1/L$ would require every i to have some row s with $D(i \mid s) > 1/L$. Each stochastic row has at most $L - 1$ such indices. But $K(L - 1) < m$. This contradiction proves the upper bound. Partition $[m]$ into at most K fibers of size at most L , retain the fiber label after R , and choose uniformly inside that fiber. On a successful report freeze $\{1\}$ when $j = 0$ and $\{0\}$ when $j = 1$; on $\perp$ freeze the empty event. The score for every j is $1/(2|\text{fiber}(i)|)$, proving equality. The event alphabet has three labels. Storing the event and its candidate indicator requires at most six labels, and the final difference has three. The chosen sensing action was separately charged, proving the displayed profile.

For the convexified upper bound use the uniform prior on (i, j) . For any one controller,

$$\frac{1}{m} \sum_i p_i = \frac{1}{m} \sum_{i,s} E(s \mid i) D(i \mid s) \leq \frac{1}{m} \sum_{i,s} D(i \mid s) = \frac{K}{m}.$$

The same upper bound holds for mixtures. For the matching mixture, choose uniformly one of the m cyclic intervals of K indices. For that interval hardcode a deterministic K -label encoder which routes each included index to its own sensing action, and assigns other indices arbitrarily. Each i belongs to exactly K intervals. A successful action is followed by the complementary singleton event and failure by the empty event, as above. Every fixed (i, j) has mean score $K/(2m)$. This proves (19). $\square$

For example, $V_{3,2} = 1/4$ whereas $\bar{V}_{3,2} = 1/3$. The mixture just constructed can be implemented with a retained interval selector and a local label, using at most mK labels at the bottleneck. It cannot be declared a K -label realization: when the displayed gap is positive, (18) proves that no such realization exists. The action buffer of size m is not part of the K coordinate and has not been omitted from the profile. The theorem is a multicut resource law, not an assertion that its uniform peak equals K .

The actions are statistically incomparable. R cannot distinguish the two values of j at any fixed i , while action a does distinguish them at $i = a$. For $a \neq b$, action a does not distinguish the pair with $i = b$, while b does. A parameter-independent garbling cannot create a distinction absent from its input experiment.

Corollary 7.3 (A strict gap with positive report probabilities). *Replace each of the two training kernels by a mixture with weight $\epsilon > 0$ of the uniform law on its report alphabet. Let $\Delta_{m,K} = K/(2m) - 1/(2\lceil m/K \rceil) > 0$. For $\epsilon < \Delta_{m,K}/8$ the resulting full-support model still has a strict gap between the genuine profile-constrained value and its free-selector convexification.*

Proof. Each training row changes by at most ϵ in total variation. Couple corresponding rows until the first disagreement. For any fixed controller or mixed controller the probability of such a disagreement is at most 2ϵ , and its conditional reward lies in $[-1, 1]$. Each of the two optimal

values changes by at most 4ϵ . Their difference is therefore at least $\Delta_{m,K} - 8\epsilon > 0$. All row arguments and register alphabets are unchanged. $\square$

The distinction between mixed and behavioral strategies without perfect recall is classical; it is not first asserted here. Contemporary polynomial formulations over behavioral simplices include [16]. Theorem 7.2 evaluates the discrepancy in this specific frozen-validation experiment for every m, K , with a matching actual register and a noisy persistence statement. In particular, a single prior followed by an unrestricted policy-tree best response cannot be substituted for (16).

8. EXACT MINIMAX LAWS FOR A MARKED PRODUCT FAMILY

The posterior-predictive dual can be solved explicitly for a continuous family containing the collision target. Throughout this section, the training and candidate validation law is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad (p, q) \in [0, 1]^2.$$

In the order 000, 001, 110, 111, let Q_γ have masses

$$((1 + \gamma)/4, (1 - \gamma)/4, (1 - \gamma)/4, (1 + \gamma)/4),$$

and zero mass elsewhere. Write $U_1(\gamma) = U_1(b, Q_\gamma)$, with exactly the training-validation convention of (8). Set

$$(20) \quad \gamma_* = 1 - 1/\sqrt{2}, \quad s_\gamma = \sqrt{\gamma^2 - 2\gamma + 5}, \quad u_\gamma = \gamma - 2 + s_\gamma, \quad \tau_\gamma = \frac{2u_\gamma}{u_\gamma + 2 - \gamma}.$$

Theorem 8.1 (Exact marked minimax family). *For every $0 \leq \gamma \leq \gamma_*$,*

$$(21) \quad U_1(\gamma) = \frac{3 + u_\gamma^2}{8}.$$

An optimal response depends on the training word only through the report count $K = X_1 + X_2$. On the four target atoms its rows are

$$(22) \quad h_0 = (\tau_\gamma, 0, 1, 1), \quad h_1 = (1, 1, 1, 1), \quad h_2 = (1, 1, 0, \tau_\gamma),$$

and all responses are zero off these atoms. A least-favorable prior is

$$(23) \quad \mu_\gamma = \frac{1}{2}\delta_{(t_\gamma, t_\gamma)} + \frac{1}{2}\delta_{(1-t_\gamma, 1-t_\gamma)}, \quad t_\gamma = \frac{1 - \sqrt{u_\gamma}}{2}.$$

The assertion optimizes over all stochastic responses to the full marked training word, not merely over count-based or symmetric responses.

Proof. First, u_γ is the positive root of

$$(24) \quad u^2 + (4 - 2\gamma)u - (1 + 2\gamma) = 0.$$

On the asserted interval, $0 < u_\gamma < 1$ and $0 < \tau_\gamma < 1$. Indeed the polynomial is negative at zero and positive at one; also $u_\gamma < 2 - \gamma$ because $\gamma < 1/2$. Substitution of $1 - 2\gamma$ into the polynomial gives $8(\gamma^2 - 2\gamma + 1/2)$, which is nonnegative precisely on the present interval. Monotonicity on $[0, \infty)$ therefore gives

$$(25) \quad u_\gamma \leq 1 - 2\gamma.$$

Put $a = (1 - p)(1 - q)$, $c = pq$, and $z = 1 - a - c$. The probabilities of the three training counts are a, z, c . The independent training mark may be discarded by the displayed construction. Let $G_\gamma(p, q)$ denote its score. With $u = (p + q - 1)^2$ and $v = (p - q)^2$, direct expansion gives

$$(26) \quad G_\gamma(p, q) - \frac{3 + u_\gamma^2}{8} = \frac{2 - \tau_\gamma}{16} \{(u - v - u_\gamma)^2 + 4v\}.$$

For clarity, this identity follows by substituting the three rows of (22) into

$$a(Q_\gamma - b)h_0 + z(Q_\gamma - b)h_1 + c(Q_\gamma - b)h_2,$$

using $a + c = (1 + u - v)/2$, $a - c = -(p + q - 1)$, and then eliminating u_γ^2 by (24). The coefficient of the resulting square is $(2 - \tau_\gamma)/16$ and that of v is four times this coefficient. Both are nonnegative. Equality holds at both points of (23), since there $v = 0$ and $u = u_\gamma$. This proves the lower bound and identifies its equality set.

It remains to bound every response, including nonsymmetric ones. Use (23) in Theorem 5.1. At $t = t_\gamma$ put $a = (1 - t)^2$, $c = t^2$, and $T = a + c = (1 + u_\gamma)/2$. For the training report pair 00, the four dual coefficients, apart from a common positive factor, are

$$(27) \quad \left(Q_H T - \frac{a^2 + c^2}{2}, \quad Q_L T - \frac{a^2 + c^2}{2}, \quad Q_L T - ac, \quad Q_H T - ac \right),$$

where $Q_H = (1 + \gamma)/4$ and $Q_L = (1 - \gamma)/4$. The first coordinate is zero by (24); the second is $-\gamma T/2 \leq 0$. The last two are positive: $T \geq 1/2$, $ac \leq 1/16$, and $\gamma < 1/2$ imply $Q_L T > 1/16 \geq ac$. Thus $(\tau_\gamma, 0, 1, 1)$ maximizes this linear functional. For the training pair 11, reversing all report bits and the mark reverses the four coordinates and gives the row h_2 .

For either training pair 01 or 10, the two prior support points have the same likelihood $t(1 - t)$. The posterior is still equally weighted, and the candidate predictive mass at each target atom is $T/4$. Every one of the four coefficients is therefore nonnegative by $Q_L - T/4 = (1 - 2\gamma - u_\gamma)/8 \geq 0$. The response h_1 is optimal. At off-diagonal validation atoms the target mass vanishes and the coefficient is nonpositive. Finally, conditioning on either training mark multiplies all these coefficients by $1/2$; it creates no further optimizing choice. We have therefore optimized every coordinate of the full marked-word response space.

The prior average of the displayed response equals its value at either support point, namely $(3 + u_\gamma^2)/8$, by (26). Every other response has prior average at most this number and hence has minimum over fixed (p, q) at most this number. Together with the lower bound this proves (21), optimality of the response, and least favorability of the prior. $\square$

Corollary 8.2 (The exact one-preparation collision value). *For $Q_* = Q_{1/4}$,*

$$(28) \quad U_1(b, Q_*) = u_{\text{ex}} := \frac{85 - 7\sqrt{73}}{64}, \quad u_{1/4} = \frac{\sqrt{73} - 7}{4}, \quad \tau_{1/4} = 2 - \frac{14}{\sqrt{73}}.$$

If only the target changes to Q , then $|U_1(b, Q) - u_{\text{ex}}| \leq \text{TV}(Q, Q_)$.*

Proof. Substitute $\gamma = 1/4$ in Theorem 8.1. For a fixed trained response the target perturbation is at most total variation, uniformly in (p, q) . Taking first the minimum and then the maximum preserves this bound in both directions. $\square$

8.1. Complementary slackness and the event alphabet. The tie in the first coordinate of (27) explains the randomization. Any value of that response coordinate is Bayes-optimal for the two-point prior. The minimax choice makes the uniform payoff stationary at its two support points; solving this condition gives τ_γ and the square identity (26). This derives the prior and response together, rather than postulating that a numerically effective prior is optimal.

Operationally, for $K = 0$ one draws once between the events with indicator vectors $(0, 0, 1, 1)$ and $(1, 0, 1, 1)$, with probabilities $1 - \tau_\gamma$ and τ_γ . For $K = 2$ one draws between $(1, 1, 0, 0)$ and $(1, 1, 0, 1)$. For $K = 1$ the event is $(1, 1, 1, 1)$. The resulting event identity is retained and used for both fresh validations. These are exactly the five events in the two-preparation construction below. The one-preparation theorem supplies their complementary-slackness interpretation;

it does not prove that the two-preparation selector is optimal. The four-point rational prior retained below is a useful rational certificate, not a least-favorable prior.

The exact response uses an atomic stochastic row. It does not receive free persistent randomness. Replacing τ_γ by a dyadic number within 2^{-b} changes each conditional response, and hence the score, by at most 2^{-b} . A sequential fair-bit implementation must retain its bit prefix and event identity. None of these implementation costs is needed by the deterministic two-preparation physical audit.

Proposition 8.3 (The zero-preparation endpoint). *For $0 \leq \gamma \leq 1$, $U_0(b, Q_\gamma) = \gamma/2$.*

Proof. The fixed event $\{000, 111\}$ has score $(1 + \gamma)/2 - (a + c)/2 \geq \gamma/2$ because $a + c \leq 1$. Conversely, the equally weighted prior on $b(0, 0)$ and $b(1, 1)$ has predictive validation law uniform on the four target atoms. Its total variation from Q_γ is $\gamma/2$. Theorem 5.1 with $N = 0$ gives the matching upper bound. $\square$

9. A SELF-CONTAINED COLLISION INTERFACE

We give the preparation, mechanical estimate, observation law and private simulator class needed for the physical specialization. The following argument does not appeal to an unpublished physical lemma. It retains the same positive-volume preparation and positive sensor noise as the earlier collision construction.

Definition 9.1 (The reset row–event architecture). A preparation supplies two sequential report slots and, afterwards, an actual mark. The source reports in the two slots are constant. A private width-one simulator has no retained information between those slots and no persistent hidden seed. It may use fresh independent coins. Its first emission is $X_1 \sim \text{Ber}(p)$; after receiving the current gate $A \in \{0, 1\}$, its second emission has a fixed Bernoulli kernel with parameter q_A . The parameters (p, q_0, q_1) are fixed throughout the audit. The simulator does not observe the actual mark. All its coins are independent of that mark. Different preparations reset all unretained variables and are independent conditional on the parameters.

The auditor observes the first report, chooses a gate using its retained state and that report, observes the second report, and then observes the mark. During training it may choose rows and stop adaptively, using at most the declared number of candidate preparations. It next freezes one gate policy and one event, with any retained random choice included in its register. It uses that same row–event continuation on one fresh candidate preparation and one independent target preparation and returns the difference of their event indicators, target minus candidate. It may not reselect the event after either validation. Independent target-only calibration is allowed but all retained calibration and randomness are charged. Peak width counts every bit, mark, event and validation-accumulator cut. Clocked schedules are declared externally; an autonomous implementation must retain its phase. An unretained transcript, seed or stopping-time value is not available to a continuation.

In the all-on row the complete private law is $b(p, q_1)$, with an independent fair actual mark. For lower bounds it suffices to restrict the adversary to $q_0 = 0$, $q_1 = q$. In this subfamily every deterministic gate policy is the pushforward

$$T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w)$$

of $b(p, q)$. The gate, when included in the transcript, is another function of these same coordinates. Randomized gate policies are covered by averaging their independent random tables. This proves both the all-on product characterization for the upper construction and the common-garbling subfamily used for the outer converse. The width-one constraint is a simulator constraint; it is not an uncharged restriction on the auditor’s register.

9.1. Preparation and elastic scattering. Consider two labeled equal-mass hard spheres in $\mathbb{R}^3$, with no walls and contact distance d . Write $r = q_1 - q_2$, $g = v_1 - v_2$, $V = (v_1 + v_2)/2$, and let the center of mass position range over any fixed rational box of positive volume. Prepare $(B, W) \in \{0, 1\}^2$ with probability $5/16$ at $B = W$ and $3/16$ at $B \neq W$. Conditional on (B, W) , use the uniform product preparation on the following boxes:

$$(29) \quad \begin{aligned} r &= (-a_0, (2B - 1)b_0, c_0), & a_0 &\in [29/10, 31/10], & b_0 &\in [9/20, 11/20], \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & & & |V_z - (2W - 1)| &\leq 1/100. \end{aligned}$$

Thus the actual mark is $W = \mathbf{1}_{\{V_z > 0\}}$, preserved by momentum conservation. Set

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector, identically zero on the incoming transverse-velocity range.

Lemma 9.2 (Uniform collision signal). *For every preparation in (29) and every $9/10 \leq d \leq 11/10$, exactly one collision occurs and its time lies in $(4/5, 3/2)$. Afterwards*

$$(2B - 1)v_{1,y} > 1/2, \quad (2B - 1)h(v_{1,y}) > 1/4.$$

For $0 < d \leq 2/5$ no collision occurs on $[0, 3]$ and the detector is zero throughout that interval.

Proof. Before contact, $r(t) = r + tg$. At $t \leq 4/5$ its x coordinate is at most $-29/10 + (4/5)(21/10) = -61/50$, so its norm exceeds $11/10$. At $t = 3/2$ its absolute coordinate bounds are $1/4$, $113/200$ and $1/40$, whose squared sum is less than $(9/10)^2$. The first contact therefore occurs in the claimed interval. At the free zero of the x coordinate, whose time is at most $31/19$, the transverse norm is less than $9/10$. Hence the entry contact has $n_x < 0$, where $n = r(t_c)/d$. Its remaining coordinates give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B - 1)n_y \geq 87/220.$$

It follows that $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$. The elastic rule gives

$$v_1^+ = V + g/2 - (g \cdot n)n, \quad g^+ = g - 2(g \cdot n)n.$$

Consequently

$$(2B - 1)v_{1,y}^+ > -3/200 + (1443/1000)(87/220) > 1/2.$$

The definition of h implies the signal bound. Since $r(t_c) \cdot g^+ > 0$, squared separation increases strictly afterwards; there is no second collision. Finally, for $d \leq 2/5$ and $0 \leq t \leq 3$, $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$. There is no contact, and the incoming $|v_{1,y}| \leq 3/200$ makes h zero. $\square$

9.2. Positive noise and the actual mark. Let $\mathcal{B}_t$ be Brownian motion independent of the preparation and set

$$Y_t = \int_0^t h(v_{1,y}(s)) ds + \sigma \mathcal{B}_t, \quad 0 < \sigma \leq 1/24.$$

The two potential all-on reports are

$$Z_1 = \mathbf{1}_{\{2(Y_2 - Y_{3/2}) > 0\}}, \quad Z_2 = \mathbf{1}_{\{2(Y_3 - Y_{5/2}) > 0\}}.$$

The actual second report is AZ_2 . The gate changes acquisition, not the mechanical path. The continuous path is implemented by the target sensor; it is not made available to the private simulator or supplied as an uncharged auditor buffer. Denote the law of (Z_1, Z_2, W) by $Q_{d,\sigma}$.

Theorem 9.3 (Uniform physical presentation). *For $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$,*

$$(30) \quad \text{TV}(Q_{d,\sigma}, Q_*) < 1/300.$$

The same estimate holds after every declared feedback gate, including its observable gate value. The all-on private product family and the common-garbling subfamily in Definition 9.1 apply to this very experiment.

Proof. Every collision occurs before $3/2$. Conditional on the preparation, the two normalized sensor windows are independent Gaussians with mean $h(v_{1,y}^+)$ and standard deviation $\sqrt{2}\sigma \leq 1/12$. Lemma 9.2 makes their means have sign $2B - 1$ and magnitude greater than $1/4$. Each potential report therefore differs from B with probability less than $\Phi(-3)$. The Gaussian tail estimate

$$\Phi(-x) \leq \frac{e^{-x^2/2}}{x\sqrt{2\pi}}$$

gives $\Phi(-3) < 1/600$: $e^{9/2} > 80$ and $\sqrt{2\pi} > 5/2$ suffice. Couple the ideal and noisy reports using the same preparation, actual mark and gate random table. Except on a set of probability less than $2\Phi(-3) < 1/300$, both potential reports equal B and the entire gated transcript agrees. The ideal marked law is precisely Q_* , by the prepared (B, W) table. This proves (30) and the feedback-uniform version. Finally, the source is blind, its actual mark is unobserved by the simulator, and the gate is its only current input at the second emission. The product and subfamily assertions follow from the two independent private-emission kernels specified above. $\square$

For $d \leq 2/5$, the two windows instead contain only independent centered Gaussian noise. Their signs are independent fair bits, independent of W , and a private width-one simulator reproduces every gated row exactly. Thus the informative regime and its no-collision control use the same detector, preparation convention and actual mark.

10. THE PHYSICAL EXPERIMENT AND ITS SHARP PREPARATION BOUNDARY

Take $\Omega = \{0, 1\}^3$, with coordinates (x_1, x_2, w) . The all-on private candidate is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad 0 \leq p, q \leq 1.$$

The mark w is the actual mark, not an invented simulator bit. Put $a = (1 - p)(1 - q)$ and $c = pq$. On the ordered atoms

$$(31) \quad 000, \quad 001, \quad 110, \quad 111,$$

let Q_* have masses $5/16, 3/16, 3/16, 5/16$, and set it to zero elsewhere. The candidate masses on these atoms are $a/2, a/2, c/2, c/2$. A target Q is allowed whenever $\text{TV}(Q, Q_*) \leq \beta$.

The original feedback operation has the form

$$(32) \quad T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w), \quad g : \{0, 1\} \longrightarrow \{0, 1\}.$$

The selected gate is observable. It may depend on earlier preparations and on the first current report. The target rows are pushforwards by (32). For the converse, restrict the adversary to the admissible private subfamily whose second emission is AX_2 , where A is the current gate and $X_2 \sim \text{Ber}(q)$ is independent of the first emission and of the fair mark. Every row is then a causal garbling of $b(p, q)$. For the construction only the all-on row is used, and every private candidate in the original width-one class has the all-on product law just specified. These two facts, proved for the microscopic interface in Definition 9.1 and Theorem 9.3, explain why using a subfamily for the lower bound does not narrow the upper construction.

The admissible audit first trains, then selects one row–event pair, then uses one fresh candidate and one fresh target validation. It reports $D = \mathbf{1}_A(Y) - \mathbf{1}_A(X)$. The event and row cannot be reselected after seeing a validation draw. All bit, mark, decision and validation cuts are charged. Define $N_c^{\text{phys}}(W, Q)$ to be the minimum number of candidate training preparations among these original-interface auditors, with peak register width at most W , whose score is uniformly strictly greater than c . Randomization, feedback and stopping are allowed. If no such audit exists, the value is $+\infty$.

Theorem 10.1 (Two-preparation physical segment). *Let u_1, m_2 be as in (2). For every $\beta \geq 0$, $\text{TV}(Q, Q_*) \leq \beta$, and*

$$W \geq 12, \quad u_1 + \beta \leq c < m_2 - \beta,$$

one has $N_c^{\text{phys}}(W, Q) = 2$. The two-preparation audit is deterministic, uses no acquired calibration and no random bits, and has the full profile

$$(33) \quad (1, 2, 3, 3, 4, 5, 5, 10, 12, 10, 10, 7, 3)$$

from the initial cut through its twelve bit/mark updates. In particular, its decision alphabet has five labels and its peak has twelve. The lower bound remains valid with no memory cap.

The proof occupies the next three subsections. The interval in the statement can be empty for large β ; no assertion is then made. For $\beta = 1/300$, it contains $c = 2/5$ strictly between its endpoints.

10.1. An exact rational certificate against one preparation.

Lemma 10.2. *Every original-interface auditor using at most one candidate training preparation has worst-case score at most $u_1 + \beta$, regardless of memory.*

Proof. By Theorem 5.2, it suffices to bound $U_1(b, Q_*)$ and then perturb the target. In (9) use the four fixed parameters

$$(p, q) = (1/8, 1/8), (1/5, 1/5), (4/5, 4/5), (7/8, 7/8)$$

with weights $1/16, 7/16, 7/16, 1/16$, respectively. This measure is not asserted to be least favorable. Since $p = q$ on its support, the training count $K = x_1 + x_2$ is sufficient. The independent training mark and the order of unequal bits contain no further information about this prior.

For $k = 0, 1, 2$ and the four validation atoms in (31), the signed coefficients in (9), after grouping by K , are the entries of

$$(34) \quad \frac{1}{40960000} \begin{pmatrix} -1977 & -1775177 & 2170423 & 3943623 \\ 1765554 & 191954 & 191954 & 1765554 \\ 3943623 & 2170423 & -1775177 & -1977 \end{pmatrix}.$$

For example, the coefficient in column j is

$$\sum_{i=1}^4 \mu_i \binom{2}{k} p_i^k (1 - p_i)^{2-k} \left(Q_{*,j} - \frac{(1 - p_i)^2 \mathbf{1}_{j \leq 2} + p_i^2 \mathbf{1}_{j \geq 3}}{2} \right).$$

This identity proves every entry of (34) by rational arithmetic. All coefficients at off-diagonal validation atoms are nonpositive. The sum of the positive entries in (34) is $4035777/10240000 = u_1$. Each response coordinate is in $[0, 1]$, so this sum bounds its prior-averaged score and hence its minimum over the four fixed candidates.

Replacing Q_* by Q changes the score of any trained response by at most β . The training law is a candidate law and is unchanged. Consequently the same prior bounds every score by $u_1 + \beta$.

A procedure with no training, or one that stops before taking its permitted sample, is covered by ignoring the generated all-on observation. $\square$

10.2. A marked five-event selector. Take two independent all-on training preparations. Let S be the total number of ones among their four report bits; discard their marks. Choose the following events, represented by their indicators on (31) and with indicator zero at every off-diagonal atom:

$$(35) \quad \begin{array}{c|cccc} S & 000 & 001 & 110 & 111 \\ \hline 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & 0 \end{array}$$

Denote these indicators by $h_0, \dots, h_4$ and report $D = h_S(Y) - h_S(X)$ on independent validation draws.

Lemma 10.3 (Exact uniform score of the selector). *Against the ideal target, the selector (35) has uniform minimum score $m_2 = (12 - 3\sqrt{3})/16$. Against Q , its score is at least $m_2 - \beta$.*

Proof. The generating polynomial of S is

$$((1-p) + pz)^2((1-q) + qz)^2.$$

Writing $g_s = (Q_* - b(p, q))h_s$, its expected score is $G(p, q) = \sum_{s=0}^4 [z^s]((1-p) + pz)^2((1-q) + qz)^2 g_s$. Substitution of the five rows in (35) and collection of terms with $x = p + q - 1$, $y = p - q$, $u = x^2$ and $v = y^2$ gives

$$(36) \quad \begin{aligned} G(p, q) = H(u, v) = & -\frac{u^3}{32} + \frac{3u^2v}{32} + \frac{3u^2}{16} - \frac{3uv^2}{32} - \frac{uv}{4} - \frac{3u}{32} \\ & + \frac{v^3}{32} + \frac{v^2}{16} + \frac{15v}{32} + \frac{7}{16}. \end{aligned}$$

This is a polynomial identity, not an interpolation over a grid. Since $0 \leq u \leq 1$ and $v \geq 0$ on the parameter square,

$$\partial_v H(u, v) = \frac{3(u-v)^2 - 8u + 4v + 15}{32} \geq \frac{7}{32}.$$

Thus $H(u, v) \geq H(u, 0) = (-u^3 + 6u^2 - 3u + 14)/32$. The derivative of the last expression is $(-3u^2 + 12u - 3)/32$. On $[0, 1]$ it changes sign from negative to positive at $u = 2 - \sqrt{3}$. Substitution gives $(12 - 3\sqrt{3})/16$. The minimizing values are feasible: take $p = q = (1 \pm \sqrt{2 - \sqrt{3}})/2$. This proves exactness of the minimum for this specified selector, not optimality of its score over all selectors with two preparations. Finally, for each s , the target expectation of $h_s \in [0, 1]^\Omega$ changes by at most β . Averaging over training proves the claimed robustness. $\square$

10.3. Causal compilation and the full register profile.

Lemma 10.4. *The selector and its two validations admit the deterministic profile (33). No selector, first-bit buffer or validation accumulator is outside the charged register. An autonomous phase-tagged realization uses at most 75 states.*

Proof. After each training report bit retain the count accumulated so far. The two marks are read and discarded. The first six updates therefore have widths 2, 3, 3, 4, 5, 5 after the initial singleton. At the decision cut the five count values select the five events.

During candidate validation, after the first bit retain the event and that bit, using at most ten states. After the second bit retain the event and the function of the unread mark that gives its candidate-event indicator. For events 0, 1, 2, 3, 4, respectively, the possible functions are

$$\{0, 1\}, \quad \{0, 1 - w, 1\}, \quad \{0, 1\}, \quad \{0, w, 1\}, \quad \{0, 1\}.$$

There are twelve event-function pairs. After the candidate mark, retain the event and its now known candidate indicator $c_0 \in \{0, 1\}$, using at most ten states.

During target validation, after the first bit discard the event name and retain c_0 and its residual function of the second bit and mark. These residual functions belong to

$$\{0, x_2, (1 - x_2)(1 - w), 1 - x_2, x_2 w\}.$$

This uses at most ten states. After the second bit, the residual *difference* as a function of the unread mark belongs to

$$\{-1, 0, 1, w, 1 - w, w - 1, -w\},$$

so seven states suffice. The mark update gives the three possible outputs $-1, 0, 1$. Each transition depends only on its retained residual and the current report. Thus it is a legal common observable row, not a family of independently optimized hidden-history maps.

For completeness, equality of two residual functions implies equality of their restrictions after either next bit, so identifying equal residuals is closed under every subsequent update. The supplied transition tables are obtained by these restrictions and can be reconstructed directly from (35). Enumerating residuals is a convenience for reproducing the tables, not the proof of their sufficiency. Finally, disjointly tag the states at the thirteen cuts by their phase and use terminal self-loops. The sum of the displayed widths is 75. This implementation charges the clock; it does not assert that 75 is optimal. $\square$

Proof of Theorem 10.1. Lemma 10.2 excludes $N \leq 1$ for a strict score greater than $c \geq u_1 + \beta$, for every architecture and every memory budget. Lemmas 10.3 and 10.4 supply a two-preparation architecture of peak twelve with score at least $m_2 - \beta > c$. Larger width budgets can use the same machine. $\square$

Corollary 10.5 (The positive-noise collision family). *For the original marked collision targets with $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$, the local microscopic estimate $\text{TV}(Q_{d,\sigma}, Q_*) < 1/300$ implies*

$$N_{2/5}^{\text{phys}}(W, Q_{d,\sigma}) = 2 \quad (W \geq 12).$$

There are four preparations in the complete audit: two candidate training, one candidate validation, and one target validation.

Proof. Apply Theorem 10.1. The lower-end margin is the exact rational number

$$\frac{2}{5} - u_1 - \frac{1}{300} = \frac{78269}{30720000} > 0.$$

For the other endpoint, $\sqrt{3} < 7/4$ implies $m_2 > 27/64$ and $27/64 - 1/300 > 2/5$. The physical total-variation estimate, including the actual mark and positive sensor noise, is proved in Theorem 9.3; the current deduction does not replace those hypotheses by a zero-noise limit. $\square$

Remark 10.6 (The part of the frontier determined here). The deterministic theorem evaluates the minimum preparation coordinate for $W \geq 12$. Theorem 11.1 and Corollary 11.2 below additionally determine the ideal maximum score with two preparations on this width range. The least peak width at $N = 2$ and the full boundary for $3 \leq W \leq 11$ remain separate questions. The earlier exact decision-width result $\kappa_{2/5}^{\text{rows}} = 3$ is unchanged: it allows longer training, whereas this shortest audit uses five decision states. The older points (399, 40602), (320000, 1604) and

$(12800 \cdot 2^{400}, 12)$ remain valid constructions with decision width three. They are dominated by $(2, 12)$ in the projection onto training preparations and peak width, but not necessarily in a resource order that also constrains the decision alphabet separately.

Corollary 10.7 (The enlarged physical preparation segment). *With the full architecture of Definition 9.1,*

$$N_c^{\text{phys}}(W, Q) = 2 \quad \text{if} \quad W \geq 12, \quad \text{TV}(Q, Q_*) \leq \beta, \quad u_{\text{ex}} + \beta \leq c < m_2 - \beta.$$

Here $u_{\text{ex}} = (85 - 7\sqrt{73})/64$ is the exact one-preparation all-on value, rather than the older rational upper bound.

Proof. Restrict the adversary to the common-garbling subfamily in Definition 9.1. Theorem 5.2 and Corollary 8.2 bound every at-most-one-preparation auditor, even with unlimited memory, by $u_{\text{ex}} + \beta$. The same deterministic two-preparation machine in Lemmas 10.3 and 10.4 has score at least $m_2 - \beta$ against the full private candidate family. These inequalities prove the assertion. The four-preparation total, peak twelve and positive-noise specialization are unchanged. $\square$

11. THE EXACT TWO-PREPARATION MARKED SADDLE

The one-preparation saddle has an exact successor. A new zero coefficient of the least-favorable predictive measure makes it necessary to randomize at the two extreme training counts. This randomization changes the value, but does not enlarge the alphabet of frozen validation events.

Let $b(p, q)$ and Q_* be as in Section 10. Let r be the unique root in $(1/3, 7/20)$ of

$$(37) \quad 2r^3 + 25r^2 - 3 = 0,$$

and define

$$(38) \quad \tau = \frac{6(-r^2 + 4r - 1)}{r(3r + 25)}, \quad v_2 = \frac{-r^3 + 6r^2 - 3r + 14}{32}, \quad p_{\pm} = \frac{1 \pm \sqrt{r}}{2}.$$

The polynomial in (37) is strictly increasing on the positive half-line and has opposite signs at the indicated endpoints. These observations prove the asserted existence, uniqueness and isolating interval. That interval also gives $0 < \tau < 1$.

Theorem 11.1 (Exact two-preparation value). *For the full marked-word response class of (8),*

$$U_2(b, Q_*) = v_2.$$

A least-favorable prior gives mass $1/2$ to each of (p_-, p_-) and (p_+, p_+) . An optimal response discards the two training marks, retains the total number S of ones in the four training report bits, and uses

$$(39) \quad \begin{array}{c|cccc} S & 000 & 001 & 110 & 111 \\ \hline 0 & \tau & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & \tau \end{array}$$

with zero response at off-diagonal validation atoms. Numerically, $r = 0.3417694155\dots$, $\tau = 0.1688236447\dots$ and $v_2 = 0.4261127787\dots$. The exact value is determined by (37)–(38), not by these decimals.

Proof. Write $a = (1 - p)(1 - q)$, $c = pq$, $u = (p + q - 1)^2$ and $v = (p - q)^2$. Both u and v belong to $[0, 1]$. The score of the deterministic five-event selector is the polynomial $H(u, v)$ in (36). Only counts zero and four are changed by (39). Their probabilities are a^2 and c^2 . Consequently the new score is

$$G(u, v) = H(u, v) + \tau J(u, v), \quad J = a^2(5/16 - a/2) + c^2(5/16 - c/2).$$

Expansion gives the polynomial identity

$$(40) \quad \begin{aligned} J(u, v) = & -\frac{u^3}{64} + \frac{3u^2v}{64} - \frac{25u^2}{128} - \frac{3uv^2}{64} + \frac{13uv}{64} \\ & + \frac{v^3}{64} - \frac{v^2}{128} - \frac{v}{32} + \frac{3}{128}. \end{aligned}$$

Thus

$$H(u, 0) = \frac{-u^3 + 6u^2 - 3u + 14}{32}, \quad J(u, 0) = \frac{3 - 25u^2 - 2u^3}{128}.$$

Equation (37) says $J(r, 0) = 0$, and the choice of τ says $\partial_u G(r, 0) = 0$. Direct factorization, using these two identities, yields

$$(41) \quad G(u, 0) - v_2 = \frac{(u - r)^2 \{75 + 12r + 6u - 37r^2 - 74ru\}}{64r(3r + 25)}.$$

The expression in braces is strictly positive for $0 \leq u \leq 1$: it is at least $75 - 37(7/20)^2 - 74(7/20) > 0$. Moreover,

$$\partial_v H = \frac{3(u - v)^2 - 8u + 4v + 15}{32} \geq \frac{7}{32}, \quad \partial_v J = \frac{3(u - v)^2 + 13u - v - 2}{64} \geq -\frac{3}{64}.$$

Because $0 < \tau < 1$, $\partial_v G \geq 11/64$ on the unit square. It follows that $G(u, v) \geq G(u, 0) \geq v_2$, with equality precisely when $v = 0$ and $u = r$. Both equality points are feasible and are the two parameters in the asserted prior. This proves the global lower bound.

It remains to prove an upper bound against every response, not only those depending on S . Under the asserted prior the four training report bits are independent Bernoulli variables with common parameter p_- or p_+ . Conditional on $S = s$, their arrangements and the two independent training marks have parameter-independent distributions. The signed predictive coefficients can therefore be grouped by S without restricting any response coordinate. For the four displayed validation atoms put

$$C_{s,j} = \frac{1}{2} \sum_{p \in \{p_-, p_+\}} \binom{4}{s} p^s (1 - p)^{4-s} \{Q_{*,j} - b_j(p, p)\}.$$

The first three rows of $256C$ are

$$\begin{aligned} s = 0 : & \quad (-(2r^3 + 25r^2 - 3), -(2r^3 + 27r^2 + 12r - 1), \\ & \quad -(2r^3 - 5r^2 - 20r - 1), -(2r^3 - 7r^2 - 32r - 3)), \\ s = 1 : & \quad 4(r - 1)(2r^2 + 7r - 3, 2r^2 + 9r - 1, \\ & \quad 2r^2 - 7r - 1, 2r^2 - 9r - 3), \\ s = 2 : & \quad -6(r - 1)^2(2r - 3, 2r - 1, 2r - 1, 2r - 3). \end{aligned}$$

Rows three and four are the reversals of rows one and zero, respectively. Their signs, using $1/3 < r < 7/20$, are

$$(0, -, +, +), \quad (+, -, +, +), \quad (+, +, +, +), \quad (+, +, -, +), \quad (+, +, -, 0).$$

Every off-diagonal validation coefficient is nonpositive because its target mass is zero. Hence (39) maximizes every signed coefficient: it is one at positive coefficients, zero at negative ones, and arbitrary at the two zeros. The lower-bound equality at both prior points shows that this Bayes maximum is exactly v_2 . The posterior-predictive dual, Theorem 5.1, now gives the matching upper bound for the full response class. $\square$

Corollary 11.2 (Exact physical score on the peak-twelve segment). *Use the atomic stochastic-row convention and the original marked interface of Definition 9.1. At two candidate training preparations, every peak budget $W \geq 12$ has exact ideal worst-case score v_2 . For a target Q with $\text{TV}(Q, Q_*) \leq \beta$, the corresponding value lies in $[v_2 - \beta, v_2 + \beta]$. In particular, with $u_{\text{ex}} = (85 - 7\sqrt{73})/64$,*

$$(42) \quad N_c^{\text{phys}}(W, Q) = 2 \quad \text{if} \quad W \geq 12, \quad u_{\text{ex}} + \beta \leq c < v_2 - \beta.$$

Proof. Let $h_0, \dots, h_4$ be the five deterministic events in (35). At count zero select h_1 with probability τ and h_0 otherwise; at count four select h_3 with probability τ and h_4 otherwise. At the other counts retain the indicated event. Discard the count and retain the selected event for both validations. This realizes (39); the event alphabet remains five. The full charged profile, now including the separate event-selection update, is

$$(1, 2, 3, 3, 4, 5, 5, 5, 10, 12, 10, 10, 7, 3).$$

The common residual-function compiler in Lemma 10.4 applies unchanged. In particular, no persistent random selector is outside the register. The all-on law for every private candidate is $b(p, q)$, so Theorem 11.1 proves the constructive lower bound. For the converse restrict to the admissible common-garbling product subfamily and apply Theorem 5.2 followed by Theorem 11.1. This bounds all architectures, including those using feedback, stopping and more memory. Target perturbation changes each frozen-event expectation by at most β . Finally the exact one-preparation theorem excludes a strict score above c at $N \leq 1$, whereas the displayed machine achieves it at $N = 2$ under (42). $\square$

The root r is larger than $2 - \sqrt{3}$. The old diagonal polynomial $H(u, 0)$ is strictly increasing between these points, whence $v_2 > m_2$. The deterministic theorem, including its zero-random-bit implementation, remains valid; the new exact score uses an additional stochastic row.

Proposition 11.3 (Finite precision for the new row). *Replacing τ by a dyadic $\tau_b = k/2^b \in [0, 1]$ with $|\tau_b - \tau| \leq 2^{-b}$ loses at most $(5/16)2^{-b}$ in uniform score. A clocked fair-bit implementation uses b extra random bits and at most $\max\{12, 6(b+1) + 5\}$ persistent states, including their phase, in addition to the unchanged preparation count.*

Proof. At each affected training word the response changes at a single atom; $|5/16 - a/2|$ and $|5/16 - c/2|$ are at most $5/16$. The two affected word probabilities sum to at most one. This gives the score estimate. To implement the dyadic coin, compare b fresh fair bits lexicographically with the fixed b -bit integer k . Retain the origin (count zero or four), one of the three comparison flags, and a phase in $\{0, \dots, b\}$. Once the comparison ends retain its chosen one of five events. Unaffected counts can wait in their event states. These states total at most $6(b+1) + 5$; the other cuts have peak twelve. The cases $k = 0, 2^b$ are deterministic. Exact atomic randomization and a finite fair-bit implementation are therefore distinct resource claims. $\square$

12. A CONTINUOUS TWO-PREPARATION MARKED FAMILY

The preceding saddle persists on an explicit interval of target laws, with exact rather than perturbative values. Use the target Q_γ from Section 8 and write $U_2(\gamma) = U_2(b, Q_\gamma)$.

Theorem 12.1 (A two-preparation saddle family). *For $6/25 \leq \gamma \leq 13/50$, let r_γ be the unique root in $(1/3, 7/20)$ of*

$$(43) \quad f_\gamma(r) = r^3 + (13 - 2\gamma)r^2 + (3 - 12\gamma)r - (1 + 2\gamma) = 0.$$

Set

$$D_\gamma(r) = 3r^2 + (26 - 4\gamma)r + 3 - 12\gamma, \quad t_\gamma = -\frac{2\{3r_\gamma^2 + (8\gamma - 14)r_\gamma + 3\}}{D_\gamma(r_\gamma)}.$$

Then $0 < t_\gamma < 1$, and

$$(44) \quad U_2(\gamma) = \frac{-r_\gamma^3 + (7 - 4\gamma)r_\gamma^2 - 3r_\gamma + 13 + 4\gamma}{32}.$$

The response (39), with $\tau = t_\gamma$, is optimal. A least-favorable prior gives equal masses to the two diagonal parameters $p = q = (1 \pm \sqrt{r_\gamma})/2$. The response has the same five-event, peak-twelve atomic realization. At $\gamma = 1/4$ these formulas reduce to Theorem 11.1.

Proof. For brevity write $r = r_\gamma$, $D = D_\gamma(r)$ and $t = t_\gamma$. The derivative D_γ is positive on the rectangle $\gamma \in [6/25, 13/50]$, $r \in [1/3, 7/20]$. For example its lower bound is $1/3 + (26 - 52/50)/3 - 3/25 > 8$. The polynomial f_γ decreases with γ at a fixed positive argument. Direct rational evaluations give $f_{6/25}(1/3) < 0$ and $f_{13/50}(7/20) > 0$. Monotonicity therefore gives the root and its uniqueness in this interval. The numerator of t is $-16\gamma r - 6r^2 + 28r - 6$. It is bounded below by $28/3 - 6 - 16(13/50)(7/20) - 6(7/20)^2 > 0$, and above by $28(7/20) - 6 < 4 < D$. Thus $0 < t < 1$.

Let H, J be the polynomials from (36) and (40). Changing the target masses linearly in γ gives

$$\begin{aligned} H_\gamma(u, v) &= H(u, v) + (\gamma - 1/4) \frac{1 - (u - v)^2}{8}, \\ J_\gamma(u, v) &= J(u, v) + (\gamma - 1/4) \frac{(u - v)^2 + 6u - 2v + 1}{32}. \end{aligned}$$

The response's full-parameter score is $G_\gamma = H_\gamma + tJ_\gamma$. One has $J_\gamma(r, 0) = -f_\gamma(r)/64 = 0$ and $\partial_u G_\gamma(r, 0) = 0$. Algebraic division gives

$$(45) \quad G_\gamma(u, 0) - H_\gamma(r, 0) = \frac{(u - r)^2 A_\gamma(r, u)}{16D},$$

where

$$\begin{aligned} A_\gamma(r, u) &= 24\gamma^2 + (3\gamma - 10)r^2 + (6\gamma - 20)ru \\ &\quad + 12\gamma r + 6\gamma u - 51\gamma + 30. \end{aligned}$$

For $0 \leq u \leq 1$ this is at least $30 - 10(7/20)^2 - 20(7/20) - 51(13/50) > 0$. The transverse derivatives satisfy

$$\begin{aligned} 32\partial_v H_\gamma &= 3(u - v)^2 + (8\gamma - 10)u + (6 - 8\gamma)v + 15 \geq 5, \\ 64\partial_v J_\gamma &= 3(u - v)^2 + (14 - 4\gamma)u + (4\gamma - 2)v - 4\gamma - 1 \geq -3 \end{aligned}$$

on $0 \leq u, v \leq 1$. Consequently $\partial_v G_\gamma \geq 7/64$. Equation (45) proves a lower bound $H_\gamma(r, 0)$, with equality precisely at the proposed prior points. This quantity is the right side of (44).

For the upper bound group the full training words by S , as in the preceding theorem. Put $L(r) = r^2 + 6r + 1$. Using $f_\gamma(r) = 0$, the first three rows of $256C$ are

$$\begin{aligned} 0 : & (0, -8\gamma L(r), 32r(r+1) - 8\gamma L(r), 32r(r+1)), \\ 1 : & 8(r-1)(r^2 + 4r - 1 - 2\gamma(r+1), \\ & \quad r^2 + 4r - 1 + 2\gamma(r+1), \\ & \quad r^2 - 4r - 1 + 2\gamma(r+1), r^2 - 4r - 1 - 2\gamma(r+1)), \\ 2 : & -12(r-1)^2(r-1-2\gamma, r-1+2\gamma, r-1+2\gamma, r-1-2\gamma). \end{aligned}$$

The other rows are their reversals. The sign pattern is again

$$(0, -, +, +), \quad (+, -, +, +), \quad (+, +, +, +), \quad (+, +, -, +), \quad (+, +, -, 0).$$

Here are explicit interval checks for the potentially small margins. In row zero the third coordinate is bounded below by $32(1/3)(4/3) - 8(13/50)((7/20)^2 + 6(7/20) + 1) > 0$. The first parenthesis of row one is at most $(7/20)^2 + 4(7/20) - 1 - 2(6/25)(4/3) < 0$; the third is at most $(7/20)^2 - 4/3 - 1 + 2(13/50)(27/20) < 0$. The second parenthesis is positive already from $r > 1/3$ and $\gamma > 0$, and the fourth is negative. Finally $r - 1 + 2\gamma \leq -13/100 < 0$. All off-diagonal validation coefficients are nonpositive. Thus the stated response is a Bayes best response for this prior, including its two zero coefficients. Its score at both supporting parameters equals the lower bound. The full posterior-predictive dual proves (44). $\square$

This interval is a uniform certificate region for the displayed saddle, not a claim that the same masks remain optimal for every mark bias. The theorem supplies a continuous family at genuinely growing training history $N = 2$; it does not infer a recurrence for every N from the one-preparation formulas.

13. AN EXACT PREPARATION-DECISION-MEMORY LAW

A second solvable family separates a genuine joint resource law from an implementation bound. Here the memory coordinate is the number of states at the training-validation cut. It is not identified with the bit-level peak width of the collision audit.

Fix $m = 2^d$, $0 < \alpha < 1$, and an unknown parameter $\theta \in \{1, \dots, m\}$. A fresh training preparation reports θ with probability α and an erasure $\perp$ otherwise. Conditional on θ , the preparations are independent. Candidate validation has law δ_θ and target validation is uniform on $\{1, \dots, m\}$. At the decision cut the auditor retains at most $K = 2^k$ states, $0 \leq k \leq d$. Its subsequent response may depend only on that state, current validation observations and fresh independent randomness. In particular, a calibration table, remembered seed or stopping time outside the retained state is not available. Upstream and downstream memory are otherwise unrestricted for the converse. Let $V_N^{\text{er}}(K)$ be the best worst-case expected signed validation score with at most N training preparations.

Theorem 13.1 (Exact joint law). *For every $N \geq 0$ and $K = 2^k \leq m$,*

$$(46) \quad V_N^{\text{er}}(K) = (1 - (1 - \alpha)^N) \left(1 - \frac{1}{K}\right).$$

The converse covers stochastic encoders and arbitrary legal stopping. The value is attained by a clocked machine with K training states. With atomic preparation and validation outcomes, the complete audit has peak at most $\max\{2K, 3\}$ and uses at most $N + 2$ preparations.

Proof. Write $q = (1 - \alpha)^N$ and pre-generate N potential training reports. A stopped auditor ignores the unused reports; this does not disclose extra information to it. On the event that all

N reports are erasures, its retained-state law $r = (r_j)_{j=1}^K$ is independent of θ . After padding any smaller register with unused states, its channel to the decision register has the form

$$(47) \quad C(j \mid \theta) = qr_j + (1 - q)S(j \mid \theta)$$

for a stochastic channel S . This identity remains valid with randomized stopping and randomized state updates.

Use the uniform prior on parameters as an upper certificate. The part qr_j has zero expected score for every response: averaging δ_θ under this prior gives the target itself. For the other part, optimizing over all responses gives a convex function of S , a sum of positive parts of linear functions. Every stochastic channel is a convex combination of deterministic channels $j = g(\theta)$; therefore its value is at most the largest deterministic-channel value. This convexification is used only to bound a channel. It does not presuppose that a mixture of width- K machines is realizable at width K .

If the fibers of g have sizes $n_1, \dots, n_K$, a nonempty fiber has posterior uniform on that fiber, so its total variation from the uniform target is $1 - n_j/m$. The prior-averaged optimal score is

$$\sum_j \frac{n_j}{m} \left(1 - \frac{n_j}{m}\right) = 1 - \sum_j (n_j/m)^2 \leq 1 - 1/K.$$

Empty fibers contribute zero. Cauchy–Schwarz proves the inequality. This is an upper bound on the average score and hence on its minimum over fixed parameters. Multiplying by $1 - q$ proves the upper bound in (46).

For attainment, partition the parameter alphabet into K equal groups and let $g(\theta)$ denote its group. Initialize the register uniformly on the K group labels. On an erasure leave it unchanged; on a revealed parameter overwrite it by $g(\theta)$. No extra “revelation occurred” flag is needed. At the decision cut its law is $q \text{Unif}[K] + (1 - q)\delta_{g(\theta)}$. For register j , freeze the event which is the complement of group j . On a revelation its score is $1 - 1/K$. Under a uniformly random initial label, the mean score is zero for each fixed θ : one of the K labels has score $1 - 1/K$ and the others have score $-1/K$. This proves the exact worst-case value.

The k initialization bits can be accumulated in the K -state register; all earlier prefix counts are at most K . Thus no persistent seed lies outside it. Candidate validation retains the group label and its event indicator, using at most $2K$ states. Target validation produces one of the three values $-1, 0, 1$. This proves the atomic-outcome peak bound. The controller uses a declared external schedule; an autonomous stopping implementation must additionally retain its phase. A phase-tagged implementation is finite, but its cost is not included in K . $\square$

Corollary 13.2 (Thresholds and matching resource scales). *For $K \geq 2$ and $0 \leq c < 1 - 1/K$, the least integer number of training preparations giving a score strictly greater than c is*

$$(48) \quad N_c^{\text{er}}(K) = \left\lceil \frac{\log(1 - c/(1 - 1/K))}{\log(1 - \alpha)} \right\rceil + 1.$$

If $c \geq 1 - 1/K$, no finite N suffices. For $0 < \varepsilon < 1$, achieving $V_N^{\text{er}}(K) \geq 1 - \varepsilon$ requires

$$K \geq \varepsilon^{-1}, \quad N \geq \frac{\log(1/\varepsilon)}{-\log(1 - \alpha)}.$$

Conversely, whenever an allowed $K \leq m$ satisfies $K \geq 2/\varepsilon$, the choice $N \geq \log(2/\varepsilon)/[-\log(1 - \alpha)]$ suffices.

Proof. The strict threshold is equivalent to $(1 - \alpha)^N < 1 - c/(1 - 1/K)$. Since the denominator logarithm is negative, the least integer satisfying this strict inequality is (48), including equality

cases of the real quotient. Also

$$1 - V_N^{\text{er}}(K) = q + (1 - q)/K.$$

This is at least q and at least $1/K$, and is at most $q + 1/K$. These three inequalities prove the necessary and sufficient bounds. $\square$

The exact law concerns an entire (N, K, α, m) family, not a single finite diagnostic. Its mechanism is a parameter-independent erasure component followed by the convex geometry of the decision channel. It demonstrates how a preparation obstruction and a memory obstruction can be matched in one theorem. It does not identify the least physical peak width at two collision preparations. Encoding the m -symbol reports bit by bit also introduces cuts not present in this atomic model; the physical bit-level accounting is given separately.

14. INTEGER MEMORY AND THE FAILURE OF A NAIVE PARTITION FORMULA

The divisibility assumptions in Theorem 13.1 can be removed, but a balanced-partition Bayes bound must not be mistaken for a minimax equality when a persistent random partition is not free. This distinction leads to another exact finite-resource transition.

Keep the revelation interface, but allow any integers $m \geq 2$ and $1 \leq K \leq m$. Put $\rho = 1 - (1 - \alpha)^N$, and denote the genuine K -label minimax value by $V_{N,m}^{\text{er}}(K)$. Write $m = Kq + s$, $0 \leq s < K$, and set

$$(49) \quad B_{m,K} = 1 - \frac{(K - s)q^2 + s(q + 1)^2}{m^2}.$$

Here and below all-erasure initialization randomness is a state initialization, not an unrecorded seed used again at validation.

Proposition 14.1 (Arbitrary integer alphabet bounds and low-reveal equality). *The optimal uniform-prior Bayes score is $\rho B_{m,K}$. The minimax value satisfies*

$$(50) \quad V_{N,m}^{\text{er}}(K) \leq \min \left\{ \rho B_{m,K}, 1 - \frac{\lceil m/K \rceil}{m} \right\}.$$

If $\rho \leq m/(m + s)$, equality with the first term holds. At perfect revelation $\rho = 1$ the exact value is the second term. In particular, when $s > 0$ the uniform-prior Bayes optimum at perfect revelation is strictly larger than the minimax value.

Proof. The all-erasure decomposition (47) holds without any divisibility condition. Its parameter-independent part has zero uniform-prior score. Optimizing the other part over responses gives a convex function of its stochastic channel. A deterministic channel with fiber sizes n_j has Bayes score $1 - \sum_j (n_j/m)^2$. If two nonempty fiber sizes differ by at least two, moving one index from the larger to the smaller strictly decreases the sum of squares. Allowing empty fibers and repeating gives sizes q and $q + 1$, proving the Bayes bound. It is attained by this partition and its complementary events, with any parameter-independent initialization. This proves the first upper bound as well as the asserted Bayes optimum.

For perfect revelation, fix any K response vectors. For each known parameter choose a response maximizing its score; this deterministic assignment cannot decrease any parameter's payoff, even when the original encoder was randomized. Some assigned fiber has size at least $\lceil m/K \rceil$. Averaging its parameters, the score of their common response is at most the total variation between the uniform target and the uniform law on that fiber, namely $1 - \lceil m/K \rceil/m$. The minimum is no larger. Balanced fibers with their complementary events attain the resulting upper bound. A perfectly informed encoder can simulate any less informative encoder, so this is an upper bound at every ρ .

For the low-reveal equality let balanced fibers have weights $w_j = n_j/m$. When $\rho < 1$, initialize their label register with probabilities

$$(51) \quad r_j = \frac{\rho w_j + (1 - 2\rho)/K}{1 - \rho}.$$

These sum to one, and are nonnegative precisely throughout the stated range: the smallest numerator is $\rho q/m + (1 - 2\rho)/K$. On a revealed parameter overwrite the label by its fiber; on erasure do nothing. No revelation flag is retained. For a parameter in fiber i , the complementary-event score is

$$\begin{aligned} g_i &= \rho(1 - w_i) + (1 - \rho) \left(r_i - \sum_j r_j w_j \right) \\ &= \rho \left(1 - \sum_j w_j^2 \right) = \rho B_{m,K}. \end{aligned}$$

This matches the upper bound uniformly in the fixed parameter. If $\rho = 1$ and $s = 0$, the deterministic revealed label already attains that value, so the singular denominator in (51) is not used. Finally, with unequal balanced fibers the weighted average of their sizes is strictly smaller than the largest size; equivalently $B_{m,K} > 1 - \lceil m/K \rceil/m$. This proves the last assertion. $\square$

Theorem 14.2 (The complete two-label revelation law). *For every $m \geq 2$, every $N \geq 0$, and $0 < \alpha < 1$, set $w = \lfloor m/2 \rfloor/m$ and $b = \lceil m/2 \rceil/m$. Then*

$$(52) \quad V_{N,m}^{\text{er}}(2) = \min\{2wb\rho, w\}, \quad \rho = 1 - (1 - \alpha)^N.$$

The same formula holds at the limiting endpoints $\rho = 0, 1$. For odd m it changes regime at $\rho = m/(m + 1)$; for even m it reduces to $\rho/2$. The attaining machine uses two training labels, a frozen two-event alphabet, and atomic peak at most four.

Proof. Proposition 14.1 gives the upper bound and its attainment for $\rho \leq 1/(2b)$. Above this threshold, partition into fibers of weights w, b and initialize deterministically in the larger fiber's label. Continue to overwrite on a reveal and leave the state unchanged on an erasure. A parameter in the larger fiber has score w ; a parameter in the smaller fiber has score $(2\rho - 1)b$. The latter is at least w exactly when $\rho \geq 1/(2b)$. This attains the second upper bound. For even m the threshold is one, so the low-reveal construction suffices. The endpoint $\rho = 0$ may instead freeze a constant event. Candidate validation retains the selected label and its indicator, with at most four states; target validation produces the difference. Initialization is atomic randomization when its probabilities are not dyadic. $\square$

For example, $m = 3, K = 2$ gives $V_{N,3}^{\text{er}}(2) = \min\{4\rho/9, 1/3\}$, rather than $4\rho/9$ for all ρ . The balanced-integer expression is therefore not the unqualified replacement of the divisible minimax formula. For $K \geq 3$ outside the low-reveal range, (50) remains a bound rather than an asserted equality. The compatible dynamic formulation of Section 7 retains the exact optimization in that regime. These results strengthen the revelation example without identifying its decision-cut coordinate with the collision model's bit-level peak.

15. QUANTITATIVE CERTIFICATES FOR A FINITE COMPATIBLE ARCHITECTURE

We return to an arbitrary finite architecture, now with a finite parameter set. It can have hidden states, controlled experiments, more than two hypotheses and registers of any size. Enumerate its distinct stochastic rows by $r = 1, \dots, R$ and write $q_r \in \Delta_{s_r}$. A reused autonomous row appears only once in this product. Assume each execution uses at most H controller rows,

and uses row r at most d_r times. Delete unused rows and put $S = \max_r s_r$. Fixed environmental kernels are not controller rows.

For an integer $m \geq 1$, intersect Δ_{s_r} with the coordinate boxes of side $1/m$. The resulting nonempty closed polytopes cover the simplex, with overlaps permitted at their boundaries. Write C for a product of such row cells and choose a vertex q_r^C in each factor. A row-cell vertex has at most one coordinate not fixed at a box boundary; consequently it has at most $v_r = s_r 2^{s_r-1}$ vertices, all on the $1/m$ grid. There are at most $m^{\sum_r s_r}$ product cells.

Write the vertices of row cell C_r as $v_{r,j}$ and parametrize $q_r = \sum_j \lambda_{r,j} v_{r,j}$ with λ_r in a simplex. A reward polynomial has degree at most d_r in block λ_r . Use the homogeneous simplex Bernstein basis of degree d_r in that block, obtaining

$$(53) \quad g_\theta(q(\lambda)) = \sum_\alpha b_{\theta,C,\alpha} B_\alpha(\lambda), \quad B_\alpha \geq 0, \quad \sum_\alpha B_\alpha = 1.$$

Degree elevation includes the factors $\sum_j \lambda_{r,j} = 1$. No independence between different visits of a shared row is imposed on its unknown row vector.

Lemma 15.1 (Causal interpretation of the coefficients). *Every coefficient in (53) satisfies*

$$|b_{\theta,C,\alpha} - g_\theta(q^C)| \leq HS/m.$$

Proof. For each row r , independently prepare an ordered list of d_r vertex labels, sampled independently with law λ_r . At its k th visit, use the stochastic row given by the k th vertex label in that list. Before conditioning, the next label is independent of the previous execution, so this experiment has exactly the row q_r at every visit and expected reward $g_\theta(q(\lambda))$. Unused labels may be sampled after termination without altering the reward.

Condition on the counts of every vertex label in every list. The probability of those counts is $B_\alpha(\lambda)$. Conditional on the counts, each list is a uniformly ordered multiset. Its expected reward is independent of λ and is therefore the corresponding Bernstein coefficient. This also proves (53) directly.

Each row used by this conditioned experiment is a vertex of C_r or a mixture of its remaining vertices. Every two vectors in C_r have coordinate differences at most $1/m$, hence total variation at most $s_r/(2m) \leq S/(2m)$. Couple the conditioned experiment to the one using q^C , using the same environmental randomness while their executions agree. At each actual controller update the probability of a first row-level disagreement is at most $S/(2m)$. There are at most H such updates on the agreeing path. The terminal laws consequently differ in total variation by at most $HS/(2m)$. A reward in $[-1, 1]$ has expectation difference at most twice that quantity. This proves the estimate. The conditioned urn is used to analyze a coefficient; it is not declared to be an admissible free-memory controller. $\square$

Theorem 15.2 (Explicit subdivision certificate). *Define*

$$\ell_C = \min_\theta g_\theta(q^C), \quad u_C = \min_\theta \max_\alpha b_{\theta,C,\alpha}, \quad L_m = \max_C \ell_C, \quad U_m = \max_C u_C.$$

For the exact all-stochastic value of the specified architecture,

$$(54) \quad L_m \leq V(A) \leq U_m \leq L_m + HS/m.$$

On each cell, a witnessing parameter θ_C supplies the nonnegative coefficient certificate

$$u_C - g_{\theta_C}(q(\lambda)) = \sum_\alpha (u_C - b_{\theta_C,C,\alpha}) B_\alpha(\lambda).$$

Its degree in row block r is d_r , and its total degree is at most $\sum_r d_r$. At most $\prod_r \binom{d_r + v_r - 1}{v_r - 1}$ coefficients per parameter and cell are needed. If the primitive data are rational, all coefficients

and bounds can be checked by rational arithmetic. Taking $m \geq HS/\varepsilon$ gives a certified interval of length at most ε .

Proof. The point q^C is feasible, so ℓ_C is a lower bound on the value restricted to C . For any $q \in C$, choose a convex representation of its rows. Equation (53) implies

$$\min_{\theta} g_{\theta}(q) \leq \min_{\theta} \max_{\alpha} b_{\theta, C, \alpha} = u_C.$$

Choose a parameter attaining the minimum defining ℓ_C and apply Lemma 15.1; this gives $u_C \leq \ell_C + HS/m$. Taking the maximum over the finite cover proves (54). Choose θ_C attaining the definition of u_C ; all displayed coefficients of its certificate are nonnegative. The degree and count statements follow from the product Bernstein basis. Cell vertices are rational grid points, and a finite execution sums finite products, proving the rational statement. $\square$

Remark 15.3. The number of cells and coefficients can be enormous. Neither polynomial running time nor a uniform degree bound for the original sum-of-squares hierarchy follows. Refinement changes the domain cover, not merely a single global SOS degree. For a finite collection of architectures obeying a specified budget, maxima of these intervals give outer intervals; an unbounded architecture class needs a separate uniform argument such as Theorem 5.2. For compact parameter families, an effective net and a verified modulus for the primitive kernels are additional inputs, not consequences of compactness alone.

The resource estimate also makes row precision explicit. With $m = 2^b$, the lower witnesses have dyadic rows and lose at most $HS2^{-b}$ in value. Thus increasing row precision provides a quantified approximation for a fixed architecture, while its random-bit implementation still has to be charged. The physical theorem avoids this cost altogether because its rows are deterministic.

16. AUTONOMOUS NOISY TESTING WITH A PRECISION AND TIME LEDGER

This section supplies a substantive autonomous comparison rather than merely identifying shared rows. The stationary optimum is classical [6]; the proof and explicit finite-time implementation are recorded to make the resource conventions unambiguous. They are not used as novelty evidence for the physical theorem.

Let reports be independent, with law $\text{Ber}(p)$ under hypothesis $-$ and $\text{Ber}(q)$ under hypothesis $+$, where $0 < p < 1/2$, $q = 1 - p$, and $r = q/p$. A W -state autonomous machine has two time-independent stochastic transition matrices T_0, T_1 . Assume their induced chains $P_- = qT_0 + pT_1$ and $P_+ = pT_0 + qT_1$ are irreducible. A final decision uses only the state, with optional fresh terminal randomization.

Lemma 16.1 (Likelihood-range bound). *If equivalent probability laws ν_+, ν_- on a finite set satisfy*

$$\frac{\max_i \nu_+(i)/\nu_-(i)}{\min_i \nu_+(i)/\nu_-(i)} \leq R,$$

then $\text{TV}(\nu_+, \nu_-) \leq (\sqrt{R} - 1)/(\sqrt{R} + 1)$.

Proof. Write $L = d\nu_+/d\nu_-$, $a = \min L$ and $b = \max L$. Since $\mathbb{E}_{\nu_-} L = 1$, one has $a \leq 1 \leq b$. The chord bound for $(L - 1)_+$ gives

$$\text{TV}(\nu_+, \nu_-) = \mathbb{E}_{\nu_-} (L - 1)_+ \leq \frac{(1 - a)(b - 1)}{b - a}.$$

For fixed b/a , maximizing the last expression gives $a = 1/\sqrt{b/a}$; substitution yields $(\sqrt{b/a} - 1)/(\sqrt{b/a} + 1)$. It is increasing in b/a , proving the assertion. The identical-law case follows directly. $\square$

Theorem 16.2 (Autonomous stationary bound and an explicit approximation). *Put $D = W - 1$, $W \geq 2$. In the irreducible autonomous class the infimum stationary minimax error is*

$$(55) \quad e_W = \frac{1}{1 + r^D}.$$

For every $\epsilon = 2^{-b}$, $b \geq 0$, there is a machine whose rows have denominator dividing 2^{b+1} , and whose minimax error at report time n from any initial state is at most

$$(56) \quad e_W + \epsilon(W - 2) + \exp \left\{ - \left(\frac{p\epsilon}{2} \right)^D \left\lfloor \frac{n}{D} \right\rfloor \right\}.$$

The stationary lower bound is not a finite-horizon lower bound.

Proof. Every positive induced transition has its likelihood ratio in $[r^{-1}, r]$, and the supports of the two induced matrices are identical. The Markov chain tree formula writes each stationary weight as a sum of products over directed spanning trees rooted at that state. Each product has D transitions; its ratio lies in $[r^{-D}, r^D]$. The ratio of the two unnormalized stationary weights lies in the same interval. Normalizing multiplies all state likelihood ratios by a common number, so their maximum-to-minimum ratio is at most r^{2D} . Lemma 16.1 therefore bounds stationary total variation by $(r^D - 1)/(r^D + 1)$. The equal-prior Bayes error is at least $1/(1 + r^D)$, and a minimax error cannot be smaller than that Bayes error. This is the equal-prior binary specialization of the Hellman–Cover bound.

For the upper approximation, take states $i = 0, \dots, D$, weights $c_0 = c_D = 1$, and $c_i = \epsilon$ at interior states. On report 1, move from i to $i + 1$ with probability $a_i = \frac{1}{2} \min(1, c_{i+1}/c_i)$, if that neighbor exists; otherwise stay. On report 0, move to $i - 1$ with probability $b_i = \frac{1}{2} \min(1, c_{i-1}/c_i)$, if it exists; otherwise stay. All other probability is assigned to staying. Detailed balance gives

$$\pi_+(i) = \frac{c_i r^i}{\sum_j c_j r^j}, \quad \pi_-(i) = \frac{c_i r^{-i}}{\sum_j c_j r^{-j}}.$$

Reflection $i \mapsto D - i$ exchanges these laws. Decide $+$ for $i > D/2$, $-$ for $i < D/2$, and use a fair terminal decision at $i = D/2$ when this state exists. The two errors agree. Conditional on being at an endpoint, the error is exactly e_W . The total interior stationary mass is at most $\epsilon(W - 2)$, since each $r^i \leq r^D$ and the denominator is at least r^D under $+$, with the reflected argument under $-$. This proves the first two terms of (56).

Every existing neighbor transition under either hypothesis has probability at least $p\epsilon/2$, and every self-loop has probability at least $1/2$. From any state, move down to zero and then stay until D steps have elapsed. This yields the common minorization $P_\pm^D(i, \{0\}) \geq \delta = (p\epsilon/2)^D$. Subtracting that common atom gives contraction in total variation by $1 - \delta$ every D steps. Comparison with the stationary law yields $\text{TV}(P_\pm^n(i, \cdot), \pi_\pm) \leq (1 - \delta)^{\lfloor n/D \rfloor} \leq e^{-\delta \lfloor n/D \rfloor}$. A bounded decision changes error by at most this amount, proving (56). Finally, letting n tend to infinity and then ϵ to zero proves that the stationary infimum equals (55). All transition probabilities belong to $\{0, 1, 1/2, \epsilon/2, 1 - 1/2, 1 - \epsilon/2\}$. $\square$

The W states in this theorem refer to an atomic stochastic update after each report. A sequential fair-bit implementation must also retain the report, the current state, a bit counter and a flag. Each nontrivial move probability is $1/2$ or $2^{-(b+1)}$; test for a string of zero bits of the required length, and pad to $b + 1$ bits if a uniform report cycle is required. At most $4W(b + 2)$ states suffice for this implementation, including its bit-phase counter and a ready phase. This conservative bound is an implementation claim, not an optimal precision–state tradeoff. The time index n in (56) is an observation time at which an external analyst reads the autonomous

machine, not a free input clock used by its transitions. The adversarial model of [10] is not covered by the independent-report proof above.

17. PHYSICAL CONSUMERS AND THE FIRST-PRINCIPLES CONNECTION

The physical preparation theorem needs no estimate of the target law as a stored input: the five events are fixed and their uniform validity uses a proved neighborhood of Q_* . This removes calibration memory from this particular consumer without asserting that target acquisition is unnecessary in an arbitrary comparison problem. The general acquired calibration theorem, with its full counter register, remains in the complete companion.

Corollary 17.1 (A raw-interface confidence construction). *Let $m = m_2 - \beta > 0$ and $0 < c < m$. Repeat the full four-preparation audit independently R times and write $\bar{D}_R$ for the mean score. Under every private candidate,*

$$\mathbb{P}(\bar{D}_R \leq c) \leq \exp\{-R(m - c)^2/2\}.$$

Under the null in which the independent candidate and target validation laws agree, with the same event chosen before both validations,

$$\mathbb{P}(\bar{D}_R > c) \leq \exp\{-Rc^2/2\}.$$

A clocked implementation uses $4R$ preparations and at most $12(2R + 1)$ persistent states, with no calibration and no random bits.

Proof. The independent block scores lie in $[-1, 1]$ and have mean at least m under a private candidate. Hoeffding's inequality gives the first bound. Under the null, conditional on the selected event, the two independent validation indicators have the same mean. Each block therefore has mean zero, and the same inequality gives the second bound. Retain the integer sum in $\{-R, \dots, R\}$ along with the current twelve-state block. At a block boundary add its terminal score and reset its register. The deterministic external schedule is part of the clocked convention. $\square$

This is an upper construction on the raw marked interface. Its proof does not transfer a lower bound from a deliberately reduced Bernoulli interface, and no optimal confidence-memory frontier is claimed.

Proposition 17.2 (Target transport without a training multiplier). *For the two-preparation audit, changing only the target from Q to Q' changes its expected score by at most $\text{TV}(Q, Q')$. Its candidate training law, register profile and event selector remain unchanged.*

Proof. Condition on the candidate training word. The expectation difference is $(Q - Q')h_S$, whose absolute value is at most $\text{TV}(Q, Q')$. Average over that unchanged training law. $\square$

In particular, every common-preparation transport estimate verified for the original positive-noise collision family can be inserted here without multiplying it by a long training horizon. If candidate kernels also change, their training and validation defects must be added by a causal coupling; the target-only statement does not remove those terms.

The first-principles route is now concrete. The positive experimental kernels and actual reset preparations give the likelihoods in (9); these generate posterior continuation states. The validation-test quotient gives the positive parts in the statistical dual. A prior then supplies an architecture-uniform impossibility certificate. On the constructive side, the five-event decision is compiled by taking its future residual functions, which are closed under observation shifts. These are compatible causal states and determine the charged profile. Thus the preparation law, predictive continuation and finite-resource realization are used in the same upper/lower theorem.

This is not a deduction of geometry or nonlinear operator compactness from the existence of finite stochastic rows.

18. A SHARPENED PHYSICAL CONSUMER AND ITS SIMULATION BUDGET

The exact two-preparation theorem has a downstream use with explicit hypotheses, rather than a merely renamed dependency arrow. It sharpens the mean margin of a raw-interface confidence test, and the causal morphism theorem transfers that test, with its full memory and calibration costs, to another executable interface.

Theorem 18.1 (Confidence and transported resource certificate). *Let Q be a physical marked target with $\text{TV}(Q, Q_*) \leq \beta$. Use the two-preparation event selector with a dyadic coin τ_b satisfying $|\tau_b - \tau| \leq 2^{-b}$, and put*

$$a_b = v_2 - \beta - \frac{5}{16}2^{-b}, \quad W_b = \max\{12, 6(b+1) + 5\}.$$

If $0 < c < a_b$, R independent four-preparation blocks, followed by the test $\bar{D}_R > c$, have error bounds

$$(57) \quad \sup_{\text{private}} \mathbb{P}(\bar{D}_R \leq c) \leq \exp\{-R(a_b - c)^2/2\},$$

$$(58) \quad \sup_{\text{equal}} \mathbb{P}(\bar{D}_R > c) \leq \exp\{-Rc^2/2\}.$$

They use $4R$ preparations, at most bR fresh random bits, and at most $W_b(2R+1)$ persistent states under a clocked schedule. With an exact atomic coin, replace a_b, W_b by $v_2 - \beta, 12$.

Suppose a source interface simulates this entire repeated audit under Theorem 3.4, uniformly over the private and equal-law families, with path defect Δ_R . Then the corresponding source error bounds are the right sides of (57)–(58) plus Δ_R , capped at one. At aligned cuts its state bound is $C_{\text{cal}}R_tM_tW_b(2R+1)$, and its preparation count is N_{cal} plus all source calls required to supply the $4R$ audit preparations. Calibration is acquired and retained, not supplied as a free real-valued table.

Proof. Corollary 11.2 and Proposition 11.3 give mean block score at least a_b under every fixed private parameter. Independent fresh preparations and independent coins give independent block scores in $[-1, 1]$. Hoeffding's inequality proves (57). Under the equal-law null, conditional on training and the frozen event, the two validation indicators have the same mean. Each block therefore has mean zero, proving (58) by the same inequality. Retain the integer sum in $\{-R, \dots, R\}$ alongside the current block register. Updating the sum and resetting the block is a deterministic boundary operation. This proves the raw resource bounds.

Apply Theorem 3.4 to the whole repeated experiment, including its stopping and validation fields. Its defect bound applies to each of the two terminal error events, whose indicators have range one; each probability therefore changes by at most Δ_R , not $2\Delta_R$. The same theorem gives the state product and preparation count. The source blocks need not be independent after a shared calibration stage: independence was used only for the reference audit, and path-law comparison transfers its final event probability. $\square$

The first part uses the exact saddle, the original microscopic bridge and the compiled residual machine. The second additionally needs the uniform acquired causal simulation theorem. A source which does not satisfy that theorem's stability and calibration hypotheses is not covered. Likewise this consumer supplies no reverse simulation and therefore does not transfer an optimal-width lower bound to an arbitrary source. It is an actual conditional theorem interface for

subsequent models, not an assertion that the independent geometric A2 chain or the unbounded B4/C2 operator obligations have already been discharged.

19. DEPENDENCIES, RETAINED RESULTS AND REMAINING DISTINCTIONS

The exact three-report binary noisy frontier, arbitrary-horizon binary-register minimax recursion and compatible Bayesian interval recursion remain as proved in the complete companion. Its exact odd-horizon majority realization is also retained; it is not a new result of this revision. Section 15 applies to registers of any finite width, controlled rows and any finite parameter set, but provides certified intervals rather than a closed minimax recursion for all such classes. The physical theorem supplies a matched preparation segment in a continuous two-parameter family with overlapping stochastic reports; its converse is not a finite parameter-grid check.

The stationary autonomous result in Section 16 has an explicit formula, but its stationary lower bound must not be applied to a finite-horizon clocked machine. Its finite-time upper bound is purposely conservative and exposes the slowing caused by finite transition precision. An arbitrary compact-family presentation still requires additional effective information to construct a kernel net. A subdivided positive-basis degree bound must not be called a degree bound for a global SOS hierarchy.

The physical result is a genuine consumer of the experimental and continuation root. It does not make the independent A2 geometric proofs dependent on this root. The original B4 normalized-resolvent, range, compactness and control-transfer obligations, and the broad C2 strict-dual, form, rigidity and optional-projection obligations, are not proved by our finite-state theorem. Their statements, derivations and current status remain in the retained historical pipeline. No aggregate closure is inferred from preserving their source files.

Finally, the rational certificate (34), polynomial identity (36), and residual transitions are independently reproducible. The program in the accompanying source checks their exact identities and includes deliberately invalid controls. These checks are not substitutes for the proofs above. Neither compilation, preservation of earlier pages nor a successful finite test establishes novelty or an editorial decision.

REFERENCES

- [1] R. D. Smallwood and E. J. Sondik, The optimal control of partially observable Markov processes over a finite horizon, *Operations Research* **21** (1973), 1071–1088.
- [2] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272.
- [3] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176.
- [4] J. Müller and G. Montúfar, The geometry of memoryless stochastic policy optimization in infinite-horizon POMDPs, arXiv:2110.07409 (2021; revised versions).
- [5] M. Putinar, Positive polynomials on compact semi-algebraic sets, *Indiana Univ. Math. J.* **42** (1993), 969–984.
- [6] M. E. Hellman and T. M. Cover, Learning with finite memory, *Ann. Math. Statist.* **41** (1970), 765–782.
- [7] M. E. Hellman and T. M. Cover, On memory saved by randomization, *Ann. Math. Statist.* **42** (1971), 1075–1078.
- [8] R. A. Flower and M. E. Hellman, Hypothesis testing with finite memory in finite time, *IEEE Trans. Inform. Theory* **18** (1972), 429–431.
- [9] T. M. Cover, M. A. Freedman and M. E. Hellman, Optimal finite memory learning algorithms for the finite sample problem, *Information and Control* **30** (1976), 49–85.
- [10] M. A. Managoli and V. M. Prabhakaran, Memory constrained adversarial hypothesis testing, arXiv:2605.12063v1 (May 12, 2026).
- [11] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28, doi:10.1524/strm.2002.20.14.1.
- [12] Q. Qi, General Theta Foundations I: Compatible statistical experiments and re-source frontiers, revision 23, September 24, 2026, complete mathematical companion, source commit 8a6043aca644043bd805668185eb227468f21170; packaged commit

- e926b783b643b3e7f275a18105528927b76cc97b. The 96-page article and 649-page development are reproduced without alteration after the present article in the two complete volumes.
- [13] H. Chernoff, Sequential design of experiments, *Ann. Math. Statist.* **30** (1959), 755–770.
 - [14] M. Naghshvar and T. Javidi, Active sequential hypothesis testing, *Ann. Statist.* **41** (2013), 2703–2738, doi:10.1214/13-AOS1144.
 - [15] S. Nitinawarat, G. K. Atia and V. V. Veeravalli, Controlled sensing for multihypothesis testing, *IEEE Trans. Automat. Control* **58** (2013), 2451–2464, arXiv:1205.0858v6.
 - [16] R. Zheng, R. Sim and A. Varvitsiotis, Solving imperfect-recall games via sum-of-squares optimization, arXiv:2602.21722v1 (2026).

PRESERVED PREDECESSOR

General Theta Foundations I, revision 25

The following predecessor volume is reproduced without alteration.
The complete current article precedes this divider.

Frozen predecessor commit:

818ca30bab506cfedaf6aee4c32bedff70c1e251

GENERAL THETA FOUNDATIONS I: CONTROLLED EXPERIMENT DUALITY AND EXACT MARKED MINIMAX LAWS

QIAN QI

ABSTRACT. We study finite-resource statistical comparison when training operations are chosen causally and a validation test is frozen before fresh candidate and target preparations. For finite controlled experiments with a compact parameter space, we prove an architecture-uniform posterior-predictive dual in which action choice remains inside a Bellman recursion and the adversarial prior is chosen only once. The actions need not be garblings of a common experiment. We solve the one-preparation minimax problem for a continuous family of marked product experiments, identifying both an optimal response and a two-point least-favorable prior by an explicit sum-of-squares identity. For the collision target the exact value is $(85 - 7\sqrt{73})/64$. Combined with a deterministic two-preparation audit, this gives an enlarged exact preparation segment at every peak width at least twelve. The microscopic collision and positive-noise bridge are proved within the article. A separate revelation family has the exact joint law $(1 - (1 - \alpha)^N)(1 - 1/K)$ for preparations and decision memory. Compatible finite-architecture certificates and a priced classical autonomous comparison complete the resource analysis.

CONTENTS

1. Introduction	2
1.1. Relation to earlier work	3
1.2. Organization and preserved development	3
2. Experiments, architectures and attainable resources	4
3. A statistical dual that is uniform over architectures	5
4. Controlled experiments without a dominating action	6
4.1. An informative action choice	8
5. Exact minimax laws for a marked product family	9
5.1. Complementary slackness and the event alphabet	11
6. A self-contained collision interface	11
6.1. Preparation and elastic scattering	12
6.2. Positive noise and the actual mark	13
7. The physical experiment and its sharp preparation boundary	13
7.1. An exact rational certificate against one preparation	14
7.2. A marked five-event selector	15
7.3. Causal compilation and the full register profile	16
8. An exact preparation–decision-memory law	17
9. Quantitative certificates for a finite compatible architecture	19
10. Autonomous noisy testing with a precision and time ledger	20
11. Physical consumers and the first-principles connection	22

Date: September 24, 2026. Revision 25.

2020 Mathematics Subject Classification. Primary 62B05, 62C20; Secondary 60J10, 68Q45, 93E20.

Key words and phrases. Statistical experiments, causal realization, predictive distributions, finite memory, minimax duality, physical certification.

12. Dependencies, retained results and remaining distinctions	23
References	24

1. INTRODUCTION

A statistical experiment determines not only which decisions can be made, but which observations and retained continuations are required to make them. In a controlled experiment, an action may change the information available at the next preparation. In a finite-memory experiment, retaining one continuation may preclude retaining another. A resource theorem must distinguish these two constraints rather than replace both by optimization within one chosen execution graph.

We consider an unknown parameter fixed throughout an audit. Training preparations are conditionally independent draws from causally selected experimental rows. The auditor then freezes a validation design and an event and compares fresh candidate and target draws through that event. All persistent randomness, report buffers and validation accumulators belong to the register. This model separates the simulator's resources from those of its auditor and keeps the mark of the actual preparation inside the statistical comparison.

The first result is a controlled posterior-predictive dual (Theorem 4.1). Its state is an unnormalized posterior measure together with the legal protocol state. At the terminal step, optimization over an event gives a positive-part functional, equivalently a posterior-predictive total variation. Earlier steps optimize the causal experimental action. One minimization over initial priors yields the exact unrestricted-memory minimax value. Every such prior, or a Bellman supersolution, gives a converse uniform over admissible finite-memory architectures. The common-dominating-row theorem becomes a specialization in which the experimental choice can be removed. The controlled formulation also applies when the available actions are statistically incomparable. A strict adaptive advantage persists under full-support perturbations (Proposition 4.4).

Our principal explicit minimax calculation concerns the product family $b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2)$ and targets Q_γ on the four marked diagonal atoms. For $0 \leq \gamma \leq 1 - 1/\sqrt{2}$, put

$$u_\gamma = \gamma - 2 + \sqrt{\gamma^2 - 2\gamma + 5}.$$

Theorem 5.1 gives

$$U_1(b, Q_\gamma) = (3 + u_\gamma^2)/8.$$

The optimal response and least-favorable prior are explicit. Their matching is certified over the entire parameter square by a square identity, not by an optimization grid. The endpoint of the displayed γ interval is the point where a posterior response coefficient changes sign. The zero-preparation value is also exact. At the physical mark bias $\gamma = 1/4$ the one-preparation value is

$$u_{\text{ex}} = (85 - 7\sqrt{73})/64.$$

This resolves the optimization problem hidden behind the earlier rational one-preparation certificate.

The physical consequence uses the same marked collision experiment, not a surrogate alphabet with a different cost convention. Define

$$(1) \quad u_1 = \frac{4035777}{10240000}, \quad m_2 = \frac{12 - 3\sqrt{3}}{16}, \quad \beta_0 = \frac{1}{300}.$$

The older rational bound u_1 remains a valid explicit certificate. The exact value u_{ex} improves it. We prove

$$(2) \quad N_c^{\text{phys}}(W, Q) = 2 \quad \text{for} \quad W \geq 12, \quad \text{TV}(Q, Q_*) \leq \beta, \quad u_{\text{ex}} + \beta \leq c < m_2 - \beta.$$

Here N counts candidate training preparations, and one candidate and one target validation are charged in addition. The upper construction is deterministic and has the complete bit-level profile $(1, 2, 3, 3, 4, 5, 5, 10, 12, 10, 10, 7, 3)$. Definition 6.1 and Theorem 6.3 prove locally every interface and physical approximation fact used in this deduction. Equation (2) is sharp in preparation count on its stated width range; it does not assert width optimality at two preparations or an exact value of U_2 .

A different theorem determines both coordinates in a revelation family. With reveal probability α , N preparations and K retained decision states, Theorem 8.1 proves

$$V_N^{\text{er}}(K) = (1 - (1 - \alpha)^N)(1 - 1/K).$$

The converse allows every stochastic encoder and legal stopping rule; the matching machine requires no uncharged revelation flag. The formula yields exact strict thresholds and matching sample and state scales. The memory coordinate here is a decision-cut width, not the collision machine's bit-level peak. Keeping those coordinates distinct is part of the resource theorem.

For a specified finite architecture, compatible occupation arrays retain all common-row constraints, including autonomous row reuse. The subdivision theorem gives an explicit HS/m certification error with controlled row-block degree. These are effective finite-architecture certificates, not a sharp universal complexity law. We also retain a resource-accounted autonomous binary comparison whose stationary formula is classical. None of these auxiliary results replaces the exact minimax or physical arguments.

1.1. Relation to earlier work. The comparison and minimax mechanisms are classical [2, 3]. Finite-horizon belief-state control and piecewise-linear value functions are established in [1]. Our controlled dual uses these mechanisms with a fixed adversarial parameter, frozen validation and explicit causal resource accounting. It is not a claim to have invented belief-state control or the minimax theorem. The exact marked-family saddle points, their equality certificate and their physical use are stated separately from these general mechanisms.

Polynomial stochastic-policy formulations and positivity certificates have substantial antecedents [4, 5]. Finite-memory testing, randomization and time-invariant rules are treated in [6, 7, 8, 9]. The stationary formula in Section 10 is explicitly attributed to Hellman and Cover. The May 2026 work of Managoli and Prabhakaran [10] treats memory-constrained adversarial binary hypothesis testing, including randomized time-invariant machines and adaptive distribution choices. Its objective and adversarial mechanism differ from the fixed-parameter, training-validation score here. No general priority for randomized state savings is asserted.

Norberg's filtered comparison [11] belongs to the surrounding comparison theory. We make no non-overlap or originality claim about that general theory without a proof-level comparison of the original paper. The novelty claims under consideration are the explicit theorems and resource realizations proved here, not an asserted absence of filtered-experiment predecessors.

1.2. Organization and preserved development. We first fix the architecture and common-row framework, then prove the controlled dual and exact marked minimax law. The physical bridge and charged two-preparation construction give (2). The joint revelation law, finite-architecture certification and autonomous comparison follow. Complete mathematical volumes retain the unaltered predecessor article and development, including the general continuation, Gaussian transport and historical pipeline proofs [12]. These remain mathematical content, not substitutes for a proof in the present article. The independent A2 geometric chain and

the original B4 and broad C2 obligations are not claimed to follow from the finite statistical theorems alone.

2. EXPERIMENTS, ARCHITECTURES AND ATTAINABLE RESOURCES

Throughout, total variation is $\text{TV}(P, Q) = \sup_{0 \leq h \leq 1} (P - Q)h = \frac{1}{2} \sum_x |P(x) - Q(x)|$. An experimental parameter is unknown and fixed throughout an execution. Controller randomization is independent of that parameter. A reset preparation is a fresh draw, independent of earlier preparations given the parameter. A controller may not recover discarded information from an uncharged tape, clock, selector or calibration table.

Definition 2.1 (Finite architecture). A finite architecture A specifies a finite causal execution graph, its legal experimental operations, the observable inputs to each controller row, and a register alphabet at every cut. It also specifies which row occurrences share a stochastic table. A clocked architecture may use different rows at different scheduled times. An autonomous architecture may use time only through a charged state. For a parameter family Θ and terminal reward in $[-1, 1]$, let

$$V(A) = \sup_{q \in \prod_r \Delta_{s_r}} \inf_{\theta \in \Theta} g_\theta(A, q).$$

If $\mathcal{A}(R)$ is the class of architectures obeying a global resource budget R , its outer value is

$$V^{\text{out}}(R) = \sup_{A \in \mathcal{A}(R)} V(A).$$

Both the architecture and the simulator parameter are fixed before the controller's random choices. In particular, a mixture of complete controllers is not automatically an admissible controller at the original register profile.

For a finite hidden execution graph, compatibility can be stated without division. Let $x_{\theta ir}$ be the incoming mass of hidden history i at observable row r , and let $f_{\theta irj}$ be the outgoing mass through choice j . Include the usual fixed-kernel flow equations, as well as

$$(3) \quad f_{\theta irj} \geq 0, \quad \sum_j f_{\theta irj} = x_{\theta ir}, \quad f_{\theta irj} x_{\theta' i' r} = f_{\theta' i' rj} x_{\theta ir}.$$

For shared rows, the indices i, i' include all their occurrences.

Proposition 2.2 (Compatible realization). *For a specified finite architecture, nonnegative flow arrays satisfying (3) are exactly those realized by a common stochastic row at every declared observable input, including at zero incoming mass.*

Proof. Necessity follows from $f_{\theta irj} = x_{\theta ir} q_r(j)$. For sufficiency, if some incoming mass at row r is positive, define $q_r(j)$ by dividing the corresponding outgoing mass by that incoming mass. Conservation makes q_r stochastic, and the cross-products force the same ratio at every other positive incoming mass. If an incoming mass vanishes, nonnegativity and conservation force all its outgoing masses to vanish. If the entire row is unreachable, choose any legal stochastic row. Induction through the finite execution graph reconstructs the prescribed flows. The same ratio argument applies to identified occurrences of an autonomous row. $\square$

This formulation is inherited from the complete companion and is included to fix the controller class. Compact semialgebraic feasibility and an Archimedean positivity theorem provide a complete semialgebraic certificate hierarchy for a *fixed finite architecture*. They do not interchange the two optimizations in the definition. Section 9 gives a quantitative alternative for the inner problem. The next section supplies a different certificate that survives the outer optimization.

3. A STATISTICAL DUAL THAT IS UNIFORM OVER ARCHITECTURES

Let Θ be compact, let Ω be finite, and let $\theta \mapsto P_\theta \in \mathcal{P}(\Omega)$ be continuous. Fix $Q \in \mathcal{P}(\Omega)$. An N -preparation reset selector observes $Z = (Z_1, \dots, Z_N) \sim P_\theta^{\otimes N}$ and chooses a response $h_Z \in [0, 1]^\Omega$. Its expected validation score is

$$(4) \quad G_\theta(h) = \sum_{z \in \Omega^N} P_\theta^{\otimes N}(z)(Q - P_\theta)h_z.$$

The response is fixed before two independent validation draws. Operationally it can be realized by choosing a random event with those coordinate marginals and retaining that event for both validations. No memory bound is imposed in the definition of $U_N(P, Q) = \max_h \min_\theta G_\theta(h)$.

Theorem 3.1 (Posterior-predictive dual). *Both extrema in the following identity are attained:*

$$(5) \quad U_N(P, Q) = \min_{\mu \in \mathcal{P}(\Theta)} \sum_{z, \omega} \left[\int P_\theta^{\otimes N}(z) \{Q(\omega) - P_\theta(\omega)\} d\mu(\theta) \right]_+$$

$$(6) \quad = \min_{\mu \in \mathcal{P}(\Theta)} \sum_z m_\mu(z) \text{TV}(Q, \bar{P}_\mu(\cdot | z)).$$

Here $m_\mu(z) = \int P_\theta^{\otimes N}(z) d\mu(\theta)$ and, when $m_\mu(z) > 0$,

$$\bar{P}_\mu(\omega | z) = \frac{\int P_\theta^{\otimes N}(z) P_\theta(\omega) d\mu(\theta)}{m_\mu(z)}.$$

A zero-mass term in (6) is defined to be zero.

Proof. The response space is a finite-dimensional compact convex box. Probability measures on the compact space Θ form a weakly compact convex set. The integral of (4) is continuous and affine in each argument. The minimum over parameters equals the minimum over probability measures. The minimax theorem [3] therefore interchanges the maximum and minimum. For fixed μ , each response coordinate is independent in the maximization and its contribution is the positive part of its coefficient, proving (5). The coefficient vector at a fixed z has zero sum. For positive $m_\mu(z)$ it equals $m_\mu(z)(Q - \bar{P}_\mu(\cdot | z))$, proving (6). If $m_\mu(z) = 0$, every nonnegative likelihood integrand vanishes μ -almost surely and all coefficients are zero. Continuity on the two compact strategy spaces gives attainment. $\square$

Theorem 3.2 (Dominating-row converse). *Suppose every permitted training row is a parameter-independent causal garbling of one fresh draw from P_θ . Suppose every permitted validation row is the same garbling of the target and candidate laws, and every selected row–event pair is fixed before validation. Then $U_N(P, Q)$ bounds the worst-case expected score of every auditor using at most N candidate training preparations, regardless of its register profile, feedback schedule, independent randomization, or stopping rule. Independent target-only calibration, whose law does not depend on θ , does not invalidate the bound. If the all-on row and all its events are permitted, the bound is attained in the class with no memory cap and N training preparations.*

Proof. Generate N independent all-on observations in advance. At the t th requested preparation, apply the declared garbling to observation t , using only previously simulated reports and independent randomization. This simulates the adaptive transcript with its exact law. A stopped execution ignores the unused observations; it does not reveal them to the auditor. Pull back its final row–event response to the all-on alphabet. Conditional on the entire generated training word and the independent random seed, this is a function in $[0, 1]^\Omega$. Averaging the seed, including any target-only calibration, produces one response h_z for each full word z . The score is consequently (4). This proves the upper bound for the union of all architectures. Conversely,

retaining the entire word and then a selected event implements any such finite response kernel when memory is unrestricted. Fresh validation gives exactly (4). $\square$

Remark 3.3 (What the dual does not convexify). The measure μ is an adversarial prior used to bound the minimum over *fixed* parameters. Its likelihood is $\int P_\theta^{\otimes N} d\mu(\theta)$, not $(\int P_\theta d\mu(\theta))^{\otimes N}$. It neither enlarges the private candidate class to mixtures nor gives the auditor a free persistent selector. Attainment of U_N without a memory cap does not assert attainment at any specified finite profile.

The state in (6) is generated by the actual preparation law, not by a geometric measure chosen independently of the experiment. For sequential continuation one retains the posterior μ_z or its future-test equivalence class; a single predictive marginal need not be closed under further observations. This is the distinction between a one-step response quotient and a future-predictive quotient.

Corollary 3.4 (Finite moment support for the product family). *For two independent Bernoulli report bits with parameters (p, q) and an independent fair mark, an optimal prior in (5) can be chosen with at most $(N + 2)^2$ atoms. The two report counts (k, ℓ) are a causally updatable sufficient statistic for computing the dual.*

Proof. Conditional on the two counts, all arrangements of report bits and all mark words have parameter-independent probabilities. Summing their coefficients reduces (5) to the $(N + 1)^2$ count pairs. Their likelihoods are

$$\binom{N}{k} p^k (1 - p)^{N-k} \binom{N}{\ell} q^\ell (1 - q)^{N-\ell}.$$

After multiplication by $Q(\omega) - P_{p,q}(\omega)$, every integrand has degree at most $N + 1$ in each variable. Thus the objective depends only on the $(N + 2)^2$ rectangular moments. Their constant coordinate is one. Carathéodory's theorem in the remaining affine space replaces an optimal moment vector by a convex combination of at most $(N + 2)^2$ point evaluations. Counts update by addition of the current bits. No claim that this count register is optimal for the decision is made. $\square$

4. CONTROLLED EXPERIMENTS WITHOUT A DOMINATING ACTION

The reduction to one dominating row is not needed for a statistical dual, provided causal action choice is retained in the dual recursion. We give the finite-action theorem with a compact parameter space and an explicit protocol state. Its equality is an unrestricted-memory statement; its upper certificates apply to every smaller-memory class.

Let Θ be compact metric and E a finite set of protocol states. At $e \in E$ let $\mathcal{A}_e$ be a finite set of training actions. Action a produces x in a finite alphabet X_a with law P_θ^a and changes the protocol state to $T(e, a, x)$. The functions $\theta \mapsto P_\theta^a(x)$ are continuous. An unknown θ is fixed throughout the experiment. Conditional on it, each requested preparation uses a fresh independent draw from the selected row. Legal validation designs form a nonempty finite set $\mathcal{B}_e$. Design b has a finite outcome alphabet Y_b , target law Q^b and continuous candidate law V_θ^b . A policy may stop after at most N training actions. It then freezes $b \in \mathcal{B}_e$ and an event on Y_b , before one candidate and one target validation. Randomization is independent of θ . No training action is assumed to dominate another.

For a finite nonnegative measure ν on Θ write $\nu_{a,x}(d\theta) = P_\theta^a(x)\nu(d\theta)$ and set

$$(7) \quad \Phi_e(\nu) = \max_{b \in \mathcal{B}_e} \sum_{y \in Y_b} \left[Q^b(y)\nu(\Theta) - \int V_\theta^b(y) d\nu \right]_+,$$

$$F_0(e, \nu) = \Phi_e(\nu),$$

$$(8) \quad F_n(e, \nu) = \max \left\{ \Phi_e(\nu), \max_{a \in \mathcal{A}_e} \sum_{x \in X_a} F_{n-1}(T(e, a, x), \nu_{a,x}) \right\}.$$

If no training action is legal, only the stopping term is retained. All the functions vanish at the zero measure, so no posterior division is required on null histories. For positive mass, (7) is the mass times the largest posterior-predictive total variation.

Theorem 4.1 (Controlled posterior-predictive dual). *Let $\mathcal{U}_N(e_0)$ be the maximum worst-case expected validation score over all randomized causal policies in the declared protocol, with at most N training preparations and no memory cap. Then*

$$(9) \quad \mathcal{U}_N(e_0) = \min_{\mu \in \mathcal{P}(\Theta)} F_N(e_0, \mu).$$

Both extrema are attained. Each $F_n(e, \cdot)$ is the maximum of finitely many continuous linear functionals on the cone of measures; in particular it is convex, continuous and positively homogeneous. For every prior μ , $F_N(e_0, \mu)$ bounds the worst-case score of every finite-memory architecture implementing this protocol.

Proof. There are finitely many histories of length at most N and finitely many event masks at each stopping history. Consequently there are finitely many deterministic policy trees, denoted by $\Pi_N(e_0)$. For $\pi \in \Pi_N(e_0)$ its payoff $g_\pi(\theta)$ is a finite sum of finite products of continuous primitive probabilities and is continuous in θ .

A randomized causal policy induces the same outcome laws as a probability distribution on these trees: sample its finite random decision table in advance and condition on it. Conversely, with no memory cap, a sampled tree can be retained and followed. Thus the unrestricted payoff vectors are exactly the convex hull of the g_π . This step is not an assertion that mixing machines preserves a specified register width.

The simplex on $\Pi_N(e_0)$ and $\mathcal{P}(\Theta)$ are compact and convex. The expected payoff is continuous and bilinear. Minimax therefore gives

$$\max_{\lambda \in \Delta(\Pi_N)} \min_{\theta} \sum_{\pi} \lambda_{\pi} g_{\pi}(\theta) = \min_{\mu} \max_{\pi \in \Pi_N} \int g_{\pi} d\mu.$$

For a fixed measure, the terminal maximum is (7), because an event includes precisely the positive coordinates of its signed predictive measure. At a training node one selects an action and, after its observed outcome, selects a continuation tree separately on that branch. The unnormalized branch measure is exactly $\nu_{a,x}$. Backward induction proves (8) and identifies the right side with (9). It also proves the finite-linear-functional representation. Compactness gives an optimizing prior and an optimizing mixed tree. Finally, the minimum over fixed parameters of any policy is at most its average under μ , which is at most $F_N(e_0, \mu)$. This remains true after restricting its memory or architecture. $\square$

The minimization in (9) occurs once, outside the recursion. It cannot be replaced by fresh adversarial choices after observations. The posterior carries dependence on one fixed unknown parameter. Also, a Bayes-optimal tree for a minimizing prior need not by itself be a minimax policy; the saddle-point mixture is supplied by the finite game in the proof.

Corollary 4.2 (Bellman upper certificates). *Suppose functions $\Psi_n(e, \nu)$ satisfy $\Psi_0 \geq \Phi$ and, for $n \geq 1$, satisfy*

$$\Psi_n(e, \nu) \geq \Phi_e(\nu), \quad \Psi_n(e, \nu) \geq \sum_x \Psi_{n-1}(T(e, a, x), \nu_{a,x}) \quad (a \in \mathcal{A}_e).$$

Then $\Psi_N(e_0, \mu)$ is an upper certificate for every admissible architecture and every prior μ . The family F_n is the least such family. An optimal prior can be chosen with at most $d + 1$ support points, where d is the affine dimension of $\{(g_\pi(\theta))_{\pi \in \Pi_N} : \theta \in \Theta\}$.

Proof. Induction in (8) gives $F_n \leq \Psi_n$. The equality case is the recursion itself. The objective depends on a prior only through its vector of integrals of the finitely many g_π . An optimal such vector belongs to the convex hull of their compact evaluation image. Carathéodory's theorem gives the stated support bound while preserving every pure-tree payoff. $\square$

Proposition 4.3 (Causal stability). *Consider two models with the same protocol, alphabets and policy class. Suppose their training rows differ uniformly in total variation by at most ϵ , their candidate validation rows by at most η , and their target validation rows by at most β . Their unrestricted values, or their values in any common specified memory class, differ by at most $2N\epsilon + \eta + \beta$.*

Proof. Use the same policy randomization and couple corresponding training rows while the histories agree. The probability of a first discrepancy is at most $N\epsilon$, even with feedback and stopping. Conditional expected validation rewards lie in $[-1, 1]$, so differing training histories contribute at most $2N\epsilon$. On agreeing histories the same frozen design and event are used, and their conditional mean rewards differ by at most $\eta + \beta$. The bound is uniform in the fixed parameter and in the policy. Taking minima and suprema preserves it, with the reverse inequality obtained by interchanging the models. $\square$

4.1. An informative action choice. The following calculation distinguishes the controlled theorem from a common-row reduction. Let $\Theta = \{(i, j) : i, j \in \{0, 1\}\}$. Action R reports i . Action L reports j when $i = 0$ and otherwise reports an independent fair bit. Action H reports j when $i = 1$ and otherwise reports an independent fair bit. The candidate validation is $\delta_{(i,j)}$ and the target is uniform on Θ .

Proposition 4.4 (Strict adaptive advantage). *With two training preparations, the unrestricted adaptive value is $3/4$ and the best nonadaptive value is $5/8$. Nonadaptive policies here include mixtures of schedules, with their random schedule retained. No available single training action dominates all the others.*

Proof. Take R first, and then L or H according to the observed i . The parameter is recovered exactly. The complement of its validation atom has score $3/4$. No policy can exceed $\text{TV}(Q, \delta_\theta) = 3/4$ even with full information.

For the converse for fixed schedules, use the uniform prior. For a pair of actions a, b its Bayes value is explicitly

$$(10) \quad \sum_{x,y,\theta} \left[\frac{1}{16} \sum_{\vartheta} P_{\vartheta}^a(x) P_{\vartheta}^b(y) - \frac{1}{4} P_{\theta}^a(x) P_{\theta}^b(y) \right]_+.$$

Substitution of the three displayed kernels gives

	R	L	H
R	$1/2$	$5/8$	$5/8$
L	$5/8$	$7/16$	$1/2$
H	$5/8$	$1/2$	$7/16$

This is an exact finite sum, not a parameter-grid estimate. It bounds any mixture of fixed schedules by $5/8$; an early stop can be padded with an ignored observation. To attain the bound uniformly in θ , choose with equal probabilities the schedules (R, L) and (R, H) . When the chosen branch-specific action matches i , the parameter is known and the score is $3/4$. Otherwise only i is known, and the complement of its two-atom fiber gives score $1/2$. Each fixed parameter therefore has mean score $5/8$.

Action R cannot distinguish parameters differing only in j . Action L cannot distinguish $(1, 0)$ and $(1, 1)$, and H cannot distinguish $(0, 0)$ and $(0, 1)$. An experiment which fails to distinguish such a pair cannot garble to one that distinguishes it. Hence no action is a common dominating row. $\square$

If every binary report is additionally passed through an independent binary symmetric channel of crossover ϵ , the primitive training rows change by at most ϵ and have full support for $0 < \epsilon < 1/2$. Proposition 4.3, applied to both policy classes, leaves an adaptive advantage at least $1/8 - 8\epsilon$, which is positive for $0 < \epsilon < 1/64$. Thus the distinction is not confined to disjoint observation supports. The finite-horizon belief recursion and its piecewise-linear structure are classical [1]. The contribution here is their explicit integration with the frozen validation score, one fixed adversarial parameter, outer certificates, and the resource conventions; no priority for Bellman recursion itself is claimed.

5. EXACT MINIMAX LAWS FOR A MARKED PRODUCT FAMILY

The posterior-predictive dual can be solved explicitly for a continuous family containing the collision target. Throughout this section, the training and candidate validation law is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad (p, q) \in [0, 1]^2.$$

In the order 000, 001, 110, 111, let Q_γ have masses

$$((1 + \gamma)/4, (1 - \gamma)/4, (1 - \gamma)/4, (1 + \gamma)/4),$$

and zero mass elsewhere. Write $U_1(\gamma) = U_1(b, Q_\gamma)$, with exactly the training-validation convention of (4). Set

$$(11) \quad \gamma_* = 1 - 1/\sqrt{2}, \quad s_\gamma = \sqrt{\gamma^2 - 2\gamma + 5}, \quad u_\gamma = \gamma - 2 + s_\gamma, \quad \tau_\gamma = \frac{2u_\gamma}{u_\gamma + 2 - \gamma}.$$

Theorem 5.1 (Exact marked minimax family). *For every $0 \leq \gamma \leq \gamma_*$,*

$$(12) \quad U_1(\gamma) = \frac{3 + u_\gamma^2}{8}.$$

An optimal response depends on the training word only through the report count $K = X_1 + X_2$. On the four target atoms its rows are

$$(13) \quad h_0 = (\tau_\gamma, 0, 1, 1), \quad h_1 = (1, 1, 1, 1), \quad h_2 = (1, 1, 0, \tau_\gamma),$$

and all responses are zero off these atoms. A least-favorable prior is

$$(14) \quad \mu_\gamma = \frac{1}{2}\delta_{(t_\gamma, t_\gamma)} + \frac{1}{2}\delta_{(1-t_\gamma, 1-t_\gamma)}, \quad t_\gamma = \frac{1 - \sqrt{u_\gamma}}{2}.$$

The assertion optimizes over all stochastic responses to the full marked training word, not merely over count-based or symmetric responses.

Proof. First, u_γ is the positive root of

$$(15) \quad u^2 + (4 - 2\gamma)u - (1 + 2\gamma) = 0.$$

On the asserted interval, $0 < u_\gamma < 1$ and $0 < \tau_\gamma < 1$. Indeed the polynomial is negative at zero and positive at one; also $u_\gamma < 2 - \gamma$ because $\gamma < 1/2$. Substitution of $1 - 2\gamma$ into the polynomial gives $8(\gamma^2 - 2\gamma + 1/2)$, which is nonnegative precisely on the present interval. Monotonicity on $[0, \infty)$ therefore gives

$$(16) \quad u_\gamma \leq 1 - 2\gamma.$$

Put $a = (1 - p)(1 - q)$, $c = pq$, and $z = 1 - a - c$. The probabilities of the three training counts are a, z, c . The independent training mark may be discarded by the displayed construction. Let $G_\gamma(p, q)$ denote its score. With $u = (p + q - 1)^2$ and $v = (p - q)^2$, direct expansion gives

$$(17) \quad G_\gamma(p, q) - \frac{3 + u_\gamma^2}{8} = \frac{2 - \tau_\gamma}{16} \{(u - v - u_\gamma)^2 + 4v\}.$$

For clarity, this identity follows by substituting the three rows of (13) into

$$a(Q_\gamma - b)h_0 + z(Q_\gamma - b)h_1 + c(Q_\gamma - b)h_2,$$

using $a + c = (1 + u - v)/2$, $a - c = -(p + q - 1)$, and then eliminating u_γ^2 by (15). The coefficient of the resulting square is $(2 - \tau_\gamma)/16$ and that of v is four times this coefficient. Both are nonnegative. Equality holds at both points of (14), since there $v = 0$ and $u = u_\gamma$. This proves the lower bound and identifies its equality set.

It remains to bound every response, including nonsymmetric ones. Use (14) in Theorem 3.1. At $t = t_\gamma$ put $a = (1 - t)^2$, $c = t^2$, and $T = a + c = (1 + u_\gamma)/2$. For the training report pair 00, the four dual coefficients, apart from a common positive factor, are

$$(18) \quad \left(Q_H T - \frac{a^2 + c^2}{2}, \quad Q_L T - \frac{a^2 + c^2}{2}, \quad Q_L T - ac, \quad Q_H T - ac \right),$$

where $Q_H = (1 + \gamma)/4$ and $Q_L = (1 - \gamma)/4$. The first coordinate is zero by (15); the second is $-\gamma T/2 \leq 0$. The last two are positive: $T \geq 1/2$, $ac \leq 1/16$, and $\gamma < 1/2$ imply $Q_L T > 1/16 \geq ac$. Thus $(\tau_\gamma, 0, 1, 1)$ maximizes this linear functional. For the training pair 11, reversing all report bits and the mark reverses the four coordinates and gives the row h_2 .

For either training pair 01 or 10, the two prior support points have the same likelihood $t(1 - t)$. The posterior is still equally weighted, and the candidate predictive mass at each target atom is $T/4$. Every one of the four coefficients is therefore nonnegative by $Q_L - T/4 = (1 - 2\gamma - u_\gamma)/8 \geq 0$. The response h_1 is optimal. At off-diagonal validation atoms the target mass vanishes and the coefficient is nonpositive. Finally, conditioning on either training mark multiplies all these coefficients by $1/2$; it creates no further optimizing choice. We have therefore optimized every coordinate of the full marked-word response space.

The prior average of the displayed response equals its value at either support point, namely $(3 + u_\gamma^2)/8$, by (17). Every other response has prior average at most this number and hence has minimum over fixed (p, q) at most this number. Together with the lower bound this proves (12), optimality of the response, and least favorability of the prior. $\square$

Corollary 5.2 (The exact one-preparation collision value). *For $Q_* = Q_{1/4}$,*

$$(19) \quad U_1(b, Q_*) = u_{\text{ex}} := \frac{85 - 7\sqrt{73}}{64}, \quad u_{1/4} = \frac{\sqrt{73} - 7}{4}, \quad \tau_{1/4} = 2 - \frac{14}{\sqrt{73}}.$$

If only the target changes to Q , then $|U_1(b, Q) - u_{\text{ex}}| \leq \text{TV}(Q, Q_)$.*

Proof. Substitute $\gamma = 1/4$ in Theorem 5.1. For a fixed trained response the target perturbation is at most total variation, uniformly in (p, q) . Taking first the minimum and then the maximum preserves this bound in both directions. $\square$

5.1. Complementary slackness and the event alphabet. The tie in the first coordinate of (18) explains the randomization. Any value of that response coordinate is Bayes-optimal for the two-point prior. The minimax choice makes the uniform payoff stationary at its two support points; solving this condition gives τ_γ and the square identity (17). This derives the prior and response together, rather than postulating that a numerically effective prior is optimal.

Operationally, for $K = 0$ one draws once between the events with indicator vectors $(0, 0, 1, 1)$ and $(1, 0, 1, 1)$, with probabilities $1 - \tau_\gamma$ and τ_γ . For $K = 2$ one draws between $(1, 1, 0, 0)$ and $(1, 1, 0, 1)$. For $K = 1$ the event is $(1, 1, 1, 1)$. The resulting event identity is retained and used for both fresh validations. These are exactly the five events in the two-preparation construction below. The one-preparation theorem supplies their complementary-slackness interpretation; it does not prove that the two-preparation selector is optimal. The four-point rational prior retained below is a useful rational certificate, not a least-favorable prior.

The exact response uses an atomic stochastic row. It does not receive free persistent randomness. Replacing τ_γ by a dyadic number within 2^{-b} changes each conditional response, and hence the score, by at most 2^{-b} . A sequential fair-bit implementation must retain its bit prefix and event identity. None of these implementation costs is needed by the deterministic two-preparation physical audit.

Proposition 5.3 (The zero-preparation endpoint). *For $0 \leq \gamma \leq 1$, $U_0(b, Q_\gamma) = \gamma/2$.*

Proof. The fixed event $\{000, 111\}$ has score $(1 + \gamma)/2 - (a + c)/2 \geq \gamma/2$ because $a + c \leq 1$. Conversely, the equally weighted prior on $b(0, 0)$ and $b(1, 1)$ has predictive validation law uniform on the four target atoms. Its total variation from Q_γ is $\gamma/2$. Theorem 3.1 with $N = 0$ gives the matching upper bound. $\square$

6. A SELF-CONTAINED COLLISION INTERFACE

We give the preparation, mechanical estimate, observation law and private simulator class needed for the physical specialization. The following argument does not appeal to an unpublished physical lemma. It retains the same positive-volume preparation and positive sensor noise as the earlier collision construction.

Definition 6.1 (The reset row–event architecture). A preparation supplies two sequential report slots and, afterwards, an actual mark. The source reports in the two slots are constant. A private width-one simulator has no retained information between those slots and no persistent hidden seed. It may use fresh independent coins. Its first emission is $X_1 \sim \text{Ber}(p)$; after receiving the current gate $A \in \{0, 1\}$, its second emission has a fixed Bernoulli kernel with parameter q_A . The parameters (p, q_0, q_1) are fixed throughout the audit. The simulator does not observe the actual mark. All its coins are independent of that mark. Different preparations reset all unretained variables and are independent conditional on the parameters.

The auditor observes the first report, chooses a gate using its retained state and that report, observes the second report, and then observes the mark. During training it may choose rows and stop adaptively, using at most the declared number of candidate preparations. It next freezes one gate policy and one event, with any retained random choice included in its register. It uses that same row–event continuation on one fresh candidate preparation and one independent target preparation and returns the difference of their event indicators, target minus candidate. It may not reselect the event after either validation. Independent target-only calibration is allowed but all retained calibration and randomness are charged. Peak width counts every bit, mark, event and validation-accumulator cut. Clocked schedules are declared externally; an autonomous implementation must retain its phase. An unretained transcript, seed or stopping-time value is not available to a continuation.

In the all-on row the complete private law is $b(p, q_1)$, with an independent fair actual mark. For lower bounds it suffices to restrict the adversary to $q_0 = 0$, $q_1 = q$. In this subfamily every deterministic gate policy is the pushforward

$$T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w)$$

of $b(p, q)$. The gate, when included in the transcript, is another function of these same coordinates. Randomized gate policies are covered by averaging their independent random tables. This proves both the all-on product characterization for the upper construction and the common-garbbling subfamily used for the outer converse. The width-one constraint is a simulator constraint; it is not an uncharged restriction on the auditor's register.

6.1. Preparation and elastic scattering. Consider two labeled equal-mass hard spheres in $\mathbb{R}^3$, with no walls and contact distance d . Write $r = q_1 - q_2$, $g = v_1 - v_2$, $V = (v_1 + v_2)/2$, and let the center of mass position range over any fixed rational box of positive volume. Prepare $(B, W) \in \{0, 1\}^2$ with probability $5/16$ at $B = W$ and $3/16$ at $B \neq W$. Conditional on (B, W) , use the uniform product preparation on the following boxes:

$$(20) \quad \begin{aligned} r &= (-a_0, (2B-1)b_0, c_0), & a_0 &\in [29/10, 31/10], & b_0 &\in [9/20, 11/20], \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & & & |V_z - (2W-1)| &\leq 1/100. \end{aligned}$$

Thus the actual mark is $W = \mathbf{1}_{\{V_z > 0\}}$, preserved by momentum conservation. Set

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector, identically zero on the incoming transverse-velocity range.

Lemma 6.2 (Uniform collision signal). *For every preparation in (20) and every $9/10 \leq d \leq 11/10$, exactly one collision occurs and its time lies in $(4/5, 3/2)$. Afterwards*

$$(2B-1)v_{1,y} > 1/2, \quad (2B-1)h(v_{1,y}) > 1/4.$$

For $0 < d \leq 2/5$ no collision occurs on $[0, 3]$ and the detector is zero throughout that interval.

Proof. Before contact, $r(t) = r + tg$. At $t \leq 4/5$ its x coordinate is at most $-29/10 + (4/5)(21/10) = -61/50$, so its norm exceeds $11/10$. At $t = 3/2$ its absolute coordinate bounds are $1/4$, $113/200$ and $1/40$, whose squared sum is less than $(9/10)^2$. The first contact therefore occurs in the claimed interval. At the free zero of the x coordinate, whose time is at most $31/19$, the transverse norm is less than $9/10$. Hence the entry contact has $n_x < 0$, where $n = r(t_c)/d$. Its remaining coordinates give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B-1)n_y \geq 87/220.$$

It follows that $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$. The elastic rule gives

$$v_1^+ = V + g/2 - (g \cdot n)n, \quad g^+ = g - 2(g \cdot n)n.$$

Consequently

$$(2B-1)v_{1,y}^+ > -3/200 + (1443/1000)(87/220) > 1/2.$$

The definition of h implies the signal bound. Since $r(t_c) \cdot g^+ > 0$, squared separation increases strictly afterwards; there is no second collision. Finally, for $d \leq 2/5$ and $0 \leq t \leq 3$, $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$. There is no contact, and the incoming $|v_{1,y}| \leq 3/200$ makes h zero. $\square$

6.2. Positive noise and the actual mark. Let $\mathcal{B}_t$ be Brownian motion independent of the preparation and set

$$Y_t = \int_0^t h(v_{1,y}(s)) ds + \sigma \mathcal{B}_t, \quad 0 < \sigma \leq 1/24.$$

The two potential all-on reports are

$$Z_1 = \mathbf{1}_{\{2(Y_2 - Y_{3/2}) > 0\}}, \quad Z_2 = \mathbf{1}_{\{2(Y_3 - Y_{5/2}) > 0\}}.$$

The actual second report is AZ_2 . The gate changes acquisition, not the mechanical path. The continuous path is implemented by the target sensor; it is not made available to the private simulator or supplied as an uncharged auditor buffer. Denote the law of (Z_1, Z_2, W) by $Q_{d,\sigma}$.

Theorem 6.3 (Uniform physical presentation). *For $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$,*

$$(21) \quad \text{TV}(Q_{d,\sigma}, Q_*) < 1/300.$$

The same estimate holds after every declared feedback gate, including its observable gate value. The all-on private product family and the common-garbling subfamily in Definition 6.1 apply to this very experiment.

Proof. Every collision occurs before $3/2$. Conditional on the preparation, the two normalized sensor windows are independent Gaussians with mean $h(v_{1,y}^+)$ and standard deviation $\sqrt{2}\sigma \leq 1/12$. Lemma 6.2 makes their means have sign $2B - 1$ and magnitude greater than $1/4$. Each potential report therefore differs from B with probability less than $\Phi(-3)$. The Gaussian tail estimate

$$\Phi(-x) \leq \frac{e^{-x^2/2}}{x\sqrt{2\pi}}$$

gives $\Phi(-3) < 1/600$: $e^{9/2} > 80$ and $\sqrt{2\pi} > 5/2$ suffice. Couple the ideal and noisy reports using the same preparation, actual mark and gate random table. Except on a set of probability less than $2\Phi(-3) < 1/300$, both potential reports equal B and the entire gated transcript agrees. The ideal marked law is precisely Q_* , by the prepared (B, W) table. This proves (21) and the feedback-uniform version. Finally, the source is blind, its actual mark is unobserved by the simulator, and the gate is its only current input at the second emission. The product and subfamily assertions follow from the two independent private-emission kernels specified above. $\square$

For $d \leq 2/5$, the two windows instead contain only independent centered Gaussian noise. Their signs are independent fair bits, independent of W , and a private width-one simulator reproduces every gated row exactly. Thus the informative regime and its no-collision control use the same detector, preparation convention and actual mark.

7. THE PHYSICAL EXPERIMENT AND ITS SHARP PREPARATION BOUNDARY

Take $\Omega = \{0, 1\}^3$, with coordinates (x_1, x_2, w) . The all-on private candidate is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad 0 \leq p, q \leq 1.$$

The mark w is the actual mark, not an invented simulator bit. Put $a = (1 - p)(1 - q)$ and $c = pq$. On the ordered atoms

$$(22) \quad 000, \quad 001, \quad 110, \quad 111,$$

let Q_* have masses $5/16, 3/16, 3/16, 5/16$, and set it to zero elsewhere. The candidate masses on these atoms are $a/2, a/2, c/2, c/2$. A target Q is allowed whenever $\text{TV}(Q, Q_*) \leq \beta$.

The original feedback operation has the form

$$(23) \quad T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w), \quad g : \{0, 1\} \longrightarrow \{0, 1\}.$$

The selected gate is observable. It may depend on earlier preparations and on the first current report. The target rows are pushforwards by (23). For the converse, restrict the adversary to the admissible private subfamily whose second emission is AX_2 , where A is the current gate and $X_2 \sim \text{Ber}(q)$ is independent of the first emission and of the fair mark. Every row is then a causal garbling of $b(p, q)$. For the construction only the all-on row is used, and every private candidate in the original width-one class has the all-on product law just specified. These two facts, proved for the microscopic interface in Definition 6.1 and Theorem 6.3, explain why using a subfamily for the lower bound does not narrow the upper construction.

The admissible audit first trains, then selects one row–event pair, then uses one fresh candidate and one fresh target validation. It reports $D = \mathbf{1}_A(Y) - \mathbf{1}_A(X)$. The event and row cannot be reselected after seeing a validation draw. All bit, mark, decision and validation cuts are charged. Define $N_c^{\text{phys}}(W, Q)$ to be the minimum number of candidate training preparations among these original-interface auditors, with peak register width at most W , whose score is uniformly strictly greater than c . Randomization, feedback and stopping are allowed. If no such audit exists, the value is $+\infty$.

Theorem 7.1 (Two-preparation physical segment). *Let u_1, m_2 be as in (1). For every $\beta \geq 0$, $\text{TV}(Q, Q_*) \leq \beta$, and*

$$W \geq 12, \quad u_1 + \beta \leq c < m_2 - \beta,$$

one has $N_c^{\text{phys}}(W, Q) = 2$. The two-preparation audit is deterministic, uses no acquired calibration and no random bits, and has the full profile

$$(24) \quad (1, 2, 3, 3, 4, 5, 5, 10, 12, 10, 10, 7, 3)$$

from the initial cut through its twelve bit/mark updates. In particular, its decision alphabet has five labels and its peak has twelve. The lower bound remains valid with no memory cap.

The proof occupies the next three subsections. The interval in the statement can be empty for large β ; no assertion is then made. For $\beta = 1/300$, it contains $c = 2/5$ strictly between its endpoints.

7.1. An exact rational certificate against one preparation.

Lemma 7.2. *Every original-interface auditor using at most one candidate training preparation has worst-case score at most $u_1 + \beta$, regardless of memory.*

Proof. By Theorem 3.2, it suffices to bound $U_1(b, Q_*)$ and then perturb the target. In (5) use the four fixed parameters

$$(p, q) = (1/8, 1/8), (1/5, 1/5), (4/5, 4/5), (7/8, 7/8)$$

with weights $1/16, 7/16, 7/16, 1/16$, respectively. This measure is not asserted to be least favorable. Since $p = q$ on its support, the training count $K = x_1 + x_2$ is sufficient. The independent training mark and the order of unequal bits contain no further information about this prior.

For $k = 0, 1, 2$ and the four validation atoms in (22), the signed coefficients in (5), after grouping by K , are the entries of

$$(25) \quad \frac{1}{40960000} \begin{pmatrix} -1977 & -1775177 & 2170423 & 3943623 \\ 1765554 & 191954 & 191954 & 1765554 \\ 3943623 & 2170423 & -1775177 & -1977 \end{pmatrix}.$$

For example, the coefficient in column j is

$$\sum_{i=1}^4 \mu_i \binom{2}{k} p_i^k (1-p_i)^{2-k} \left(Q_{*,j} - \frac{(1-p_i)^2 \mathbf{1}_{j \leq 2} + p_i^2 \mathbf{1}_{j \geq 3}}{2} \right).$$

This identity proves every entry of (25) by rational arithmetic. All coefficients at off-diagonal validation atoms are nonpositive. The sum of the positive entries in (25) is $4035777/10240000 = u_1$. Each response coordinate is in $[0, 1]$, so this sum bounds its prior-averaged score and hence its minimum over the four fixed candidates.

Replacing Q_* by Q changes the score of any trained response by at most β . The training law is a candidate law and is unchanged. Consequently the same prior bounds every score by $u_1 + \beta$. A procedure with no training, or one that stops before taking its permitted sample, is covered by ignoring the generated all-on observation. $\square$

7.2. A marked five-event selector. Take two independent all-on training preparations. Let S be the total number of ones among their four report bits; discard their marks. Choose the following events, represented by their indicators on (22) and with indicator zero at every off-diagonal atom:

$$(26) \quad \begin{array}{c|cccc} S & 000 & 001 & 110 & 111 \\ \hline 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & 0 \end{array}$$

Denote these indicators by $h_0, \dots, h_4$ and report $D = h_S(Y) - h_S(X)$ on independent validation draws.

Lemma 7.3 (Exact uniform score of the selector). *Against the ideal target, the selector (26) has uniform minimum score $m_2 = (12 - 3\sqrt{3})/16$. Against Q , its score is at least $m_2 - \beta$.*

Proof. The generating polynomial of S is

$$((1-p) + pz)^2((1-q) + qz)^2.$$

Writing $g_s = (Q_* - b(p, q))h_s$, its expected score is $G(p, q) = \sum_{s=0}^4 [z^s]((1-p) + pz)^2((1-q) + qz)^2 g_s$. Substitution of the five rows in (26) and collection of terms with $x = p + q - 1$, $y = p - q$, $u = x^2$ and $v = y^2$ gives

$$(27) \quad \begin{aligned} G(p, q) = H(u, v) = & -\frac{u^3}{32} + \frac{3u^2v}{32} + \frac{3u^2}{16} - \frac{3uv^2}{32} - \frac{uv}{4} - \frac{3u}{32} \\ & + \frac{v^3}{32} + \frac{v^2}{16} + \frac{15v}{32} + \frac{7}{16}. \end{aligned}$$

This is a polynomial identity, not an interpolation over a grid. Since $0 \leq u \leq 1$ and $v \geq 0$ on the parameter square,

$$\partial_v H(u, v) = \frac{3(u-v)^2 - 8u + 4v + 15}{32} \geq \frac{7}{32}.$$

Thus $H(u, v) \geq H(u, 0) = (-u^3 + 6u^2 - 3u + 14)/32$. The derivative of the last expression is $(-3u^2 + 12u - 3)/32$. On $[0, 1]$ it changes sign from negative to positive at $u = 2 - \sqrt{3}$. Substitution gives $(12 - 3\sqrt{3})/16$. The minimizing values are feasible: take $p = q = (1 \pm \sqrt{2 - \sqrt{3}})/2$. This proves exactness of the minimum for this specified selector, not optimality of its score over all

selectors with two preparations. Finally, for each s , the target expectation of $h_s \in [0, 1]^\Omega$ changes by at most β . Averaging over training proves the claimed robustness. $\square$

7.3. Causal compilation and the full register profile.

Lemma 7.4. *The selector and its two validations admit the deterministic profile (24). No selector, first-bit buffer or validation accumulator is outside the charged register. An autonomous phase-tagged realization uses at most 75 states.*

Proof. After each training report bit retain the count accumulated so far. The two marks are read and discarded. The first six updates therefore have widths 2, 3, 3, 4, 5, 5 after the initial singleton. At the decision cut the five count values select the five events.

During candidate validation, after the first bit retain the event and that bit, using at most ten states. After the second bit retain the event and the function of the unread mark that gives its candidate-event indicator. For events 0, 1, 2, 3, 4, respectively, the possible functions are

$$\{0, 1\}, \quad \{0, 1 - w, 1\}, \quad \{0, 1\}, \quad \{0, w, 1\}, \quad \{0, 1\}.$$

There are twelve event–function pairs. After the candidate mark, retain the event and its now known candidate indicator $c_0 \in \{0, 1\}$, using at most ten states.

During target validation, after the first bit discard the event name and retain c_0 and its residual function of the second bit and mark. These residual functions belong to

$$\{0, x_2, (1 - x_2)(1 - w), 1 - x_2, x_2 w\}.$$

This uses at most ten states. After the second bit, the residual *difference* as a function of the unread mark belongs to

$$\{-1, 0, 1, w, 1 - w, w - 1, -w\},$$

so seven states suffice. The mark update gives the three possible outputs $-1, 0, 1$. Each transition depends only on its retained residual and the current report. Thus it is a legal common observable row, not a family of independently optimized hidden-history maps.

For completeness, equality of two residual functions implies equality of their restrictions after either next bit, so identifying equal residuals is closed under every subsequent update. The supplied transition tables are obtained by these restrictions and can be reconstructed directly from (26). Enumerating residuals is a convenience for reproducing the tables, not the proof of their sufficiency. Finally, disjointly tag the states at the thirteen cuts by their phase and use terminal self-loops. The sum of the displayed widths is 75. This implementation charges the clock; it does not assert that 75 is optimal. $\square$

Proof of Theorem 7.1. Lemma 7.2 excludes $N \leq 1$ for a strict score greater than $c \geq u_1 + \beta$, for every architecture and every memory budget. Lemmas 7.3 and 7.4 supply a two-preparation architecture of peak twelve with score at least $m_2 - \beta > c$. Larger width budgets can use the same machine. $\square$

Corollary 7.5 (The positive-noise collision family). *For the original marked collision targets with $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$, the local microscopic estimate $\text{TV}(Q_{d,\sigma}, Q_*) < 1/300$ implies*

$$N_{2/5}^{\text{phys}}(W, Q_{d,\sigma}) = 2 \quad (W \geq 12).$$

There are four preparations in the complete audit: two candidate training, one candidate validation, and one target validation.

Proof. Apply Theorem 7.1. The lower-end margin is the exact rational number

$$\frac{2}{5} - u_1 - \frac{1}{300} = \frac{78269}{30720000} > 0.$$

For the other endpoint, $\sqrt{3} < 7/4$ implies $m_2 > 27/64$ and $27/64 - 1/300 > 2/5$. The physical total-variation estimate, including the actual mark and positive sensor noise, is proved in Theorem 6.3; the current deduction does not replace those hypotheses by a zero-noise limit. $\square$

Remark 7.6 (The part of the frontier determined here). The theorem evaluates the minimum preparation coordinate for $W \geq 12$. It does not determine $N_c^{\text{phys}}(W, Q)$ for $3 \leq W \leq 11$, the maximum score with two preparations, or the least peak width at $N = 2$. The earlier exact decision-width result $\kappa_{2/5}^{\text{rows}} = 3$ is unchanged: it allows longer training, whereas this shortest audit uses five decision states. The older points (399, 40602), (320000, 1604) and $(12800 \cdot 2^{400}, 12)$ remain valid constructions with decision width three. They are dominated by (2, 12) in the projection onto training preparations and peak width, but not necessarily in a resource order that also constrains the decision alphabet separately.

Corollary 7.7 (The enlarged physical preparation segment). *With the full architecture of Definition 6.1,*

$$N_c^{\text{phys}}(W, Q) = 2 \quad \text{if} \quad W \geq 12, \quad \text{TV}(Q, Q_*) \leq \beta, \quad u_{\text{ex}} + \beta \leq c < m_2 - \beta.$$

Here $u_{\text{ex}} = (85 - 7\sqrt{73})/64$ is the exact one-preparation all-on value, rather than the older rational upper bound.

Proof. Restrict the adversary to the common-garbling subfamily in Definition 6.1. Theorem 3.2 and Corollary 5.2 bound every at-most-one-preparation auditor, even with unlimited memory, by $u_{\text{ex}} + \beta$. The same deterministic two-preparation machine in Lemmas 7.3 and 7.4 has score at least $m_2 - \beta$ against the full private candidate family. These inequalities prove the assertion. The four-preparation total, peak twelve and positive-noise specialization are unchanged. $\square$

8. AN EXACT PREPARATION–DECISION-MEMORY LAW

A second solvable family separates a genuine joint resource law from an implementation bound. Here the memory coordinate is the number of states at the training-validation cut. It is not identified with the bit-level peak width of the collision audit.

Fix $m = 2^d$, $0 < \alpha < 1$, and an unknown parameter $\theta \in \{1, \dots, m\}$. A fresh training preparation reports θ with probability α and an erasure $\perp$ otherwise. Conditional on θ , the preparations are independent. Candidate validation has law δ_θ and target validation is uniform on $\{1, \dots, m\}$. At the decision cut the auditor retains at most $K = 2^k$ states, $0 \leq k \leq d$. Its subsequent response may depend only on that state, current validation observations and fresh independent randomness. In particular, a calibration table, remembered seed or stopping time outside the retained state is not available. Upstream and downstream memory are otherwise unrestricted for the converse. Let $V_N^{\text{er}}(K)$ be the best worst-case expected signed validation score with at most N training preparations.

Theorem 8.1 (Exact joint law). *For every $N \geq 0$ and $K = 2^k \leq m$,*

$$(28) \quad V_N^{\text{er}}(K) = (1 - (1 - \alpha)^N) \left(1 - \frac{1}{K}\right).$$

The converse covers stochastic encoders and arbitrary legal stopping. The value is attained by a clocked machine with K training states. With atomic preparation and validation outcomes, the complete audit has peak at most $\max\{2K, 3\}$ and uses at most $N + 2$ preparations.

Proof. Write $q = (1 - \alpha)^N$ and pre-generate N potential training reports. A stopped auditor ignores the unused reports; this does not disclose extra information to it. On the event that all N reports are erasures, its retained-state law $r = (r_j)_{j=1}^K$ is independent of θ . After padding any smaller register with unused states, its channel to the decision register has the form

$$(29) \quad C(j \mid \theta) = qr_j + (1 - q)S(j \mid \theta)$$

for a stochastic channel S . This identity remains valid with randomized stopping and randomized state updates.

Use the uniform prior on parameters as an upper certificate. The part qr_j has zero expected score for every response: averaging δ_θ under this prior gives the target itself. For the other part, optimizing over all responses gives a convex function of S , a sum of positive parts of linear functions. Every stochastic channel is a convex combination of deterministic channels $j = g(\theta)$; therefore its value is at most the largest deterministic-channel value. This convexification is used only to bound a channel. It does not presuppose that a mixture of width- K machines is realizable at width K .

If the fibers of g have sizes $n_1, \dots, n_K$, a nonempty fiber has posterior uniform on that fiber, so its total variation from the uniform target is $1 - n_j/m$. The prior-averaged optimal score is

$$\sum_j \frac{n_j}{m} \left(1 - \frac{n_j}{m}\right) = 1 - \sum_j (n_j/m)^2 \leq 1 - 1/K.$$

Empty fibers contribute zero. Cauchy–Schwarz proves the inequality. This is an upper bound on the average score and hence on its minimum over fixed parameters. Multiplying by $1 - q$ proves the upper bound in (28).

For attainment, partition the parameter alphabet into K equal groups and let $g(\theta)$ denote its group. Initialize the register uniformly on the K group labels. On an erasure leave it unchanged; on a revealed parameter overwrite it by $g(\theta)$. No extra “revelation occurred” flag is needed. At the decision cut its law is $q \text{Unif}[K] + (1 - q)\delta_{g(\theta)}$. For register j , freeze the event which is the complement of group j . On a revelation its score is $1 - 1/K$. Under a uniformly random initial label, the mean score is zero for each fixed θ : one of the K labels has score $1 - 1/K$ and the others have score $-1/K$. This proves the exact worst-case value.

The k initialization bits can be accumulated in the K -state register; all earlier prefix counts are at most K . Thus no persistent seed lies outside it. Candidate validation retains the group label and its event indicator, using at most $2K$ states. Target validation produces one of the three values $-1, 0, 1$. This proves the atomic-outcome peak bound. The controller uses a declared external schedule; an autonomous stopping implementation must additionally retain its phase. A phase-tagged implementation is finite, but its cost is not included in K . $\square$

Corollary 8.2 (Thresholds and matching resource scales). *For $K \geq 2$ and $0 \leq c < 1 - 1/K$, the least integer number of training preparations giving a score strictly greater than c is*

$$(30) \quad N_c^{\text{er}}(K) = \left\lceil \frac{\log(1 - c/(1 - 1/K))}{\log(1 - \alpha)} \right\rceil + 1.$$

If $c \geq 1 - 1/K$, no finite N suffices. For $0 < \varepsilon < 1$, achieving $V_N^{\text{er}}(K) \geq 1 - \varepsilon$ requires

$$K \geq \varepsilon^{-1}, \quad N \geq \frac{\log(1/\varepsilon)}{-\log(1 - \alpha)}.$$

Conversely, whenever an allowed $K \leq m$ satisfies $K \geq 2/\varepsilon$, the choice $N \geq \log(2/\varepsilon)/[-\log(1 - \alpha)]$ suffices.

Proof. The strict threshold is equivalent to $(1 - \alpha)^N < 1 - c/(1 - 1/K)$. Since the denominator logarithm is negative, the least integer satisfying this strict inequality is (30), including equality cases of the real quotient. Also

$$1 - V_N^{\text{er}}(K) = q + (1 - q)/K.$$

This is at least q and at least $1/K$, and is at most $q + 1/K$. These three inequalities prove the necessary and sufficient bounds. $\square$

The exact law concerns an entire (N, K, α, m) family, not a single finite diagnostic. Its mechanism is a parameter-independent erasure component followed by the convex geometry of the decision channel. It demonstrates how a preparation obstruction and a memory obstruction can be matched in one theorem. It does not identify the least physical peak width at two collision preparations. Encoding the m -symbol reports bit by bit also introduces cuts not present in this atomic model; the physical bit-level accounting is given separately.

9. QUANTITATIVE CERTIFICATES FOR A FINITE COMPATIBLE ARCHITECTURE

We return to an arbitrary finite architecture, now with a finite parameter set. It can have hidden states, controlled experiments, more than two hypotheses and registers of any size. Enumerate its distinct stochastic rows by $r = 1, \dots, R$ and write $q_r \in \Delta_{s_r}$. A reused autonomous row appears only once in this product. Assume each execution uses at most H controller rows, and uses row r at most d_r times. Delete unused rows and put $S = \max_r s_r$. Fixed environmental kernels are not controller rows.

For an integer $m \geq 1$, intersect Δ_{s_r} with the coordinate boxes of side $1/m$. The resulting nonempty closed polytopes cover the simplex, with overlaps permitted at their boundaries. Write C for a product of such row cells and choose a vertex q_r^C in each factor. A row-cell vertex has at most one coordinate not fixed at a box boundary; consequently it has at most $v_r = s_r 2^{s_r - 1}$ vertices, all on the $1/m$ grid. There are at most $m^{\sum_r s_r}$ product cells.

Write the vertices of row cell C_r as $v_{r,j}$ and parametrize $q_r = \sum_j \lambda_{r,j} v_{r,j}$ with λ_r in a simplex. A reward polynomial has degree at most d_r in block λ_r . Use the homogeneous simplex Bernstein basis of degree d_r in that block, obtaining

$$(31) \quad g_\theta(q(\lambda)) = \sum_\alpha b_{\theta,C,\alpha} B_\alpha(\lambda), \quad B_\alpha \geq 0, \quad \sum_\alpha B_\alpha = 1.$$

Degree elevation includes the factors $\sum_j \lambda_{r,j} = 1$. No independence between different visits of a shared row is imposed on its unknown row vector.

Lemma 9.1 (Causal interpretation of the coefficients). *Every coefficient in (31) satisfies*

$$|b_{\theta,C,\alpha} - g_\theta(q^C)| \leq HS/m.$$

Proof. For each row r , independently prepare an ordered list of d_r vertex labels, sampled independently with law λ_r . At its k th visit, use the stochastic row given by the k th vertex label in that list. Before conditioning, the next label is independent of the previous execution, so this experiment has exactly the row q_r at every visit and expected reward $g_\theta(q(\lambda))$. Unused labels may be sampled after termination without altering the reward.

Condition on the counts of every vertex label in every list. The probability of those counts is $B_\alpha(\lambda)$. Conditional on the counts, each list is a uniformly ordered multiset. Its expected reward is independent of λ and is therefore the corresponding Bernstein coefficient. This also proves (31) directly.

Each row used by this conditioned experiment is a vertex of C_r or a mixture of its remaining vertices. Every two vectors in C_r have coordinate differences at most $1/m$, hence total variation

at most $s_r/(2m) \leq S/(2m)$. Couple the conditioned experiment to the one using q^C , using the same environmental randomness while their executions agree. At each actual controller update the probability of a first row-level disagreement is at most $S/(2m)$. There are at most H such updates on the agreeing path. The terminal laws consequently differ in total variation by at most $HS/(2m)$. A reward in $[-1, 1]$ has expectation difference at most twice that quantity. This proves the estimate. The conditioned urn is used to analyze a coefficient; it is not declared to be an admissible free-memory controller. $\square$

Theorem 9.2 (Explicit subdivision certificate). *Define*

$$\ell_C = \min_{\theta} g_{\theta}(q^C), \quad u_C = \min_{\theta} \max_{\alpha} b_{\theta, C, \alpha}, \quad L_m = \max_C \ell_C, \quad U_m = \max_C u_C.$$

For the exact all-stochastic value of the specified architecture,

$$(32) \quad L_m \leq V(A) \leq U_m \leq L_m + HS/m.$$

On each cell, a witnessing parameter θ_C supplies the nonnegative coefficient certificate

$$u_C - g_{\theta_C}(q(\lambda)) = \sum_{\alpha} (u_C - b_{\theta_C, C, \alpha}) B_{\alpha}(\lambda).$$

Its degree in row block r is d_r , and its total degree is at most $\sum_r d_r$. At most $\prod_r \binom{d_r + v_r - 1}{v_r - 1}$ coefficients per parameter and cell are needed. If the primitive data are rational, all coefficients and bounds can be checked by rational arithmetic. Taking $m \geq HS/\varepsilon$ gives a certified interval of length at most ε .

Proof. The point q^C is feasible, so ℓ_C is a lower bound on the value restricted to C . For any $q \in C$, choose a convex representation of its rows. Equation (31) implies

$$\min_{\theta} g_{\theta}(q) \leq \min_{\theta} \max_{\alpha} b_{\theta, C, \alpha} = u_C.$$

Choose a parameter attaining the minimum defining ℓ_C and apply Lemma 9.1; this gives $u_C \leq \ell_C + HS/m$. Taking the maximum over the finite cover proves (32). Choose θ_C attaining the definition of u_C ; all displayed coefficients of its certificate are nonnegative. The degree and count statements follow from the product Bernstein basis. Cell vertices are rational grid points, and a finite execution sums finite products, proving the rational statement. $\square$

Remark 9.3. The number of cells and coefficients can be enormous. Neither polynomial running time nor a uniform degree bound for the original sum-of-squares hierarchy follows. Refinement changes the domain cover, not merely a single global SOS degree. For a finite collection of architectures obeying a specified budget, maxima of these intervals give outer intervals; an unbounded architecture class needs a separate uniform argument such as Theorem 3.2. For compact parameter families, an effective net and a verified modulus for the primitive kernels are additional inputs, not consequences of compactness alone.

The resource estimate also makes row precision explicit. With $m = 2^b$, the lower witnesses have dyadic rows and lose at most $HS2^{-b}$ in value. Thus increasing row precision provides a quantified approximation for a fixed architecture, while its random-bit implementation still has to be charged. The physical theorem avoids this cost altogether because its rows are deterministic.

10. AUTONOMOUS NOISY TESTING WITH A PRECISION AND TIME LEDGER

This section supplies a substantive autonomous comparison rather than merely identifying shared rows. The stationary optimum is classical [6]; the proof and explicit finite-time implementation are recorded to make the resource conventions unambiguous. They are not used as novelty evidence for the physical theorem.

Let reports be independent, with law $\text{Ber}(p)$ under hypothesis $-$ and $\text{Ber}(q)$ under hypothesis $+$, where $0 < p < 1/2$, $q = 1 - p$, and $r = q/p$. A W -state autonomous machine has two time-independent stochastic transition matrices T_0, T_1 . Assume their induced chains $P_- = qT_0 + pT_1$ and $P_+ = pT_0 + qT_1$ are irreducible. A final decision uses only the state, with optional fresh terminal randomization.

Lemma 10.1 (Likelihood-range bound). *If equivalent probability laws ν_+, ν_- on a finite set satisfy*

$$\frac{\max_i \nu_+(i)/\nu_-(i)}{\min_i \nu_+(i)/\nu_-(i)} \leq R,$$

then $\text{TV}(\nu_+, \nu_-) \leq (\sqrt{R} - 1)/(\sqrt{R} + 1)$.

Proof. Write $L = d\nu_+/d\nu_-$, $a = \min L$ and $b = \max L$. Since $\mathbb{E}_{\nu_-} L = 1$, one has $a \leq 1 \leq b$. The chord bound for $(L - 1)_+$ gives

$$\text{TV}(\nu_+, \nu_-) = \mathbb{E}_{\nu_-} (L - 1)_+ \leq \frac{(1 - a)(b - 1)}{b - a}.$$

For fixed b/a , maximizing the last expression gives $a = 1/\sqrt{b/a}$; substitution yields $(\sqrt{b/a} - 1)/(\sqrt{b/a} + 1)$. It is increasing in b/a , proving the assertion. The identical-law case follows directly. $\square$

Theorem 10.2 (Autonomous stationary bound and an explicit approximation). *Put $D = W - 1$, $W \geq 2$. In the irreducible autonomous class the infimum stationary minimax error is*

$$(33) \quad e_W = \frac{1}{1 + r^D}.$$

For every $\epsilon = 2^{-b}$, $b \geq 0$, there is a machine whose rows have denominator dividing 2^{b+1} , and whose minimax error at report time n from any initial state is at most

$$(34) \quad e_W + \epsilon(W - 2) + \exp \left\{ - \left(\frac{p\epsilon}{2} \right)^D \left\lfloor \frac{n}{D} \right\rfloor \right\}.$$

The stationary lower bound is not a finite-horizon lower bound.

Proof. Every positive induced transition has its likelihood ratio in $[r^{-1}, r]$, and the supports of the two induced matrices are identical. The Markov chain tree formula writes each stationary weight as a sum of products over directed spanning trees rooted at that state. Each product has D transitions; its ratio lies in $[r^{-D}, r^D]$. The ratio of the two unnormalized stationary weights lies in the same interval. Normalizing multiplies all state likelihood ratios by a common number, so their maximum-to-minimum ratio is at most r^{2D} . Lemma 10.1 therefore bounds stationary total variation by $(r^D - 1)/(r^D + 1)$. The equal-prior Bayes error is at least $1/(1 + r^D)$, and a minimax error cannot be smaller than that Bayes error. This is the equal-prior binary specialization of the Hellman–Cover bound.

For the upper approximation, take states $i = 0, \dots, D$, weights $c_0 = c_D = 1$, and $c_i = \epsilon$ at interior states. On report 1, move from i to $i + 1$ with probability $a_i = \frac{1}{2} \min(1, c_{i+1}/c_i)$, if that neighbor exists; otherwise stay. On report 0, move to $i - 1$ with probability $b_i = \frac{1}{2} \min(1, c_{i-1}/c_i)$, if it exists; otherwise stay. All other probability is assigned to staying. Detailed balance gives

$$\pi_+(i) = \frac{c_i r^i}{\sum_j c_j r^j}, \quad \pi_-(i) = \frac{c_i r^{-i}}{\sum_j c_j r^{-j}}.$$

Reflection $i \mapsto D - i$ exchanges these laws. Decide $+$ for $i > D/2$, $-$ for $i < D/2$, and use a fair terminal decision at $i = D/2$ when this state exists. The two errors agree. Conditional on being

at an endpoint, the error is exactly e_W . The total interior stationary mass is at most $\epsilon(W - 2)$, since each $r^i \leq r^D$ and the denominator is at least r^D under $+$, with the reflected argument under $-$. This proves the first two terms of (34).

Every existing neighbor transition under either hypothesis has probability at least $p\epsilon/2$, and every self-loop has probability at least $1/2$. From any state, move down to zero and then stay until D steps have elapsed. This yields the common minorization $P_{\pm}^D(i, \{0\}) \geq \delta = (p\epsilon/2)^D$. Subtracting that common atom gives contraction in total variation by $1 - \delta$ every D steps. Comparison with the stationary law yields $\text{TV}(P_{\pm}^n(i, \cdot), \pi_{\pm}) \leq (1 - \delta)^{\lfloor n/D \rfloor} \leq e^{-\delta \lfloor n/D \rfloor}$. A bounded decision changes error by at most this amount, proving (34). Finally, letting n tend to infinity and then ϵ to zero proves that the stationary infimum equals (33). All transition probabilities belong to $\{0, 1, 1/2, \epsilon/2, 1 - 1/2, 1 - \epsilon/2\}$. $\square$

The W states in this theorem refer to an atomic stochastic update after each report. A sequential fair-bit implementation must also retain the report, the current state, a bit counter and a flag. Each nontrivial move probability is $1/2$ or $2^{-(b+1)}$; test for a string of zero bits of the required length, and pad to $b + 1$ bits if a uniform report cycle is required. At most $4W(b + 2)$ states suffice for this implementation, including its bit-phase counter and a ready phase. This conservative bound is an implementation claim, not an optimal precision–state tradeoff. The time index n in (34) is an observation time at which an external analyst reads the autonomous machine, not a free input clock used by its transitions. The adversarial model of [10] is not covered by the independent-report proof above.

11. PHYSICAL CONSUMERS AND THE FIRST-PRINCIPLES CONNECTION

The physical preparation theorem needs no estimate of the target law as a stored input: the five events are fixed and their uniform validity uses a proved neighborhood of Q_* . This removes calibration memory from this particular consumer without asserting that target acquisition is unnecessary in an arbitrary comparison problem. The general acquired calibration theorem, with its full counter register, remains in the complete companion.

Corollary 11.1 (A raw-interface confidence construction). *Let $m = m_2 - \beta > 0$ and $0 < c < m$. Repeat the full four-preparation audit independently R times and write $\overline{D}_R$ for the mean score. Under every private candidate,*

$$\mathbb{P}(\overline{D}_R \leq c) \leq \exp\{-R(m - c)^2/2\}.$$

Under the null in which the independent candidate and target validation laws agree, with the same event chosen before both validations,

$$\mathbb{P}(\overline{D}_R > c) \leq \exp\{-Rc^2/2\}.$$

A clocked implementation uses $4R$ preparations and at most $12(2R + 1)$ persistent states, with no calibration and no random bits.

Proof. The independent block scores lie in $[-1, 1]$ and have mean at least m under a private candidate. Hoeffding's inequality gives the first bound. Under the null, conditional on the selected event, the two independent validation indicators have the same mean. Each block therefore has mean zero, and the same inequality gives the second bound. Retain the integer sum in $\{-R, \dots, R\}$ along with the current twelve-state block. At a block boundary add its terminal score and reset its register. The deterministic external schedule is part of the clocked convention. $\square$

This is an upper construction on the raw marked interface. Its proof does not transfer a lower bound from a deliberately reduced Bernoulli interface, and no optimal confidence-memory frontier is claimed.

Proposition 11.2 (Target transport without a training multiplier). *For the two-preparation audit, changing only the target from Q to Q' changes its expected score by at most $\text{TV}(Q, Q')$. Its candidate training law, register profile and event selector remain unchanged.*

Proof. Condition on the candidate training word. The expectation difference is $(Q - Q')h_S$, whose absolute value is at most $\text{TV}(Q, Q')$. Average over that unchanged training law. $\square$

In particular, every common-preparation transport estimate verified for the original positive-noise collision family can be inserted here without multiplying it by a long training horizon. If candidate kernels also change, their training and validation defects must be added by a causal coupling; the target-only statement does not remove those terms.

The first-principles route is now concrete. The positive experimental kernels and actual reset preparations give the likelihoods in (5); these generate posterior continuation states. The validation-test quotient gives the positive parts in the statistical dual. A prior then supplies an architecture-uniform impossibility certificate. On the constructive side, the five-event decision is compiled by taking its future residual functions, which are closed under observation shifts. These are compatible causal states and determine the charged profile. Thus the preparation law, predictive continuation and finite-resource realization are used in the same upper/lower theorem. This is not a deduction of geometry or nonlinear operator compactness from the existence of finite stochastic rows.

12. DEPENDENCIES, RETAINED RESULTS AND REMAINING DISTINCTIONS

The exact three-report binary noisy frontier, arbitrary-horizon binary-register minimax recursion and compatible Bayesian interval recursion remain as proved in the complete companion. Its exact odd-horizon majority realization is also retained; it is not a new result of this revision. Section 9 applies to registers of any finite width, controlled rows and any finite parameter set, but provides certified intervals rather than a closed minimax recursion for all such classes. The physical theorem supplies a matched preparation segment in a continuous two-parameter family with overlapping stochastic reports; its converse is not a finite parameter-grid check.

The stationary autonomous result in Section 10 has an explicit formula, but its stationary lower bound must not be applied to a finite-horizon clocked machine. Its finite-time upper bound is purposely conservative and exposes the slowing caused by finite transition precision. An arbitrary compact-family presentation still requires additional effective information to construct a kernel net. A subdivided positive-basis degree bound must not be called a degree bound for a global SOS hierarchy.

The physical result is a genuine consumer of the experimental and continuation root. It does not make the independent A2 geometric proofs dependent on this root. The original B4 normalized-resolvent, range, compactness and control-transfer obligations, and the broad C2 strict-dual, form, rigidity and optional-projection obligations, are not proved by our finite-state theorem. Their statements, derivations and current status remain in the retained historical pipeline. No aggregate closure is inferred from preserving their source files.

Finally, the rational certificate (25), polynomial identity (27), and residual transitions are independently reproducible. The program in the accompanying source checks their exact identities and includes deliberately invalid controls. These checks are not substitutes for the proofs above. Neither compilation, preservation of earlier pages nor a successful finite test establishes novelty or an editorial decision.

REFERENCES

- [1] R. D. Smallwood and E. J. Sondik, The optimal control of partially observable Markov processes over a finite horizon, *Operations Research* **21** (1973), 1071–1088.
- [2] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272.
- [3] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176.
- [4] J. Müller and G. Montúfar, The geometry of memoryless stochastic policy optimization in infinite-horizon POMDPs, arXiv:2110.07409 (2021; revised versions).
- [5] M. Putinar, Positive polynomials on compact semi-algebraic sets, *Indiana Univ. Math. J.* **42** (1993), 969–984.
- [6] M. E. Hellman and T. M. Cover, Learning with finite memory, *Ann. Math. Statist.* **41** (1970), 765–782.
- [7] M. E. Hellman and T. M. Cover, On memory saved by randomization, *Ann. Math. Statist.* **42** (1971), 1075–1078.
- [8] R. A. Flower and M. E. Hellman, Hypothesis testing with finite memory in finite time, *IEEE Trans. Inform. Theory* **18** (1972), 429–431.
- [9] T. M. Cover, M. A. Freedman and M. E. Hellman, Optimal finite memory learning algorithms for the finite sample problem, *Information and Control* **30** (1976), 49–85.
- [10] M. A. Managoli and V. M. Prabhakaran, Memory constrained adversarial hypothesis testing, arXiv:2605.12063v1 (May 12, 2026).
- [11] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28, doi:10.1524/strm.2002.20.14.1.
- [12] Q. Qi, General Theta Foundations I: Compatible statistical experiments and resource frontiers, revision 23, September 24, 2026, complete mathematical companion, source commit 8a6043aca644043bd805668185eb227468f21170; packaged commit e926b783b643b3e7f275a18105528927b76cc97b. The 96-page article and 649-page development are reproduced without alteration after the present article in the two complete volumes.

PRESERVED PREDECESSOR

General Theta Foundations I, revision 24

The following volume is reproduced without alteration.

The current revision precedes this divider.

Predecessor commit:

ad1336e48e414b2c01e928b844c796ab4765178b

GENERAL THETA FOUNDATIONS I: STATISTICAL RESOURCE DUALITY AND TWO-PREPARATION PHYSICAL CERTIFICATION

QIAN QI

ABSTRACT. We study the resources needed to distinguish a marked experiment from a family of private causal simulators. We separate the optimization of one finite architecture from the optimization over all admissible architectures. For reset experiments whose feedback rows are garblings of a common row, we express the unrestricted-memory preparation value as a minimum of posterior-predictive total variations. The resulting dual bounds every adaptive finite-memory architecture, without selecting an architecture in advance. For the positive-noise marked collision experiment, a four-point rational dual certificate excludes all auditors with at most one training preparation. A deterministic auditor with two training preparations attains a strictly larger uniform score. Its complete bit-level register profile has peak twelve, proving that two training preparations are necessary and sufficient at every peak budget at least twelve on an explicit interval of score thresholds. For general finite architectures we give subdivided positive-basis certificates of explicitly bounded degree and error, including reused stochastic rows. We also give a fully priced autonomous binary-testing construction, with the stationary bound attributed to Hellman and Cover. The earlier compatible-occupation, noisy-testing, continuation and physical-comparison developments are retained in the accompanying complete mathematical volumes.

CONTENTS

1. Introduction	2
1.1. Relation to the literature	2
1.2. The mathematical volumes	3
2. Experiments, architectures and attainable resources	3
3. A statistical dual that is uniform over architectures	4
4. The physical experiment and its sharp preparation boundary	6
4.1. An exact rational certificate against one preparation	6
4.2. A marked five-event selector	7
4.3. Causal compilation and the full register profile	8
5. Quantitative certificates for a finite compatible architecture	9
6. Autonomous noisy testing with a precision and time ledger	11
7. Physical consumers and the first-principles connection	12
8. Dependencies, retained results and remaining distinctions	13
References	14

Date: September 24, 2026. Revision 24.

2020 Mathematics Subject Classification. Primary 62B05, 62C20; Secondary 60J10, 68Q45, 93E20.

Key words and phrases. Statistical experiments, causal realization, predictive distributions, finite memory, minimax duality, physical certification.

1. INTRODUCTION

An exact description of finite stochastic controllers does not by itself determine a resource frontier. There are two distinct questions. The first asks for the best controller within a specified architecture. The second asks whether any architecture satisfying a resource budget can perform a specified statistical task. A lower bound for the first question need not answer the second. We develop certificates that respect this distinction.

The principal application is a marked collision experiment. Its ideal all-on law is supported on four atoms, but neither the simulator nor the auditor is restricted to a four-atom transcript. The simulator produces two independent report bits and an independent actual mark; the target has a correlation between its reports and that mark. The auditor may use the original feedback rows, arbitrary stochastic transitions and stopping rules, subject to the reset and validation interface. Our upper certificate for the auditor is valid simultaneously for all such architectures. The matching construction uses the mark during validation and not during training. This distinction permits a much shorter preparation schedule than an audit based on unmarked diagonal events.

Put

$$(1) \quad u_1 = \frac{4035777}{10240000}, \quad m_2 = \frac{12 - 3\sqrt{3}}{16}, \quad \beta_0 = \frac{1}{300}.$$

The main resource conclusion is

$$(2) \quad N_c^{\text{phys}}(W, Q) = 2 \quad \text{if} \quad W \geq 12, \quad \text{TV}(Q, Q_*) \leq \beta, \quad u_1 + \beta \leq c < m_2 - \beta.$$

Here N counts candidate training preparations; two additional, fresh validation preparations are charged separately. The objective is a strict expected signed score greater than c . In particular, (2) applies to $c = 2/5$ and $\beta = \beta_0$. The lower bound permits unlimited memory, whereas the construction has peak width twelve. This is an exact preparation boundary on a high-width segment, not a claim that twelve is the least possible peak width.

Three mechanisms organize the proof. First, a dominating experimental row removes the outer choice of feedback schedules from the converse. Second, minimax duality turns a prior on fixed simulator parameters into a posterior-predictive upper certificate. Third, residual continuation functions compile the winning decision into a compatible small-memory machine. The general algebraic encoding is useful for checking that the compiled machine belongs to the intended class, but no generic Positivstellensatz is needed to prove (2).

For finite architectures we also supply quantitative information not contained in eventual sum-of-squares completeness. A partition of the stochastic-row simplices into cells of diameter $O(m^{-1})$ gives positive-basis certificates with error at most HS/m , where H bounds the number of row uses and S bounds the number of destinations in a row. The block degree is the maximum number of times the corresponding row can be used. The argument works with autonomous row reuse. It gives an effective, generally exponential, certification procedure; it is not a uniform degree bound for the unmodified Putinar hierarchy.

1.1. Relation to the literature. Comparison of experiments and minimax theory provide the classical background [1, 2]. Polynomial feasible-frequency formulations for partially observed stochastic control are treated by Müller and Montúfar [3]; the rank-one compatibility constraints below are not presented as a new general conditional-independence device. Likewise, strict positivity certificates use established real algebraic geometry [4]. Our quantitative subdivision theorem is a positive-basis construction with a causal error estimate, rather than a new general Positivstellensatz.

Finite-state testing has a substantial earlier theory. Hellman and Cover [5, 6], Flower and Hellman [7], and Cover, Freedman and Hellman [8] distinguish time-invariant rules, finite horizons

and randomization. We explicitly identify the stationary result in Section 6 as a specialization of that theory. Managoli and Prabhakaran [9], posted in May 2026, study randomized time-invariant finite-state binary testing against adaptive adversarial distribution choices and establish matching state exponents, with exact bounds for a class of problems. Our physical theorem has a different objective: a finite training–validation signed score against a fixed private simulator, with bit-level memory and preparations charged. We make no priority claim for the general phenomenon that randomization can save states.

Norberg’s filtered comparison of experiments [10] is pertinent to the broader causal comparison program. The original proof-level comparison remains incomplete: the accessible publisher record verifies the bibliographic identity but does not justify an assertion that its results exclude the present constructions. None of our proofs uses such an exclusion.

1.2. The mathematical volumes. The present article gives the new proof chain. The accompanying *Complete Mathematical Manuscript* contains this article followed by the unaltered 96-page revision 23. The *Complete Development* contains this article followed by the unaltered 649-page development. Thus the earlier general continuation, occupation, noisy binary, physical, Gaussian and operator interfaces remain available with their full hypotheses and proofs [11]. They are not silently replaced by (2). In particular, the independent A2 geometric chain and the unresolved historical B4 and broad C2 obligations remain distinct. The mathematical role of the present first foundation is an experimental and resource-comparison root; a universal derivation of every program branch is not assumed.

2. EXPERIMENTS, ARCHITECTURES AND ATTAINABLE RESOURCES

Throughout, total variation is $\text{TV}(P, Q) = \sup_{0 \leq h \leq 1} (P - Q)h = \frac{1}{2} \sum_x |P(x) - Q(x)|$. An experimental parameter is unknown and fixed throughout an execution. Controller randomization is independent of that parameter. A reset preparation is a fresh draw, independent of earlier preparations given the parameter. A controller may not recover discarded information from an uncharged tape, clock, selector or calibration table.

Definition 2.1 (Finite architecture). A finite architecture A specifies a finite causal execution graph, its legal experimental operations, the observable inputs to each controller row, and a register alphabet at every cut. It also specifies which row occurrences share a stochastic table. A clocked architecture may use different rows at different scheduled times. An autonomous architecture may use time only through a charged state. For a parameter family Θ and terminal reward in $[-1, 1]$, let

$$V(A) = \sup_{q \in \prod_r \Delta_{sr}} \inf_{\theta \in \Theta} g_\theta(A, q).$$

If $\mathcal{A}(R)$ is the class of architectures obeying a global resource budget R , its outer value is

$$V^{\text{out}}(R) = \sup_{A \in \mathcal{A}(R)} V(A).$$

Both the architecture and the simulator parameter are fixed before the controller’s random choices. In particular, a mixture of complete controllers is not automatically an admissible controller at the original register profile.

For a finite hidden execution graph, compatibility can be stated without division. Let $x_{\theta ir}$ be the incoming mass of hidden history i at observable row r , and let $f_{\theta irj}$ be the outgoing mass through choice j . Include the usual fixed-kernel flow equations, as well as

$$(3) \quad f_{\theta irj} \geq 0, \quad \sum_j f_{\theta irj} = x_{\theta ir}, \quad f_{\theta irj} x_{\theta' i' r} = f_{\theta' i' rj} x_{\theta ir}.$$

For shared rows, the indices i, i' include all their occurrences.

Proposition 2.2 (Compatible realization). *For a specified finite architecture, nonnegative flow arrays satisfying (3) are exactly those realized by a common stochastic row at every declared observable input, including at zero incoming mass.*

Proof. Necessity follows from $f_{\theta irj} = x_{\theta ir} q_r(j)$. For sufficiency, if some incoming mass at row r is positive, define $q_r(j)$ by dividing the corresponding outgoing mass by that incoming mass. Conservation makes q_r stochastic, and the cross-products force the same ratio at every other positive incoming mass. If an incoming mass vanishes, nonnegativity and conservation force all its outgoing masses to vanish. If the entire row is unreachable, choose any legal stochastic row. Induction through the finite execution graph reconstructs the prescribed flows. The same ratio argument applies to identified occurrences of an autonomous row. $\square$

This formulation is inherited from the complete companion and is included to fix the controller class. Compact semialgebraic feasibility and an Archimedean positivity theorem provide a complete semialgebraic certificate hierarchy for a *fixed finite architecture*. They do not interchange the two optimizations in the definition. Section 5 gives a quantitative alternative for the inner problem. The next section supplies a different certificate that survives the outer optimization.

3. A STATISTICAL DUAL THAT IS UNIFORM OVER ARCHITECTURES

Let Θ be compact, let Ω be finite, and let $\theta \mapsto P_\theta \in \mathcal{P}(\Omega)$ be continuous. Fix $Q \in \mathcal{P}(\Omega)$. An N -preparation reset selector observes $Z = (Z_1, \dots, Z_N) \sim P_\theta^{\otimes N}$ and chooses a response $h_Z \in [0, 1]^\Omega$. Its expected validation score is

$$(4) \quad G_\theta(h) = \sum_{z \in \Omega^N} P_\theta^{\otimes N}(z) (Q - P_\theta) h_z.$$

The response is fixed before two independent validation draws. Operationally it can be realized by choosing a random event with those coordinate marginals and retaining that event for both validations. No memory bound is imposed in the definition of $U_N(P, Q) = \max_h \min_\theta G_\theta(h)$.

Theorem 3.1 (Posterior-predictive dual). *Both extrema in the following identity are attained:*

$$(5) \quad U_N(P, Q) = \min_{\mu \in \mathcal{P}(\Theta)} \sum_{z, \omega} \left[\int P_\theta^{\otimes N}(z) \{Q(\omega) - P_\theta(\omega)\} d\mu(\theta) \right]_+$$

$$(6) \quad = \min_{\mu \in \mathcal{P}(\Theta)} \sum_z m_\mu(z) \text{TV}(Q, \bar{P}_\mu(\cdot | z)).$$

Here $m_\mu(z) = \int P_\theta^{\otimes N}(z) d\mu(\theta)$ and, when $m_\mu(z) > 0$,

$$\bar{P}_\mu(\omega | z) = \frac{\int P_\theta^{\otimes N}(z) P_\theta(\omega) d\mu(\theta)}{m_\mu(z)}.$$

A zero-mass term in (6) is defined to be zero.

Proof. The response space is a finite-dimensional compact convex box. Probability measures on the compact space Θ form a weakly compact convex set. The integral of (4) is continuous and affine in each argument. The minimum over parameters equals the minimum over probability measures. The minimax theorem [2] therefore interchanges the maximum and minimum. For fixed μ , each response coordinate is independent in the maximization and its contribution is the positive part of its coefficient, proving (5). The coefficient vector at a fixed z has zero sum. For positive $m_\mu(z)$ it equals $m_\mu(z)(Q - \bar{P}_\mu(\cdot | z))$, proving (6). If $m_\mu(z) = 0$, every nonnegative likelihood integrand vanishes μ -almost surely and all coefficients are zero. Continuity on the two compact strategy spaces gives attainment. $\square$

Theorem 3.2 (Dominating-row converse). *Suppose every permitted training row is a parameter-independent causal garbling of one fresh draw from P_θ . Suppose every permitted validation row is the same garbling of the target and candidate laws, and every selected row–event pair is fixed before validation. Then $U_N(P, Q)$ bounds the worst-case expected score of every auditor using at most N candidate training preparations, regardless of its register profile, feedback schedule, independent randomization, or stopping rule. Independent target-only calibration, whose law does not depend on θ , does not invalidate the bound. If the all-on row and all its events are permitted, the bound is attained in the class with no memory cap and N training preparations.*

Proof. Generate N independent all-on observations in advance. At the t th requested preparation, apply the declared garbling to observation t , using only previously simulated reports and independent randomization. This simulates the adaptive transcript with its exact law. A stopped execution ignores the unused observations; it does not reveal them to the auditor. Pull back its final row–event response to the all-on alphabet. Conditional on the entire generated training word and the independent random seed, this is a function in $[0, 1]^\Omega$. Averaging the seed, including any target-only calibration, produces one response h_z for each full word z . The score is consequently (4). This proves the upper bound for the union of all architectures. Conversely, retaining the entire word and then a selected event implements any such finite response kernel when memory is unrestricted. Fresh validation gives exactly (4). $\square$

Remark 3.3 (What the dual does not convexify). The measure μ is an adversarial prior used to bound the minimum over *fixed* parameters. Its likelihood is $\int P_\theta^{\otimes N} d\mu(\theta)$, not $(\int P_\theta d\mu(\theta))^{\otimes N}$. It neither enlarges the private candidate class to mixtures nor gives the auditor a free persistent selector. Attainment of U_N without a memory cap does not assert attainment at any specified finite profile.

The state in (6) is generated by the actual preparation law, not by a geometric measure chosen independently of the experiment. For sequential continuation one retains the posterior μ_z or its future-test equivalence class; a single predictive marginal need not be closed under further observations. This is the distinction between a one-step response quotient and a future-predictive quotient.

Corollary 3.4 (Finite moment support for the product family). *For two independent Bernoulli report bits with parameters (p, q) and an independent fair mark, an optimal prior in (5) can be chosen with at most $(N + 2)^2$ atoms. The two report counts (k, ℓ) are a causally updatable sufficient statistic for computing the dual.*

Proof. Conditional on the two counts, all arrangements of report bits and all mark words have parameter-independent probabilities. Summing their coefficients reduces (5) to the $(N + 1)^2$ count pairs. Their likelihoods are

$$\binom{N}{k} p^k (1 - p)^{N-k} \binom{N}{\ell} q^\ell (1 - q)^{N-\ell}.$$

After multiplication by $Q(\omega) - P_{p,q}(\omega)$, every integrand has degree at most $N + 1$ in each variable. Thus the objective depends only on the $(N + 2)^2$ rectangular moments. Their constant coordinate is one. Carathéodory’s theorem in the remaining affine space replaces an optimal moment vector by a convex combination of at most $(N + 2)^2$ point evaluations. Counts update by addition of the current bits. No claim that this count register is optimal for the decision is made. $\square$

4. THE PHYSICAL EXPERIMENT AND ITS SHARP PREPARATION BOUNDARY

Take $\Omega = \{0, 1\}^3$, with coordinates (x_1, x_2, w) . The all-on private candidate is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad 0 \leq p, q \leq 1.$$

The mark w is the actual mark, not an invented simulator bit. Put $a = (1 - p)(1 - q)$ and $c = pq$. On the ordered atoms

$$(7) \quad 000, \quad 001, \quad 110, \quad 111,$$

let Q_* have masses $5/16, 3/16, 3/16, 5/16$, and set it to zero elsewhere. The candidate masses on these atoms are $a/2, a/2, c/2, c/2$. A target Q is allowed whenever $\text{TV}(Q, Q_*) \leq \beta$.

The original feedback operation has the form

$$(8) \quad T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w), \quad g : \{0, 1\} \rightarrow \{0, 1\}.$$

The selected gate is observable. It may depend on earlier preparations and on the first current report. The target rows are pushforwards by (8). For the converse, restrict the adversary to the admissible private subfamily whose second emission is AX_2 , where A is the current gate and $X_2 \sim \text{Ber}(q)$ is independent of the first emission and of the fair mark. Every row is then a causal garbling of $b(p, q)$. For the construction only the all-on row is used, and every private candidate in the original width-one class has the all-on product law just specified. These two facts, proved for the microscopic interface in [11, physical-complexity, feedback-completion lemma], explain why using a subfamily for the lower bound does not narrow the upper construction.

The admissible audit first trains, then selects one row–event pair, then uses one fresh candidate and one fresh target validation. It reports $D = \mathbf{1}_A(Y) - \mathbf{1}_A(X)$. The event and row cannot be reselected after seeing a validation draw. All bit, mark, decision and validation cuts are charged. Define $N_c^{\text{phys}}(W, Q)$ to be the minimum number of candidate training preparations among these original-interface auditors, with peak register width at most W , whose score is uniformly strictly greater than c . Randomization, feedback and stopping are allowed. If no such audit exists, the value is $+\infty$.

Theorem 4.1 (Two-preparation physical segment). *Let u_1, m_2 be as in (1). For every $\beta \geq 0$, $\text{TV}(Q, Q_*) \leq \beta$, and*

$$W \geq 12, \quad u_1 + \beta \leq c < m_2 - \beta,$$

one has $N_c^{\text{phys}}(W, Q) = 2$. The two-preparation audit is deterministic, uses no acquired calibration and no random bits, and has the full profile

$$(9) \quad (1, 2, 3, 3, 4, 5, 5, 10, 12, 10, 10, 7, 3)$$

from the initial cut through its twelve bit/mark updates. In particular, its decision alphabet has five labels and its peak has twelve. The lower bound remains valid with no memory cap.

The proof occupies the next three subsections. The interval in the statement can be empty for large β ; no assertion is then made. For $\beta = 1/300$, it contains $c = 2/5$ strictly between its endpoints.

4.1. An exact rational certificate against one preparation.

Lemma 4.2. *Every original-interface auditor using at most one candidate training preparation has worst-case score at most $u_1 + \beta$, regardless of memory.*

Proof. By Theorem 3.2, it suffices to bound $U_1(b, Q_*)$ and then perturb the target. In (5) use the four fixed parameters

$$(p, q) = (1/8, 1/8), (1/5, 1/5), (4/5, 4/5), (7/8, 7/8)$$

with weights $1/16, 7/16, 7/16, 1/16$, respectively. This measure is not asserted to be least favorable. Since $p = q$ on its support, the training count $K = x_1 + x_2$ is sufficient. The independent training mark and the order of unequal bits contain no further information about this prior.

For $k = 0, 1, 2$ and the four validation atoms in (7), the signed coefficients in (5), after grouping by K , are the entries of

$$(10) \quad \frac{1}{40960000} \begin{pmatrix} -1977 & -1775177 & 2170423 & 3943623 \\ 1765554 & 191954 & 191954 & 1765554 \\ 3943623 & 2170423 & -1775177 & -1977 \end{pmatrix}.$$

For example, the coefficient in column j is

$$\sum_{i=1}^4 \mu_i \binom{2}{k} p_i^k (1-p_i)^{2-k} \left(Q_{*,j} - \frac{(1-p_i)^2 \mathbf{1}_{j \leq 2} + p_i^2 \mathbf{1}_{j \geq 3}}{2} \right).$$

This identity proves every entry of (10) by rational arithmetic. All coefficients at off-diagonal validation atoms are nonpositive. The sum of the positive entries in (10) is $4035777/10240000 = u_1$. Each response coordinate is in $[0, 1]$, so this sum bounds its prior-averaged score and hence its minimum over the four fixed candidates.

Replacing Q_* by Q changes the score of any trained response by at most β . The training law is a candidate law and is unchanged. Consequently the same prior bounds every score by $u_1 + \beta$. A procedure with no training, or one that stops before taking its permitted sample, is covered by ignoring the generated all-on observation. $\square$

4.2. A marked five-event selector. Take two independent all-on training preparations. Let S be the total number of ones among their four report bits; discard their marks. Choose the following events, represented by their indicators on (7) and with indicator zero at every off-diagonal atom:

$$(11) \quad \begin{array}{c|cccc} S & 000 & 001 & 110 & 111 \\ \hline 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & 0 \end{array}$$

Denote these indicators by $h_0, \dots, h_4$ and report $D = h_S(Y) - h_S(X)$ on independent validation draws.

Lemma 4.3 (Exact uniform score of the selector). *Against the ideal target, the selector (11) has uniform minimum score $m_2 = (12 - 3\sqrt{3})/16$. Against Q , its score is at least $m_2 - \beta$.*

Proof. The generating polynomial of S is

$$((1-p) + pz)^2((1-q) + qz)^2.$$

Writing $g_s = (Q_* - b(p, q))h_s$, its expected score is $G(p, q) = \sum_{s=0}^4 [z^s]((1-p) + pz)^2((1-q) + qz)^2 g_s$. Substitution of the five rows in (11) and collection of terms with $x = p + q - 1$, $y = p - q$, $u = x^2$ and $v = y^2$ gives

$$(12) \quad \begin{aligned} G(p, q) = H(u, v) = & -\frac{u^3}{32} + \frac{3u^2v}{32} + \frac{3u^2}{16} - \frac{3uv^2}{32} - \frac{uv}{4} - \frac{3u}{32} \\ & + \frac{v^3}{32} + \frac{v^2}{16} + \frac{15v}{32} + \frac{7}{16}. \end{aligned}$$

This is a polynomial identity, not an interpolation over a grid. Since $0 \leq u \leq 1$ and $v \geq 0$ on the parameter square,

$$\partial_v H(u, v) = \frac{3(u-v)^2 - 8u + 4v + 15}{32} \geq \frac{7}{32}.$$

Thus $H(u, v) \geq H(u, 0) = (-u^3 + 6u^2 - 3u + 14)/32$. The derivative of the last expression is $(-3u^2 + 12u - 3)/32$. On $[0, 1]$ it changes sign from negative to positive at $u = 2 - \sqrt{3}$. Substitution gives $(12 - 3\sqrt{3})/16$. The minimizing values are feasible: take $p = q = (1 \pm \sqrt{2 - \sqrt{3}})/2$. This proves exactness of the minimum for this specified selector, not optimality of its score over all selectors with two preparations. Finally, for each s , the target expectation of $h_s \in [0, 1]^\Omega$ changes by at most β . Averaging over training proves the claimed robustness. $\square$

4.3. Causal compilation and the full register profile.

Lemma 4.4. *The selector and its two validations admit the deterministic profile (9). No selector, first-bit buffer or validation accumulator is outside the charged register. An autonomous phase-tagged realization uses at most 75 states.*

Proof. After each training report bit retain the count accumulated so far. The two marks are read and discarded. The first six updates therefore have widths 2, 3, 3, 4, 5, 5 after the initial singleton. At the decision cut the five count values select the five events.

During candidate validation, after the first bit retain the event and that bit, using at most ten states. After the second bit retain the event and the function of the unread mark that gives its candidate-event indicator. For events 0, 1, 2, 3, 4, respectively, the possible functions are

$$\{0, 1\}, \quad \{0, 1 - w, 1\}, \quad \{0, 1\}, \quad \{0, w, 1\}, \quad \{0, 1\}.$$

There are twelve event–function pairs. After the candidate mark, retain the event and its now known candidate indicator $c_0 \in \{0, 1\}$, using at most ten states.

During target validation, after the first bit discard the event name and retain c_0 and its residual function of the second bit and mark. These residual functions belong to

$$\{0, x_2, (1 - x_2)(1 - w), 1 - x_2, x_2 w\}.$$

This uses at most ten states. After the second bit, the residual *difference* as a function of the unread mark belongs to

$$\{-1, 0, 1, w, 1 - w, w - 1, -w\},$$

so seven states suffice. The mark update gives the three possible outputs $-1, 0, 1$. Each transition depends only on its retained residual and the current report. Thus it is a legal common observable row, not a family of independently optimized hidden-history maps.

For completeness, equality of two residual functions implies equality of their restrictions after either next bit, so identifying equal residuals is closed under every subsequent update. The supplied transition tables are obtained by these restrictions and can be reconstructed directly from (11). Enumerating residuals is a convenience for reproducing the tables, not the proof of their sufficiency. Finally, disjointly tag the states at the thirteen cuts by their phase and use terminal self-loops. The sum of the displayed widths is 75. This implementation charges the clock; it does not assert that 75 is optimal. $\square$

Proof of Theorem 4.1. Lemma 4.2 excludes $N \leq 1$ for a strict score greater than $c \geq u_1 + \beta$, for every architecture and every memory budget. Lemmas 4.3 and 4.4 supply a two-preparation architecture of peak twelve with score at least $m_2 - \beta > c$. Larger width budgets can use the same machine. $\square$

Corollary 4.5 (The positive-noise collision family). *For the original marked collision targets with $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$, the inherited microscopic estimate $\text{TV}(Q_{d,\sigma}, Q_*) < 1/300$ implies*

$$N_{2/5}^{\text{phys}}(W, Q_{d,\sigma}) = 2 \quad (W \geq 12).$$

There are four preparations in the complete audit: two candidate training, one candidate validation, and one target validation.

Proof. Apply Theorem 4.1. The lower-end margin is the exact rational number

$$\frac{2}{5} - u_1 - \frac{1}{300} = \frac{78269}{30720000} > 0.$$

For the other endpoint, $\sqrt{3} < 7/4$ implies $m_2 > 27/64$ and $27/64 - 1/300 > 2/5$. The physical total-variation estimate, including the actual mark and positive sensor noise, is proved in the retained microscopic-composition and physical-complexity modules [11]; the current deduction does not replace those hypotheses by a zero-noise limit. $\square$

Remark 4.6 (The part of the frontier determined here). The theorem evaluates the minimum preparation coordinate for $W \geq 12$. It does not determine $N_c^{\text{phys}}(W, Q)$ for $3 \leq W \leq 11$, the maximum score with two preparations, or the least peak width at $N = 2$. The earlier exact decision-width result $\kappa_{2/5}^{\text{rows}} = 3$ is unchanged: it allows longer training, whereas this shortest audit uses five decision states. The older points (399, 40602), (320000, 1604) and $(12800 \cdot 2^{400}, 12)$ remain valid constructions with decision width three. They are dominated by (2, 12) in the projection onto training preparations and peak width, but not necessarily in a resource order that also constrains the decision alphabet separately.

5. QUANTITATIVE CERTIFICATES FOR A FINITE COMPATIBLE ARCHITECTURE

We return to an arbitrary finite architecture, now with a finite parameter set. It can have hidden states, controlled experiments, more than two hypotheses and registers of any size. Enumerate its distinct stochastic rows by $r = 1, \dots, R$ and write $q_r \in \Delta_{s_r}$. A reused autonomous row appears only once in this product. Assume each execution uses at most H controller rows, and uses row r at most d_r times. Delete unused rows and put $S = \max_r s_r$. Fixed environmental kernels are not controller rows.

For an integer $m \geq 1$, intersect Δ_{s_r} with the coordinate boxes of side $1/m$. The resulting nonempty closed polytopes cover the simplex, with overlaps permitted at their boundaries. Write C for a product of such row cells and choose a vertex q_r^C in each factor. A row-cell vertex has at most one coordinate not fixed at a box boundary; consequently it has at most $v_r = s_r 2^{s_r - 1}$ vertices, all on the $1/m$ grid. There are at most $m^{\sum_r s_r}$ product cells.

Write the vertices of row cell C_r as $v_{r,j}$ and parametrize $q_r = \sum_j \lambda_{r,j} v_{r,j}$ with λ_r in a simplex. A reward polynomial has degree at most d_r in block λ_r . Use the homogeneous simplex Bernstein basis of degree d_r in that block, obtaining

$$(13) \quad g_\theta(q(\lambda)) = \sum_\alpha b_{\theta,C,\alpha} B_\alpha(\lambda), \quad B_\alpha \geq 0, \quad \sum_\alpha B_\alpha = 1.$$

Degree elevation includes the factors $\sum_j \lambda_{r,j} = 1$. No independence between different visits of a shared row is imposed on its unknown row vector.

Lemma 5.1 (Causal interpretation of the coefficients). *Every coefficient in (13) satisfies*

$$|b_{\theta,C,\alpha} - g_\theta(q^C)| \leq HS/m.$$

Proof. For each row r , independently prepare an ordered list of d_r vertex labels, sampled independently with law λ_r . At its k th visit, use the stochastic row given by the k th vertex label in that list. Before conditioning, the next label is independent of the previous execution, so this experiment has exactly the row q_r at every visit and expected reward $g_\theta(q(\lambda))$. Unused labels may be sampled after termination without altering the reward.

Condition on the counts of every vertex label in every list. The probability of those counts is $B_\alpha(\lambda)$. Conditional on the counts, each list is a uniformly ordered multiset. Its expected reward is independent of λ and is therefore the corresponding Bernstein coefficient. This also proves (13) directly.

Each row used by this conditioned experiment is a vertex of C_r or a mixture of its remaining vertices. Every two vectors in C_r have coordinate differences at most $1/m$, hence total variation at most $s_r/(2m) \leq S/(2m)$. Couple the conditioned experiment to the one using q^C , using the same environmental randomness while their executions agree. At each actual controller update the probability of a first row-level disagreement is at most $S/(2m)$. There are at most H such updates on the agreeing path. The terminal laws consequently differ in total variation by at most $HS/(2m)$. A reward in $[-1, 1]$ has expectation difference at most twice that quantity. This proves the estimate. The conditioned urn is used to analyze a coefficient; it is not declared to be an admissible free-memory controller. $\square$

Theorem 5.2 (Explicit subdivision certificate). *Define*

$$\ell_C = \min_{\theta} g_{\theta}(q^C), \quad u_C = \min_{\theta} \max_{\alpha} b_{\theta, C, \alpha}, \quad L_m = \max_C \ell_C, \quad U_m = \max_C u_C.$$

For the exact all-stochastic value of the specified architecture,

$$(14) \quad L_m \leq V(A) \leq U_m \leq L_m + HS/m.$$

On each cell, a witnessing parameter θ_C supplies the nonnegative coefficient certificate

$$u_C - g_{\theta_C}(q(\lambda)) = \sum_{\alpha} (u_C - b_{\theta_C, C, \alpha}) B_{\alpha}(\lambda).$$

Its degree in row block r is d_r , and its total degree is at most $\sum_r d_r$. At most $\prod_r \binom{d_r + v_r - 1}{v_r - 1}$ coefficients per parameter and cell are needed. If the primitive data are rational, all coefficients and bounds can be checked by rational arithmetic. Taking $m \geq HS/\varepsilon$ gives a certified interval of length at most ε .

Proof. The point q^C is feasible, so ℓ_C is a lower bound on the value restricted to C . For any $q \in C$, choose a convex representation of its rows. Equation (13) implies

$$\min_{\theta} g_{\theta}(q) \leq \min_{\theta} \max_{\alpha} b_{\theta, C, \alpha} = u_C.$$

Choose a parameter attaining the minimum defining ℓ_C and apply Lemma 5.1; this gives $u_C \leq \ell_C + HS/m$. Taking the maximum over the finite cover proves (14). Choose θ_C attaining the definition of u_C ; all displayed coefficients of its certificate are nonnegative. The degree and count statements follow from the product Bernstein basis. Cell vertices are rational grid points, and a finite execution sums finite products, proving the rational statement. $\square$

Remark 5.3. The number of cells and coefficients can be enormous. Neither polynomial running time nor a uniform degree bound for the original sum-of-squares hierarchy follows. Refinement changes the domain cover, not merely a single global SOS degree. For a finite collection of architectures obeying a specified budget, maxima of these intervals give outer intervals; an unbounded architecture class needs a separate uniform argument such as Theorem 3.2. For compact parameter families, an effective net and a verified modulus for the primitive kernels are additional inputs, not consequences of compactness alone.

The resource estimate also makes row precision explicit. With $m = 2^b$, the lower witnesses have dyadic rows and lose at most $HS2^{-b}$ in value. Thus increasing row precision provides a quantified approximation for a fixed architecture, while its random-bit implementation still has to be charged. The physical theorem avoids this cost altogether because its rows are deterministic.

6. AUTONOMOUS NOISY TESTING WITH A PRECISION AND TIME LEDGER

This section supplies a substantive autonomous comparison rather than merely identifying shared rows. The stationary optimum is classical [5]; the proof and explicit finite-time implementation are recorded to make the resource conventions unambiguous. They are not used as novelty evidence for the physical theorem.

Let reports be independent, with law $\text{Ber}(p)$ under hypothesis $-$ and $\text{Ber}(q)$ under hypothesis $+$, where $0 < p < 1/2$, $q = 1 - p$, and $r = q/p$. A W -state autonomous machine has two time-independent stochastic transition matrices T_0, T_1 . Assume their induced chains $P_- = qT_0 + pT_1$ and $P_+ = pT_0 + qT_1$ are irreducible. A final decision uses only the state, with optional fresh terminal randomization.

Lemma 6.1 (Likelihood-range bound). *If equivalent probability laws ν_+, ν_- on a finite set satisfy*

$$\frac{\max_i \nu_+(i)/\nu_-(i)}{\min_i \nu_+(i)/\nu_-(i)} \leq R,$$

then $\text{TV}(\nu_+, \nu_-) \leq (\sqrt{R} - 1)/(\sqrt{R} + 1)$.

Proof. Write $L = d\nu_+/d\nu_-$, $a = \min L$ and $b = \max L$. Since $\mathbb{E}_{\nu_-} L = 1$, one has $a \leq 1 \leq b$. The chord bound for $(L - 1)_+$ gives

$$\text{TV}(\nu_+, \nu_-) = \mathbb{E}_{\nu_-} (L - 1)_+ \leq \frac{(1 - a)(b - 1)}{b - a}.$$

For fixed b/a , maximizing the last expression gives $a = 1/\sqrt{b/a}$; substitution yields $(\sqrt{b/a} - 1)/(\sqrt{b/a} + 1)$. It is increasing in b/a , proving the assertion. The identical-law case follows directly. $\square$

Theorem 6.2 (Autonomous stationary bound and an explicit approximation). *Put $D = W - 1$, $W \geq 2$. In the irreducible autonomous class the infimum stationary minimax error is*

$$(15) \quad e_W = \frac{1}{1 + r^D}.$$

For every $\epsilon = 2^{-b}$, $b \geq 0$, there is a machine whose rows have denominator dividing 2^{b+1} , and whose minimax error at report time n from any initial state is at most

$$(16) \quad e_W + \epsilon(W - 2) + \exp \left\{ - \left(\frac{p\epsilon}{2} \right)^D \left\lfloor \frac{n}{D} \right\rfloor \right\}.$$

The stationary lower bound is not a finite-horizon lower bound.

Proof. Every positive induced transition has its likelihood ratio in $[r^{-1}, r]$, and the supports of the two induced matrices are identical. The Markov chain tree formula writes each stationary weight as a sum of products over directed spanning trees rooted at that state. Each product has D transitions; its ratio lies in $[r^{-D}, r^D]$. The ratio of the two unnormalized stationary weights lies in the same interval. Normalizing multiplies all state likelihood ratios by a common number, so their maximum-to-minimum ratio is at most r^{2D} . Lemma 6.1 therefore bounds stationary total variation by $(r^D - 1)/(r^D + 1)$. The equal-prior Bayes error is at least $1/(1 + r^D)$,

and a minimax error cannot be smaller than that Bayes error. This is the equal-prior binary specialization of the Hellman–Cover bound.

For the upper approximation, take states $i = 0, \dots, D$, weights $c_0 = c_D = 1$, and $c_i = \epsilon$ at interior states. On report 1, move from i to $i + 1$ with probability $a_i = \frac{1}{2} \min(1, c_{i+1}/c_i)$, if that neighbor exists; otherwise stay. On report 0, move to $i - 1$ with probability $b_i = \frac{1}{2} \min(1, c_{i-1}/c_i)$, if it exists; otherwise stay. All other probability is assigned to staying. Detailed balance gives

$$\pi_+(i) = \frac{c_i r^i}{\sum_j c_j r^j}, \quad \pi_-(i) = \frac{c_i r^{-i}}{\sum_j c_j r^{-j}}.$$

Reflection $i \mapsto D - i$ exchanges these laws. Decide $+$ for $i > D/2$, $-$ for $i < D/2$, and use a fair terminal decision at $i = D/2$ when this state exists. The two errors agree. Conditional on being at an endpoint, the error is exactly e_W . The total interior stationary mass is at most $\epsilon(W - 2)$, since each $r^i \leq r^D$ and the denominator is at least r^D under $+$, with the reflected argument under $-$. This proves the first two terms of (16).

Every existing neighbor transition under either hypothesis has probability at least $p\epsilon/2$, and every self-loop has probability at least $1/2$. From any state, move down to zero and then stay until D steps have elapsed. This yields the common minorization $P_\pm^D(i, \{0\}) \geq \delta = (p\epsilon/2)^D$. Subtracting that common atom gives contraction in total variation by $1 - \delta$ every D steps. Comparison with the stationary law yields $\text{TV}(P_\pm^n(i, \cdot), \pi_\pm) \leq (1 - \delta)^{\lfloor n/D \rfloor} \leq e^{-\delta \lfloor n/D \rfloor}$. A bounded decision changes error by at most this amount, proving (16). Finally, letting n tend to infinity and then ϵ to zero proves that the stationary infimum equals (15). All transition probabilities belong to $\{0, 1, 1/2, \epsilon/2, 1 - 1/2, 1 - \epsilon/2\}$. $\square$

The W states in this theorem refer to an atomic stochastic update after each report. A sequential fair-bit implementation must also retain the report, the current state, a bit counter and a flag. Each nontrivial move probability is $1/2$ or $2^{-(b+1)}$; test for a string of zero bits of the required length, and pad to $b + 1$ bits if a uniform report cycle is required. At most $4W(b + 2)$ states suffice for this implementation, including its bit-phase counter and a ready phase. This conservative bound is an implementation claim, not an optimal precision–state tradeoff. The time index n in (16) is an observation time at which an external analyst reads the autonomous machine, not a free input clock used by its transitions. The adversarial model of [9] is not covered by the independent-report proof above.

7. PHYSICAL CONSUMERS AND THE FIRST-PRINCIPLES CONNECTION

The physical preparation theorem needs no estimate of the target law as a stored input: the five events are fixed and their uniform validity uses a proved neighborhood of Q_* . This removes calibration memory from this particular consumer without asserting that target acquisition is unnecessary in an arbitrary comparison problem. The general acquired calibration theorem, with its full counter register, remains in the complete companion.

Corollary 7.1 (A raw-interface confidence construction). *Let $m = m_2 - \beta > 0$ and $0 < c < m$. Repeat the full four-preparation audit independently R times and write $\bar{D}_R$ for the mean score. Under every private candidate,*

$$\mathbb{P}(\bar{D}_R \leq c) \leq \exp\{-R(m - c)^2/2\}.$$

Under the null in which the independent candidate and target validation laws agree, with the same event chosen before both validations,

$$\mathbb{P}(\bar{D}_R > c) \leq \exp\{-Rc^2/2\}.$$

A clocked implementation uses $4R$ preparations and at most $12(2R + 1)$ persistent states, with no calibration and no random bits.

Proof. The independent block scores lie in $[-1, 1]$ and have mean at least m under a private candidate. Hoeffding's inequality gives the first bound. Under the null, conditional on the selected event, the two independent validation indicators have the same mean. Each block therefore has mean zero, and the same inequality gives the second bound. Retain the integer sum in $\{-R, \dots, R\}$ along with the current twelve-state block. At a block boundary add its terminal score and reset its register. The deterministic external schedule is part of the clocked convention. $\square$

This is an upper construction on the raw marked interface. Its proof does not transfer a lower bound from a deliberately reduced Bernoulli interface, and no optimal confidence-memory frontier is claimed.

Proposition 7.2 (Target transport without a training multiplier). *For the two-preparation audit, changing only the target from Q to Q' changes its expected score by at most $\text{TV}(Q, Q')$. Its candidate training law, register profile and event selector remain unchanged.*

Proof. Condition on the candidate training word. The expectation difference is $(Q - Q')h_S$, whose absolute value is at most $\text{TV}(Q, Q')$. Average over that unchanged training law. $\square$

In particular, every common-preparation transport estimate verified for the original positive-noise collision family can be inserted here without multiplying it by a long training horizon. If candidate kernels also change, their training and validation defects must be added by a causal coupling; the target-only statement does not remove those terms.

The first-principles route is now concrete. The positive experimental kernels and actual reset preparations give the likelihoods in (5); these generate posterior continuation states. The validation-test quotient gives the positive parts in the statistical dual. A prior then supplies an architecture-uniform impossibility certificate. On the constructive side, the five-event decision is compiled by taking its future residual functions, which are closed under observation shifts. These are compatible causal states and determine the charged profile. Thus the preparation law, predictive continuation and finite-resource realization are used in the same upper/lower theorem. This is not a deduction of geometry or nonlinear operator compactness from the existence of finite stochastic rows.

8. DEPENDENCIES, RETAINED RESULTS AND REMAINING DISTINCTIONS

The exact three-report binary noisy frontier, arbitrary-horizon binary-register minimax recursion and compatible Bayesian interval recursion remain as proved in the complete companion. Its exact odd-horizon majority realization is also retained; it is not a new result of this revision. Section 5 applies to registers of any finite width, controlled rows and any finite parameter set, but provides certified intervals rather than a closed minimax recursion for all such classes. The physical theorem supplies a matched preparation segment in a continuous two-parameter family with overlapping stochastic reports; its converse is not a finite parameter-grid check.

The stationary autonomous result in Section 6 has an explicit formula, but its stationary lower bound must not be applied to a finite-horizon clocked machine. Its finite-time upper bound is purposely conservative and exposes the slowing caused by finite transition precision. An arbitrary compact-family presentation still requires additional effective information to construct a kernel net. A subdivided positive-basis degree bound must not be called a degree bound for a global SOS hierarchy.

The physical result is a genuine consumer of the experimental and continuation root. It does not make the independent A2 geometric proofs dependent on this root. The original B4 normalized-resolvent, range, compactness and control-transfer obligations, and the broad C2 strict-dual, form, rigidity and optional-projection obligations, are not proved by our finite-state theorem. Their statements, derivations and current status remain in the retained historical pipeline. No aggregate closure is inferred from preserving their source files.

Finally, the rational certificate (10), polynomial identity (12), and residual transitions are independently reproducible. The program in the accompanying source checks their exact identities and includes deliberately invalid controls. These checks are not substitutes for the proofs above. Neither compilation, preservation of earlier pages nor a successful finite test establishes novelty or an editorial decision.

REFERENCES

- [1] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272.
- [2] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176.
- [3] J. Müller and G. Montúfar, The geometry of memoryless stochastic policy optimization in infinite-horizon POMDPs, arXiv:2110.07409 (2021; revised versions).
- [4] M. Putinar, Positive polynomials on compact semi-algebraic sets, *Indiana Univ. Math. J.* **42** (1993), 969–984.
- [5] M. E. Hellman and T. M. Cover, Learning with finite memory, *Ann. Math. Statist.* **41** (1970), 765–782.
- [6] M. E. Hellman and T. M. Cover, On memory saved by randomization, *Ann. Math. Statist.* **42** (1971), 1075–1078.
- [7] R. A. Flower and M. E. Hellman, Hypothesis testing with finite memory in finite time, *IEEE Trans. Inform. Theory* **18** (1972), 429–431.
- [8] T. M. Cover, M. A. Freedman and M. E. Hellman, Optimal finite memory learning algorithms for the finite sample problem, *Information and Control* **30** (1976), 49–85.
- [9] M. A. Managoli and V. M. Prabhakaran, Memory constrained adversarial hypothesis testing, arXiv:2605.12063v1 (May 12, 2026).
- [10] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28, doi:10.1524/strm.2002.20.14.1.
- [11] Q. Qi, General Theta Foundations I: Compatible statistical experiments and resource frontiers, revision 23, September 24, 2026, complete mathematical companion, source commit 8a6043aca644043bd805668185eb227468f21170; packaged commit e926b783b643b3e7f275a18105528927b76cc97b. The 96-page article and 649-page development are reproduced without alteration after the present article in the two complete volumes.

General Theta Foundations I

Complete Mathematical Manuscript

Part II: preserved revision 23

The following 96 pages reproduce the predecessor volume without alteration.

Part I contains the new revision 24 proof chain. Earlier theorem numbers in Part II retain their original meaning. The source modules, proof qualifications and historical references remain available in the accompanying repository and source archive.

GENERAL THETA FOUNDATIONS I: COMPATIBLE STATISTICAL EXPERIMENTS AND RESOURCE FRONTIERS

QIAN QI

ABSTRACT. We study statistical experiments subject to a register constraint at each cut of a finite causal protocol. Their feasible occupation arrays are characterized by linear flow equations and quadratic compatibility equations, including at zero-probability histories. This yields constructive realizations and a complete polynomial upper-certificate hierarchy for the full-profile minimax value, without convexifying the class of machines. For binary experiments we obtain an exact posterior partition recursion for Bayesian decisions and an exact receiver-curve recursion for minimax decisions with binary registers. In a three-report experiment with overlapping binary noise, we compute the entire register-profile frontier for both objectives; the resulting strict gap shows that a free mixture of machines changes the minimax value. The characterization is connected to finite-precision response-cover auditors, acquired calibration, and positive-noise physical experiments, with preparations, intermediate registers and clocks explicitly charged. The marked continuation, residual, microscopic and Gaussian comparison results are developed with their proofs and with their distinct hypotheses retained.

CONTENTS

1. Introduction	3
1.1. The full-profile problem	3
1.2. Comparison with existing methods	4
1.3. Organization and scope of the applications	5
2. Compatible occupation arrays	5
2.1. The finite protocol and its observable rows	5
2.2. A complete polynomial converse	7
3. Continuation polytopes and noisy posterior partitions	8
3.1. One common continuation matrix	8
3.2. Binary posterior geometry	9
3.3. An exact minimax recursion for binary registers	11
4. An exact noisy frontier and the cost of hidden mixing	12
5. Acquisition, clocks, and the resource order	15
5.1. Comparable resource points	15
5.2. Acquiring a calibration rather than assuming it	16
6. Consumers of the compatible frontier	17
6.1. Reset scores and physical protocols	17
6.2. An exact confidence consumer	18
6.3. The foundation and the longer program	19
7. Intrinsic causal deficiency	19
7.1. Experiments and executable signatures	19

Date: September 24, 2026.

2020 Mathematics Subject Classification. Primary 62B05, 93E20; Secondary 60G44, 68Q45, 82C40.

Key words and phrases. Statistical experiments, finite memory, causal realization, minimax testing, likelihood-ratio quantization, polynomial certificates.

7.2.	The finite comparison theorem	20
7.3.	A delayed channel and a strict signature separation	22
7.4.	Optimal resource sets and decision certificates	24
8.	Continuation geometry and an exact memory obstruction	24
8.1.	Normalized continuation laws	24
8.2.	An erasure cut	26
9.	Decision continuation spectra	26
9.1.	A charged training-validation interface	27
9.2.	Several feedback rows	29
10.	Finite precision and charged randomization	29
11.	Approximate continuation profiles and resource transfer	30
11.1.	A causal comparison norm	30
11.2.	The full value and its effective approximation	31
12.	From a decision cover to an end-to-end machine	33
12.1.	The whole-schedule continuation value	33
13.	Finite-horizon sample-memory bounds	35
14.	An exact statistical frontier at all cuts	37
15.	Transport priced by active updates	39
16.	Weighted continuation obstructions	40
17.	Joint residual budgets and growing families	42
17.1.	Actual physical word weights	43
17.2.	A family with matching end-to-end growth	43
17.3.	A sharp simultaneous storage profile	44
18.	Finite-data testing on the original behavior image	44
18.1.	The reset interface	44
18.2.	A matching uniform sampling obstruction	46
18.3.	Transport of one fixed audit	47
19.	Continuation complexity of a finite-data test	47
19.1.	The width required to attain the binary reset value	48
20.	Polynomial testing of the original resource image	49
20.1.	A sharp two-time nonconvex example	51
21.	Adaptive testing on the observable behavior image	52
21.1.	The testing game	52
21.2.	A size bound in behavior dimension	53
21.3.	A three-weight certificate invisible to constant tests	55
21.4.	Strict certificates and effective physical input	56
22.	An exact algorithm for the observable chart	56
23.	A marked resource gap generated by a binary collision	57
23.1.	The exact marked two-report experiment	58
23.2.	A full-dimensional collision preparation	59
23.3.	Uniformly effective finite presentations	60
24.	Uniform stability of optional projections	61
24.1.	A microscopic sufficient condition	63
25.	Changing trajectories, likelihoods, and conditional consumers	64
25.1.	A common preparation, not a common flow	64
25.2.	The projected stochastic exponential	65
25.3.	A bounded entropic backward-equation consumer	66
25.4.	Resource transport is not the posterior coupling	67
26.	Observation densities and coupled process versions	68

27.	Global transport of the backward equation	69
28.	A uniform certificate through a moving collision	71
29.	Sharp decision memory for the collision experiment	74
29.1.	The marked family and its first two spectral values	74
29.2.	A three-response construction	76
29.3.	Simultaneous exact widths and the full acquisition schedule	77
29.4.	What the two exact complexity statements measure	79
30.	Full-profile certification of the collision family	79
30.1.	The row-indexed decision invariant	79
31.	A third resource regime for the same physical task	81
32.	Confidence-level certification	82
33.	Confidence tests with priced finite-state amplification	84
34.	A finite-register certificate transported through the microscopic limit	85
35.	Explicit phase transitions for the two-count certificate	87
Appendix A.	The proof dependency chain	88
A.1.	The full-profile resource chain	88
Appendix B.	A namespaced map of the eleven-component program	90
Appendix C.	Preserved organizing statements	90
C.1.	Inherited baseline	92
Appendix D.	Unrestricted-horizon multicut realization	93
Appendix E.	Dependencies of the statistical resource frontier	94
Appendix F.	The constructive resource theorem	94
References		95

1. INTRODUCTION

An experiment is used through a causal observer. Information acquired at one time may be lost at a later checkpoint, and a decision rule that is economical after training may be expensive to train. The statistical value of an observer therefore depends on its entire register profile, not only on the dimension of its last statistic. A further difficulty is that two feasible observers need not be mixable at the same profile: the variable that selects an observer has to be retained wherever their future actions differ.

We study this problem from positive experimental kernels, preparations, causal interventions and bounded executable tests. These objects do not presuppose a convex parameter image, an information metric, or a finite-dimensional closure of the underlying dynamics. The present paper develops their finite resource layer and its comparison maps. The word “finite” refers to a declared protocol and its observable alphabets, not to a removal of the microscopic or limiting questions in the larger theory.

1.1. The full-profile problem. Fix a finite schedule σ , a register profile $\mathbf{K}$, a family of environments B , and a terminal payoff $G \in [-1, 1]$. Let

$$V_{\sigma, \mathbf{K}}^{\text{clk}}(B, G) = \sup_A \inf_{b \in B} \mathbb{E}_b^A G,$$

where every persistent data-dependent variable of A passes through the specified registers. The clocked model allows a predetermined transition table at each epoch; its clock carries no discarded data. An autonomous controller instead reuses its declared transition rows and must retain any phase variable needed to select a different row. Refining an atomic operation into input-bit or random-bit epochs changes σ and its resource problem. These conventions are part of the theorem statements.

The central object is a compatible occupation array. Its coordinates record the probability mass entering and leaving each observable controller row, under every environment and hidden history. Ordinary flow conservation is necessary but not sufficient. All hidden histories that present the same observable row must use the same stochastic transition. This requirement is a collection of quadratic proportionality equations. Unlike a division-based parametrization, these equations also retain every zero-occupation boundary case.

Theorem 1.1 (Compatible statistical resource frontiers). *The following statements hold for the indicated finite protocols.*

- (i) *For a finite environment family, compatible occupation arrays are exactly the arrays of all stochastic machines of the full profile. Their minimax value is attained. It equals the decreasing limit of polynomial sum-of-squares upper certificates, and every feasible array reconstructs a machine. Shared-row equations give the corresponding autonomous characterization. Finite-precision controller witnesses and compact-family kernel nets give matching two-sided approximations.*
- (ii) *For binary, conditionally independent finite observations, possibly with a finite controlled choice of experiment, the optimal clocked Bayesian value at any finite register profile is given by a recursion of compatible interval partitions of incoming posterior atoms. For uncontrolled observations with one- or two-state registers, a receiver-curve recursion gives the exact clocked minimax value at every finite horizon. These are optimizations over all stochastic machines, not implementation bounds for a chosen statistic.*
- (iii) *With three independent binary reports of error probability $p \in (0, 1/2)$, put $d = p(1-p)(1-2p)$. At the atomic profile $(2, 2, 2)$ the exact fair-prior and minimax successes are respectively $1 - p + d/2$ and $1 - p + d/3$. At profile $(2, 3, 2)$ both equal $1 - p + d$. A free mixture of the two Bayesian optimal machines would raise the first profile's minimax value by $d/6$, so that convexification is strictly incorrect for that resource problem.*
- (iv) *A fixed K -response calibration can be acquired from $M = 2^s$ target preparations, with uniform error at most ρ except with probability β , whenever $M \geq (2\rho^2)^{-1} \log(2K/\beta)$. Its $(M + 1)^K$ -state counter vector is retained and priced through the subsequent auditor. The resulting response-cover constructions and their physical consumers belong to the same full-profile classes as the occupation converse.*

Proof. Part (i) is Theorems 2.1 and 2.2 and Corollary 2.3. Part (ii) is Theorems 3.2 and 3.3. Part (iii) is Theorem 4.1 and Corollary 4.3. Part (iv) is Theorem 5.2 and Corollary 6.1. $\square$

The certificate hierarchy in (i) uses a classical positivity theorem. Its role is to give an upper bound on every complete machine, paired with the reverse reconstruction from feasible occupations. It is not an efficiency or finite-termination assertion. The noisy recursions in (ii) and the closed formulas in (iii) supply structural information that a generic polynomial encoding alone would not provide. In particular, the distinction between Bayesian and minimax objectives survives in an experiment whose two laws have common support at every observation.

1.2. Comparison with existing methods. Comparison of experiments is classical [1, 2, 13]. Filtered randomization is also an existing subject. Weisshaupt's filtered criterion [10, Lemma 1 and Theorem 5] concerns compact convex classes of positive operators and a risk criterion for filtered randomization. Our feasible object is a specified finite register execution, whose behavior set need not be convex. We also cite Norberg [3]; the original proof text has not been obtained for a definitive theorem-level comparison, and no priority claim rests on that gap in access.

Finite-memory statistical learning and the role of time-dependent transition rules are treated by Cover and by Hellman and Cover [24, 26]. Likelihood-ratio quantizer optimality is a classical mechanism [27, Section III]. We use it both for compatible posterior partitions and for binary-channel domination. The receiver-curve and equal-error arguments are not introduced as new

statistical principles. Their role here is to solve one full-profile optimization, including its stochastic minimax part and the selector cost exposed by the explicit three-report frontier.

Polynomial descriptions of partially observed controller occupations are likewise established. Müller and Montúfar [28, Example 15, Theorem 16 and Remark 19] give semialgebraic feasible-frequency descriptions and polynomial planning for infinite-horizon memoryless POMDPs, with their stated positivity assumptions. The finite-protocol formulation here makes the observable row equivalences, changing register alphabets and zero-mass reconstruction explicit. Its upper-certificate completeness applies Putinar’s theorem and its rational version [29, 30]. The combined contribution is the exact compatible frontier, its noisy evaluations, and its connections to fully priced experimental consumers; neither polynomial optimization nor the vanishing-minor identity is claimed as a new invention.

1.3. Organization and scope of the applications. Sections 2–6 contain the principal proof chain. The marked continuation and decision spectra then identify the underlying experimental interfaces. The finite precision, response-cover tournaments, multicut revelation formula, weighted residuals, nonlinear tests and adaptive tests are developed with their proofs. The microscopic moving-collision construction and the bounded Gaussian comparison produce physical consumers of the same interfaces. The original short, polynomial-horizon and long-clock physical resource points are all retained.

Those three points are constructions, not an exact physical Pareto curve. The general converse now applies to each of their fixed protocols, but its numerical evaluation across all physical budgets is a separate task. Likewise a confidence lower bound at a specified retained-coin interface is not automatically a lower bound for an auditor that uses richer raw observations. These distinctions are mathematical typing conditions on the applications, not changes to the physical or operator targets.

The article retains the full development of its supporting results; versioned derivation and review records are kept outside the exposition. The dependency appendix records the namespaced eleven-component program. The primary A2 geometric chain remains independent, while B4 kinetic and broader C2 operator conclusions require their own stated hypotheses and proofs. No artificial dependence on a finite-state theorem is imposed on those branches of the program.

2. COMPATIBLE OCCUPATION ARRAYS

A fixed terminal statistic and an optimal statistic under a sequence of memory constraints are different objects. We begin with a characterization of the second object. All the spaces in this section are finite. No strict positivity assumption is imposed on a probability coordinate.

2.1. The finite protocol and its observable rows. Fix a finite schedule σ and a register profile $\mathbf{K} = (K_t)_{t \in \sigma}$. A cut is part of the specification of σ : subdividing an atomic observation or a randomization operation inserts additional register constraints. The clocked model permits a different transition table at each scheduled operation. It does not permit access to a discarded observation. We write $\mathcal{M}_\sigma^{\text{clk}}(\mathbf{K})$ for this class. Initial randomization, when allowed, is a transition of the same kind as all other randomizations.

Let Θ be a finite set of environments. Under θ , all preparation and experimental kernels are specified. One may include the complete physical and public past in a finite hidden index i . This index is a variable in the analysis, not an observer register. An elementary observer operation has an accessible row r and chooses j from a nonempty finite set J_r . The row contains exactly the current register and the current atomic input, if there is one. The choice j can specify a new register and a public action. A public action that is subsequently needed by the observer must be

retained in that register. Every permissible transition has the form

$$u(j \mid r), \quad u(j \mid r) \geq 0, \quad \sum_{j \in J_r} u(j \mid r) = 1. \quad (2.1)$$

Neither θ nor the unobserved part of i is an argument of u .

Known environment operations act linearly on conditional occupation masses. For example, if $z(\theta, h, m)$ is the mass before an observation, then its incoming mass at the observer is

$$x(\theta, h', (m, y)) = \sum_h z(\theta, h, m) L^\theta(h', y \mid h, m). \quad (2.2)$$

Dependence on a previously issued action is included in h or in the retained part of m . After an observer choice, the public history is updated and discarded indices are summed out, again by a known linear map. Alternating these two kinds of operation represents a finite controlled experiment, including feedback, calibration, validation, and a terminal readout. The cut alphabets and the allowed rows are fixed before optimization. In particular, this representation does not silently grant the observer the hidden history used to define the experimental kernels.

For each observer operation introduce outgoing masses $f(\theta, i, r, j)$. Impose the conservation and compatibility equations

$$f(\theta, i, r, j) \geq 0, \quad \sum_{j \in J_r} f(\theta, i, r, j) = x(\theta, i, r), \quad (2.3)$$

$$f(\theta, i, r, j) x(\theta', i', r) = f(\theta', i', r, j) x(\theta, i, r) \quad (\theta, i, \theta', i', r, j). \quad (2.4)$$

All incoming masses are nonnegative and are linked to the preceding operation by the known linear maps. Preparations are prescribed and have conditional mass one for each θ . We call the resulting set of arrays $\mathcal{O}_\sigma^{\text{clk}}(\mathbf{K})$.

Theorem 2.1 (Exact full-profile realization). *The occupation arrays of $\mathcal{M}_\sigma^{\text{clk}}(\mathbf{K})$ are exactly $\mathcal{O}_\sigma^{\text{clk}}(\mathbf{K})$. This set is compact and is defined by linear equations, nonnegativity, and the quadratic equations (2.4). Every feasible array determines an implementing machine, with arbitrary legal rows only where the incoming mass vanishes under every environment.*

More generally, suppose specified occurrences of observable rows must use the same transition distribution, with their choices identified. For each such class impose (2.4) also between different occurrences. The resulting arrays are exactly those realized by machines with these shared rows. In particular this gives an exact formulation for an autonomous controller with a fixed state alphabet and a single reused transition table.

Proof. A machine gives $f(\theta, i, r, j) = x(\theta, i, r) u(j \mid r)$, so conservation and every minor equation follow. Conversely, fix an occurrence of r . If some $x(\theta_0, i_0, r) > 0$, define

$$u(j \mid r) = \frac{f(\theta_0, i_0, r, j)}{x(\theta_0, i_0, r)}.$$

Conservation makes this a stochastic row. The minor with (θ_0, i_0) shows that the same formula reproduces f at every positive incoming mass. At a zero incoming mass, nonnegativity and conservation force each outgoing mass to be zero. If all incoming masses vanish, choose any legal stochastic row. Thus the recovered row is independent of both the environment and the hidden index.

Perform this construction at every observer operation. Induction along the schedule, using (2.2) and the corresponding linear outgoing maps, proves that the recovered machine has exactly the given occupation arrays. Its registers are the declared alphabets; the proof has not introduced a retained selector. For a shared row class, choose one positive incoming mass anywhere in that class and use the cross-occurrence equations. The same argument reconstructs one common row. A class with no positive incoming mass is treated arbitrarily.

Each coordinate is a conditional probability or a subprobability mass and is therefore at most one. Adding these redundant upper bounds makes boundedness explicit. The equations and inequalities are closed, and there are finitely many coordinates. Compactness follows. $\square$

The zero-mass provision is essential. Cancelling an occupation coordinate without a case distinction would remove valid boundary machines. Nor would separate low-rank descriptions at successive cuts imply (2.2)–(2.4). The latter equations refer to one execution, not to a list of independently selected encoders. For an autonomous machine, the phase cannot be supplied through a row name unless it is either an actual input or part of the charged state. A clocked machine can be embedded autonomously by retaining (t, m) ; Theorem 2.1 also permits direct optimization without making that possibly wasteful construction.

2.2. A complete polynomial converse. Let $G \in [-1, 1]$ be a terminal payoff and let $L_\theta(z)$ denote its expectation as a linear function of the complete occupation array z . The terminal decision is included among the observer operations. Set

$$V_{\sigma, \mathbf{K}}^{\text{clk}}(\Theta, G) = \max_{A \in \mathcal{M}_\sigma^{\text{clk}}(\mathbf{K})} \min_{\theta \in \Theta} \mathbb{E}_\theta^A G. \quad (2.5)$$

The admissible row sets are assumed nonempty. Introduce $v \in [-1, 1]$ and impose $L_\theta(z) - v \geq 0$ for every θ . Write $h_a = 0$ for all the linear and quadratic equalities and $g_b \geq 0$ for the remaining inequalities. Include the redundant ball inequality

$$R - \|(z, v)\|^2 \geq 0,$$

where the positive integer R is larger than the number of coordinates. Let $\mathcal{F}$ be this basic closed feasible set. For $d \geq 1$, define U_d as the infimum of $u \in \mathbb{R}$ for which the polynomial identity

$$u - v = \sigma_0 + \sum_b \sigma_b g_b + \sum_a q_a h_a \quad (2.6)$$

holds, with each σ_b a sum of squares, each q_a a polynomial, and every displayed summand of degree at most $2d$. An empty certificate class has infimum $+\infty$.

Theorem 2.2 (Complete full-profile certificates). *For the finite protocol above,*

$$V_{\sigma, \mathbf{K}}^{\text{clk}}(\Theta, G) = \max_{(z, v) \in \mathcal{F}} v = \inf_{d \geq 1} U_d, \quad U_{d+1} \leq U_d. \quad (2.7)$$

A feasible occupation array is a constructive lower certificate. An identity (2.6) is an upper certificate for every stochastic machine of the specified full profile. For rational experimental data, every rational $u > V_{\sigma, \mathbf{K}}^{\text{clk}}$ admits such a certificate with rational coefficients. The same conclusions hold with shared rows, including the autonomous case.

Let ℓ_r be the best worst-case payoff among the actual controller tables whose entries are dyadic with denominator 2^r . If transition t has at most s_t choices per row, then

$$\ell_r \leq V_{\sigma, \mathbf{K}}^{\text{clk}} \leq \min \left\{ U_d, \ell_r + 2 \sum_t (s_t - 1) 2^{-r} \right\}. \quad (2.8)$$

Here repeated uses of a shared row are counted repeatedly in the sum. Thus there are matching constructive and converse certificates to any positive accuracy, without a minimax interchange or a convexification of fixed-width machines.

Proof. The first equality follows from Theorem 2.1 and the inequalities $v \leq L_\theta(z)$. Compactness gives attainment. Evaluating (2.6) at any feasible point gives $v \leq u$. Increasing d only enlarges the certificate class.

For completeness, take $u > \max_{\mathcal{F}} v$. The polynomial $u - v$ is strictly positive on $\mathcal{F}$. Apply Putinar's positivity theorem [29] to the inequalities $g_b \geq 0$ and the pairs $h_a \geq 0, -h_a \geq 0$. The explicitly included ball makes the quadratic module Archimedean. The resulting identity has

finite degree. Collecting the two multipliers of h_a gives (2.6). Letting u decrease to the maximum proves the second equality. For rational data and rational u , the rational version with an explicit ball generator [30, Theorem 6] gives rational sums of squares and rational multipliers. This is an application of those positivity theorems, not a new Positivstellensatz.

Finally round each stochastic row to the dyadic simplex, preserving its causal zeros, at total-variation cost at most $(s_t - 1)2^{-r}$. Round a shared row once and use that rounded row at every occurrence. Coupling the original and rounded executions until their first discrepancy bounds the probability of a discrepancy by the sum in (2.8), capped at one. The payoff range is at most two. This estimate is uniform in θ , so rounding a maximizing machine proves the rightmost bound. The dyadic machines are actual machines, which proves the left bound. Their finite enumeration is a construction, even though its cost may be large. $\square$

A numerical semidefinite solution is not by itself an identity (2.6); coefficients and nonnegativity must be certified. The theorem proves convergence, not a uniform degree bound, finite termination at $u = V$, or an efficient algorithm. Its necessity statement is nevertheless different from a resource upper construction: every machine obeys every certified upper bound. Dropping compatibility constraints produces a relaxation. It does not preserve the exact value in general, as Theorem 4.1 will demonstrate explicitly.

Corollary 2.3 (Compact families and finite experimental adapters). *Suppose B is a compact family of finite experimental kernels on a fixed schedule with N environment calls. Let $B_j \subset B$ be a finite ω_j -net in the maximum conditional total-variation metric, with $\omega_j \rightarrow 0$. Use the same bounded terminal payoff throughout. Then*

$$\ell_r(B_j) - 2N\omega_j \leq V_{\sigma, \mathbf{K}}^{\text{clk}}(B, G) \leq U_d(B_j). \quad (2.9)$$

As j, r, d tend to infinity in a suitable order, the two sides have common limit $V_{\sigma, \mathbf{K}}^{\text{clk}}(B, G)$. A certified adapter changing call i by at most δ_i contributes $2\sum_i \delta_i$ to a payoff comparison. A simulator with profile $\mathbf{R}$ is charged by the product profile $(R_t K_t)$.

Proof. For any one controller, couple corresponding calls until the first unequal outcome. Its expected payoff varies by at most $2N\omega_j$ between a kernel and its net representative. Hence $V(B_j) - 2N\omega_j \leq V(B) \leq V(B_j)$. Apply Theorem 2.2 to B_j and choose the dyadic and certificate accuracies after choosing j . The same coupling with unequal δ_i gives the adapter bound. Retaining the ordered pair of simulator and controller states proves the product assertion. $\square$

An effective net is additional data; abstract compactness is not an algorithm for an arbitrary presentation. The finite-alphabet, compact parameter collision experiments considered below have this interface. Consequently their fixed-schedule, full-profile values admit the same all-auditor certificates. This implication does not evaluate those certificates numerically for every physical sample budget, nor does it identify a minimal autonomous implementation.

3. CONTINUATION POLYTOPES AND NOISY POSTERIOR PARTITIONS

We next obtain a finite exact recursion, and then an explicit solution, for noisy experiments. Unlike a revelation channel, these experiments need never disclose the parameter at any time.

3.1. One common continuation matrix. Let Θ and the terminal action set A be finite. Conditional on θ , report t has distribution p_t^θ on Y_t and reports are independent. A clocked observer starts in one state, retains at most K_t states after report t , and receives payoff $G(\theta, a)$. For the moment there are no prescribed zeros in its update tables. For a fixed prior π , denote its optimal Bayesian payoff by $B_{n, \mathbf{K}}^{\text{clk}}(\pi, G)$. The terminal readout has no further observation of a discarded report.

Define finite sets of continuation matrices, with one column per retained state, backwards by

$$\mathcal{C}_n = \{(G(\theta, a_s))_{\theta, s} : (a_s)_s \in A^{K_n}\}, \quad (3.1)$$

$$\mathcal{C}_{t-1} = \bigcup_{f: [K_{t-1}] \times Y_t \rightarrow [K_t]} T_{t,f}(\mathcal{C}_t), \quad (3.2)$$

$$(T_{t,f}C)_{\theta, s} = \sum_{y \in Y_t} p_t^\theta(y) C_{\theta, f(s, y)}, \quad K_0 = 1. \quad (3.3)$$

The same matrix C occurs in every term of (3.3). Different incoming states do not receive independently selected future machines. If there are finitely many controlled experiments, include an action choice a_s in the union and replace $p_t^\theta(y)$ by $p_t^\theta(y \mid a_s)$.

Proposition 3.1 (Exact Bayesian continuation recursion). *Every matrix in $\mathcal{C}_t$ is implemented by one deterministic suffix of the prescribed profile, and every such suffix has a matrix in $\mathcal{C}_t$. Moreover,*

$$B_{n, \mathbf{K}}^{\text{clk}}(\pi, G) = \max_{C \in \mathcal{C}_0} \sum_{\theta} \pi_{\theta} C_{\theta, 1}. \quad (3.4)$$

For Bayesian optimization the sets can be replaced by their joint convex hulls $\mathcal{P}_t = \text{conv } \mathcal{C}_t$, using $\mathcal{P}_{t-1} = \text{conv } \bigcup_f T_{t,f}(\mathcal{P}_t)$. This operation does not authorize a free shared random seed. Replacing $\mathcal{P}_t$ by the product of its separate column projections is not valid in general.

Proof. At the terminal time, each column records the selected terminal action. At an earlier time, a deterministic update f feeds the new state into one fixed suffix. Conditional expectation gives (3.3). Backward induction proves the realization and converse statements, including recovery of the update tables.

For a fixed prior the payoff is a multiaffine function of the finitely many time-indexed stochastic rows: an execution visits any such row at most once. A maximum exists. Holding all but one row fixed, replace that row by a maximizing vertex of its simplex. Doing this finitely many times gives a deterministic maximum without changing the budget. The same argument includes an experiment-selection row. This proves (3.4) for all stochastic machines. Linearity of $T_{t,f}$ gives the asserted convex-hull recursion. A maximizing vertex of the final finite hull has a deterministic witness; the hull is used only to calculate a support value.

For the last assertion, consider a suffix with two possible initial states, a compulsory next cut of width one, and no informative subsequent report. Its two continuation columns must be equal. Individually each column may implement either constant binary decision. The product of the two column projections also contains the matrix that makes opposite decisions in the two initial states, which the compulsory one-state cut cannot implement. Under a fair preparation where the two initial states reveal the binary parameter, this false matrix gives success one instead of the true value one half. $\square$

3.2. Binary posterior geometry. For $\Theta = \{0, 1\}$ write a current occupation list as $Z = (z_s)_s$, where

$$z_s = (\mathbb{P}(\theta = 0, M = s), \mathbb{P}(\theta = 1, M = s)) \in \mathbb{R}_+^2.$$

Zero columns can be discarded and unused labels can always be padded. After report t , its incoming atoms are

$$b_{s,y} = (z_{s,0} p_t^0(y), z_{s,1} p_t^1(y)). \quad (3.5)$$

Order the nonzero atoms by $q(b) = b_1/(b_0 + b_1) \in [0, 1]$. An interval partition divides that ordered list into at most K_t consecutive nonempty blocks; summing the atoms in each block gives a new list $\mathcal{I}_t(Z)$. Equal-posterior atoms may first be merged. No nestedness of the partitions at different times is required.

Set

$$\begin{aligned} F_n(Z) &= \sum_s \max_{a \in A} \{z_{s,0}G(0, a) + z_{s,1}G(1, a)\}, \\ F_{t-1}(Z) &= \max_{\mathcal{I}_t} F_t(\mathcal{I}_t(Z)), \quad t = n, \dots, 1. \end{aligned} \tag{3.6}$$

For controlled observations, first maximize over the finite vector of experiment choices $(a_s)_s$ and then form (3.5) with those choices. Only lists reachable from the initial list are needed in evaluating this recursion.

Theorem 3.2 (Exact noisy full-profile recursion). *For arbitrary binary finite experiments of the conditionally independent form above, any prior, any bounded terminal payoff, and any finite cardinality profile,*

$$B_{n, \mathbf{K}}^{\text{clk}}(\pi, G) = F_0((\pi_0, \pi_1)). \tag{3.7}$$

There is an optimal deterministic machine whose update at every time is an interval partition of the current incoming posterior atoms. The recursion produces such a machine and optimizes over all stochastic machines, not just over a preselected statistic. It also holds with a finite choice of controlled experiments at each state.

Proof. Deterministic optimality follows from Proposition 3.1. Fix a deterministic global maximum, and consider one of its update times. Hold its prefix, experiment choices at this time, and entire suffix fixed. If the suffix starting at label j has continuation rewards c_{0j}, c_{1j} , assigning incoming atom b to j contributes

$$b_0 c_{0j} + b_1 c_{1j} = (b_0 + b_1)((1 - q(b))c_{0j} + q(b)c_{1j}).$$

It is therefore optimal to select a maximizing affine function of q . The upper envelope of finitely many affine functions on $[0, 1]$ has interval regions. Choose tied affine functions consistently, ordering active slopes and discarding identical duplicates. Each selected label then occupies an interval. At an identical posterior, all atoms have the same maximizing set, so they can be assigned to the same label. Unused labels cause no difficulty. Relabel the active suffix states in interval order, including their future experiment choices.

Now perform these replacements in chronological order. A replacement does not alter any earlier incoming occupation list or earlier update. It cannot decrease the payoff and thus preserves global optimality. Later incoming lists may change; they are recomputed before their own replacement. Hence the final machine has the interval property at all times simultaneously. This forward argument is why independently optimizing a partition at each cut would not prove the theorem.

Conversely, every sequence of choices in (3.6) defines causal finite update tables and therefore one admissible machine. The terminal formula is exactly its optimal readout. The existence just proved shows that restricting the finite search to these choices loses nothing. This proves (3.7). With controlled observations, first keep the current deterministic experiment choices fixed in the replacement argument; the same affine-envelope reasoning applies to their incoming atoms. Maximizing over those finite choices completes the proof. $\square$

This theorem uses the likelihood-ratio quantizer principle [27]; the affine-envelope mechanism is not claimed as new. The point of the recursion is the common, changing occupation list and the simultaneous realizability of all its partitions. The posterior pairs are computed by the designer. The implemented machine stores only a finite label and uses a fixed finite lookup table; it does not store a real-valued posterior. The number of reachable lists can be large. No polynomial-time claim is part of the theorem.

The result is Bayesian and clocked. A maximum of a minimum over environments is not multiaffine, and a reused autonomous row can occur several times in one execution. Neither of those cases is covered by the purification argument. They remain covered by Theorems 2.1 and

2.2. The following example solves the minimax problem directly and shows why the distinction is substantive.

3.3. An exact minimax recursion for binary registers. There is also a structural recursion that does not purify the minimax objective. In this subsection observations are conditionally independent and uncontrolled, and every retained alphabet has size one or two. Represent a binary prefix experiment by its pair $r = (r_0, r_1)$, where $r_\theta = \mathbb{P}_\theta(M = 1)$. Its labels may be exchanged at no cost. Say that s dominates r if a single stochastic binary channel, independent of θ , maps s to r .

For a pair s and the next report law, let $E_t(s)$ be the experiment on $\{0, 1\} \times Y_t$ with masses

$$e_\theta(m, y) = s_\theta^m (1 - s_\theta)^{1-m} p_t^\theta(y).$$

At endpoints this formula has its direct Bernoulli meaning. For any finite binary experiment $E = (e_0, e_1)$ define its upper receiver curve by the finite linear program

$$R_E(a) = \max \left\{ \sum_x e_1(x) \phi(x) : 0 \leq \phi(x) \leq 1, \sum_x e_0(x) \phi(x) = a \right\}, \quad 0 \leq a \leq 1. \quad (3.8)$$

Ordering atoms by likelihood ratio solves this program: accept the largest ratios first, randomizing at at most one ratio level. Atoms with zero e_0 are treated by their direct contribution to the objective; no division by zero is needed.

Starting with $\mathcal{R}_0 = \{(0, 0)\}$, define

$$\mathcal{R}_t = \begin{cases} \{(0, 0)\}, & K_t = 1, \\ \{(a, R_{E_t(s)}(a)) : s \in \mathcal{R}_{t-1}, 0 \leq a \leq 1\}, & K_t = 2. \end{cases} \quad (3.9)$$

This is a recursion of attainable channels, not of their convex hulls. Its continuous parameters are design parameters of finite stochastic rows, not stored posterior coordinates. In particular, it need not have a polygonal reachable set after several stages.

Theorem 3.3 (Binary-register minimax frontier). *Every channel in $\mathcal{R}_t$ is attainable with the prefix profile, and every stochastic prefix experiment with that profile is dominated by some member of $\mathcal{R}_t$. For a bounded payoff G and finite terminal action set, let $\Gamma_G(r)$ be the optimal minimum of the two expected payoffs obtained by a terminal stochastic readout of the binary experiment r . Then*

$$V_{n, \mathbf{K}}^{\text{clk}}(\{0, 1\}, G) = \max_{r \in \mathcal{R}_n} \Gamma_G(r). \quad (3.10)$$

For correct-decision payoff this reduces to

$$V_{n, \mathbf{K}}^{\text{clk}} = \max_{r \in \mathcal{R}_n} \left\{ \frac{1}{2} + \frac{|r_1 - r_0|}{2(1 + |1 - r_0 - r_1|)} \right\}. \quad (3.11)$$

An optimizer can be implemented by likelihood-ratio quantizers with at most one randomized ratio level per update and a terminal readout. These assertions hold at every finite horizon, with overlapping noise and with arbitrary placement of one-state cuts.

Proof. First consider a binary summary of any finite binary experiment, with output probabilities (r_0, r_1) . Exchange its labels if necessary so that $r_1 \geq r_0$. The receiver-curve optimizer at $a = r_0$ has output probabilities (r_0, b) with $b \geq r_1$. It dominates (r_0, r_1) . Indeed, if $b > r_0$, put

$$\lambda = \frac{r_1 - r_0}{b - r_0} \in [0, 1].$$

The postprocessing probabilities $\mathbb{P}(M = 1 \mid S = 0) = r_0(1 - \lambda)$ and $\mathbb{P}(M = 1 \mid S = 1) = r_0(1 - \lambda) + \lambda$ are in $[0, 1]$ and give the required pair. If $b = r_0$, the desired summary is uninformative and can be generated by a constant stochastic row. Undoing the exchange of labels covers the original summary. The receiver program is solved by an ordered threshold because moving an equal amount

of e_0 -mass from a smaller likelihood ratio to a larger one cannot decrease its objective. Sorting the finite atoms and filling the prescribed mass a proves the stated one-level randomization rule, including zero-mass atoms.

Proceed by induction on t . Each member of (3.9) is attained by an attaining previous channel and a receiver-curve quantizer. Conversely, suppose a previous prefix channel r is dominated by an attainable s in $\mathcal{R}_{t-1}$. Conditional independence of the next observation implies that $E_t(s)$ dominates $E_t(r)$ by applying the same postprocessing to the state coordinate. Composing this postprocessing with any proposed new summary gives a binary quantizer of $E_t(s)$. The preceding argument dominates its output by a receiver-curve point, which belongs to $\mathcal{R}_t$. This composition can be incorporated in one update table and introduces no retained selector. At a one-state cut every prefix is uninformative and the reset to $(0, 0)$ is exact. This proves both inductive assertions.

A terminal readout can compose any dominating channel's postprocessing, so Γ_G is monotone under domination. The original finite machine class has a maximizer by compactness of its row simplices. The inductive domination statement supplies an attainable member of $\mathcal{R}_n$ with at least that value. No attainable member can exceed the optimum. This proves the maximum and attainment in (3.10), without needing a convexity claim about $\mathcal{R}_n$.

For the last formula, exchange labels to arrange $r_1 \geq r_0$. If $r_0 + r_1 \leq 1$, accept hypothesis one on state one and randomize on state zero with probability $(1 - r_0 - r_1)/(2 - r_0 - r_1)$. The two successes then equal $(1 - r_0)/(2 - r_0 - r_1)$. If $r_0 + r_1 \geq 1$, reject on state zero and accept on state one with probability $1/(r_0 + r_1)$; both successes equal $r_1/(r_0 + r_1)$. These are likelihood-ratio tests and are minimax by the same equal-error supporting-prior argument as for a finite receiver curve. Both expressions equal (3.11); uninformative endpoints give $1/2$ directly. Choosing the attaining prefix quantizers proves the final implementation assertion. $\square$

Equation (3.9) is a genuine minimax whole-schedule characterization for arbitrary finite horizons in this binary-register class. It uses the ordinary comparison order of binary experiments, not a minimax interchange for complete machines. The restriction to uncontrolled, conditionally independent reports is used exactly when the dominating prefix is tensored with the next report. General controlled experiments and larger minimax registers use the occupation theorem instead. The three-report result that follows is a closed-form evaluation, with a strict mixing gap, of this same noisy resource problem.

4. AN EXACT NOISY FRONTIER AND THE COST OF HIDDEN MIXING

Let $\theta \in \{0, 1\}$, and let Y_1, Y_2, Y_3 be conditionally independent binary reports with

$$\mathbb{P}_\theta(Y_t \neq \theta) = p, \quad 0 < p < 1/2.$$

Every report has overlapping support under the two hypotheses. A clocked machine has at most K_t labels after report t and makes a binary terminal decision. Updates are atomic stochastic operations; additional random-bit cuts are discussed below. Let $B_{\mathbf{K}}(p)$ be its optimal success under the fair prior, and let $V_{\mathbf{K}}(p)$ be its optimal minimum of the two conditional success probabilities. Both optimize over all stochastic machines of this profile. Put

$$d(p) = p(1 - p)(1 - 2p) > 0.$$

Theorem 4.1 (Complete three-report profile frontier). *For every positive integer profile (K_1, K_2, K_3) ,*

<i>profile condition</i>	$B_{\mathbf{K}}(p)$	$V_{\mathbf{K}}(p)$	(4.1)
$K_3 = 1$	$1/2$	$1/2$	
$K_3 \geq 2, \min(K_1, K_2) = 1$	$1 - p$	$1 - p$	
$K_1, K_3 \geq 2, K_2 = 2$	$1 - p + \frac{1}{2}d(p)$	$1 - p + \frac{1}{3}d(p)$	
$K_1, K_3 \geq 2, K_2 \geq 3$	$1 - p + d(p)$	$1 - p + d(p)$	

In the third line a minimax optimal machine retains Y_1 , then retains Y_1Y_2 (the Boolean AND), and after the third report decides one with probability

$$\frac{(M_2, Y_3)}{\mathbb{P}(\hat{\theta} = 1 \mid M_2, Y_3)} \mid \begin{array}{cccc} (0, 0) & (0, 1) & (1, 0) & (1, 1) \\ 0 & 2/3 & 1 & 1. \end{array} \quad (4.2)$$

The two conditional success probabilities are equal. In particular, for $p = 1/4$ and profile $(2, 2, 2)$ the exact Bayesian and minimax values are $51/64$ and $25/32$, whereas profile $(2, 3, 2)$ has value $27/32$ for both objectives.

We prove the minimax assertion without passing to a convex hull of machines. The following elementary binary-channel domination is useful.

Lemma 4.2 (Reduction of a two-report binary register). *Every stochastic binary summary of (Y_1, Y_2) is, up to exchanging its two output labels, a stochastic postprocessing of a summary with*

$$\mathbb{P}(M = 1 \mid 00) = 0, \quad \mathbb{P}(M = 1 \mid 01) = \mathbb{P}(M = 1 \mid 10) = w, \quad \mathbb{P}(M = 1 \mid 11) = 1 \quad (4.3)$$

for some $w \in [0, 1]$. This domination holds simultaneously under both hypotheses and continues to hold after appending an independent third report.

Proof. Write u, v for the summary probabilities on $00, 11$, and write b for the average of its probabilities on $01, 10$. Those two words have the same probability under either hypothesis, so their average contains all relevant information. Exchange output labels if necessary to arrange $v \geq u$, and put $\alpha = v - u \in [0, 1]$ and $c = 2p(1 - p) \in (0, 1)$. Its output-one probabilities satisfy

$$r_0 = u + c(b - u) + p^2\alpha, \quad r_1 - r_0 = (1 - 2p)\alpha.$$

For the summary (4.3) these probabilities are $s_0 = p^2 + cw$, $s_1 = (1 - p)^2 + cw$. Seek a postprocessing that sends input zero to one with probability a and input one to one with probability $a + \alpha$. It is stochastic precisely when $0 \leq a \leq 1 - \alpha$, and it reproduces r_0, r_1 if

$$a = u + c(b - u) - c\alpha w.$$

As w ranges over $[0, 1]$, the right side ranges from $u + c(b - v)$ to $u + c(b - u)$. The upper endpoint is nonnegative. The lower endpoint is at most $1 - \alpha$, since

$$1 - \alpha - u - c(b - v) = (1 - v) + c(v - b) \geq (1 - v)(1 - c) \geq 0.$$

Consequently this interval meets $[0, 1 - \alpha]$. This proves the claim, including $\alpha = 0$. Applying the postprocessing independently of the third report proves the last statement. $\square$

Proof of Theorem 4.1. If the last register has one state, its terminal decision is independent of the reports. The fair-prior value is one half, and a fair terminal randomization gives minimax value one half. If an earlier register has one state, at most the last two reports remain informative. The unrestricted optimal success from two equally reliable reports is $1 - p$ under either criterion: use their common value on agreement and a fair decision on disagreement. One report already gives $1 - p$. These rules respect the stated profiles and prove the second line.

Suppose henceforth that $K_1, K_3 \geq 2$. There is no loss in retaining Y_1 exactly. Indeed any stochastic first update followed by a second update induces one binary channel from (Y_1, Y_2) to M_2 , and that channel can be implemented directly when Y_1 is retained. Allowing more than two first-stage labels gives no larger set of such channels. At the last stage any randomized binary decision based on (M_2, Y_3) can be placed directly in a two-label register. Thus, when $K_2 = 2$, the unrestricted optimization is exactly over a binary summary of the first two reports and a randomized readout of it together with Y_3 .

For the Bayesian problem, Theorem 3.2 reduces the middle update to intervals of the three distinct likelihood-ratio classes $00, \{01, 10\}, 11$. A nonconstant two-interval update isolates 00 or

isolates 11. These are the Boolean OR and AND, up to labels. With AND, the Bayesian readout is $\hat{\theta} = Y_3 \vee (Y_1 \wedge Y_2)$. Its conditional successes are

$$c_0 = 1 - p - p^2 + p^3, \quad c_1 = 1 - 2p^2 + p^3.$$

Their average is $1 - p + d(p)/2$. The complementary rule interchanges c_0, c_1 . A constant middle update gives only $1 - p$, proving the Bayesian third line. This also covers updates that split the tied middle words: the theorem permits assigning the entire tied class to one maximizing suffix label without decreasing the Bayesian objective.

For the minimax problem, Lemma 4.2 reduces the search, without loss, to (4.3). Complementing all reports and output labels replaces w by $1 - w$ and interchanges the hypotheses, so it suffices to take $0 \leq w \leq 1/2$. Write

$$r_0 = p^2 + 2p(1 - p)w, \quad r_1 = (1 - p)^2 + 2p(1 - p)w.$$

In the experiment (M_2, Y_3) the likelihood ratios are ordered as 00, 01, 10, 11. Only the ordering of the middle two needs verification; after multiplying by their positive denominators it is equivalent to

$$r_1(1 - r_0)p^2 - r_0(1 - r_1)(1 - p)^2 = p^2(1 - p)^2(1 - 2p)(1 - 2w)^2 \geq 0.$$

Consider the test that accepts hypothesis one on 10, 11, rejects it on 00, and accepts it on 01 with probability

$$a(w) = \frac{1 - r_0 - r_1}{(1 - r_0)p + (1 - r_1)(1 - p)} = \frac{2(1 - 2w)}{3 - 2w}. \quad (4.4)$$

This number is in $[0, 1]$. Its two conditional success probabilities are equal, with common value

$$R(w) = 1 - r_0 - a(w)(1 - r_0)p. \quad (4.5)$$

It is a likelihood-ratio test, possibly randomized at one threshold. Choose the prior odds equal to that threshold. The test is then Bayesian optimal for those odds. Since its two successes are equal, no test can have a larger minimum: such a test would also have a larger payoff for this prior. Thus $R(w)$ is the exact minimax success for this summary, not merely an equal-error attainable value.

Direct differentiation gives

$$R'(w) = -\frac{2p(1 - p)(1 - 2p)(5 - 2w)(1 - 2w)}{(3 - 2w)^2} \leq 0. \quad (4.6)$$

Equivalently, an explicit nonnegative gap certificate is

$$R(0) - R(w) = \frac{2p(1 - p)(1 - 2p)w(5 - 6w)}{3(3 - 2w)} \geq 0, \quad 0 \leq w \leq 1/2. \quad (4.7)$$

The denominator is positive and every factor in the numerator is nonnegative on the stated domain. Therefore the best w is zero. Equations (4.4)–(4.5) give $a(0) = 2/3$ and

$$R(0) = 1 - \frac{2}{3}p - p^2 + \frac{2}{3}p^3 = 1 - p + \frac{1}{3}d(p).$$

The postprocessing in Lemma 4.2 can be composed with any final readout and does not need a retained selector. It follows that this upper bound applies to every original stochastic machine. The rule (4.2) attains it.

Finally, if $K_2 \geq 3$, retain the number of ones in the first two reports and output the majority of all three. Both successes are $1 - 3p^2 + 2p^3 = 1 - p + d(p)$. The likelihood-ratio rule for the full three reports is majority under the fair prior, so no machine can have a larger Bayesian value or a larger minimum. This proves the last line and completes all cases. $\square$

Corollary 4.3 (A strict fixed-profile mixing gap). *For profile $(2, 2, 2)$ a free initial selector that mixes the two complementary Bayesian optimal machines would give worst-case success $1 - p + d(p)/2$. Without that uncharged selector the exact value is $1 - p + d(p)/3$. The excess $d(p)/6$ is strictly positive.*

Proof. Mix the two payoff vectors (c_0, c_1) and (c_1, c_0) from the proof with equal weights. Each coordinate becomes their average. Theorem 4.1 gives the smaller all-stochastic value at the original profile. To implement the mixture literally, retain the selector jointly with the machine state; a profile $(4, 4, 2)$ suffices. The corollary does not claim that this product construction minimizes the expanded profile. $\square$

The gap identifies a specific failure of a convexified all-machine converse. A terminal output law can be a convex combination of laws while the machines producing those laws cannot be mixed within their original memory profile. In contrast, the quadratic compatibility constraints of Theorem 2.1 retain exactly the required coupling. The noisy loss $2d(p)/3$ between two and three middle states is also a strict full-schedule minimax loss; it is not an erasure probability or the cost of remembering a label that was already known.

Remark 4.4 (Atomic randomization and finite precision). The fraction $2/3$ in (4.2) is an exact rational row of the all-stochastic model. A fixed finite string of fair bits cannot realize this fraction exactly. Replacing it by a dyadic number within 2^{-r} gives a genuine finite-precision rule whose minimum success is at least $V_{(2,2,2)}(p) - 2^{-r}$. If each random-bit epoch is a charged cut, retain (M_2, Y_3) while executing the dyadic sampler; the three-status comparison sampler of Section 10 gives a conservative peak of 12 for this refined execution. The theorem's exact two-state value refers to its stated atomic-update profile, not to this finer profile. The latter has its own exact occupation formulation. Neither an unbounded rejection sampler nor a free fair-bit clock is hidden in the two-state assertion.

5. ACQUISITION, CLOCKS, AND THE RESOURCE ORDER

A statistical bound is ambiguous unless the protocol and the resource coordinates being compared are specified. The superscript clk in the preceding theorems is therefore part of their notation, not a qualification confined to an implementation paragraph.

5.1. Comparable resource points. On one declared schedule, profiles are ordered coordinatewise. Enlarging a register alphabet preserves every previously feasible machine by padding unused states. Inserting observations is comparable only through an explicit embedding that says which new inputs are ignored and which registers survive the inserted cuts. Atomic and bit-serial schedules are different schedules, even when they have the same number of experimental preparations.

For a finite-precision implementation we record

$$\mathbf{r} = (N_B, N_Q, \mathbf{K}, r, D, \mathbf{C}). \quad (5.1)$$

Here N_B, N_Q count candidate and target preparations, r bounds fair bits per declared row sampler, D bounds a specified read-only description, and $\mathbf{C}$ specifies clocked or autonomous execution. A description is measured in a fixed code, not by an unspecified notion of compressibility. For example, with a known table layout, store the first $s - 1$ numerators of a 2^{-r} stochastic row in $(s - 1)(r + 1)$ bits; its last numerator is the remainder. The layout, response dictionary and any program used to expand repeated phases have their own fixed-code costs. Sample-dependent calibration is not part of that read-only description.

For the same code and schedule, increasing numerical resource caps or register alphabets gives an inclusion of machine classes. Changing $\mathbf{C}$ does not mean increasing one numerical coordinate. A clocked execution with T update epochs and profile $\mathbf{K}$ has an autonomous realization on the

disjoint union of its phase registers, using at most $\sum_{t=0}^T K_t$ states. This is an attainable conversion, not an equality of optimal frontiers. Shared-row optimization in Theorem 2.1 can be strictly smaller than that conversion.

Proposition 5.1 (Joint finite resource optimization). *Fix a finite experimental protocol and a finite table code. With finite caps on description and precision, the feasible machines form a finite set and their optimal worst-case payoff is exactly the maximum over that set. If the precision restriction is removed on a fixed table architecture, Theorems 2.1 and 2.2 characterize its exact value. A finite collection of permitted architectures is handled by taking the maximum of their values. Calibration calls can be included as an initial part of the same protocol, with all their retained information passed through the same profile constraints.*

Proof. There are finitely many descriptions of bounded length and finitely many allowed dyadic rows. Retain exactly the descriptions that decode to legal tables of the specified finite schedule and budgets. Each has a well-defined payoff. For unrestricted stochastic rows on a fixed architecture, use its observable row sets in the occupation formulation. Optimizing over a finite union is a maximum over its components. A calibration prefix uses known experimental kernels at the unknown environment in exactly the same way as any other prefix; its sample-dependent state is an ordinary retained register. Thus it introduces neither a parameter oracle nor a new exception to the compatibility constraints. $\square$

The proposition supplies a joint optimization model rather than merely a post hoc resource ledger. It does not claim a short closed formula for arbitrary description budgets. For compact continuous families the net condition in Corollary 2.3 is still required for effective finite certificates.

5.2. Acquiring a calibration rather than assuming it. The next result gives a fully priced constructive calibration prefix. Let $h_1, \dots, h_K : \Omega \rightarrow [0, 1]$ be a fixed known response list, chosen independently of the calibration observations, and suppose its values have denominator 2^r . Write $d = |\Omega|$ and $q_i = Qh_i$. The target Q need not be given as a table: independent preparations from Q are available. Let $M = 2^s$ and $0 < \beta < 1$.

Theorem 5.2 (Sampled calibration with persistent-register pricing). *For every $\rho > 0$ with*

$$M \geq \frac{1}{2\rho^2} \log \frac{2K}{\beta}, \quad (5.2)$$

there is a clocked finite-precision calibration protocol using M target preparations and MKr fair bits. It returns $\hat{q}_i \in \{0, 1/M, \dots, 1\}$ with

$$\mathbb{P} \left(\max_i |\hat{q}_i - q_i| \leq \rho \right) \geq 1 - \beta.$$

The complete counter register has at most $C = (M + 1)^K$ states; a word buffer and the three-status dyadic sampler give peak at most $3dC$. If a subsequent auditor has peak at most W for any exogenously fixed calibration vector, retaining the acquired calibration gives the bound $C \max(3d, W)$ for the combined execution.

Let $v \in [-1, 1]$ and $\epsilon \geq 0$. Suppose, uniformly for calibration vectors $\tilde{q}$, the auditor's true expected score is at least $v - \epsilon - 2\|\tilde{q} - q\|_\infty$, and its score is in $[-1, 1]$. With fresh training and validation after this prefix, the combined expected score is at least

$$v - \epsilon - 2\rho - 2\beta. \quad (5.3)$$

Proof. On target trial j , retain its word Y_j . For each i , generate a Bernoulli variable B_{ij} of parameter $h_i(Y_j)$ with an independent r -bit dyadic sampler, and increment counter i . After all

K updates, discard Y_j . For fixed i , the variables B_{ij} over different trials are independent, lie in $\{0, 1\}$, and have mean q_i . Their averages $\hat{q}_i$ therefore satisfy

$$\mathbb{P}(|\hat{q}_i - q_i| > \rho) \leq 2e^{-2M\rho^2}.$$

This bounded-variable exponential estimate follows from the exponential moment bound used in Theorem 9.2. A union bound proves the probability assertion. Independence between different counters is not required.

The K counters each have $M + 1$ possible values. The word buffer has d values, and the running comparison of a uniform r -bit word with a fixed dyadic threshold has three comparison statuses. These yield the stated state counts at every sampler cut. Trial, response and bit indices are in the declared clocked schedule, not in an unmentioned work tape. The persistent counter vector remains in the register of the subsequent auditor and multiplies its width by at most C . All its values are dyadic because $M = 2^s$.

Conditional on the calibration vector, later observations have their original independent laws. On the good calibration event the score bound is $v - \epsilon - 2\rho$. On the complementary event the score is at least -1 . Since $v \leq 1$ and $\epsilon, \rho \geq 0$, a loss of at most two times that event's probability suffices. This proves (5.3). $\square$

For the response-cover tournament of Theorem 13.2, one may take

$$N_B = n(K - 1), \quad W = 3dK(2m + 1), \quad \epsilon = 4(K - 1)\theta + 2(K - 1)(e^{-4\theta m} + e^{-2n\theta^2})$$

when v is the envelope of the already dyadic response list. Starting with an arbitrary list and rounding it to r bits incurs the additional 2^{1-r} term of that theorem. The new prefix prices target acquisition and its persistent calibration state. It is an explicit construction, not an optimal calibration-memory claim. Joint optimality at fixed finite budgets is the distinct problem in Proposition 5.1.

6. CONSUMERS OF THE COMPATIBLE FRONTIER

The preceding characterizations and the constructive continuation results below have the same object of optimization: one experiment, one schedule and one whole register profile. This section records the connections before developing the underlying marked and physical experiments in detail.

6.1. Reset scores and physical protocols. In the reset interface of Section 9, a fixed candidate supplies the training and candidate-validation kernels. The target supplies its own preparation and validation kernels. The final score is a function of the two retained validation bits. Include the response selection, all input-bit buffers, and all randomization cuts in σ . This produces precisely the finite protocol of Section 2. The image of the candidate family is not replaced by its convex hull.

Corollary 6.1 (Common constructive and converse interface). *At any fixed finite reset protocol and full profile, the best worst-case score has the occupation characterization of Theorem 2.1; for a finite family it has the complete dual of Theorem 2.2. For a compact family with effective finite kernel nets it has the converging upper and lower certificates of Corollary 2.3. The response-cover machines of Theorems 12.2 and 13.2 supply feasible lower certificates in these very classes. Any certified dual bound applies to those machines and to every other stochastic machine with the same protocol and profile.*

Proof. Augment the hidden index by the fixed candidate parameter and the finite public trial history. Selection and memory updates are parameter-independent observer rows. Experimental draws are the known linear environment operations. Terminal scoring is linear in the terminal

occupation array. Thus the realization theorem applies in both directions. The constructive machines specify legal rows at every cut and hence feasible occupation arrays. Compact-family and dual assertions are exactly the corresponding general theorems. $\square$

For the positive-noise marked collision experiment, the finite row-report alphabet is fixed and the private row family is compact. On compact subfamilies with the stated continuous kernel data, finite kernel nets give the preceding certificates. The microscopic and Gaussian comparison theorems below then transfer a certified payoff with their stated common-preparation error. If training kernels differ, their errors must also be charged; equality of the training source cannot be inferred from equality of the terminal validation task. The active-state estimate in Section 15 can replace a worst-case telescoping estimate for a fixed retained execution, but is not a bound on a full discarded transcript.

Thus the physical resource constructions (399, 40602), (320000, 1604) and $(12800 \cdot 2^{400}, 12)$ are now accompanied by a general all-auditor converse procedure on each fixed full profile. They remain attainable resource points, not a list of proved Pareto optima. In particular, the numerical statement $3 \leq W_{2/5}^{\text{phys}} \leq 12$ is not changed into an equality without an additional certificate or construction.

6.2. An exact confidence consumer. Let a reset audit return $D \in \{-1, 0, 1\}$. Suppose its null mean is zero and its alternative mean is at least g , where $0 < g \leq 1$ is known. On each independent repetition, use a fresh coin with

$$\mathbb{P}(Z = 1 \mid D) = \frac{1}{2} + \frac{D - g/2}{4}. \quad (6.1)$$

Under the null its success probability is $p = 1/2 - g/8$, and under the alternative it is at least $1 - p$. Consider the declared coin interface consisting of the null Bernoulli law p and all alternative Bernoulli laws with parameter in $[1 - p, 1]$, independently repeated.

Corollary 6.2 (Exact three-repetition confidence frontier). *At the atomic retained-coin profile (2, 2, 2), the minimum possible maximum of type-I and type-II error over this coin interface is*

$$p - \frac{1}{3}p(1 - p)(1 - 2p). \quad (6.2)$$

The rule (4.2) attains it. Increasing only the middle retained-coin alphabet to three changes this value to $p - p(1 - p)(1 - 2p)$. If one underlying audit uses b candidate training trials, two validation trials, and peak W , the three-repetition construction uses $3(b + 2)$ preparations. Its inner audit, coin sampler, and outer state are charged jointly, rather than identifying an outer two-state register with the entire physical register.

Proof. The alternative family contains the endpoint $1 - p$, so Theorem 4.1 gives the lower bound on the maximum error. For the attaining test, its probability of deciding one, as a function of the three input bits, is coordinatewise nondecreasing: the middle state is AND, and the four final probabilities are 0, 2/3, 1, 1. Couple Bernoulli inputs with common independent uniforms. The power is then nondecreasing in their parameter, so the largest type-II error occurs at the endpoint. This proves attainment for the whole declared alternative interval. The majority test proves the assertion with three middle states in the same way. Preparation counts follow by repeating the base experiment independently three times. Product registers implement the outer transitions together with each inner repetition. $\square$

If g is dyadic, the coin in (6.1) has a finite dyadic sampler. At its internal cuts an explicit conservative bound is $2 \max(W, 9)$ for the outer state and inner audit/coin sampler; the additional 2/3 terminal row is either an atomic rational operation or is replaced by the finite-precision realization

in Remark 4.4. A fully fair-bit-serial realization has the conservative peak $\max\{2\max(W, 9), 12\}$ and the indicated 2^{-r} error allowance. This does not erase its clock or its bits from the schedule.

The exact lower bound concerns the declared coin interface. A physical subfamily inherits that lower bound only if it contains its endpoint experiment (or an appropriate certified limiting experiment). A raw physical auditor that uses more of D or of the original report is not restricted to this interface. For such auditors the general occupation converse, with those richer inputs retained, is the relevant statement. The confidence result is therefore a proved consumer, not a relabeling of an expected-score lower bound as a hypothesis-testing lower bound.

6.3. The foundation and the longer program. The initial objects remain positive experimental kernels, preparations, causal interventions and executable continuations. Finite protocols are a resource-controlled subcategory, not a replacement for the standard Borel formulation or its limiting targets. The new occupation and continuation results supply a precise G3-type finite resource layer. The marked deficiency and Gaussian transport results supply the comparison maps used above. The finite resource theorems do not supply a missing kinetic generator domain, a compactness theorem for collision measures, or an optional-projection theorem on a general operator domain.

The full historical dependency graph is retained in Appendix B. The geometric A2 chain has its own premises. The B4 state-dependent control-transfer, range and compactness targets, and the broader C2 operator and rigidity targets, retain their original scope. Their relation to this paper is through explicit experiment and simulator interfaces when those have been established, not through a declaration that every manuscript is a consequence of a finite-state comparison theorem.

7. INTRINSIC CAUSAL DEFICIENCY

7.1. Experiments and executable signatures. Fix a finite parameter set Θ , a horizon T , and finite control alphabets $A_0, \dots, A_{T-1}$. An experiment has a report at checkpoints $0, \dots, T$ and an unobserved terminal mark W in a common finite set. It is specified by finite causal instruments: for each parameter and control history they give the joint law of the report path and the actual mark. Equivalently, a finite latent-state realization may be expanded into its history tree. The parameter and mark are unavailable to the encoder. The mark may depend on the physical controlled path. We compare experiments with the same control interface and mark space.

Write Y_t for the source report and Z_t for the target report. A private simulator of width K has stochastic rows

$$\begin{aligned} q_0(z, m \mid y), \quad m \in [K_0], \\ q_t(z, m \mid m', a, y), \quad m' \in [K_{t-1}], \quad m \in [K_t], \quad 1 \leq t \leq T. \end{aligned} \tag{7.1}$$

A row consumes the current source report, emits the target report, and retains the new state. The previous target reports are not available to that row unless encoded in m' . The clock and current control are part of the declared interface. Fresh private randomness is permitted, but no other persistent random variable is free. The product of all row simplexes is denoted $\mathcal{K}_K$.

A deterministic feedback tree g chooses a_t from $z_{0:t}$ and its own previous controls. It is the same controller on both sides of a comparison. Let Ω_g be the finite marked transcript space, including reports, controls, and W . Put

$$P_{\theta,g}^q = \text{Law}_\theta(E \xrightarrow{q} g), \quad Q_{\theta,g} = \text{Law}_\theta(F \rightarrow g).$$

Here W on the left is the source's actual mark. The total variation convention is $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$. Private deficiency is

$$\delta_K^p(E, F) = \min_{q \in \mathcal{K}_K} \max_{\theta, g} \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}. \tag{7.2}$$

The minimum will be justified below. Randomized feedback need not be added to the maximum: conditioning on its independent random table expresses its law as a mixture of deterministic feedback laws. The same argument permits arbitrary independent side randomness in the tester.

A *hidden selector* $U \in [r]$, independent of the source, chooses one simulator q^u . It is not visible to the feedback controller or the final test. A *visible selector* is revealed before feedback begins. The two costs are

$$\delta_{K,r}^h(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, g} \left\| \sum_u \lambda_u P_{\theta, g}^{q^u} - Q_{\theta, g} \right\|_{\text{TV}}, \quad (7.3)$$

$$\delta_{K,r}^v(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, (g_u)} \left\| \sum_u \lambda_u \delta_u \otimes P_{\theta, g_u}^{q^u} - \sum_u \lambda_u \delta_u \otimes Q_{\theta, g_u} \right\|_{\text{TV}}. \quad (7.4)$$

In both expressions $\lambda \in \Delta_r$ and $q^u \in \mathcal{K}_K$. The second formula compares the joint seed–marked interface. Its target seed has the same distribution and is independent of the target experiment. A hidden selector can be stored privately by retaining (u, m) , at width rK_t . A visible selector is separate side information and is charged by its own alphabet size. Neither interpretation identifies an arbitrary real-valued design with a free finite register.

Definition 7.1 (Deficiency spectrum). The private resource spectrum of E relative to F is the upward-closed set of pairs (K, ε) satisfying $\delta_K^p(E, F) \leq \varepsilon$. The hidden spectrum consists of triples (K, r, ε) satisfying $\delta_{K,r}^h(E, F) \leq \varepsilon$, and the visible spectrum uses $\delta_{K,r}^v$. Width profiles are ordered coordinatewise; in the seeded spectra the seed bound and error are ordered as well. Unused states or selector values may be adjoined. The hidden convexification is $\delta_K^{\text{mix}} = \inf_{r \geq 1} \delta_{K,r}^h$.

The spectrum, not a presumed unique coordinatewise minimum profile, is the intrinsic object. For example one may minimize $\max_t K_t$, with a specified seed signature and tolerance, over its feasible profiles. Different checkpoints can have incomparable optimal tradeoffs.

7.2. The finite comparison theorem. Concatenate all probabilities $P_{\theta, g}^q(\omega)$ into $b(q) \in \mathbb{R}^N$, retaining their row indices; put $b_F = (Q_{\theta, g}(\omega))$. Let $\mathcal{D}_K$ be the finite set of deterministic choices of all rows in (7.1). Empty or unreachable rows remain harmless coordinates. Set

$$a_K = \dim \text{aff}\{b(q) : q \in \mathcal{K}_K\}, \quad \mathcal{I} = \{(\theta, g, B) : B \subseteq \Omega_g\}.$$

For $i = (\theta, g, B)$ and $d \in \mathcal{D}_K$, define

$$A_{i,d} = Q_{\theta, g}(B) - P_{\theta, g}^d(B). \quad (7.5)$$

Including both an event and its complement includes both signs.

Theorem 7.2 (Intrinsic deficiency, priced converse, and visibility). *For the finite marked causal experiments above the following assertions hold.*

(i) *Private deficiency and both fixed- r deficiencies are attained. The private problem has the exact polynomial formulation*

$$\min\{\varepsilon : q \in \mathcal{K}_K, 0 \leq \varepsilon \leq 1, Q_{\theta, g}(B) - P_{\theta, g}^q(B) \leq \varepsilon \text{ for all } (\theta, g, B) \in \mathcal{I}\}. \quad (7.6)$$

In particular its zero set is exactly the nonnegative causal factorization of the target marked behavior by the rows (7.1). At $K_t = 1$ for every t , this is precisely the same factorization with rows $q_t(z \mid a, y)$ and $q_0(z \mid y)$.

(ii) *Hidden convexification has the attained dual*

$$\delta_K^{\text{mix}}(E, F) = \min_{\lambda \in \Delta(\mathcal{D}_K)} \max_{i \in \mathcal{I}} \sum_d \lambda_d A_{i,d} = \max_{w \in \Delta(\mathcal{I})} \min_{d \in \mathcal{D}_K} \sum_i w_i A_{i,d}. \quad (7.7)$$

An optimal simulation uses at most $a_K + 1$ selector values. Thus every weighted causal test has optimal discrepancy at most ε if and only if a hidden mixture of K -state simulators has error at most ε ; it has a private realization of width at most $(a_K + 1)K$. More specifically,

$$\delta_{rK}^p \leq \delta_{K,r}^h, \quad \delta_K^{\text{mix}} = \delta_{K,a_K+1}^h \leq \delta_K^p. \quad (7.8)$$

(iii) Put $e_\theta(q) = \max_g \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}$. Then

$$\delta_{K,r}^v = \min_{\lambda, q^1, \dots, q^r} \max_{\theta} \sum_u \lambda_u e_\theta(q^u). \quad (7.9)$$

Writing $a_K^v = \dim \text{aff}\{(e_\theta(q))_\theta : q \in \mathcal{K}_K\}$, the infimum over all finite visible selectors is attained using $a_K^v + 1 \leq |\Theta| + 1$ values and equals

$$\max_{v \in \Delta(\Theta)} \min_{q \in \mathcal{K}_K} \sum_{\theta} v_\theta e_\theta(q). \quad (7.10)$$

For every $r \geq 1$, and also after optimizing over all r ,

$$\delta_{K,r}^v(E, F) = 0 \iff \delta_K^p(E, F) = 0. \quad (7.11)$$

No corresponding equivalence with $\delta_K^{\text{mix}} = 0$ holds in general.

(iv) For compatible experiments E, F, G and profiles K, L ,

$$\begin{aligned} \delta_{KL}^p(E, G) &\leq \delta_K^p(E, F) + \delta_L^p(F, G), \\ \delta_{KL,rs}^s(E, G) &\leq \delta_{K,r}^s(E, F) + \delta_{L,s}^s(F, G), \quad s \in \{h, v\}. \end{aligned} \quad (7.12)$$

Here KL is the pointwise product, and the second line uses independent selectors. If an F -consumer has widths S_t , composing it with a private simulator gives widths $K_t S_t$ and changes every $[0, 1]$ marked risk by at most $\delta_K^p(E, F)$. A hidden selector is stored at the additional price r , whereas visible selector alphabets multiply. The same bounds hold for optimized upper risk sets and their lower support functions with the indicated costs.

Proof. Expand a probability of a complete source path, simulator state path, and transcript. It is a product of fixed experiment probabilities and one chosen row entry per time. Summing latent and state paths gives a polynomial in the finitely many row entries. No row belonging to a fixed time can occur twice along one path. Consequently the polynomial is multiaffine in the groups of stochastic rows. Continuity and compactness of the product of closed finite simplexes prove attainment in (7.2); the same proof includes λ and the r policies in (7.3). For the visible problem there are only finitely many collections (g_u) and transcript events, so its objective is again continuous on a compact set. The finite-event formula for total variation proves (7.6). Setting $\varepsilon = 0$ forces equality of every transcript probability, in both directions of each event inequality. This proves the factorization and stateless claims, including their necessity.

For each stochastic row of q , sample its deterministic output once, independently of all other rows, before the experiment. This constructs a probability Λ_q on $\mathcal{D}_K$. On any fixed path the row variables encountered have different time indices, so the expectation of their product is the product of the row probabilities. Summing paths shows, simultaneously for every θ, g , that

$$b(q) = \sum_{d \in \mathcal{D}_K} \Lambda_q(d) b(d). \quad (7.13)$$

This is a behavior identity, not a same-width deterministic purification of minimax error. It implies $\text{conv } b(\mathcal{K}_K) = \text{conv}\{b(d) : d \in \mathcal{D}_K\}$. The latter is a compact polytope. Minimizing the continuous convex maximum of event discrepancies over this polytope gives the first expression in (7.7). Finite-dimensional minimax gives the second, with both optima attained. A point of this polytope is a convex combination of at most $a_K + 1$ of its generating points by Carathéodory's theorem. For completeness, if a representation uses more points, their affine dependence gives

coefficients c_d with $\sum c_d = 0$ and $\sum c_d b(d) = 0$. Subtract the largest permissible multiple of c from the positive mixing weights; a weight becomes zero without changing the represented point. Repeating gives the stated bound. Store its selector together with the machine state to obtain (7.8). The dual inequality for every w is equivalent to the maximum over w being at most ε , which proves the priced testing converse.

For disjoint seed fibers, total variation is the sum of the fiber total variations weighted by λ_u . For a fixed θ , the tester can choose a separate maximizing tree on each revealed fiber. Therefore its maximum is $\sum_u \lambda_u e_\theta(q^u)$, proving (7.9). The map $q \mapsto e(q)$ is continuous, being a finite maximum of continuous functions. Its image is compact. Convexifying it in $\mathbb{R}^{|\Theta|}$ and applying the preceding finite-dimensional separation and affine-dependence argument proves (7.10) and its selector bound. Notice that the inner minimum is over stochastic q , not over deterministic tables: the nonlinear function e_θ does not commute with (7.13). If the attained visible optimum is zero, then for every u of positive weight and every θ one has $e_\theta(q^u) = 0$, since every summand is nonnegative. Any such q^u is a private exact simulator. The converse uses a selector of one value. This also proves the assertion for the optimized visible selector, whose support is finite by the bound just established.

Finally, cascade a K -state E -to- F transducer and an L -state F -to- G transducer. Within a step, consume the intermediate report before discarding it. The retained pair has at most $K_t L_t$ values. For any terminal tester on G , the second transducer together with that tester is an admissible randomized feedback system on F . The first error bound applies to this system, and total variation contracts under the remaining channel. The second error bound applies to the genuine F system. The triangle inequality gives (7.12). With selectors, condition on the independent selectors in the appropriate places; in the visible case keep both selectors in the compared transcript. The same argument is valid because the target reference selectors are independent. Storing a hidden selector or exposing both visible selectors gives the stated costs. The minima established above may be used; equivalently one can use nearly optimal simulators and let the extra tolerance decrease.

A consumer is another causal transducer with task-specific readout and, where allowed, task-specific control, but with its required common encoder. The simulator is task blind, so its product with that encoder is still common. Condition on the consumer's independent random table to apply the feedback bound. Integration of any $[0, 1]$ marked loss changes by at most the total variation error. Taking the same paired consumer for every row proves the risk-set assertion, rather than a separate choice of simulator for each row. Infima of nonnegative weighted risks give the lower support inequalities. $\square$

Remark 7.3 (What the converse says). The dual (7.7) uses mixtures of indexed causal tests and optimizes over the entire common transducer. It is not a task-by-task choice of encoder. Its conclusion is an explicit hidden-selector simulation, with the selector subsequently charged if private storage is required. It does not replace the nonconvex private set by its convex hull at unchanged resource. The next example shows why that replacement would change the zero set.

7.3. A delayed channel and a strict signature separation. Let the source reveal $i \in [n]$ at checkpoint zero and reveal only a constant at checkpoint one. There is no control and the mark is constant. The target reports a constant at checkpoint zero and at checkpoint one draws $j \in [m]$ from a specified row-stochastic matrix $V = (V_{ij})$. The unknown parameter is i . The first target alphabet has one element: the simulator cannot use an early public output as unpriced storage. Only the intermediate register of size K matters.

Proposition 7.4 (The nonnegative-rank slice). *For this experiment,*

$$\delta_K^p(E, F_V) = \min_{A, B} \max_i \frac{1}{2} \sum_j |V_{ij} - (AB)_{ij}|, \quad (7.14)$$

where A is an n -by- K row-stochastic matrix and B is a K -by- m row-stochastic matrix. In particular

$$\delta_K^p(E, F_V) = 0 \iff \text{rank}_+(V) \leq K, \quad \delta_1^p(E, F_V) = 0 \iff V \text{ has identical rows.}$$

For $1 \leq K \leq n$ the identity channel has the sharp laws

$$\delta_K^p(E, F_{I_n}) = 1 - \frac{1}{\lceil n/K \rceil}, \quad \delta_K^{\text{mix}}(E, F_{I_n}) = 1 - \frac{K}{n}. \quad (7.15)$$

Thus, at tolerance $0 \leq \varepsilon < 1$, the least private width is $\lceil n/(1-\varepsilon)^{-1} \rceil$, whereas the least within-mode width with optimized hidden selection is $\max\{1, \lceil n(1-\varepsilon) \rceil\}$. The latter still pays for its selector. In particular, for $n = m = 3$ and $K = 2$,

$$\delta_2^p(E, F_{I_3}) = \frac{1}{2}, \quad \delta_2^{\text{mix}}(E, F_{I_3}) = \frac{1}{3}. \quad (7.16)$$

For $V_* = (1/2)I_3 + (1/6)\mathbf{1}\mathbf{1}^\top$,

$$\delta_{2,3}^h(E, F_{V_*}) = 0, \quad \delta_2^p(E, F_{V_*}) \geq \frac{1}{4\sqrt{3}} > 0, \quad \inf_r \delta_{2,r}^v(E, F_{V_*}) > 0. \quad (7.17)$$

Proof. The encoder maps the consumed i into its retained label by A ; the terminal decoder maps that label into j by B . This proves (7.14) including its converse implementation. If $V = CD$ is any nonnegative factorization, let $s_k = \sum_j D_{kj}$. Delete zero rows of D , define $B_{kj} = D_{kj}/s_k$ and $A_{ik} = C_{ik}s_k$. Since V is stochastic, the rows of A sum to one. Padding with unused states gives the claimed equivalence with nonnegative rank. The one-state statement follows immediately.

For $V = I_n$, row error is $1 - (AB)_{ii}$. If the least correct probability is c , for each i select a row $k(i)$ of B with $B_{k(i),i} \geq c$. Some row is selected at least $\lceil n/K \rceil$ times, so stochasticity gives $c \leq 1/\lceil n/K \rceil$. A partition into K groups of sizes at most $\lceil n/K \rceil$, each decoded uniformly, attains that bound. Also $\sum_i (AB)_{ii} \leq K$, by the calculation below with K in place of two. It remains valid under mixing, giving hidden error at least $1 - K/n$. Start with a deterministic channel that reproduces K input symbols exactly and sends all others to those K outputs. Average all simultaneous permutations of its input and output labels. Every diagonal entry is then K/n , proving the second formula. Inverting the two monotone formulas gives the stated least widths.

For $V = I_3$, total variation in row i is $1 - (AB)_{ii}$. If all three diagonal entries of AB were greater than $1/2$, then for each i some row of B would have its i th entry greater than $1/2$. A stochastic row has at most one such entry, so two rows cannot serve three indices. This proves the private lower bound $1/2$. Merge two input symbols into a retained label that decodes uniformly to that pair, and retain the third symbol exactly. This attains it. For every stochastic factorization with two intermediate states,

$$\sum_{i=1}^3 (AB)_{ii} = \sum_{k=1}^2 \sum_{i=1}^3 A_{ik} B_{ki} \leq \sum_{k=1}^2 \sum_{i=1}^3 B_{ki} = 2.$$

The same inequality holds under hidden mixtures, giving maximal row error at least $1/3$. Mix uniformly the three preceding pair-merge channels. The resulting diagonal entries are $2/3$ and the off-diagonal entries are $1/6$. This is exactly V_* and has error $1/3$ from I_3 . It proves the hidden claims in both displays.

The smallest singular value of V_* is $1/2$. Any AB with two intermediate states has a unit vector x in its right null space, so $\|V_* - AB\|_F \geq \|(V_* - AB)x\|_2 \geq 1/2$. If the maximal row total variation error is ε , each row has Euclidean error at most 2ε , and hence the Frobenius error is at most $2\sqrt{3}\varepsilon$. This proves the explicit private bound. The optimized visible value is attained by Theorem 7.2; if it were zero, its zero-set assertion would contradict that bound. This proves its strict positivity. $\square$

7.4. Optimal resource sets and decision certificates. For a class $\mathcal{C}_{S,q}(F)$ of common consumers, with all other information conventions fixed, let $\mathcal{R}_{S,q}(F)$ be its vector of attainable $[0, 1]$ risks. For a nonnegative row weight w summing to one, write

$$h_F(S, q; w) = \inf_{r \in \mathcal{R}_{S,q}(F)} \langle w, r \rangle.$$

If E privately simulates F with cost K and error ε , then

$$\mathcal{R}_{S,q}(F) \subseteq \mathcal{R}_{KS,q}(E) + [-\varepsilon, \varepsilon]^J, \quad h_E(KS, q; w) \leq h_F(S, q; w) + \varepsilon. \quad (7.18)$$

For nonfinite row sets the first expression means pointwise paired risk functions within uniform distance ε ; no finite coordinate box is needed. Upper sets give the equivalent one-sided order formulation. Hidden selectors multiply private width by r ; visible selectors multiply the separate public alphabet by r . These are direct consequences of the final part of Theorem 7.2, with the optimal intrinsic error in place of an externally estimated simulator error.

When two stateless comparison directions have error at most ε , paired risk sets and their support values at unchanged width are within ε . If each is centered by a paired unrestricted benchmark whose error is also at most ε , the centered values are within 2ε . Therefore a gap $g > 0$ between zero excess and infeasible widths is stable under $\varepsilon < g/4$: a value below $g/2$ in the approximation certifies feasibility in the exact finite problem, and a value above $g/2$ certifies infeasibility. The benchmark and the class being optimized must be specified; this assertion does not identify different information signatures.

Proposition 7.5 (Intrinsic finite-dimensional selector approximation). *Let $\mathcal{R}$ be a compact risk image in $C(J)$, where J is compact. Suppose a bounded linear operator $L : C(J) \rightarrow C(J)$ has rank at most m and $\sup_{r \in \mathcal{R}} \|r - Lr\|_\infty \leq \eta$. Every barycenter $\bar{r}$ of a probability on $\mathcal{R}$ has a barycenter r_* of at most $m + 1$ members of $\mathcal{R}$ with $\|\bar{r} - r_*\|_\infty \leq 2\eta$. If $\mathcal{R}$ lies in a finite affine space of dimension m , the error is zero with at most $m + 1$ members. Thus the selector price can be controlled by the dimension of attainable risks, rather than by a covering number of the row space.*

Proof. The compact set $L\mathcal{R}$ lies in an m -dimensional space. Its convex hull is compact, and $L\bar{r}$ belongs to it. The finite-dimensional affine-dependence argument in the proof of Theorem 7.2 gives $r_1, \dots, r_{m+1} \in \mathcal{R}$ and weights λ such that $L\bar{r} = \sum_i \lambda_i Lr_i$. Put $r_* = \sum_i \lambda_i r_i$. The two residual terms $\bar{r} - L\bar{r}$ and $r_* - Lr_*$ have norms at most η , proving the estimate. Apply the same argument after translation in the exact affine case. The implementation stores the selected design index whenever it is not provided as visible side information. $\square$

Remark 7.6 (Finite computation and general carriers). For rational finite instruments, (7.6) and the fixed- r formulations are finite polynomial feasibility problems; (7.7) is a finite linear program. These are exact variational characterizations, not assertions of polynomial running time in a succinct history description. The number of feedback trees, events, or deterministic tables can be large. On standard Borel carriers the same definitions use suprema over measurable feedback and tests and infima over executable kernels. Their product-state composition and decision transport remain valid by the same channel argument. Attainment and the finite testing dual are asserted here only for the stated finite class. The average-risk and physical results below give separate nonfinite theorems with their own hypotheses.

8. CONTINUATION GEOMETRY AND AN EXACT MEMORY OBSTRUCTION

8.1. Normalized continuation laws. We first separate the transducer from the source to which it is attached. There are L epochs with finite input alphabets X_t and output alphabets Z_t . A causal kernel $p(z_{1:L} \mid x_{1:L})$ is nonnegative, normalized for every input word, and has output-prefix marginals independent of later inputs. The current input can include both a source report and the

current control. It never includes an unavailable parameter, a terminal mark, or an earlier output not retained by the machine.

Let $\mathcal{C}_t$, $0 \leq t \leq L$, be the finite-dimensional polytope of normalized causal kernels from $x_{t+1:L}$ to $z_{t+1:L}$; put $\mathcal{C}_L = \{1\}$. For $v \in \mathcal{C}_t$ define the shift

$$(S_{x,z}v)(z_{t+2:L} \mid x_{t+2:L}) = v(z, z_{t+2:L} \mid x, x_{t+2:L}).$$

Its total mass $v(z \mid x)$ is independent of the future input word. A zero shift is allowed.

Definition 8.1 (Compatible continuation polytopes). A continuation cover of width $K = (K_0, \dots, K_L)$ consists of lists $V_t = (v_{t,m})_{m \leq K_t}$ in $\mathcal{C}_t$ such that, for every m, x, z ,

$$S_{x,z}v_{t,m} \in \text{cone } V_{t+1}. \quad (8.1)$$

It covers p when $p \in \text{conv } V_0$. Lists may repeat generators and may contain unreachable generators. Thus unused states can be adjoined.

Theorem 8.2 (Continuation realization with a width profile). *A causal kernel p has a stochastic transducer realization with an initial K_0 -valued register and K_t retained states after epoch t if and only if it has a compatible continuation cover of width K . Initial randomization is part of the charged initial register. With a deterministic initial state the condition is $p = v_{0,1}$ and $K_0 = 1$.*

Consequently the set of feasible width profiles depends only on the causal kernel and its declared input interface, not on a row presentation or a labeling of the hidden states. Every matrix of conditional continuation probabilities at cut t has nonnegative rank at most K_t . Individual rank bounds do not replace the simultaneous shift conditions (8.1).

Proof. Given a transducer, let $v_{t,m}$ be the suffix response when its register at time t equals m . The remaining rows define this response even for an unreachable state. If the next row is $q_{t+1}(z, m' \mid m, x)$, conditioning on its output and new state gives

$$S_{x,z}v_{t,m} = \sum_{m'} q_{t+1}(z, m' \mid m, x) v_{t+1,m'}.$$

The initial state distribution expresses p as a convex combination of V_0 .

Conversely choose nonnegative coefficients in every cone membership in (8.1). There are only finitely many choices to make. Since every generator on the right has mass one for each future input word, the sum of the coefficients for a fixed m, x, z is $v_{t,m}(z \mid x)$. Summing over z gives one. The coefficients are therefore stochastic rows $q_{t+1}(z, m' \mid m, x)$. Choose an initial distribution representing p in $\text{conv } V_0$. Backward induction from $\mathcal{C}_L = \{1\}$ proves that the constructed machine has exactly the response $v_{t,m}$ from each register state. In particular its initial response is p . Equality for open-loop input words implies equality under adaptive input selection: at each node of an input–output history tree, the same causal conditional probability is used on the two sides.

For the rank assertion, condition on any positive-probability observed history at the cut. Its continuation response is a convex combination of $v_{t,m}$ with weights equal to the posterior register distribution. Stack these responses as rows, and use all future input–output pairs as columns. This is a nonnegative factorization through K_t generators. The converse construction required the same generator families at adjacent cuts and all their shifts; an unrelated factorization of each cut matrix provides no such compatibility. The characterization, being stated entirely in terms of these response polytopes, also proves the asserted invariance. $\square$

Attach such a causal kernel to a marked source experiment. The output law is obtained by contraction with the source’s causal instruments, while its actual mark is retained. Hence zero private deficiency is equivalently the existence of a compatible continuation cover whose contraction equals the target marked behavior in every parameter and feedback row. This is the geometric form of the factorization in Theorem 7.2; no conditional dependence on the mark is introduced.

A hidden selector chooses one such cover independently of the source. It is a convex combination of complete realizations, not a convexification inside every retained-state simplex at the original width. Its private implementation uses the disjoint union of the covers and keeps the selector in the state. A visible selector keeps those fibers in the compared law. The nonnegative fiber argument in Theorem 7.2 therefore continues to give equality of the visible and private zero sets.

8.2. An erasure cut. A simple family makes the width obstruction exact, including its error. Let X take values in $[N]$ with law π . Before a specified cut the source reveals X . After the cut it has no reports correlated with X . The controller has no feedback channel carrying X across the cut, and all earlier emitted reports are prescribed constants. The only permitted cross-cut information of the simulator is its K -valued register and any independent selector explicitly included in its signature. The terminal mark is the actual $W = X$. The target's terminal report is X . All memory at other checkpoints is unrestricted.

Theorem 8.3 (Exact cut spectrum). *For this experiment, for private, hidden and visible signatures alike,*

$$\delta_K = 1 - \sum_{i=1}^{\min(K,N)} \pi_{(i)}. \quad (8.2)$$

For uniform X , $\delta_K = (1 - K/N)_+$. Thus an erasure cut for m unbiased bits requires $K \geq (1 - \epsilon)2^m$ whenever its error is at most ϵ . The bound holds for every surrounding width profile; equality is attained when the other checkpoints are unrestricted. If every $\pi_x > 0$, exact transmission has nonnegative cut rank N .

Proof. With all post-cut randomness averaged into the decoder, any private machine gives kernels $e(m | x)$ and $d(z | m)$. Its joint terminal law is $\pi_x \sum_m e(m | x) d(z | m)$. Since the target law is $\pi_x \mathbf{1}_{z=x}$, total variation is exactly one minus the correct-decoding probability. Put $u_x = \max_m d(x | m)$. Then

$$0 \leq u_x \leq 1, \quad \sum_x u_x \leq K, \quad \mathbb{P}(Z = X) \leq \sum_x \pi_x u_x \leq \sum_{i=1}^{\min(K,N)} \pi_{(i)}.$$

The last inequality follows by moving mass in u from a smaller weight to a larger one until the K largest weights are filled. To attain it, retain distinct labels for those letters and decode them deterministically; all other letters are mapped to any retained label. Before the cut one may use the unrestricted memory to compute this encoding.

For a hidden selector, condition on its value. It is independent of X , so every conditional correct-decoding probability obeys the same bound; averaging cannot improve it. For a visible selector, variation is the average of the fiber variations, with the same conclusion. The private construction works for both signatures. Finally, the exact joint cut matrix is diagonal with its N positive diagonal entries π_x . Its ordinary rank is N , so its nonnegative rank is at least N and its diagonal factorization attains N . $\square$

The absence of an auxiliary cross-cut feedback channel is essential. An unpriced controller storing X and returning it as a later control would change the experiment, not disprove the cut bound. Likewise a selector correlated with X is not an independent hidden selector. These distinctions are the same ones needed in the collision certificate below.

9. DECISION CONTINUATION SPECTRA

The memory at an experimental cut has a different role from the memory needed to reach that cut. We first identify its exact asymptotic value. This will give both an obstruction and

a construction for the same physical audit, rather than a lower bound for an auxiliary binary problem.

9.1. A charged training-validation interface. Let Ω be a finite marked alphabet, $d = |\Omega|$, let $B \subset \Delta(\Omega)$ be nonempty and compact, and fix $Q \in \Delta(\Omega)$. A candidate $b \in B$ is fixed throughout the experiment. Training consists of n independent observations from b . At the end of training the auditor retains a register $M \in [K]$. Everything depending on the training observations or on an earlier random seed must pass through this register. In particular, there is no external copy of the training history and no uncharged persistent selector.

After this cut, fresh independent randomness may select an event A using M . The event is chosen before independent candidate and target validation observations $X \sim b$ and $Y \sim Q$ are drawn. The score is $\mathbf{1}_A(Y) - \mathbf{1}_A(X)$. The downstream register needed to keep A and to compute this score is distinct from the register at the specified cut. The upstream and downstream widths will be stated separately in concrete implementations. This is the reset interface of Section 18, with one additional charged checkpoint. The actual mark is part of a completed observation; it is not an input to the private candidate's online transition rows.

Write $\mathcal{H} = [0, 1]^\Omega$ and

$$f_h(b) = \sum_{w \in \Omega} (Q(w) - b(w))h(w), \quad h \in \mathcal{H}.$$

Every randomized event has a response $h(w) = \Pr(w \in A)$. Conversely, every $h \in \mathcal{H}$ is the response of a random event: include each atom independently with its specified probability. This is a finite probability distribution on 2^Ω . Only its mean response matters for expected signed score. The event itself must still be retained during validation.

Definition 9.1 (Decision continuation spectrum). The K -state decision continuation value is

$$\mathfrak{V}_K(B, Q) = \max_{h_1, \dots, h_K \in \mathcal{H}} \min_{b \in B} \max_{m \leq K} f_{h_m}(b). \quad (9.1)$$

Let $v_{n,K}(B, Q)$ be the supremum of the worst-case expected score of n -sample auditors at the preceding interface. Only the designated cut has width K in this definition; all other widths are finite but unrestricted. For $c \in \mathbb{R}$, let $\kappa_c(B, Q)$ be the least K for which $\mathfrak{V}_K(B, Q) > c$, with value $+\infty$ when no such K exists.

The definition uses only the marked behavior image and its target, not a parameterization of B or a labeling of candidate states. No convex hull of B is taken. A response list is a list of conditional validation continuations; it is not a mixture of feasible private candidates.

Theorem 9.2 (Operational classification at a decision cut). *All extrema in (9.1) are attained. For every $n \geq 1$,*

$$0 \leq \mathfrak{V}_K(B, Q) - v_{n,K}(B, Q) \leq \min \left\{ \sqrt{\frac{d-1}{n}}, \sqrt{\frac{2 \log(2K)}{n}} \right\}. \quad (9.2)$$

Consequently $v_{n,K} \rightarrow \mathfrak{V}_K$. Every auditor with K states at the designated cut, at any finite training length, has worst-case score at most $\mathfrak{V}_K$. If $\mathfrak{V}_K > c$, some finite-sample auditor with K states at that cut has worst-case score greater than c . Thus κ_c is exactly the minimum cut width permitting such a strict certificate with sufficiently many training samples.

Equivalently, κ_c is the least number of response half-spaces

$$\{b \in B : f_h(b) > c\}, \quad h \in \mathcal{H}, \quad (9.3)$$

that cover B . If two parameterized behavior families satisfy, uniformly in their common parameter q ,

$$\|b(q) - \tilde{b}(q)\|_{\text{TV}} \leq \alpha, \quad \|Q - \tilde{Q}\|_{\text{TV}} \leq \beta,$$

then

$$|\mathfrak{V}_K(B, Q) - \mathfrak{V}_K(\tilde{B}, \tilde{Q})| \leq \alpha + \beta. \quad (9.4)$$

Proof. For a fixed response list, the inner objective is continuous in b . Its variation when the list changes is bounded uniformly over b by $2 \max_m \|h_m - \tilde{h}_m\|_\infty$. Its minimum over compact B is therefore continuous in the list. Compactness of $\mathcal{H}^K$ proves attainment.

Condition on the retained state of any admissible auditor. Its conditional post-cut event law depends only on that state and fresh randomness, and has a response h_m . Independence of validation from training gives

$$g(b) = \sum_{m=1}^K \Pr_b(M = m) f_{h_m}(b) \leq \max_m f_{h_m}(b).$$

Taking the minimum over b , and then the supremum over auditors, proves the upper bound. This argument permits arbitrary stochastic training and arbitrary finite upstream memory. An extra training-dependent decoder would invalidate its first conditioning step and is excluded by the interface, not silently supplied.

For the reverse bound fix an optimal list. Let $\hat{b}$ be the empirical measure and choose the first index maximizing $f_{h_m}(\hat{b})$. Retain that index and discard the empirical table. If m_* maximizes the true score, then pointwise

$$\max_m f_{h_m}(b) - f_{h_{m_*}}(b) \leq 2 \max_{m \leq K} |(\hat{b} - b)h_m|. \quad (9.5)$$

The maximum is at most $\|\hat{b} - b\|_{\text{TV}}$, since $0 \leq h_m \leq 1$. Cauchy–Schwarz and the multinomial variances give

$$\mathbb{E} \|\hat{b} - b\|_{\text{TV}}^2 \leq \frac{d}{4n} \left(1 - \sum_w b(w)^2 \right) \leq \frac{d-1}{4n}.$$

This proves the first rate.

For completeness, if V takes values in an interval of length one, then $\log \mathbb{E} e^{s(V - \mathbb{E}V)} \leq s^2/8$. Indeed the second derivative of the logarithmic moment generating function is the variance under the exponentially tilted law, at most $1/4$; its value and first derivative vanish at zero. Integrating twice proves the inequality for both signs of s . Independence consequently gives $\mathbb{E} e^{s(b-b)h_m} \leq e^{s^2/(8n)}$. For $s > 0$,

$$\mathbb{E} \max_m |(\hat{b} - b)h_m| \leq \frac{\log(2K)}{s} + \frac{s}{8n}.$$

Minimizing in s gives $\sqrt{\log(2K)/(2n)}$ and proves the second rate in (9.2). The empirical table has finitely many values; the construction is therefore a finite-state auditor, with only the chosen index surviving the specified cut. Real response coefficients are allowed here; the physical construction below has deterministic rational transitions.

A finite cover in (9.3) says that the continuous function $b \mapsto \max_m f_{h_m}(b)$ is greater than c everywhere on compact B . Its attained minimum is then greater than c . This proves the covering characterization in both directions. It also proves the strict-threshold assertion; no claim about finite-sample attainment at the boundary $c = \mathfrak{V}_K$ is needed.

Finally $|(Q - b)h - (\tilde{Q} - \tilde{b})h| \leq \alpha + \beta$ for every response. Taking maxima, minima over the common parameter set, and maxima over lists preserves this uniform bound, proving (9.4). $\square$

Corollary 9.3 (Compression by a finite decision dictionary). *Suppose responses $h_1, \dots, h_K$ satisfy $\min_{b \in B} \max_m f_{h_m}(b) \geq c + \eta$ for $\eta > 0$. An empirical selector with $n > 2 \log(2K)/\eta^2$ achieves score greater than c and retains only K states at the decision cut. If these responses depend on a finite statistic $T : \Omega \rightarrow \Xi$, the training selector can count $T(X)$ instead of X . A sufficient*

upstream register has at most $\binom{n+|\Xi|-1}{|\Xi|-1}$ states at the last training checkpoint, and the corresponding time-dependent count registers suffice at earlier checkpoints.

Proof. Use the second bound in (9.2). Empirical expectations of h_m depend only on the histogram of $T(X)$. At time t , that histogram has $\binom{t+|\Xi|-1}{|\Xi|-1}$ possibilities. Its update and the final maximization are deterministic. No statistical sufficiency for estimating all of b is required; sufficiency for the specified responses is enough. $\square$

This corollary supplies an intermediate principle between a full marked histogram and a problem-specific compressed counter. The invariant is the number and geometry of validation continuations needed to cover the original candidate image at the desired score. It is not the algebraic dimension of a chosen parameterization. The empirical upper bound need not be sample-optimal, and a minimum decision-cut width is not a minimum of the largest width over the entire acquisition schedule.

9.2. Several feedback rows. The classification does not require that the declared experiment have only one feedback row. For the row-indexed interface of Section 18, let $\mathcal{I} = \{(j, A) : A \subseteq \Omega_j\}$ and define

$$f_\lambda(b) = \sum_{i \in \mathcal{I}} \lambda_i f_i(b), \quad \lambda \in \Delta(\mathcal{I}).$$

A post-cut continuation is now a probability distribution on row–event pairs; it is not an arbitrary test on a joint product of validation rows. Put

$$\mathfrak{V}_K^{\text{rows}}(B, Q) = \max_{\lambda_1, \dots, \lambda_K \in \Delta(\mathcal{I})} \min_{b \in B} \max_m f_{\lambda_m}(b).$$

Corollary 9.4 (Row-indexed classification). *With n training samples per row, the row-indexed decision-cut value converges to $\mathfrak{V}_K^{\text{rows}}$, and its deficit is at most*

$$\min\{\sqrt{D/n}, \sqrt{2 \log(2K)/n}\}, \quad D = \sum_j (|\Omega_j| - 1).$$

The strict half-space cover characterization holds with f_λ . The upper bound by $\mathfrak{V}_K^{\text{rows}}$ applies also to any other finite training schedule, including adaptive row selection.

Proof. Conditioning on the cut state gives the same convex-mixture upper bound as before; its proof places no restriction on how that state was reached. For empirical selection among an optimal list, $|(\hat{b} - b)\lambda_m| \leq \max_j \|\hat{b}_j - b_j\|_{\text{TV}}$. Sum the rowwise second moments to obtain the first rate. To obtain the second, align the r th independent training sample from each row into a tuple and write its response as $\sum_{j,A} \lambda_{m,j,A} \mathbf{1}_A(X_{j,r})$. This random variable belongs to $[0, 1]$, and its sample mean is the empirical candidate mean for that continuation. The logarithmic moment-generating argument in Theorem 9.2 therefore gives the same finite-list rate. Each chosen row–event distribution can be drawn using fresh randomness after the cut. Compactness and the covering argument are unchanged. $\square$

10. FINITE PRECISION AND CHARGED RANDOMIZATION

We distinguish the number of persistent labels from the description of a transition and from the clock that selects it. An observation in a finite alphabet is initially an atomic input. Every random bit used after that input is a separate, charged epoch. Section 11 gives the additional cost of splitting the observation itself.

Lemma 10.1 (Dyadic rows and finite randomizers). *A probability vector on s points has a dyadic approximation with denominator 2^r at total variation distance at most $(s-1)2^{-r}$. It can be sampled using exactly r independent fair bits and at most $2s - 1$ persistent labels at each intermediate bit*

epoch. The label count uses the bit position supplied by the schedule. No partial random integer is retained outside these labels.

Proof. Round down the first $s - 1$ coordinates and assign the remaining mass to the last coordinate. The positive and negative parts of the change have equal mass, at most $(s - 1)2^{-r}$. For sampling, partition $\{0, \dots, 2^r - 1\}$ into consecutive intervals of the prescribed integer lengths. A binary prefix describes a dyadic interval. If that interval is contained in one output interval, keep the output label and ignore the remaining bits. Otherwise keep the prefix interval. At a fixed depth, every unresolved interval contains a partition boundary in its interior; there are at most $s - 1$ such intervals. Together with at most s resolved outputs this gives $2s - 1$ labels. The children and their resolved labels are fixed by the read-only partition and the depth. Thus the construction is an actual finite transition table, not a real-valued random seed. Zero-length intervals cause no difficulty. $\square$

Proposition 10.2 (Rational decision continuations). *Let $d = |\Omega|$. A list of K responses in $[0, 1]^\Omega$ can be replaced by dyadic responses of denominator 2^r , losing at most 2^{1-r} in its uniform score envelope. After a decision cut retaining K labels, the randomized event for the chosen dyadic response can be drawn using r fair bits, at most $K(2d + 1)$ labels during this draw, and at most $K(d + 1)$ labels afterwards. Keeping the event through one candidate and one target validation requires at most $2K(d + 1)$ labels at the intervening atomic-observation cuts. Its final signed score has three values.*

Proof. Round every coordinate down. The sup-norm error is at most 2^{-r} , and $\sum_w |Q(w) - b(w)| \leq 2$. This proves the envelope estimate. For a fixed rounded response h , draw a uniform dyadic integer U and choose $A_U = \{w : U < 2^r h(w)\}$. These events are nested and have at most $d + 1$ distinct values. Their mean response is exactly h . Apply Lemma 10.1 to their interval lengths, retaining the chosen response index during sampling. Afterwards the event label and the candidate-event indicator suffice; the target indicator emits their difference. All event masks are read-only data fixed before observing the candidate. $\square$

For K responses, their rounded coordinates require $Kd(r + 1)$ bits in a direct read-only representation. These bits are not charged as persistent sample-dependent labels, but are not declared costless precision. A construction using them records both quantities. In particular, the proposition turns every strict response-cover margin into a finite-randomness implementation; it says nothing about exact attainment of an irrational boundary value.

11. APPROXIMATE CONTINUATION PROFILES AND RESOURCE TRANSFER

A cut profile records the whole causal schedule, rather than its final compression. We give both a realization certificate and an operational value. These notions keep the compatibility condition of Theorem 8.2; separate nonnegative factorizations at individual cuts are not a substitute for that condition.

11.1. A causal comparison norm. For two normalized finite causal suffix kernels p, q at cut t , put

$$d_t^{\text{ca}}(p, q) = \sup_{\pi} \|P_p^\pi - P_q^\pi\|_{\text{TV}},$$

where π ranges over deterministic causal input policies on the remaining schedule and P_p^π is the law of its complete output transcript. Randomized policies give the same supremum, by conditioning on their independent random tables. In particular this distance includes feedback; the supremum over open-loop input words alone is not used.

For stochastic coefficients $q_t(z, j \mid m, x)$ and suffix kernels $v_{t+1,j}$, denote by $\mathcal{Q}_{t+1,m}$ the suffix kernel which, upon input x , emits z and chooses j with these coefficients, then uses $v_{t+1,j}$. The notation denotes one first-step operation, not a conditioning on an event of zero probability.

Theorem 11.1 (Approximate compatible realization). *Let $K_0, \dots, K_L$ be positive integers. Suppose normalized causal kernels $v_{t,m}$, $m \leq K_t$, and stochastic rows q_t satisfy*

$$d_t^{\text{ca}}(v_{t,m}, \mathcal{Q}_t v_{t+1,m}) \leq \eta_t \quad (t < L, m \leq K_t). \quad (11.1)$$

At the terminal cut the suffix kernels are the unit empty kernel. If $\rho \in \Delta_{K_0}$ and $d_0^{\text{ca}}(\rho, \sum_m \rho_m v_{0,m}) \leq \eta_{-1}$, then the machine with initial law ρ and transition rows q_t has the whole profile (K_t) and realizes a kernel $\hat{p}$ satisfying

$$d_0^{\text{ca}}(\rho, \hat{p}) \leq \min\{1, \eta_{-1} + \sum_{t=0}^{L-1} \eta_t\}. \quad (11.2)$$

Conversely, a machine with this profile within distance ε of p supplies such a certificate with $\eta_{-1} = \varepsilon$ and all local errors zero. If a deterministic initial state is required, use $K_0 = 1$ throughout both statements.

Proof. Let $\hat{v}_{t,m}$ be the actual continuation of the rows q from state m . Fix a future policy. Conditional on the first output and next state, its remaining actions form an admissible suffix policy. Replacing $v_{t+1,j}$ by $\hat{v}_{t+1,j}$ therefore changes its transcript law by at most $\max_j d_{t+1}^{\text{ca}}(v_{t+1,j}, \hat{v}_{t+1,j})$. Mixtures and pushforwards contract total variation. The triangle inequality and (11.1) imply, backwards from L ,

$$\max_m d_t^{\text{ca}}(v_{t,m}, \hat{v}_{t,m}) \leq \sum_{s=t}^{L-1} \eta_s.$$

Mixing with ρ proves (11.2). The converse takes the actual normalized continuations of the given machine, including arbitrary valid continuations for unreachable states. Their one-step equations are identities. There is no assertion that independently optimal approximate lists at successive cuts have compatible updates. $\square$

11.2. The full value and its effective approximation. Fix a finite schedule with finitely many observation and action symbols, N calls to an experiment, and a bounded terminal payoff $G \in [-1, 1]$. An environment $b \in B$ specifies the conditional observation kernels at all possible visible histories and actions. Assume B is a nonempty compact subset of the resulting finite product of probability simplexes. A controller A has a specified persistent profile $\mathbf{K}$; the clock has no access to sample history. All action choices and all retained randomness factor through the current register. Define

$$V_{N,\mathbf{K}}(B, G) = \max_{A \in \mathcal{M}(\mathbf{K})} \min_{b \in B} \mathbb{E}_b^A G. \quad (11.3)$$

Unused states pad smaller machines, so $\mathcal{M}(\mathbf{K})$ is a finite product of stochastic-row simplexes, with any prescribed causal zeros imposed. The terminal readout is included among its rows.

Theorem 11.2 (Full-profile resource transfer). *The maximum in (11.3) is attained. Its value is nondecreasing in each K_t . Removing a charged intermediate cut gives a relaxation, whereas inserting identity-holding cuts preserves every implementation with the same held register.*

Suppose two environments at each of their N calls differ in conditional total variation by at most δ_i , uniformly over corresponding histories, actions and parameters. Replace the controller rows at its L transitions by dyadic rows of denominator 2^r , with at most s_t possible next-state/output pairs at transition t . Any fixed controller loses at most

$$2 \min \left\{ 1, \sum_{i=1}^N \delta_i + \sum_{t=1}^L (s_t - 1) 2^{-r} \right\} \quad (11.4)$$

in expected payoff. The same bound compares optimal values on the same profile, with the appropriate dyadic approximation on one side. More generally an approximate continuation certificate contributes its error in (11.2) inside the braces.

If $B_0 \subset B$ is a finite ω -net in the maximum conditional-TV metric and v_0 is the maximum over the dyadic controller grid of the minimum over B_0 , then

$$v_0 - 2N\omega \leq V_{N,\mathbf{K}}(B, G) \leq v_0 + 2 \sum_{t=1}^L (s_t - 1) 2^{-r}. \quad (11.5)$$

The bounds are certificates whenever the net and the finite evaluations are certified. Compactness alone supplies existence of nets, not an algorithm for an arbitrary presentation of B .

Proof. For fixed b , each path probability is a finite product of controller and environment coordinates. Summing the finitely many paths gives a jointly continuous payoff. Its minimum over compact B is continuous in the controller rows; the controller space is compact. This proves attainment. Padding proves monotonicity. Removing a constraint enlarges the feasible set, and identity transitions prove the insertion assertion.

For the perturbation bound, couple two executions until their first mismatch. On equal histories the conditional observation laws can be maximally coupled, with mismatch probability at most δ_i . Couple the controller transitions similarly, using Lemma 10.1. The probability of any mismatch is at most the sum in (11.4), capped at one. The payoff changes by at most two on that event. This coupling is recursive on finite histories and remains valid for feedback controls. Alternatively, approximate continuation kernels are compared in the causal distance and then composed, as in Theorem 11.1.

For a fixed grid controller, a minimizing point of B has a point of B_0 within ω ; the corresponding payoffs differ by at most $2N\omega$. Thus the grid optimum on all of B is at least $v_0 - 2N\omega$. Conversely approximate a maximizing unrestricted controller row by row. Its minimum over B decreases by at most the row-rounding bound, and its minimum over B_0 is no smaller. Taking the grid maximum proves the upper bound. No minimax interchange or convexification of the fixed-width behavior image occurs. $\square$

The formula records preparation count, persistent labels, calibration error, causal simulation error and transition precision in one execution. The training-count factor is present when training kernels change. It is absent only when those kernels are identical. Failure probabilities are also transported, with factor one rather than two, since their indicators have range $[0, 1]$.

Proposition 11.3 (Composition and schedule refinement). *A causal simulator of profile $\mathbf{R}$ followed on the same synchronized schedule by a controller of profile $\mathbf{K}$ has a realization with profile $(R_t K_t)$. Causal simulation errors add under successive composition. Two disjoint phases with a genuine reset and no retained bridge use the maximum of their peak widths. If a bridge with H labels must survive the second phase, its direct width bound is H times the second-phase width.*

If one atomic input x is encoded by ℓ symbols, let D_j be the number of possible length- j prefixes. Holding the pre-input state and the prefix realizes the refined acquisition cuts with widths $K_{\text{before}} D_j$, followed by the original update. A time-dependent machine running for T scheduled epochs becomes autonomous by retaining its clock, with at most $(T + 1) \max_t K_t$ states. These are implementation bounds, not claims of optimality for every refinement.

Proof. Keep the ordered pair of simulator and controller registers and compose their transition kernels. The simulator's emitted observation is the controller's current input. Conditional coupling until the first simulation discrepancy gives the error statement. A genuine reset permits reusing the alphabet, whereas a surviving bridge is another register. A partial encoded input is determined by its prefix, giving the stated refinement construction. Finally the augmented state (t, m) implements the time-indexed transition without an external clock. These constructions also show why neither hidden selectors nor an external transcript may be omitted from the register being counted. $\square$

12. FROM A DECISION COVER TO AN END-TO-END MACHINE

The unrestricted empirical table in Theorem 9.2 can be replaced by a bounded-state training procedure. This replacement trades preparation count against memory, rather than assuming that the training computation has already been performed. The elementary block mechanism belongs to the classical finite-statistics theory of Cover [24]; the result below tracks a finite horizon, a uniform indifference band, rational precision, the whole validation phase, and a compact nonconvex candidate image.

Lemma 12.1 (A finite-horizon four-state comparison). *Let Z_i be independent Bernoulli observations of unknown mean u . For $0 < \theta < 1/2$ and $0 < \delta < 1$, put*

$$L = \left\lceil \frac{\log(2/\delta)}{4\theta} \right\rceil, \quad R = \left\lceil 2^L \log(2/\delta) \right\rceil. \quad (12.1)$$

There is a deterministic clocked machine with four states at every observation cut, using exactly $2LR$ observations, that decides the sign of $u - 1/2$ with error at most δ whenever $|u - 1/2| \geq \theta$. No assertion is imposed within the indifference band.

Proof. Partition the observations into R pairs of blocks, each block of length L . A positive block succeeds when all its observations are one; the following negative block succeeds when all are zero. Stop updating the decision at the first successful block and retain its sign. If none succeeds, decide negative at the end. The prescribed remaining observations are still consumed, but ignored.

Two states are absorbing positive and negative decisions. The other two mean undecided with the current block still alive or already failed. At a block boundary the clock resets the undecided flag to alive. Its polarity and position are supplied by the clock; they are not inferred from an unrecorded sample. The last observation of each block either latches the sign or starts the next block. Thus four states suffice even at the block boundaries.

Write $x = u^L$, $y = (1 - u)^L$ and $s = x + (1 - x)y$. The probability of a positive decision is exactly

$$H_{L,R}(u) = \frac{x}{s} \{1 - (1 - s)^R\}. \quad (12.2)$$

Here $s > 0$, including at $u = 0, 1$. For $u \geq 1/2 + \theta$, the error is at most $y/x + (1 - s)^R$. For $u \leq 1/2 - \theta$, it is at most x/y ; the default negative decision is then correct. Since

$$\log \frac{1/2 + \theta}{1/2 - \theta} \geq 4\theta, \quad s \geq \max(x, y) \geq 2^{-L},$$

both errors are bounded by $e^{-4\theta L} + e^{-R2^{-L}} \leq \delta$. The logarithmic inequality follows by differentiation from its equality at zero. This also proves uniformity over the two composite hypotheses. $\square$

12.1. The whole-schedule continuation value. In the one-row reset interface of Section 9, let $W_c(B, Q)$ be the least peak width over all finite training lengths and all auditors whose worst-case expected signed score is strictly greater than c . Every training-observation cut, the decision cut, every fresh random-bit cut, and both validation cuts are charged. Observations in Ω are atomic for this definition. The event is selected before the two validations, exactly as in that interface. The schedule clock and read-only constants are recorded as separate resources. The class includes randomized machines; its rational subclass uses only finite fair-bit randomizers. Write W_c^{rat} for the corresponding minimum in the latter class. Empty feasible sets have value $+\infty$.

Theorem 12.2 (Uniform end-to-end realization of a decision cover). *Let $B \subset \Delta(\Omega)$ be nonempty and compact, Q fixed, and $d = |\Omega|$. For every strict threshold c ,*

$$\kappa_c(B, Q) \leq W_c(B, Q) \leq W_c^{\text{rat}}(B, Q) \leq 12d \kappa_c(B, Q). \quad (12.3)$$

The last inequality is interpreted only for finite κ_c ; when $\kappa_c = +\infty$ all three quantities are infinite.

More explicitly, fix $K \geq 2$ and a response list with uniform envelope at least v . Round its coordinates to denominator 2^r , and choose $q_i \in [0, 1] \cap 2^{-r}\mathbb{Z}$ with $|q_i - Qh_i^{(r)}| \leq \rho$. For θ, δ as in Lemma 12.1, a rational auditor uses

$$N = 2LR(K - 1) \quad (12.4)$$

candidate training observations and two validation observations. Its expected score is at least

$$v - 2^{1-r} - 2\rho - 4(K - 1)\theta - 2(K - 1)\delta. \quad (12.5)$$

Its completed-training-observation cuts have width at most $4K$, its decision cut has width K , and its entire random-bit-refined schedule has width at most $12Kd$. No empirical histogram is constructed. The response and comparison thresholds have a direct read-only representation of size $O(K^2d(r + 3) + \log L + \log R + \log K)$ bits.

Proof. Use a sequential tournament. Initially response 1 is incumbent. At stage $j = 2, \dots, K$, compare incumbent i with challenger j using fresh candidate observations. For one observation $w \sim b$, generate a Bernoulli variable with conditional success probability

$$\frac{1}{2} + \frac{q_j - q_i - h_j^{(r)}(w) + h_i^{(r)}(w)}{4}. \quad (12.6)$$

This lies in $[0, 1]$ and is dyadic of denominator 2^{r+2} . Its unconditional mean is $1/2 + (\tilde{f}_j(b) - \tilde{f}_i(b))/4$, where $\tilde{f}_i(b) = q_i - bh_i^{(r)}$. Conditional on the incumbent at the start of this stage, all these variables are independent and have this mean. The fixed candidate b is not changed by the comparison. Run Lemma 12.1 and replace i by j exactly when its decision is positive.

At each comparison, conditionally on all earlier stages, the probability of choosing the inferior response when their $\tilde{f}$ values differ by at least 4θ is at most δ . With probability at least $1 - (K - 1)\delta$, no such error occurs. On this event each stage loses at most 4θ relative to the better of its two responses. Induction on j gives

$$\tilde{f}_i(b) \geq \max_i \tilde{f}_i(b) - 4(K - 1)\theta.$$

Replacing the calibrated means by their true means loses at most 2ρ . Rounding the original list loses at most 2^{1-r} in its envelope, by Proposition 10.2. Any response payoff belongs to $[-1, 1]$, so failure of the good event loses at most $2(K - 1)\delta$ in expectation. This proves (12.5), uniformly over b .

During a comparison keep the incumbent index and one of the four block states. The challenger and block positions are fixed by the schedule. At the end of a comparison the sign updates the incumbent and is forgotten. This gives the $4K$ bound at completed-observation cuts. While drawing the coin in (12.6), keep this register and the current observation, and use the three-label binary sampler of Lemma 10.1. Thus at most $12Kd$ labels occur at any intervening random-bit cut. At the final decision cut only the incumbent remains. Proposition 10.2 gives at most $K(2d + 1)$ labels during event selection and $2K(d + 1)$ during validation; both are bounded by $12Kd$. The three-valued terminal output fits this bound as well. These are compatible transitions over the entire schedule.

The coin thresholds can be precomputed for all triples (i, j, w) and stored as $r + 3$ -bit integers. The event thresholds and L, R, K give the stated description bound; no arbitrary real coefficient is used by the executed machine. Formula (12.4) counts every training observation, including ignored observations after a block decision has been latched.

For (12.3), a width- W auditor has at most W labels at its decision cut. The operational upper bound of Theorem 9.2 implies $\kappa_c \leq W$. If $K = \kappa_c < \infty$, its optimal list has a positive margin over c . Take r large, $\rho \leq 2^{-r}$, and then θ, δ small enough that the loss in (12.5) is smaller than that margin. If $K = 1$, no training is needed: round the single response and use Proposition 10.2.

Rational auditors are a subclass of all auditors, proving the remaining inequality. Finally a finite-width successful machine would give finite κ_c , proving the assertion at infinity. $\square$

The dependence on the schedule is essential. The theorem is a constant-factor classification for fixed observation alphabet, not an optimal sample-complexity theorem. In particular R is exponential in L . For an autonomous implementation, a coarse bound on the number of epochs is $T \leq N(r+3) + r + 4$; Proposition 11.3 then gives at most $12Kd(T+1)$ states. The free-clock and autonomous statements are different resource bounds. Cover's theorem already exhibits the importance of this distinction [24, Theorem 2 and Section 4]; bounded-state learning itself is not a novelty claim of the present theorem.

Corollary 12.3 (Finite-precision, calibrated certificates). *Suppose the true training kernels differ from those used in an ideal auditor by at most α per training call, and the candidate and target validation laws differ by at most β_c and β_t . The right side of (12.5) is decreased by at most*

$$2 \min\{1, N\alpha\} + \beta_c + \beta_t$$

for the same implemented machine. A calibration error ρ in the known response means is already accounted for separately in that formula. When the training source is unchanged, $\alpha = 0$, regardless of how large N is.

Proof. Couple the training executions until their first mismatch, as in Theorem 11.2. The selected-event payoff has range $[-1, 1]$, giving the factor two. Conditional on a common selected event, its two validation means change by at most the two total variation errors. This uses no independence between a failed coupling and its terminal event. $\square$

13. FINITE-HORIZON SAMPLE-MEMORY BOUNDS

The minimum W_c in Theorem 12.2 permits arbitrarily long finite training. To retain the preparation constraint, let $V_{N,K}(B, Q)$ denote the supremum of the worst-case expected score over the same reset auditors, using at most N training trials and satisfying the specified register bound at each acquisition, random-bit, decision and validation cut. The observation order and the external clock are fixed parts of this interface. Write $V_{N,W}$ when every bound equals W . Padding an audit with ignored trials shows monotonicity in N . This value does not identify machines having only the same terminal dictionary.

A second realization of a response cover gives a polynomial preparation bound. Its counter is larger than the four-state block counter, and both constructions remain useful. The elementary random-walk argument below is classical; the purpose of retaining its proof is to price its use in the complete, calibrated continuation machine.

Lemma 13.1 (A truncated absorbing comparison). *Let Z_i be independent Bernoulli variables with mean u . Fix integers $m, n \geq 1$ and $0 < \theta < 1/2$. Start at $S_0 = 0$, add $2Z_i - 1$ while $|S_{i-1}| < m$, and retain the state forever after either boundary is reached. At time n declare positive exactly when $S_n \geq 0$. The machine has $2m + 1$ states. Whenever $|u - 1/2| \geq \theta$, its error probability is at most*

$$e_{m,n}(\theta) = e^{-4\theta m} + e^{-2n\theta^2}. \quad (13.1)$$

If τ is the boundary hitting time without truncation, then, with $x = 2u - 1$,

$$\mathbb{E}_u \tau = \begin{cases} m \tanh(m \operatorname{arctanh} x) / x, & 0 < |x| < 1, \\ m^2, & x = 0, \\ m, & |x| = 1. \end{cases} \quad (13.2)$$

In particular $\mathbb{E}_u(\tau \wedge n) \leq \min(n, m^2)$, and outside the indifference band it is at most $\min(n, m^2, m/(2\theta))$.

Proof. First suppose $1/2 < u < 1$ and put $r = (1 - u)/u$. For the unabsorbed walk $T_j = \sum_{i \leq j} (2Z_i - 1)$, the process r^{T_j} is a nonnegative martingale. Stopping on $\{-m, m\}$ gives

$$\mathbb{P}_u(T_\tau = -m) = \frac{r^m}{1 + r^m} \leq r^m.$$

For completeness, absorption has finite mean: in each block of $2m$ steps, a run of $2m$ positive increments has probability $u^{2m} > 0$ and forces absorption from every interior state. Thus the unabsorbed survival time is bounded by a geometric number of such blocks. Stopping first at $\tau \wedge j$, then using bounded convergence on the stopped exponential martingale, justifies the displayed identity. The estimate $\log((1 + 2\theta)/(1 - 2\theta)) \geq 4\theta$ gives $r^m \leq e^{-4\theta m}$ when $u \geq 1/2 + \theta$.

If the walk has not been absorbed, a negative final answer requires $T_n < 0$. Hoeffding's inequality [25] for variables in $[-1, 1]$ gives

$$\mathbb{P}_u(T_n \leq 0) \leq e^{-2n(u-1/2)^2} \leq e^{-2n\theta^2}.$$

Partitioning a wrong answer according to absorption proves (13.1). Reflection proves the lower-mean case; its nonabsorbed error includes $T_n = 0$ and satisfies the same bound. At $u = 0, 1$ the assertion follows directly.

The drift martingale $T_j - xj$ stopped at $\tau \wedge j$ gives $\mathbb{E}_u T_\tau = x\mathbb{E}_u \tau$, by integrability of τ and boundedness of the stopped walk. The hitting probabilities yield $\mathbb{E}_u T_\tau = m(1 - r^m)/(1 + r^m)$. Since $r = \exp(-2 \operatorname{arctanh} x)$, this is (13.2) when $x \neq 0$. For $x = 0$, stop $T_j^2 - j$ to obtain $\mathbb{E} \tau = m^2$. The endpoint formula is deterministic. Finally $\tanh(my) \leq m \tanh y$ for integer $m \geq 1$ and $y \geq 0$: induction follows from the addition formula for $\tanh$. This proves $\mathbb{E} \tau \leq m^2$; the bound $m/|x|$ follows from $|T_\tau| \leq m$. Truncation can only decrease the mean. $\square$

Theorem 13.2 (A preparation-width family of realizations). *Let $B \subset \Delta(\Omega)$ be compact, $d = |\Omega|$, and let $h_1, \dots, h_K \in [0, 1]^\Omega$, $K \geq 2$, have envelope $v = \min_{b \in B} \max_i (Q - b)h_i$. Round their coordinates to dyadic precision r , obtaining $h_i^{(r)}$, and choose known dyadic numbers $q_i \in [0, 1]$ with $|q_i - Qh_i^{(r)}| \leq \rho$. Suppose q_i also have denominator 2^r ; enlarging r accommodates any specified dyadic calibration. For $m, n \geq 1$ and $0 < \theta < 1/2$, an explicitly clocked fair-bit audit satisfies*

$$N = n(K - 1), \quad W \leq 3dK(2m + 1), \quad (13.3)$$

$$\inf_{b \in B} \mathbb{E}_b D \geq v - 2^{1-r} - 2\rho - 4(K - 1)\theta - 2(K - 1)e_{m,n}(\theta). \quad (13.4)$$

At completed training-trial cuts its width is at most $K(2m + 1)$; at the decision cut it is at most K . Every random-bit cut, the selected event, and both validations are included in W . In particular one can replace $e_{m,n}(\theta)$ by δ by taking

$$m = \left\lceil \frac{\log(2/\delta)}{4\theta} \right\rceil, \quad n = \left\lceil \frac{\log(2/\delta)}{2\theta^2} \right\rceil. \quad (13.5)$$

Proof. For a fixed unknown b , set $g_i(b) = q_i - bh_i^{(r)}$. Compare the current incumbent i with the next index j on fresh observations. Conditional on the observation w , generate a coin with probability

$$\frac{1}{2} + \frac{q_j - q_i - h_j^{(r)}(w) + h_i^{(r)}(w)}{4}. \quad (13.6)$$

This is in $[0, 1]$, is dyadic with denominator 2^{r+2} , and has mean $1/2 + (g_j(b) - g_i(b))/4$. Apply Lemma 13.1, selecting j for a positive answer and i otherwise. Conditional on all earlier stages, the observations and fair bits for this comparison are fresh. The probability of selecting the worse index when $|g_j - g_i| \geq 4\theta$ is at most $e_{m,n}(\theta)$.

Outside these exceptional comparisons each tournament stage loses at most 4θ relative to the better of its two contestants. Induction shows a total loss at most $4(K - 1)\theta$ relative to $\max_i g_i$.

The union probability of exceptional comparisons is at most $(K-1)e_{m,n}(\theta)$ even though the incumbent is random. All g_i lie in $[-1, 1]$, so their range is at most two. Consequently

$$\mathbb{E}_b g_i(b) \geq \max_i g_i(b) - 4(K-1)\theta - 2(K-1)e_{m,n}(\theta).$$

The calibration changes selection and evaluation by at most 2ρ . For each coordinate rounding, $\|h_i - h_i^{(r)}\|_\infty \leq 2^{-r}$, so $|(Q-b)(h_i - h_i^{(r)})| \leq 2^{1-r}$. A nested random event with mean $h_i^{(r)}$ has exactly that rounded payoff; hence (13.4).

At a completed trial retain the incumbent and the absorbing walk state. The stage number j is supplied by the predetermined clock, not by a sample-dependent register. During coin generation additionally retain w and at most three sampler labels, by Proposition 10.2. This gives $3dK(2m+1)$ labels. The terminal event sampler, held event, and validation storage use at most $K(2d+1)$, $K(d+1)$, and $2K(d+1)$ labels, respectively, as in Theorem 12.2; all are bounded by the stated W . The stage boundary discards the walk and initializes the next one. After absorption, remaining observations and bits in that stage are ignored, but are still charged to the fixed schedule. Thus the count N includes padding. Each active coin uses $r+2$ fair bits. A coarse bound for the full number of epochs is $T \leq N(r+3) + r + 4$.

The finite transition description consists of the dyadic coin table, the boundary and horizon integers, and the nested-event thresholds. It has $O(K^2d(r+3) + \log m + \log n + \log K)$ bits under binary encoding; the constants depend on a fixed encoding convention. No unknown b occurs in that description. The autonomous conversion stores the clock and has width at most $(T+1)W$, rather than W . Substitution of (13.5) proves the last assertion. $\square$

Corollary 13.3 (Two general preparation regimes). *For a strict K -response cover and a prescribed positive score margin, there are whole-schedule rational realizations of that margin with width at most $12dK$ and exponential block preparation count, and with width at most $3dK(2m+1)$ and preparation count $n(K-1)$ as in (13.5). Both constrain every training cut. Their resource pairs are upper bounds for $\mathbb{V}_{N,W}$, not a claim of an optimal frontier for every compact image.*

Proof. Choose rounding, calibration, band and error budgets whose sum is smaller than the strict margin. Apply Theorem 12.2 or Theorem 13.2. A one-response cover needs no training and is implemented directly by the rational event sampler. $\square$

14. AN EXACT STATISTICAL FRONTIER AT ALL CUTS

The preceding realizations are uniform upper bounds. A family with an exact answer illustrates what an all-algorithm statistical lower bound requires. Unlike exact computation of a prescribed word function, the objective here is the best decision risk among all rules, including stochastic rules, under a fixed preparation schedule and full profile.

Let $\Theta = \{1, \dots, M\}$ and fix a known prior $\pi_1 \geq \dots \geq \pi_M \geq 0$, $\sum_j \pi_j = 1$. At time t , an independent indicator E_t has known mean $q_t \in [0, 1]$. The report is $Y_t = \vartheta$ if $E_t = 1$, and $Y_t = *$ otherwise. All indicators are independent of the parameter and the machine's private randomization. The unknown parameter ϑ is fixed during the n observations. The machine starts in a parameter-independent state, has at most K_t labels after report t , and finally guesses ϑ . Its initial and final randomization is part of the same finite causal machine. The objective is Bayesian probability of a correct answer under π . Parameter-dependent loss is specified here and is not identified with the one-row physical signed-score problem.

Write $P_k = \sum_{j=1}^k \pi_j$ and put

$$k_t = \min\{M, K_t, K_{t+1}, \dots, K_n\}, \quad a = \prod_{s=1}^n (1 - q_s), \quad w_t = q_t \prod_{s=t+1}^n (1 - q_s). \quad (14.1)$$

Thus $a + \sum_t w_t = 1$ and k_t is nondecreasing in t .

Theorem 14.1 (Exact all-cut revelation frontier). *Over all stochastic causal machines satisfying the specified profile, the optimal Bayesian success probability is*

$$S_{\mathbf{q}, \mathbf{K}}(\pi) = a\pi_1 + \sum_{t=1}^n w_t P_{k_t}. \quad (14.2)$$

A single deterministic machine attains this value simultaneously for all $\mathbf{q} \in [0, 1]^n$. The upper bound therefore applies to all approximate statistical decision rules, not just to machines implementing a chosen selector exactly.

Proof. Let L be the last time with $E_L = 1$, with $L = 0$ when no revelation occurs. Then $\mathbb{P}(L = 0) = a$ and $\mathbb{P}(L = t) = w_t$. These events, and every particular pattern of indicators, are independent of ϑ .

First consider any stochastic machine. Realize all of its transition and output kernels by independent random variables indexed by time, state and input, and also realize its initial randomization. This is possible on a product space independent of ϑ and the indicators. Conditional on the entire random table the machine is deterministic. This conditioning is an analysis device, not a free persistent random seed in an implementation.

Fix that table and an indicator pattern with last revelation $t > 0$. The map from ϑ to the terminal guess factors through the state at every time $s \geq t$: all reports after t are the same symbol $*$, so the remaining transitions and output are fixed maps. The set of possible terminal guesses therefore has cardinality at most $\min_{s \geq t} K_s$, and at most M guesses can be correct. Its prior success probability is bounded by the mass of the k_t largest prior atoms, namely P_{k_t} . Earlier revelations and any parameter information encoded before t do not evade this future bottleneck. On the no-revelation pattern, the output is independent of ϑ and its success probability is at most π_1 . Integrate these bounds over the random table and indicator patterns to obtain the upper bound in (14.2). In particular no extremality claim about a nonconvex minimax set has been used.

For attainment, use the labels $\{1, \dots, k_t\}$ at time t , start with label one, and use the following deterministic updates. On $*$, hold the label. On a revelation $j \leq k_t$, replace it by j ; on a revelation $j > k_t$, replace it by one. The holding update is legal because $k_{t-1} \leq k_t$. At termination output the label. Conditional on $L = t > 0$, exactly the parameters $1, \dots, k_t$ are correct; on $L = 0$, exactly parameter one is correct. This proves equality. The updates use only the prescribed profile and prior ordering, not the probabilities q_t , establishing simultaneous attainment. $\square$

Corollary 14.2 (Sharp preparation–peak tradeoff). *For the uniform prior, a constant reveal probability $0 < q < 1$, and constant width $1 \leq K \leq M$, write $a_n = (1 - q)^n$. Then*

$$S_{n, K} = \frac{a_n + K(1 - a_n)}{M}. \quad (14.3)$$

For $n \geq 1$ and target success $s \in [1/M, 1]$, the least feasible width is

$$K_{n, s} = \max \left\{ 1, \left\lceil \frac{Ms - a_n}{1 - a_n} \right\rceil \right\}, \quad (14.4)$$

provided this integer is at most M ; otherwise the target is infeasible at that preparation count. If $s > 1/M$ and $K > Ms$, the least preparation count for width K is

$$n_{K, s} = \left\lceil \frac{\log((K - 1)/(K - Ms))}{-\log(1 - q)} \right\rceil. \quad (14.5)$$

If $K \leq Ms$ and $s > 1/M$, no finite n attains the target, except that equality can be attained when revelations are certain, a case excluded by $q < 1$. Width one gives success exactly $1/M$ for all n .

Proof. Here $P_k = k/M$ and $k_t = K$. Since $\sum_t w_t = 1 - a_n$, Theorem 14.1 gives (14.3). Solving $a_n + K(1 - a_n) \geq Ms$ first for K and then for n proves the formulas. For $K > 1$, the limiting value K/M is approached strictly from below when $0 < q < 1$, which proves the boundary assertion. $\square$

Corollary 14.3 (Exact cost of a changed checkpoint). *Replacing a profile $\mathbf{K}$ by a tighter profile $\widetilde{\mathbf{K}}$ loses exactly*

$$\sum_{t=1}^n w_t (P_{k_t} - P_{\widetilde{k}_t}).$$

Inserting a charged holding cut of width H immediately after time s replaces k_t by $\min(k_t, H)$ for $t \leq s$, and leaves the later terms unchanged. In particular a single narrow early cut does not constrain information acquired after that cut.

Proof. Subtract (14.2) for the two profiles. A holding cut is a further bottleneck exactly for revelations that precede it. The same deterministic construction realizes the refined schedule. $\square$

For example, with $M = 4$, $q_1 = q_2 = q_3 = 1/2$, and $(K_1, K_2, K_3) = (4, 1, 4)$, the effective profile is $(1, 1, 4)$ and the optimal uniform-prior success is $5/8$. Four labels at the terminal cut alone would instead give $29/32$. These are risks of complete experiments, not counts of residual classes of one word function. For a nonuniform prior the same theorem weights every surviving label by its actual mass. The nested optimal label sets are what makes all cut bounds jointly attainable. For a general loss or observation channel such nesting need not hold; compatible continuation updates, not independently optimized dictionaries, remain necessary.

The exact formula is Bayesian. Its upper bound also obstructs any uniform worst-case guarantee exceeding that Bayesian value, but the attaining prior-ordered machine is not asserted to be minimax. In particular an uncharged persistent random permutation cannot be used to turn its Bayesian success into a minimax statement. The prior, observation channel and all register cuts have been declared explicitly.

15. TRANSPORT PRICED BY ACTIVE UPDATES

A changed training source need not affect a latched computation at every subsequent scheduled preparation. The relevant quantity is the defect of its induced retained-state transition, not the number of raw reports that happen to be present. This distinction gives a sharper transfer bound without assuming equality of the training laws.

Theorem 15.1 (Occupation-weighted retained-law transfer). *Consider a fixed finite execution with common initial state law μ and retained-state transition kernels P_t, Q_t on the same finite state spaces, $1 \leq t \leq T$. The kernels may depend on a fixed unknown parameter; all inequalities below hold separately for each parameter. Let*

$$d_t(x) = \|P_t(x, \cdot) - Q_t(x, \cdot)\|_{\text{TV}}, \quad \mu_{t-1} = \mu P_1 \cdots P_{t-1}.$$

For the same terminal readout kernel F ,

$$\|\mu P_1 \cdots P_T F - \mu Q_1 \cdots Q_T F\|_{\text{TV}} \leq \min \left\{ 1, \sum_{t=1}^T \int d_t d\mu_{t-1} \right\}. \quad (15.1)$$

If $d_t(x) \leq \alpha_t \mathbf{1}_{A_t}(x) + \zeta_t(x)$, the right side can be replaced by the truncated sum $\sum_t \alpha_t \mathbb{P}_P(X_{t-1} \in A_t) + \sum_t \mathbb{E}_P \zeta_t(X_{t-1})$. For a readout in $[-1, 1]$ multiply the bound by two; for an indicator no factor two is needed. The reverse bound with P, Q exchanged is also valid.

Proof. For signed measures, expand the product difference in the order

$$\mu P_1 \cdots P_T - \mu Q_1 \cdots Q_T = \sum_{t=1}^T \mu P_1 \cdots P_{t-1} (P_t - Q_t) Q_{t+1} \cdots Q_T.$$

Integration of a row difference against μ_{t-1} has total variation at most $\int d_t d\mu_{t-1}$. All suffix kernels and F contract total variation. The triangle inequality and the bound one for probability-law

distance prove (15.1). Substitute the rowwise estimate for its occupation version. The expectation bounds follow from the oscillation of the readout. $\square$

Here the retained state must support the displayed transition kernels. For independent reset observations and a time-inhomogeneous finite machine this is automatic, even when the selected row depends on the state. For a non-reset physical process one must include the necessary analysis state or use its history kernels; ignoring a report need not remove its influence on subsequent physical dynamics. No Markov closure of an arbitrary compressed physical state is assumed.

Corollary 15.2 (Calibrated training with absorption). *Run the audit of Theorem 13.2 under an ideal reset source and a perturbed reset source whose row laws differ by at most α per active call. Use the same implemented coin and event randomizers. If A is the number of active comparison updates under the ideal execution, the change in its selected-response score is at most*

$$2 \min(1, \alpha \mathbb{E}A) + \beta_c + \beta_t, \quad \mathbb{E}A \leq (K - 1) \min(n, m^2), \quad (15.2)$$

where β_c, β_t bound candidate and target validation-law changes. A comparison with conditional coin mean u has the sharper active expectation $\mathbb{E}_u(\tau \wedge n)$ in (13.2). Uniform gaps $|u - 1/2| \geq \eta$ give the bound $\min(n, m^2, m/(2\eta))$ for that comparison.

Proof. At a completed comparison trial, marginalize the fresh coin bits into its transition kernel. Before absorption, data processing bounds its row defect by the source defect α . After absorption its retained transition is the same identity under both sources; its defect is zero. At predetermined stage boundaries the common reset of the walk is also defect-free. Theorem 15.1 therefore bounds the terminal incumbent-law distance by $\alpha \mathbb{E}A$. The score associated with an incumbent lies in $[-1, 1]$. Conditional on a common incumbent and common nested event, the two validation expectations change by at most $\beta_c + \beta_t$. This proves the score estimate. Condition on the incumbent at each stage and apply Lemma 13.1; summing proves the active-count bound. Marginalizing the coin here is a proof of a law estimate, not a waiver of its implementation states, which remain charged in (13.3). $\square$

The estimate is for the output or retained computation, not for the complete raw-observation transcript. A padded sequence of N changed reports can still have total variation of order $N\alpha$ even after the machine has stopped using them. Conversely, all N preparations remain in the resource ledger even when only A updates influence the answer. Finite row-rounding defects can be inserted as the terms ζ_t in Theorem 15.1; a changed initial law adds its total variation separately. Thus source-preserving transport is the case $\alpha = 0$, while the active bound also has content for $\alpha > 0$ and does not assert generic sample-count-free stability.

16. WEIGHTED CONTINUATION OBSTRUCTIONS

We next give a distribution-sensitive version of the wordwise obstruction. It applies to finite-valued terminal decisions and is useful even when a word of small probability should not be assigned the same weight as a likely one.

Let $f : \Sigma_1 \times \cdots \times \Sigma_L \rightarrow \mathcal{A}$ have finite range. The residual of a prefix x at cut t is the function $v \mapsto f(xv)$ on the suffix alphabet. Write $r_t(f)$ for the number of distinct residuals. A stochastic machine has a charged register at every cut; all persistent randomness is included in this register. Its wordwise error is $e(w) = \Pr(\hat{f}(w) \neq f(w))$.

Theorem 16.1 (Finite-valued realization and weighted cut bound). *Exact stochastic computation of f requires $K_t \geq r_t(f)$ at every cut. The whole residual profile is simultaneously attained by a deterministic machine.*

Fix a cut, prefixes $x_1, \dots, x_r$, and a graph on $[r]$. For each edge ij choose a suffix v_{ij} such that $f(x_i v_{ij}) \neq f(x_j v_{ij})$. Let $\mu(w) \geq 0$ be arbitrary word weights. Choose edge weights $\lambda_{ij} \geq 0$ satisfying the word capacity constraints

$$\sum_{ij: w=x_i v_{ij} \text{ or } w=x_j v_{ij}} \lambda_{ij} \leq \mu(w) \quad \text{for every word } w. \quad (16.1)$$

Every machine with K states at this cut satisfies

$$\sum_w \mu(w) e(w) \geq \min_{\phi: [r] \rightarrow [K]} \sum_{ij: \phi(i)=\phi(j)} \lambda_{ij}. \quad (16.2)$$

The right side is a weighted monochromatic-edge obstruction. Maximizing it over admissible witnesses and weights gives a finite, distribution-sensitive lower bound. The bound can be an equality with positive error.

Proof. Let $a_i(m)$ be the state distribution after prefix x_i . For a fixed edge and a common state m , the future output distribution on suffix v_{ij} is the same for both prefixes. Since their correct outputs are different, the sum of their two conditional error probabilities is at least one. Multiplying by the smaller state mass and summing gives

$$e(x_i v_{ij}) + e(x_j v_{ij}) \geq \sum_{m=1}^K \min\{a_i(m), a_j(m)\}. \quad (16.3)$$

Weight the inequalities by λ_{ij} and use (16.1). The resulting lower bound is

$$F(a_1, \dots, a_r) = \sum_{ij} \lambda_{ij} \sum_m \min\{a_i(m), a_j(m)\}.$$

This is a concave continuous function on $(\Delta_K)^r$. Every point of that product is a convex combination of its deterministic vertices; concavity implies that its value is at least the minimum over those vertices. A vertex is a labeling $\phi: [r] \rightarrow [K]$, and its value is exactly the right side of (16.2). This proves the bound without treating stochastic encodings as deterministic encodings.

For zero error, choose one distinguishing suffix for every two different residuals. Equation (16.3) forces the corresponding state supports to be disjoint. Hence at least $r_t(f)$ states are needed. For the converse, keep the residual class of the observed prefix. Equal residuals have equal shifts after every next input, so the next class is well defined. At the last epoch the residual is the output label itself. These quotient transitions realize all widths simultaneously.

For a positive-error equality, let the only relevant words be two one-letter prefixes with different required labels, weighted p and $1-p$, where $0 < p < 1$. Put $K = 1$ and $\lambda_{12} = \min(p, 1-p)$. The cut obstruction is $\min(p, 1-p)$, attained by always emitting the more likely label. Thus the weighted constant is sometimes exact; universal sharpness for all graphs or input distributions is not asserted. With unit weights on the same two words the total-error lower bound is one, also attained by a constant output; this is equality in the earlier $(r-K)/(r-1)$ bound at $r = 2$, $K = 1$. $\square$

Corollary 16.2 (A Bayes-regret obstruction for constrained audits). *Fix a finite action set $\mathcal{A}$, candidate class B , and a probability measure ν on B . For a training word w , let $p_w(b)$ be its likelihood and let $f_a(b)$ be the expected validation payoff. Put*

$$R_w(a) = \int_B p_w(b) f_a(b) d\nu(b), \quad S_\nu = \sum_w \max_a R_w(a).$$

Choose a maximizing label $f(w)$. Set $\mu(w) = R_w(f(w)) - \max_{a \neq f(w)} R_w(a)$, taking zero when a maximum is tied. Let $\Gamma_{t,K}(f, \mu)$ be any lower bound supplied by (16.2). Then every width- K audit

at cut t has worst-case expected payoff at most

$$S_\nu - \Gamma_{t,K}(f, \mu). \quad (16.4)$$

Proof. The conditional distribution of the auditor's output for a fixed word is independent of the unknown candidate. Its Bayes payoff is therefore $\sum_{w,a} R_w(a) \Pr(\hat{f}(w) = a)$. Subtracting this from S_ν gives at least $\sum_w \mu(w)e(w)$. Apply the theorem, and use that a minimum over candidates is at most the average under ν . $\square$

The corollary does not exchange a maximum and minimum over a nonconvex fixed-width machine class. The prior is a lower-bound device, not a randomization resource supplied to the candidate. With a unique maximizing label and strictly positive margins at every word, equality with S_ν forces exact wordwise computation. This makes explicit the uniqueness step used in the odd-sample binary theorem below. For even sample size, the balanced-count words have zero likelihood gap. A rule can differ on those ties without losing value; equality of values then does not force one prescribed total Boolean rule, and the odd-sample width profile cannot simply be asserted for that case.

17. JOINT RESIDUAL BUDGETS AND GROWING FAMILIES

We now put lower bounds on several cuts into one statistical budget. Summing separate cut bounds without controlling their common words would count the same regret more than once.

Theorem 17.1 (A shared-capacity multicut obstruction). *Let $f : \prod_{i=1}^L \Sigma_i \rightarrow \mathcal{A}$ and $\mu(w) \geq 0$ be as in Theorem 16.1. At each chosen cut t , choose a finite prefix graph with an edge $e = (u, v)$ carrying a distinguishing suffix s_e , and a weight $\lambda_{t,e} \geq 0$. Impose the single family of constraints*

$$\sum_{t,e: w=us_e \text{ or } w=vs_e} \lambda_{t,e} \leq \mu(w) \quad \text{for every complete word } w. \quad (17.1)$$

For a graph at cut t , write

$$\Gamma_{t,K_t}(\lambda) = \min_{\phi: \text{prefixes} \rightarrow [K_t]} \sum_{e=(u,v): \phi(u)=\phi(v)} \lambda_{t,e}.$$

Every stochastic machine of profile $\mathbf{K}$ satisfies

$$\sum_w \mu(w) \Pr(\hat{f}(w) \neq f(w)) \geq \sum_t \Gamma_{t,K_t}(\lambda). \quad (17.2)$$

Consequently, with R_w, S_ν, μ defined in Corollary 16.2, every audit of this profile and the fixed training schedule has worst-case payoff at most

$$S_\nu - \sup_{\lambda \text{ satisfying (17.1)}} \sum_t \Gamma_{t,K_t}(\lambda). \quad (17.3)$$

For fixed graphs the supremum is the value of a finite linear program.

Proof. For each edge use the overlap inequality (16.3). Multiply it by $\lambda_{t,e}$ and sum over all cuts and edges. The left side is bounded above by the left side of (17.2), by the shared capacity constraint. At each cut the resulting overlap expression is concave in its prefix state distributions, hence at least its minimum over deterministic labelings, as proved in Theorem 16.1. Adding these bounds proves (17.2). Although prefix distributions at different cuts are compatible, minimizing them separately only weakens the bound; it does not assume independent realizability. The Bayes-regret argument of Corollary 16.2 proves (17.3).

For the last assertion introduce variables z_t and impose, for each labeling ϕ of each finite graph,

$$z_t \leq \sum_{e=(u,v): \phi(u)=\phi(v)} \lambda_{t,e}.$$

Maximize $\sum_t z_t$ subject to these inequalities, nonnegative edge weights, and (17.1). There are finitely many constraints. This is a computable obstruction, not a claim that every multicut statistical optimum equals this relaxation. $\square$

17.1. Actual physical word weights. For n all-on trials of the private marked collision subfamily, a complete training word w specifies the two report bits and the mark of each trial. With N_{ab} its report-pair counts,

$$p_w(p, q) = 2^{-n} p^{N_{10}+N_{11}} (1-p)^{N_{00}+N_{01}} q^{N_{01}+N_{11}} (1-q)^{N_{00}+N_{10}}. \quad (17.4)$$

The formula has the continuous endpoint interpretation $0^0 = 1$. For any finitely supported prior on actual pairs $(p, q) \in [0, 1]^2$, use this likelihood in $R_w(A) = \int p_w(p, q)(Q - b(p, q))\mathbf{1}_A d\nu$, with all marked events $A \subseteq \Omega$ as actions. Randomized events are included by linearity. Equations (17.1) and (17.3) then give a finite all-algorithm obstruction for that full training profile, with the true physical word probabilities. Changing only Q by total variation β changes every audit's payoff by at most β .

This specialization bounds all stochastic selectors on the fixed all-on training schedule, not merely exact implementations of a majority rule. It does not assert a new numerical lower bound above three for every adaptive physical audit. Adaptive training has controller-dependent word likelihoods and must instead be included in the full experiment of (11.3); using (17.4) unchanged for such a likelihood would be incorrect. The feedback-uniform lower bound used below remains Lemma 29.2.

17.2. A family with matching end-to-end growth.

Theorem 17.2 (Point-source decision spectra). *Let $d \geq 2$, $B_d = \{\delta_1, \dots, \delta_d\}$ on $[d]$, and let Q_d be uniform. For every $K \geq 1$,*

$$\mathfrak{V}_K(B_d, Q_d) = 1 - \frac{\lceil d/K \rceil}{d}. \quad (17.5)$$

For $1 - 2/d < c < 1 - 1/d$, the decision complexity is exactly d , and the whole-schedule peak satisfies

$$d \leq W_c(B_d, Q_d) \leq W_c^{\text{rat}}(B_d, Q_d) \leq 2d.$$

The upper bound uses one training observation and two validations, with no randomization. Thus the end-to-end width grows linearly in this nontrivial sequence of compact behavior images.

Proof. Assign each point source to a response that maximizes its score. Some response is assigned a set S of at least $\lceil d/K \rceil$ indices. For $s = |S| < d$, the mean of its scores on S equals

$$\frac{d-s}{d} \left(\frac{1}{d-s} \sum_{i \notin S} h_i - \frac{1}{s} \sum_{i \in S} h_i \right) \leq 1 - \frac{s}{d}.$$

For $s = d$ the mean is zero. At least one assigned source has score no larger than this mean, proving the upper bound. Partition $[d]$ into at most K groups of size at most $\lceil d/K \rceil$ and use their complements as events. This attains the lower bound.

In the stated interval the formula forces $K \geq d$. A single training observation identifies the point source i exactly. Keep i and choose the complement of $\{i\}$. During validation retain i and the candidate indicator, at most $2d$ states, then emit the target indicator minus it. This computes the prescribed signed score on every input word, including words that have zero probability under the point-source family. Its expected score is $1 - 1/d$. The lower bound follows from the necessary decision-cut width. $\square$

17.3. A sharp simultaneous storage profile. A second class separates overlapping retention obligations. Independent uniform labels $X_j \in [q_j]$ are revealed at scheduled epochs a_j and must be reported at epochs $b_j > a_j$. No later input reveals an active label, and previous outputs cannot be read as external memory. Let $A_t = \{j : a_j \leq t < b_j\}$ and $D_t = \prod_{j \in A_t} q_j$.

Theorem 17.3 (Overlapping retention intervals). *Every stochastic machine whose probability of correctly reporting all labels is at least $1 - \varepsilon$ has*

$$K_t \geq (1 - \varepsilon)D_t \quad \text{at every cut } t. \quad (17.6)$$

The exact profile $K_t = D_t$ is attained simultaneously by a deterministic machine. Thus the minimum peak number of bits lies between $\log_2 \max_t D_t + \log_2(1 - \varepsilon)$ and $\lceil \log_2 \max_t D_t \rceil$ for $\varepsilon < 1$.

Proof. Fix a cut and give the decoder, as additional information, all labels outside A_t , whether past or future. They are independent of the uniform active vector. Conditional on them, every future prediction of that vector factors through the K_t -valued retained state and fresh randomness. If $a(m | x)$ is its conditional encoder and $d(x | m)$ the induced joint prediction law, its success probability is at most

$$\frac{1}{D_t} \sum_m \sum_x a(m | x) d(x | m) \leq \frac{1}{D_t} \sum_m \sum_x d(x | m) = \frac{K_t}{D_t}.$$

Correctness of all reports implies correctness of the active reports, so averaging over the independent additional information proves (17.6). For the upper bound, keep precisely the vector of active labels, append a label when revealed, and delete it when reported. Every update is deterministic and respects the schedule. The bit bounds follow by taking logarithms and rounding for an actual binary register. The lower bound is a counting argument, not an independence assumption about labels after conditioning on a compressed state. $\square$

18. FINITE-DATA TESTING ON THE ORIGINAL BEHAVIOR IMAGE

18.1. The reset interface. Let $B \subset \prod_{j=1}^J \Delta(\Omega_j)$ be compact, and let $Q = (Q_j)$ be fixed. Set $d_j = |\Omega_j|$, $D = \sum_j (d_j - 1)$, $\mathcal{I} = \{(j, A) : A \subseteq \Omega_j\}$, and $f_{j,A}(b) = Q_j(A) - b_j(A)$. Include the empty event. No convexity of B is assumed.

A candidate b is fixed for the whole audit. The auditor obtains n independent samples of each b_j , with independence between rows. In a causal experiment, row j means a specified parameter and deterministic feedback tree; the reset interface supplies that row. It is an additional experimental interface, not information available to the online simulator. A complete sample contains the terminal mark. Private randomness and any per-run hidden selector are reset between runs. The design of the candidate is not resampled.

Write $C_{j,w}$ for the counts, and $\mathcal{N}_n$ for all count arrays with $\sum_w C_{j,w} = n$. Their probability is the polynomial

$$p_C(b) = \prod_j \frac{n!}{\prod_w C_{j,w}!} \prod_w b_j(w)^{C_{j,w}}.$$

A randomized audit is an array $c = (c_{C,i})$ with $c_C \in \Delta(\mathcal{I})$. After training it chooses $i = (j, A)$ according to c_C . Independent candidate and target validation runs give the signed score $\mathbf{1}_A(\text{target}) - \mathbf{1}_A(\text{candidate})$. Its expected value is

$$g_c(b) = \sum_{C,i} p_C(b) c_{C,i} f_i(b), \quad v_n^{\text{aud}}(B, Q) = \max_c \min_{b \in B} g_c(b). \quad (18.1)$$

Thus every audit payoff is polynomial of total degree at most $Jn + 1$ in the behavior coordinates. The probability of a large observed score in a single audit is not asserted to equal its expectation; additional independent validation would be required for a specified confidence level.

Theorem 18.1 (Reset-audit duality and finite-sample approximation). *Let $\delta = \min_{b \in B} \max_i f_i(b)$. Then all the extrema below are attained, and*

$$v_n^{\text{aud}}(B, Q) = \min_{\mu \in \mathcal{P}(B)} \sum_{C \in \mathcal{N}_n} \max_{i \in \mathcal{I}} \int_B p_C(b) f_i(b) d\mu(b), \quad (18.2)$$

$$0 \leq \delta - v_n^{\text{aud}}(B, Q) \leq \sqrt{D/n}. \quad (18.3)$$

The values are nondecreasing in n . If $a = \dim \text{aff } B$, a minimizing measure in (18.2) can be supported on at most $\binom{a+Jn+1}{a}$ points.

An explicit feasible audit attaining the lower bound $\delta - \sqrt{D/n}$ chooses an event maximizing $Q_j(A) - C_j(A)/n$, with a fixed deterministic tie rule. If the rational target probabilities are given, this is a rationally specified finite rule. With a free schedule clock, a conservative retained-state bound is

$$H(|\mathcal{I}| + 1)(n + 1) \sum_j d_j,$$

where H is the number of states needed to execute and record one causal transcript. This budget is charged to the auditor, not to the simulator.

Proof. The coefficient arrays form a compact convex product of finite simplexes. The integrand is continuous in b and affine in c . The probability measures on compact B form a compact convex set for weak convergence. The minimum over b equals the minimum over such measures for fixed c . Finite-dimensional minimax (or its compact measure version) therefore exchanges the two extrema. Maximization over each independent coefficient simplex selects its largest coordinate integral, giving (18.2). Continuity and compactness give attainment. This step uses measures on the original B , not feasible private realizations of their barycenters.

For any audit, $g_c(b) \leq F(b) = \max_i f_i(b)$. At a minimizer of F this gives $v_n^{\text{aud}} \leq \delta$. The zero event gives $v_n^{\text{aud}} \geq 0$. For the stated explicit rule put $\hat{b}_j = C_j/n$ and $\rho = \max_j \|\hat{b}_j - b_j\|_{\text{TV}}$. Comparing the empirical maximizer with a true maximizing event gives the pointwise regret bound

$$F(b) - f_{i(C)}(b) \leq 2\rho.$$

For a multinomial empirical measure,

$$\mathbb{E} \|\hat{b}_j - b_j\|_{\text{TV}}^2 \leq \frac{d_j}{4} \sum_w \mathbb{E} (\hat{b}_j(w) - b_j(w))^2 = \frac{d_j(1 - \sum_w b_j(w)^2)}{4n} \leq \frac{d_j - 1}{4n}.$$

Consequently $\mathbb{E} \rho \leq (\sum_j \mathbb{E} \|\hat{b}_j - b_j\|_{\text{TV}}^2)^{1/2} \leq \sqrt{D/n}/2$. Averaging the regret uniformly over B proves (18.3).

From $n + 1$ unordered samples in each row, discard one uniformly at random. The remaining counts have exactly the n -sample law, independently across rows. This realizes every n -sample audit as a randomized $(n + 1)$ -sample audit and proves monotonicity.

Choose affine coordinates on $\text{aff } B$. All the functions $p_C f_i$ lie in the span of monomials of total degree at most $Jn + 1$, a space of dimension at most $s = \binom{a+Jn+1}{a}$ including the constant. The moment vector of a probability measure is in the convex hull of the evaluation vectors; their constant coordinate is one, so their affine dimension is at most $s - 1$. Eliminating affine dependencies gives a representation by at most s points. It preserves every integral in (18.2), and hence preserves the objective.

Finally, each count is in $\{0, \dots, n\}$. The indicated product register stores all counts, the within-run state, and the selected event for validation. Only the latter event and the within-run buffer need survive once training ends, so the product bound is conservative. Randomizing an event uses fresh randomness. The schedule, reset access and terminal mark readout were declared before the theorem. No private simulator state is read by this implementation. $\square$

The dual measure chooses a candidate once, after which its samples are independent conditional on that candidate. Replacing it by $\bar{b} = \int b d\mu$ would replace $\int p_C(b) f_i(b) d\mu$ by $p_C(\bar{b}) f_i(\bar{b})$, generally a different quantity. It would also replace the joint replicated law by a product of its marginals. Neither operation is part of the theorem. This is the operational meaning of the moment side, rather than a same-budget convexification theorem.

Proposition 18.2 (Both sides sampled). *Suppose the target reference Q^b varies with the candidate, as it does when visible-selector weights are design variables. Supply independent training samples from both b and Q^b , and use their empirical difference. If their row dimensions are D_P, D_Q , the same construction has regret at most $\sqrt{D_P/n} + \sqrt{D_Q/n}$. Minimax and attainment hold on the compact image of pairs (b, Q^b) . In particular a variable reference law must not be treated as known fixed input to Theorem 18.1.*

Proof. The error in any event discrepancy is at most the sum of the maximal row variations of the two empirical tables. Apply the preceding second moment bound separately and then the empirical-maximizer regret inequality. Count probabilities are products for the two tables. They are continuous polynomials on the compact image of pairs, so the same minimax argument applies without changing the simulator signature. $\square$

18.2. A matching uniform sampling obstruction.

Theorem 18.3 (Binary reset lower bound). *Take one binary row with target $\text{Ber}(1/2)$ and candidate class $B_h = \{\text{Ber}(1/2 - h), \text{Ber}(1/2 + h)\}$, where $0 < h < 1/2$. Then*

$$\delta = h, \quad v_n^{\text{aud}} = h \|\text{Ber}(1/2 + h)^{\otimes n} - \text{Ber}(1/2 - h)^{\otimes n}\|_{\text{TV}}. \quad (18.4)$$

For every $n \geq 1$, choosing $h = 1/(8\sqrt{n})$ gives

$$\delta - v_n^{\text{aud}} \geq \frac{1}{16\sqrt{n}}.$$

Thus the exponent in (18.3) is sharp uniformly over compact candidate classes, already at affine dimension one. This does not assert a degree lower bound for revealed-behavior polynomials, nor that this two-point class equals an unrestricted private-row image.

Proof. Only the two complementary singleton events and the zero event matter. A randomized choice is a statistic d in $[-1, 1]$. The two expected payoffs are $h\mathbb{E}_+d$ and $-h\mathbb{E}_-d$. Their minimum is at most half their sum, which is at most $h\|P_+^n - P_-^n\|_{\text{TV}}$. Conversely take the sign of the likelihood difference, assigning zero at a tie. Complementing every sample interchanges P_+^n and P_-^n and changes that sign. The two payoffs are equal, and their average attains the variation bound. The statistic depends only on the count, so it is an admissible audit. The symmetric measure on the two candidates attains the dual value. Every Dirac measure has dual objective h , whereas for finite n the two product laws are equivalent and their variation is strictly less than one. Thus no dual minimizer is a Dirac measure. Their barycenter is the target law, but replacing the dual by that barycenter would incorrectly give value zero instead of the strictly positive value in (18.4).

For $h \leq 1/8$, elementary integration of $(1 - u^2)^{-1}$ gives

$$\text{KL}(\text{Ber}(1/2 + h) \parallel \text{Ber}(1/2 - h)) = 2h \log \frac{1 + 2h}{1 - 2h} \leq 16h^2.$$

The product relative entropy is at most $16nh^2$. Pinsker's inequality bounds its total variation by $\sqrt{8nh^2}$. At the specified h this is less than $1/2$. Substituting in (18.4) proves the claim. The inequality uses the classical two-point testing mechanism; the restriction being measured is the reset budget of the present game. $\square$

18.3. Transport of one fixed audit.

Proposition 18.4 (Reset cost under model perturbation). *Let q range over the same compact set of executable row designs in two models. Suppose uniformly in q, j that*

$$\|P_j^q - \tilde{P}_j^q\|_{\text{TV}} \leq \alpha, \quad \|Q_j - \tilde{Q}_j\|_{\text{TV}} \leq \beta.$$

For the same coefficient rule c , evaluated with each model's own target validation,

$$|g_c(q) - \tilde{g}_c(q)| \leq (2Jn + 1)\alpha + \beta. \quad (18.5)$$

The deficiencies themselves differ by at most $\alpha + \beta$, at the same resource budget. If the source behavior is unchanged, a fixed audit loses at most β , independently of its reset count.

Proof. A product coupling of the Jn training runs fails with probability at most $Jn\alpha$. Taking counts and choosing the event are contractions. A score with fixed target discrepancies lies in $[-1, 1]$, so the change in its expectation from this training-law perturbation is at most $2Jn\alpha$. At each event the discrepancy changes by at most $\alpha + \beta$. This proves (18.5). For deficiency, every row variation changes by at most $\alpha + \beta$; taking the maximum and then the minimum preserves that bound. No new simulator is inserted. $\square$

For a rational polynomial image, a rational rule and a rational threshold, strict positivity of $g_c(q)$ above that threshold is a first-order formula over the original row simplexes. Real quantifier elimination decides it. The same is true of an existential rule followed by this universal condition. This is an exact finite verification statement, not an efficient algorithm. The explicit empirical rule below will be verified by a uniform analytic inequality, rather than by storing its enormous complete table.

19. CONTINUATION COMPLEXITY OF A FINITE-DATA TEST

The reset budget does not by itself bound the auditor's retained memory. We now impose both constraints. At scheduled epoch t a streaming auditor reads one letter x_t from a finite alphabet X_t , updates a register, and emits no information to any auxiliary store. It makes its decision on reading the last letter. Transition rows may depend on the epoch; the clock is free. All other information crossing a cut belongs to the register. Fresh randomization is allowed. An independent persistent selector, when allowed, selects a complete auditor before any letters are read.

For a Boolean decision $\psi : X_1 \times \cdots \times X_n \rightarrow \{-1, 1\}$, let $u \sim_t v$ mean that

$$\psi(us) = \psi(vs) \quad \text{for every } s \in X_{t+1} \times \cdots \times X_n.$$

Write $r_t(\psi)$ for the number of equivalence classes of prefixes of length t , for $0 \leq t < n$. These are the deterministic continuation responses. The terminal decision is emitted on the last transition, so no terminal register is needed. In particular, $r_0 = 1$. The equivalence is a finite-horizon version of the classical residual-state construction [16]. The following quantitative statement will also measure the loss when a statistical auditor has fewer states than this construction.

Theorem 19.1 (Exact continuation width and a quantitative cut obstruction). *A stochastic streaming auditor computes ψ exactly on every word with width profile $(K_t)_{0 \leq t < n}$ if and only if $K_t \geq r_t(\psi)$ at every cut. The profiles are attained simultaneously by a deterministic auditor. Independent persistent randomization does not reduce these widths for exact computation.*

More generally, at a cut with $r = r_t(\psi) \geq 2$, every stochastic auditor of width K satisfies

$$\sum_{w \in X_1 \times \cdots \times X_n} e_\psi(w) \geq \frac{(r - K)_+}{r - 1}, \quad (19.1)$$

where $e_\psi(w)$ is its probability of an incorrect Boolean decision on input w . The inequality also holds after averaging over an independent selector whose conditional auditors each have width at most K .

Proof. From a prefix u let $e_u(m)$ be the distribution of the retained state. From state m and suffix s let $a_m(s)$ be the probability of terminal answer $+1$. Exact computation says

$$\sum_m e_u(m) a_m(s) = \frac{1 + \psi(us)}{2}.$$

The right side is zero or one. Every state of positive e_u -mass must therefore have precisely that zero-one continuation response for all s . Distinct prefix classes have disjoint positive supports. Hence $K_t \geq r_t$. Conversely, retain the prefix class. Appending the same letter to two equivalent prefixes gives equivalent new prefixes; thus every transition is well defined. At the last transition the class and the last letter determine ψ . This realizes every width r_t simultaneously. For a persistent independent selector, a zero average error on each of the finitely many words implies zero conditional error on all words outside a single null set of selector values. The preceding bound applies there.

For the quantitative claim choose representative prefixes $u_1, \dots, u_r$, one per class, and put $e_i(m) = e_{u_i}(m)$. Define

$$o_{ij} = \sum_{m=1}^K \min\{e_i(m), e_j(m)\}.$$

For nonnegative numbers $a_1, \dots, a_r$, ordering them shows that $\sum_{i < j} \min(a_i, a_j) \geq \sum_i a_i - \max_i a_i$. Apply this at each state. Since every e_i has mass one and each $\max_i e_i(m) \leq 1$, it follows that

$$\sum_{i < j} o_{ij} \geq r - K. \quad (19.2)$$

Choose a suffix s_{ij} distinguishing u_i, u_j . Conditional on a common state and this suffix, the two desired answers are opposite. Their error probabilities, weighted by $e_i(m), e_j(m)$, have sum at least $\min\{e_i(m), e_j(m)\}$. Thus

$$e_\psi(u_i s_{ij}) + e_\psi(u_j s_{ij}) \geq o_{ij}.$$

On summing, each complete word on the left occurs at most $r - 1$ times: its prefix identifies a single representative, which participates in $r - 1$ pairs. Nonnegativity of all other word errors and (19.2) prove (19.1). When $r \leq K$ the claim is just nonnegativity. Finally apply the bound conditional on a selector and integrate. Independence ensures that conditioning has not changed the input experiment. $\square$

The exact part is also the extremal zero-one case of Theorem 8.2. Its content is a simultaneous width profile, rather than separate unrelated factorizations at different cuts. The error bound does not assert that its constant is optimal for every Boolean function. Its use below is to convert a continuation obstruction into a strict loss of the attainable statistical value.

19.1. The width required to attain the binary reset value. Fix $0 < h < 1/2$ and the binary candidate class B_h of Theorem 18.3. A terminal decision $+1$ chooses the singleton event appropriate to the $+$ candidate, and -1 the complementary event. A zero-score event can equivalently be replaced by an independent fair choice of these decisions. The expected payoffs are therefore $h\mathbb{E}_+d$ and $-h\mathbb{E}_-d$, where $d \in [-1, 1]$ is the conditional mean decision and $P_\pm = \text{Ber}(1/2 \pm h)^{\otimes n}$. Let $v_{n,K}$ be the supremum of the smaller of these two payoffs over streaming auditors with the declared width profile. The final validation interface is unchanged; the widths here count training and selection only.

Theorem 19.2 (Exact memory threshold for optimal binary auditing). *Let $n = 2m + 1$ be odd. The unconstrained reset value is $v_n = h\|P_+ - P_-\|_{\text{TV}}$. It is attained at the profile*

$$r_t = \min\{t + 1, n - t + 2\}, \quad 0 \leq t < n, \quad (19.3)$$

and $v_{n,K} = v_n$ if and only if $K_t \geq r_t$ for all these cuts. For every cut with $r_t \geq 2$ one has the explicit deficit

$$v_n - v_{n,K} \geq 2h^2(1/4 - h^2)^m \frac{(r_t - K_t)_+}{r_t - 1}. \quad (19.4)$$

These conclusions remain valid when an independent persistent selector chooses auditors of the same conditional widths. For $n \geq 3$ the least peak training width attaining v_n is $m + 2$; for $n = 1$ it is one.

Proof. The sign of $P_+(w) - P_-(w)$ is the majority decision $\psi(w)$, with no zero difference since n is odd. Complementing every bit interchanges the two laws and reverses the decision. Consequently majority has equal payoffs and attains $h\|P_+ - P_-\|_{\text{TV}}$, as also proved in Theorem 18.3.

A prefix with k ones requires at least $m + 1 - k$ ones in its remaining $n - t$ places. Thresholds at most zero give the constant positive response; thresholds greater than $n - t$ give the constant negative response; the remaining thresholds give different nonconstant responses. Counting them for $0 \leq k \leq t$ gives (19.3). The exact realization in Theorem 19.1 implements majority at all these widths.

For an arbitrary randomized decision, its smaller payoff is bounded by the equal-prior average. The deficit of this average from the majority value is exactly

$$h \sum_w |P_+(w) - P_-(w)| e_\psi(w). \quad (19.5)$$

Put $a = 1/2 + h$ and $b = 1/2 - h$. A word with $k \geq m + 1$ ones has absolute likelihood difference $(ab)^{n/2} \{(a/b)^{k-n/2} - (a/b)^{-(k-n/2)}\}$, which increases with k . Symmetry treats the other half of the words. The smallest difference is therefore attained at $k = m, m + 1$ and equals

$$a^{m+1}b^m - b^{m+1}a^m = 2h(1/4 - h^2)^m.$$

Substitution of (19.1) into (19.5) gives (19.4). Any deficient cut gives a strictly positive loss; all sufficiently wide profiles implement majority. Maximizing (19.3) proves the peak-width assertion. The same average-payoff argument works conditional on an independent selector and hence after averaging. $\square$

The result identifies the exact memory needed to attain the optimal value, not the memory needed for a prescribed approximate value uniformly in h . The quantitative deficit can be small for large n . In particular it is not a sharp joint asymptotic tradeoff in sample size, state size and error. It complements, rather than replaces, the uniform sampling exponent in Theorem 18.3. No minimax interchange on the generally nonconvex set of fixed-width auditors has been used.

20. POLYNOMIAL TESTING OF THE ORIGINAL RESOURCE IMAGE

The private behavior image is usually nonconvex. Its testing criterion can nevertheless be written as a limit of finite algebraic lower certificates, without replacing that image by randomized designs. The testing weights in this section depend on the candidate simulator. Unlike Theorem 21.1, this identity is a certificate transform, not an independent variational dual. They are universal proof functions, not a hidden selector supplied to an executed experiment.

Use any one of the finite signatures of Theorem 7.2, with rational instrument data. Write its objective as $F(q) = \max_{1 \leq i \leq m} f_i(q)$, where f_i are all its marked event discrepancies, including the empty event $f_1 = 0$. For visible selectors the event and feedback rule include their selector

fibers, and the target selector law remains in f_i . Thus $-1 \leq f_i \leq 1$ for all three signatures; the polynomials may be different.

Parameterize a row with d outcomes by $d - 1$ variables in $[0, 1]$:

$$q_1 = x_1, \quad q_j = x_j \prod_{l < j} (1 - x_l) \quad (1 < j < d), \quad q_d = \prod_{l < d} (1 - x_l).$$

This map is onto the row simplex, including its boundary. Indeed, divide each desired probability by the remaining mass while that mass is positive; when it is zero choose the remaining variables arbitrarily. Taking the product of these maps, including a selector row when needed, gives an onto polynomial map $q(x)$ from $[0, 1]^D$ to the feasible row product. Put $h_i(x) = f_i(q(x))$ and $g_i = (1 + h_i)/2$. All h_i are rational polynomials, $0 \leq g_i \leq 1$, and $g_1 = 1/2$. The case $D = 0$ is a single feasible table and is interpreted directly.

Theorem 20.1 (Strict polynomial lower-certificate hierarchy). *For $p \geq 1$ define*

$$A_p(x) = \frac{\sum_{i=1}^m g_i(x)^p h_i(x)}{\sum_{i=1}^m g_i(x)^p}, \quad a_p = \min_{x \in [0, 1]^D} A_p(x), \quad \delta = \min_q F(q).$$

Then the denominators are positive, a_p is nondecreasing, and

$$\delta - 2(1 - m^{-1/(p+1)}) \leq a_p \leq \delta, \quad \delta = \sup_{p \geq 1} a_p. \quad (20.1)$$

For rational L let

$$P_{p,L}(x) = \sum_{i=1}^m g_i(x)^p (h_i(x) - L). \quad (20.2)$$

A nonnegative tensor Bernstein expansion of $P_{p,L}$ on $[0, 1]^D$ certifies $\delta \geq L$. If $L < \delta$, some p and some finite Bernstein degree have strictly positive coefficients. Conversely, a strictly positive such expansion certifies $L < \delta$. These are lower certificates at the original state and selector budgets. No convexification of the feasible behavior image occurs.

Proof. Fix x and put $b = \max_i g_i(x) \geq 1/2$ and $r_i = g_i(x)/b$. At least one r_i is one. Write $S_j = \sum_i r_i^j$. Hölder's inequality gives

$$S_p \leq m^{1/(p+1)} S_{p+1}^{p/(p+1)}, \quad S_{p+1} \geq 1.$$

Consequently $S_{p+1}/S_p \geq m^{-1/(p+1)}$. Since $A_p = 2bS_{p+1}/S_p - 1$ and $\max_i h_i = 2b - 1$,

$$0 \leq \max_i h_i - A_p \leq 2(1 - m^{-1/(p+1)}).$$

Cauchy-Schwarz gives $S_{p+2}S_p \geq S_{p+1}^2$, so A_p is nondecreasing in p . Its denominator is at least 2^{-p} and it is continuous on the cube. Taking minima proves (20.1). The bound is also at most $2 \log m/(p+1)$, by $1 - e^{-u} \leq u$.

Every tensor Bernstein basis function is nonnegative on the cube, and the basis functions sum to one. Nonnegative coefficients therefore imply $P_{p,L} \geq 0$. Dividing by the positive denominator gives $F(q(x)) \geq A_p(x) \geq L$ for every x , which proves soundness. For completeness choose p so that the error in (20.1) is less than $\delta - L$. Then

$$P_{p,L}(x) \geq 2^{-p} \{\delta - L - 2(1 - m^{-1/(p+1)})\} > 0.$$

Lemma 20.2 below gives a finite strictly positive Bernstein expansion. Conversely strictly positive coefficients have a positive minimum, and the denominator is at most m , so they give a strict positive uniform gap above L . $\square$

Positive Bernstein certificates and their degree elevation are classical; see de Klerk-Laurent [20]. The next elementary bound is retained for a self-contained exact verifier, not as a new general hypercube positivity theorem.

Lemma 20.2 (An explicit positivity degree). *Let $P(x) = \sum_k c_k x^k$ be a rational polynomial in D variables and $n \geq \max_{k,j} k_j$. In its tensor Bernstein expansion of degree n in each coordinate the coefficient at $\alpha \in \{0, \dots, n\}^D$ is*

$$b_{\alpha,n} = \sum_k c_k \prod_{j=1}^D \frac{\binom{\alpha_j}{k_j}}{\binom{n}{k_j}}, \quad |b_{\alpha,n} - P(\alpha/n)| \leq \frac{C(P)}{n}, \quad C(P) = \sum_k |c_k| \sum_j \binom{k_j}{2}. \quad (20.3)$$

In particular, $\min P > 0$ implies positivity of every coefficient whenever $n > C(P)/\min P$ and the degree restriction holds.

Proof. For a binomial random variable B with parameters (n, x) , $\mathbb{E} \binom{B}{k} = \binom{n}{k} x^k$. Expanding this expectation proves the univariate monomial formula, and products prove the displayed coefficient identity. The convention is $\binom{a}{k} = 0$ for $a < k$. To compare $\binom{a}{k}/\binom{n}{k}$ with $(a/n)^k$, draw k indices uniformly from $\{1, \dots, n\}$. Conditional on no repetition this is sampling without replacement. The probability of a repetition is at most $\binom{k}{2}/n$. Apply this total variation bound to the event that all drawn indices lie in a prescribed a -element subset. Hence the difference of the two factors is at most $\binom{k}{2}/n$. All factors lie in $[0, 1]$. A product telescoping bound followed by the triangle inequality proves (20.3). The final assertion follows by taking the minimum of P . $\square$

Corollary 20.3 (Rational lower witnesses without policy cells). *At every fixed finite rational signature, strict infeasibility above a rational level L is equivalent to the existence of the following finite rational certificate: an integer p , an integer n , and a strictly positive coefficient array for (20.2) whose identity is checked by (20.3). Dovetailing these certificates with rational original-budget upper witnesses produces nested intervals converging to δ . If a subfamily of the original tests gives a strict gap, the same construction applies to it after adjoining the empty test.*

Proof. Soundness and completeness are Theorem 20.1. Each finite candidate is checked by rational polynomial arithmetic. Enumerate all (p, n) and rational levels, and retain the largest verified lower level so far, together with zero. Rational row tables are dense in the feasible product, and their continuous objectives converge to the minimum, so enumeration and retention of the best upper table converge from above. This is a dovetailed enumeration, not a claim that any fixed (p, n) test search terminates. For a requested positive interval width, convergence of both endpoints proves termination. The reduced family conclusion uses only the same pointwise maximum argument. $\square$

Remark 20.4 (Meaning and size of the strict certificate transform). A constant distribution on tests generally detects only the convex hull of the behavior image. The functions $g_i^p / \sum_j g_j^p$ are instead functions of a proposed row table. Their universally quantified weighted inequality is an algebraic certificate; the tester need not observe a private state in an experiment. The displayed certificate identity is not an equality obtained by exchanging a private minimum with a fixed mixed test. Power means and Bernstein degree elevation are classical algebraic approximation devices, proved here in the precise form needed. The claim here is strict-level completeness for the original marked causal resource image, including visible selectors, and its physical transfer below. The coefficient count $(n+1)^D$, the number of original tests, and the algebraic degrees may be large. No polynomial complexity or intrinsic-dimension bound on this row-cube coefficient count is asserted. The independent adaptive test has the separate behavior-size bound of Theorem 21.3; its verification is a different cost.

20.1. A sharp two-time nonconvex example. The source has singleton reports, the control alphabet and actual mark are singleton, and the target emits $(0, 0)$ or $(1, 1)$ with probability $1/2$ each. At width one, a private simulator emits two independent bits with probabilities p, q . It cannot reread its first emitted bit.

Proposition 20.5 (A gap invisible to constant testing mixtures). *For this delayed experiment,*

$$\delta_{(1,1)}^p = \sqrt{2} - 1, \quad \delta_{(1,1),2}^h = 0, \quad \delta_{(1,1),r}^v = \sqrt{2} - 1 \quad (r \geq 1).$$

The private objective is the maximum of the four polynomials

$$0, \quad \frac{1}{2} - pq, \quad \frac{1}{2} - (1-p)(1-q), \quad p + q - 2pq.$$

Every constant mixture of these tests has minimum at most zero, whereas Corollary 20.3 certifies every rational $L < \sqrt{2} - 1$ at private width one.

Proof. Put $u = pq$ and $v = (1-p)(1-q)$. Direct total variation calculation gives $F = 1 - \min(u, 1/2) - \min(v, 1/2)$ and the displayed maximum. By Cauchy–Schwarz, $\sqrt{u} + \sqrt{v} \leq 1$. If $u \geq 1/2$, then $v \leq (1 - 1/\sqrt{2})^2$, so the sum of the two capped terms is at most $2 - \sqrt{2}$; the case $v \geq 1/2$ is symmetric. Otherwise set $a = \sqrt{u}$, $b = \sqrt{v}$. On $0 \leq a, b \leq 1/\sqrt{2}$, $a + b \leq 1$, the maximum of $a^2 + b^2$ is at a vertex of this polygon and equals $2 - \sqrt{2}$. Equality is attained at $p = q = 1/\sqrt{2}$. A hidden fair bit selects the two deterministic constant sequences and exactly reproduces the target. If it is exposed, variation is averaged over the selector fibers; each fiber has error at least the private minimum, which is itself attainable without a selector. Finally the target lies in the convex hull of the two private deterministic behaviors. The average discrepancy of any fixed test mixture at those behaviors is zero, so at least one has discrepancy at most zero. The strict polynomial certificates therefore detect information lost by constant convex separation. $\square$

21. ADAPTIVE TESTING ON THE OBSERVABLE BEHAVIOR IMAGE

The obstruction to linear randomization criteria at a fixed private register is the nonconvexity of the behavior image, not a failure of compactness. We give a variational formulation in which a test is an independent function of an observable candidate law. The construction does not give the simulator an extra random variable.

21.1. The testing game. Let B be a nonempty compact subset of a finite-dimensional affine space, and let $f_1, \dots, f_m$ be affine real functions, including the zero function. Put

$$F(b) = \max_{i \leq m} f_i(b), \quad \delta = \min_{b \in B} F(b).$$

For the finite comparison problem of Section 7, b is the array of discrepancies between the simulated and target marked laws, indexed by parameters, feedback trees and transcripts; f_i ranges over their signed event evaluations. In the visible case the array includes the selector and its target weights. Thus the target coordinate is part of the array when those weights vary. Private, fixed-hidden and fixed-visible signatures each give their own compact set B . No convex hull is taken.

An *adaptive test* is a continuous map $\kappa : B \rightarrow \Delta_m$, chosen before a candidate b . Its score against that candidate is

$$G(b, \kappa) = \sum_i \kappa_i(b) f_i(b).$$

This is a verification game: the selected candidate law is revealed to the evaluator of the already chosen function κ . It is not a one-observation statistical test, and does not make an unavailable parameter or mark observable to a physical controller.

Theorem 21.1 (Revealed-behavior adaptive representation). *For every such compact behavior image,*

$$\begin{aligned} \delta &= \sup_{\kappa \in C(B, \Delta_m)} \min_{b \in B} G(b, \kappa) \\ &= \min_{\mu \in \mathcal{P}(B)} \sup_{\kappa \in C(B, \Delta_m)} \int_B G(b, \kappa) d\mu(b). \end{aligned} \tag{21.1}$$

The minimization over probability measures is attained. If Borel adaptive tests are allowed, the first supremum is attained as well. The continuous-test assertion is a supremum; continuous attainment is not needed or asserted.

Proof. Fix $\eta > 0$. The sets $U_i = \{b : f_i(b) > F(b) - \eta\}$ form an open cover of the compact metric space B . Choose a continuous partition of unity subordinate to this finite cover. The resulting test has $F(b) - \eta < G(b, \kappa) \leq F(b)$ for all b . Consequently the first supremum equals δ . The same uniform inequality shows, for every probability μ ,

$$\sup_{\kappa} \int G(b, \kappa) d\mu(b) = \int F d\mu.$$

Its minimum is δ , attained by a point mass at a minimizer of F . This also proves the displayed minimax exchange without an assumption that B is convex. Equivalently one can apply the usual compact-convex minimax theorem to $\mathcal{P}(B)$ and the convex class of continuous tests, with the pairing $\int G d\mu$. Finally choose the smallest index attaining $F(b)$. Its level sets are Borel, and its indicator vector is a Borel test with score exactly F . $\square$

The probability μ here describes a lottery over *revealed candidate laws*. The score is $\int G(b, \kappa) d\mu(b)$, not $G(\int b d\mu, \kappa)$. Hiding the candidate and replacing it by its barycenter is a different game; in general it lowers the value. This distinction is essential for a finite persistent-state budget.

21.2. A size bound in behavior dimension. Factor out directions annihilated by every $f_i - f_i(b_0)$, so that on the resulting difference space

$$\|v\|_* = \max_i |f_i(b_0 + v) - f_i(b_0)|$$

is a norm. Denote its dimension by a and let $R = \text{diam}_*(B)$. All conclusions below are unaffected by this observable quotient. The case $a = 0$ is a single constant comparison and requires one test. Suppose henceforth that $a \geq 1$, so $R > 0$.

Lemma 21.2 (A bounded behavior chart). *There exist $b_0 \in B$ and vectors $v_j \in B - b_0$, $1 \leq j \leq a$, forming a basis, such that $\|v_j\|_* \leq R$ and every $b \in B$ has coordinates $b = b_0 + \sum_j t_j v_j$ with $|t_j| \leq 2$. If B is a rational polynomial image of a product of simplexes, the points can be chosen rational and can be found effectively.*

Proof. Fix $b_0 \in B$. In any fixed coordinates on the difference space, let $D_0 > 0$ be the maximum absolute determinant of a vectors in $B - b_0$. Compactness and spanning give existence and positivity. Choose a tuple with determinant greater than $D_0/2$. Replacing its j th vector by $b - b_0$ multiplies the determinant by t_j ; maximality gives $|t_j| < 2$. The norm bound is the definition of R . For a rational polynomial image, take b_0 at rational rows. Rational row images are dense. An affine basis can be found by enumerating rational images and testing rational determinants; the true affine span and the maximum determinant are semialgebraic data. Quantifier elimination decides whether a proposed nonzero determinant exceeds half the maximum. Density therefore makes the search terminate. This is an effectiveness assertion, not a bit-complexity bound for finding the chart. $\square$

Set $b(x) = b_0 + \sum_j (4x_j - 2)v_j$ and $C = \{x \in [0, 1]^a : b(x) \in B\}$. The ambient cube is a coordinate box, not an enlargement of the feasible behavior set. Write $\ell_i(x) = f_i(b(x))$. Then

$$|\ell_i(x) - \ell_i(y)| \leq 4R \sum_j |x_j - y_j|. \quad (21.2)$$

For $n \geq 0$ let $\beta_{\alpha,n}$ be the tensor Bernstein basis of coordinatewise degree n . A feasible coefficient array has $c_{\alpha i} \geq 0$ and $\sum_i c_{\alpha i} = 1$ for every $\alpha \in \{0, \dots, n\}^a$. It specifies the independent test

$$\kappa_i^c(x) = \sum_{\alpha} c_{\alpha i} \beta_{\alpha,n}(x).$$

Define

$$d_n = \max_c \min_{x \in C} \sum_{\alpha, i} c_{\alpha i} \beta_{\alpha,n}(x) \ell_i(x). \quad (21.3)$$

Theorem 21.3 (Intrinsic adaptive degree and support bounds). *The maxima in (21.3) are attained and $d_n \leq d_{n+1} \leq \delta$. For $n \geq 1$,*

$$0 \leq \delta - d_n \leq \frac{4Ra}{\sqrt{n}}. \quad (21.4)$$

There is an array with at most $(n+1)^a$ nonzero coefficients, all equal to one, whose minimum score is at least $\delta - 4Ra/\sqrt{n}$. Its test has coordinatewise degree n , and its score degree at most $n+1$. Furthermore

$$d_n = \min_{\mu \in \mathcal{P}(C)} \sum_{\alpha} \max_i \int_C \beta_{\alpha,n}(x) \ell_i(x) d\mu(x), \quad (21.5)$$

and a minimizing measure can be supported on at most $(n+2)^a$ points. In particular a strict level $L < \delta$ admits an adaptive test of positive uniform margin with $n > (8Ra/(\delta - L))^2$ and at most $(n+1)^a$ occupied coefficient entries. These bounds depend on the observable image and its metric, not on the number of stochastic rows in a realization.

Proof. The coefficient space is a finite product of compact simplexes, and the minimum of a jointly continuous function over compact C is continuous in its coefficients. This proves attainment. Bernstein degree elevation replaces each coefficient by convex combinations of adjacent coefficients, preserving every simplex constraint and the represented function. It proves monotonicity. All scores are convex combinations of the $\ell_i(x)$, proving the upper bound by δ .

At each grid point α/n choose an index i_{α} maximizing ℓ_i there. The choice is made even at infeasible grid points; only the final score is minimized over C . Give that index coefficient one and the others zero. For fixed x let A_j be independent $\text{Bin}(n, x_j)$ variables. If i_* maximizes at x ,

$$\begin{aligned} \max_i \ell_i(x) - \mathbb{E} \ell_{i_A}(x) &\leq \mathbb{E} \{ \ell_{i_*}(x) - \ell_{i_*}(A/n) + \ell_{i_A}(A/n) - \ell_{i_A}(x) \} \\ &\leq 8R \sum_j \mathbb{E} |A_j/n - x_j| \leq \frac{4Ra}{\sqrt{n}}. \end{aligned}$$

Here the omitted middle difference is nonpositive, and the last bound uses $\text{Var}(A_j/n) \leq 1/(4n)$. This proves the rate and the sparse coefficient assertion.

For the last formula replace the minimum over C by a minimum over $\mathcal{P}(C)$. The pairing is bilinear and continuous, and both feasible sets are compact convex, so the compact minimax theorem [18] exchanges the extrema. Optimizing each coefficient simplex gives (21.5). The integrands are polynomials of coordinatewise degree at most $n+1$. Preserve all monomial moments of that degree by Carathéodory's theorem. There are $(n+2)^a$ monomials including the constant; their moment vectors lie in an affine hyperplane of dimension at most $(n+2)^a - 1$. A measure with at most $(n+2)^a$ atoms therefore has exactly the same objective. The strict-margin assertion follows from the rate with error less than $(\delta - L)/2$. $\square$

The score is a nonlinear function of behavior although each event functional is affine. This is the reason the measure in (21.5) cannot be replaced by its behavior barycenter. For $n = 0$ it can: d_0 is precisely the constant-test convex relaxation. In the two-time unmarked example of Proposition 20.5, the target belongs to the convex hull of private behaviors, so $d_0 = 0$, whereas $\lim_n d_n = \sqrt{2} - 1$. Thus the hierarchy recovers a gap which every constant test misses.

Remark 21.4 (What the size bound measures). The bound counts the nonzero entries of the independent adaptive test and controls its behavior-coordinate degree. It does not bound the cost of verifying its minimum on the generally nonconvex set C , the bit lengths of a rational chart, or the size of a description of C . Distinct row presentations of the same metric behavior image satisfy the same existence bound, although their coefficient arrays need not coincide. For the private fixed-target law array one may use $R \leq 1$ with the maximum total-variation metric. For the visible signed-discrepancy array the uniform bound $R \leq 2$ is sufficient. The old row-cube positivity certificate is a separate verification engine, with its own coordinate-dependent complexity.

21.3. A three-weight certificate invisible to constant tests.

Proposition 21.5 (A finite intrinsic nonlinear witness). *Consider a private width-one simulator producing two independent bits, with probabilities p, q , for the target which assigns probability $1/2$ to each of 00, 11. Put $u = pq$, $v = (1-p)(1-q)$ and $J(t) = 3t^2 - 2t^3$. The three behavior-dependent weights*

$$\kappa_1 = J(v), \quad \kappa_2 = J(u), \quad \kappa_3 = 1 - J(u) - J(v)$$

are nonnegative and sum to one. Against the event discrepancies $1/2 - u, 1/2 - v, 1 - u - v$, respectively, their score G satisfies

$$G \geq \frac{2}{5} + \frac{9}{8960}. \quad (21.6)$$

The certificate uses cubic weights in observable probabilities, a nine-term nonnegative derivative identity, and seventeen positive rational Bernstein coefficients in one variable. Every constant mixture of event discrepancies has minimum at most zero.

Proof. We have $u + v \leq 1$. Since J is increasing on $[0, 1]$ and $J(1-t) = 1 - J(t)$, the weights are feasible. Their score is

$$G = 1 - u - v + J(u)(u - 1/2) + J(v)(v - 1/2).$$

Set $a = (p + q - 1)^2$ and $b = (p - q)^2$. Then $a, b \geq 0$ and $a + b \leq 1$. Direct expansion gives

$$\begin{aligned} 64G &= 27 - 12a + 38a^2 - 20a^3 - a^4 + 36b - 60ab + 36a^2b + 4a^3b \\ &\quad + 6b^2 - 12ab^2 - 6a^2b^2 - 4b^3 + 4ab^3 - b^4. \end{aligned}$$

Let $c = 1 - a - b$. Another polynomial identity, valid when $a + b + c = 1$, is

$$\begin{aligned} 16\partial_b G &= 4a^3 + 6a^2c + 12ab^2 + 24abc + 12ac^2 \\ &\quad + 8b^3 + 30b^2c + 30bc^2 + 9c^3 \geq 0. \end{aligned}$$

Thus $G(a, b) \geq G(a, 0)$. The univariate polynomial

$$G(a, 0) - \frac{2}{5} = \frac{7}{320} - \frac{3}{16}a + \frac{19}{32}a^2 - \frac{5}{16}a^3 - \frac{1}{64}a^4$$

has degree-16 Bernstein coefficients $z_k/349440$, where

$$\begin{aligned} (z_0, \dots, z_{16}) &= (7644, 3549, 1183, 351, 855, 2494, 5064, 8358, \\ &\quad 12166, 16275, 20469, 24529, 28233, 31356, 33670, 34944, 34944). \end{aligned}$$

The Bernstein basis is nonnegative and sums to one. Its smallest coefficient is $351/349440 = 9/8960$, proving (21.6). All identities are rational polynomial identities; no numerical minimization is used. Finally the target is the barycenter of the two feasible deterministic laws 00 and 11. An affine mixture of signed event discrepancies has average zero at these two laws and hence minimum at most zero. The empty event supplies the reverse constant-test bound zero. $\square$

For comparison, the affine weights $v, u, 1 - u - v$ already give the smaller bound $3/8$, with the exact identity

$$G_{\text{affine}} - \frac{3}{8} = \frac{1}{2}(p - q)^2 + 2(p - 1/2)^2(q - 1/2)^2.$$

Neither certificate changes the private feasible set. The cubic certificate is an independent test chosen once for the whole behavior image, not a mixture chosen anew by an optimizer's unknown row values. Its evaluation does use the revealed behavior, as specified in the verification game. The 17-coefficient positivity check is classical; the distinction from constant tests is the behavior-dependent feasible weight family and its resource-sensitive minimum.

The infinite-dimensional representation above holds for continuous f_i on any compact metric space: its proof uses a partition of unity, not causal factorization. The measure minimum is attained at a Dirac mass at a minimizer of F . The finite-level polynomial theorem has a different content. It is an existence-size bound for revealed-law verification; the reset-audit theorem of Section 18 imposes an explicit finite-data restriction.

21.4. Strict certificates and effective physical input. Given rational finite instruments, a rational chart and the sparse array constructed above, its strict minimum can be certified by real quantifier elimination on the original row constraints. Alternatively, pull the score back through the original polynomial row map and certify its strict positivity using degree-elevated Bernstein coefficients. The pullback changes the verification degree and is not covered by the behavior-coordinate bound. Generic quantifier elimination also decides the exact finite semialgebraic boundary; this classical observation does not assert that a finite nonnegative Bernstein expansion exists at $L = \delta$.

Proposition 21.6 (Uniform effective presentation input). *Suppose a single algorithm, on input j , returns rational finite instruments E_j, F_j and nonnegative rational error enclosures a_j, c_j for two-sided stateless marked feedback presentations of E, F . Suppose $a_j + c_j \rightarrow 0$. All alphabets, widths, selector laws and finite subroutines must be computable uniformly from j and the requested signature. Then, for every positive rational η , an algorithm returns a rational interval of width below η containing $\delta_b(E, F)$ at the same signature.*

Proof. Enumerate j until $a_j + c_j < \eta/8$. This terminates without a known rate because the rational enclosures themselves tend to zero. For the selected finite problem, enumerate rational same-budget upper tables and strict rational lower certificates, dovetailing all finite checks; include the always valid lower bound zero. Compactness, density and strict completeness provide an interval of width below $\eta/2$. Enlarge it by $a_j + c_j$ at both ends using stateless comparison. The enlarged width is below $3\eta/4$. Every enumeration and comparison is uniform by the hypotheses. A mere sequence of individually computable presentations without a uniform enumeration is not an input satisfying this proposition. $\square$

22. AN EXACT ALGORITHM FOR THE OBSERVABLE CHART

The chart in Lemma 21.2 can be found without taking its dimension as an oracle. We state the input and output explicitly. Rational polynomials are encoded by finite lists of monomial exponent vectors and rational coefficients in reduced integer form. A product of simplexes is encoded by the number of entries in each row. Eliminating the last entry of every nontrivial row leaves a full-dimensional rational polytope $\Omega \subset \mathbb{R}^r$. Zero-dimensional factors are omitted.

Proposition 22.1 (Effective observable quotient and pullback). *The input consists of the preceding simplex description, a rational polynomial map $b : \Omega \rightarrow \mathbb{R}^N$, and finitely many rational affine functionals f_i on $\mathbb{R}^N$. There is a terminating algorithm which returns its observable affine*

dimension a , a rational basis and coordinate map for the observable quotient, and, when $a > 0$, rational parameter points $q_0, q_1, \dots, q_a$ with the following properties. With

$$z(q) = (f_i(b(q)) - f_i(b(q_0)))_i, \quad v_j = z(q_j),$$

the vectors v_j are a basis of the observable difference space, and all coordinates $t(q)$ in that basis satisfy $|t_j(q)| < 2$. The output also contains the rational polynomial cube map $x(q) = (t(q) + 2)/4$, a rational upper bound for the observable diameter, and the exact pullback of any subsequently supplied rational polynomial in x . The algorithm outputs the universal determinant formula it has decided. It makes no polynomial-time or coefficient-height claim.

Proof. Choose a rational relative interior point q_0 of the input simplexes. Expand the vector polynomial $z(q)$ in the free row coordinates. Let V be the rational span of all its coefficient vectors, including its constant coefficient. This equals $\text{span } z(\Omega)$: a linear functional vanishes on $z(\Omega)$ exactly when its polynomial vanishes on the nonempty interior of Ω , hence identically, hence on every coefficient. Gaussian elimination therefore gives $a = \dim V$ and a rational basis matrix U . Choosing independent rows gives a rational left inverse L with $LU = I_a$. Thus $y(q) = Lz(q)$ provides exact quotient coordinates. If $a = 0$, all discrepancies are constant and no chart is required.

For $a > 0$ let

$$D_0 = \max_{u_1, \dots, u_a \in \Omega} |\det(y(u_1), \dots, y(u_a))|.$$

Compactness gives a maximum, and spanning gives $D_0 > 0$. Enumerate rational tuples $(q_1, \dots, q_a)$ in Ω^a . Membership and their determinant d are decided by rational arithmetic. For $d \neq 0$, decide the first-order real formula

$$\forall u_1, \dots, u_a \in \Omega : \quad \det(y(u_1), \dots, y(u_a))^2 < 4d^2. \quad (22.1)$$

Quantifier elimination terminates for each formula. Density of rational points and continuity of determinant imply that some enumerated tuple has $|d| > D_0/2$. For such a tuple the strict formula is true, so the search terminates. No computation of an exact algebraic maximizer is needed.

Invert the rational matrix with columns $y(q_j)$ to obtain $t(q)$. Cramer's rule and (22.1) give $|t_j(q)| < 2$ for every q . In the observable norm $\|z\|_* = \max_i |z_i|$, each v_j has norm at most the diameter of $z(\Omega)$. An upper bound for that diameter can be produced, for example, by bounding each free coordinate in $[0, 1]$ and summing the absolute polynomial coefficients for $f_i(b(q)) - f_i(b(q'))$. This bound is rational and computable. Rational matrix multiplication and substitution give $x(q)$ and every requested pullback. They also give its full sparse coefficient encoding. All searches have now been specified by finite encodings and terminating operations. $\square$

There is a small but important distinction between the quotient and a projection onto a smaller set of simulator rows. The former identifies exactly the directions invisible to the chosen event family; the original row constraints remain present in every universal positivity test. The chart need not identify equivalent tasks after a change of marks or of information signature. Those are changes to the comparison problem.

23. A MARKED RESOURCE GAP GENERATED BY A BINARY COLLISION

We now realize a private–hidden–visible separation in a physical experiment whose information vanishes when the collision is removed. The controller chooses a later sensor gate after its first report. The gate controls acquisition, not the mechanical collision law. Both the source and the target carry the same actual, unobserved mark.

23.1. The exact marked two-report experiment. Let $B, W \in \{0, 1\}$ have the joint distribution

$$\mathbb{P}(B = b, W = w) = \begin{cases} 5/16, & b = w, \\ 3/16, & b \neq w. \end{cases} \quad (23.1)$$

The blind source E_* has two constant reports and actual mark W . The target F_* reports $Y_1 = B$, receives $A \in \{0, 1\}$ after that report, and reports $Y_2 = AB$. The feedback tester may choose A as any function of its earlier report. A width-one private simulator cannot reread its own first emission, except through the current gate A . Hidden selectors persist but are not revealed; visible selectors are included in the joint target law independently of (B, W) . Put

$$d_* = \sqrt{5/2} - 9/8.$$

Theorem 23.1 (An exact marked causal separation). *At the width profile $(1, 1)$,*

$$\delta^P(E_*, F_*) = d_*, \quad \delta_2^h(E_*, F_*) = 1/8, \quad \delta_r^v(E_*, F_*) = d_* \quad (r \geq 1). \quad (23.2)$$

The hidden upper bound is realized by two deterministic tables. The private upper bound is realized by two independent Bernoulli variables with parameter $\sqrt{5/8}$, followed by the gate. These bounds hold against every feedback rule, not just open-loop gates.

Proof. First choose the constant gate $A = 1$. The two private emissions are independent Bernoulli variables with parameters p, q and are independent of W . Let

$$u = pq, \quad v = (1 - p)(1 - q), \quad a = 3/8, \quad b = 5/8.$$

The overlap of the target and simulated marked laws is $H(u) + H(v)$, where

$$H(t) = \frac{1}{2} \{\min(t, a) + \min(t, b)\}.$$

Indeed the target is supported on equal report pairs, and its two W -masses at each equal pair are $3/16$ and $5/16$. Total variation is one minus this overlap.

Cauchy-Schwarz gives $\sqrt{u} + \sqrt{v} \leq 1$. By symmetry suppose $u \geq v$. If $u \leq 1/4$ the overlap is at most $1/2$. If $u \geq 1/4$, then $v \leq (1 - \sqrt{u})^2 \leq 1/4 < a$, so the overlap is at most

$$H(u) + (1 - \sqrt{u})^2.$$

This function is convex on each of $[1/4, a]$ and $[a, b]$ and nonincreasing on $[b, 1]$. Its maximum is therefore attained at $1/4, a$ or b . The values at a and b are $1 + 2a - 2\sqrt{a}$ and $3/2 + b - 2\sqrt{b}$ respectively. The second is larger, since

$$(3/2 + b - 2\sqrt{b}) - (1 + 2a - 2\sqrt{a}) = 3/8 - \frac{1/2}{\sqrt{a} + \sqrt{b}} > 0,$$

using $\sqrt{a} > 3/5$, $\sqrt{b} > 3/4$. It also exceeds $1/2$. Taking $p = q = \sqrt{b}$ attains this maximum, and hence the minimum error is $1 - (3/2 + b - 2\sqrt{b}) = d_*$.

For any hidden selector, the all-on report pair is still independent of W . If its probabilities of 11 and 00 are u, v , then $u + v \leq 1$. Concavity and monotonicity of H give

$$H(u) + H(v) \leq 2H((u + v)/2) \leq 2H(1/2) = 7/8.$$

A fair hidden bit J , emitting J twice, attains this overlap. This requires precisely two deterministic selector tables and gives error $1/8$. In particular hiding a seed does not erase the actual-mark obstruction.

These all-on bounds are lower bounds for the feedback problem. For the upper bounds first generate the corresponding all-on pair (X_1, X_2) , emit X_1 , and after receiving A emit AX_2 . For a deterministic feedback $A = g(X_1)$ the full gated marked transcript is the pushforward of the all-on marked pair. Applying the same map to the target pair cannot increase total variation. The construction is executable at width one: the second fresh Bernoulli variable is independent in the

private case, and the bit J is the priced persistent hidden selector in the hidden case. Randomized feedback follows by conditioning on its independent random table.

Finally, in the visible case the two compared laws are

$$\sum_j \lambda_j \delta_j \otimes P_{g_j}^{q_j} \quad \text{and} \quad \sum_j \lambda_j \delta_j \otimes Q_{g_j}.$$

The target selector is independent of (B, W) . The all-on gate in every fiber has error at least d_* in every private fiber. Joint variation is the weighted sum of these fiber errors. Ignoring the visible selector and using the private optimizer attains d_* . $\square$

23.2. A full-dimensional collision preparation. Consider two labeled equal-mass hard spheres in $\mathbb{R}^3$ with contact distance d . The center of mass $Q = (q_1 + q_2)/2$ ranges over any fixed rational position box. Write $r = q_1 - q_2$, $g = v_1 - v_2$, and $V = (v_1 + v_2)/2$. Conditional on (B, W) in (23.1), take the uniform product preparation on

$$\begin{aligned} r &= (-a_0, (2B - 1)b_0, c_0), \quad a_0 \in [29/10, 31/10], & b_0 &\in [9/20, 11/20], \quad (23.3) \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & |V_z - (2W - 1)| &\leq 1/100. \end{aligned}$$

The boxes have positive volume in the full phase space. The actual mark is $W = \mathbf{1}_{\{V_z > 0\}}$; total momentum conservation retains it. The informative quantity will instead be the transverse velocity created at a collision. Put

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector which is identically zero on the incoming transverse-velocity range.

Lemma 23.2 (Uniform transverse collision signal). *For every preparation in (23.3) and every $d \in [9/10, 11/10]$, exactly one collision occurs before time 3. Its time t_c lies in $(4/5, 3/2)$, and afterwards*

$$(2B - 1)v_{1,y} > 1/2, \quad (2B - 1)h(v_{1,y}) > 1/4. \quad (23.4)$$

For $0 < d \leq 2/5$ no collision occurs before time 3, and the detector is zero throughout that interval.

Proof. Up to a collision $r(t) = r + tg$. For $t \leq 4/5$, its x coordinate is at most $-29/10 + (4/5)(21/10) = -61/50$, so its norm exceeds $11/10$. At time $3/2$ the absolute coordinate bounds are $1/4$, $113/200$ and $1/40$, whose squared sum is less than $(9/10)^2$. Thus a first contact occurs in the stated interval. At the free zero of the x coordinate, whose time is at most $31/19$, the transverse norm is less than $9/10$ as well. The first contact therefore has $n_x < 0$, where $n = r(t_c)/d$. The entry contact is strictly incoming. The bounds above give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B - 1)n_y \geq \frac{87}{220}.$$

It follows that $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$. The equal-mass elastic collision rule is

$$v'_1 = V + g/2 - (g \cdot n)n, \quad g' = g - 2(g \cdot n)n.$$

Consequently

$$(2B - 1)v'_{1,y} > -3/200 + \frac{1443}{1000} \frac{87}{220} > 1/2.$$

The relative velocity is outgoing, $r(t_c) \cdot g' > 0$, so the squared separation increases strictly for all later times. There are only two spheres and no walls; no second collision is possible. This proves (23.4). For $d \leq 2/5$ and $0 \leq t \leq 3$, $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$. Thus there is no collision. The incoming $|v_{1,y}| \leq 3/200$ then makes h zero. $\square$

At time 2 the physical target $F_{d,\sigma}$ reports

$$Y_1 = \mathbf{1}_{\{h(v_{1,y}(2)) + \sigma\xi_1 > 0\}}.$$

It then receives a gate $A \in \{0, 1\}$, and at time 3 reports

$$Y_2 = A\mathbf{1}_{\{h(v_{1,y}(3)) + \sigma\xi_2 > 0\}},$$

where the ξ_i are independent standard normal variables, independent of the preparation. The source E_d has the same preparation, evolution, control interface and actual mark, but emits two constant reports. The gates do not change the particle motion. This is a feedback acquisition experiment: taking $A = 0$ suppresses later information and taking $A = 1$ exposes another noisy observation of a collision-generated quantity.

Theorem 23.3 (Collision-sensitive physical resource separation). *Let $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/12$. At retained width $(1, 1)$ the physical deficiencies obey, uniformly in d ,*

$$\begin{aligned} |\delta^P(E_d, F_{d,\sigma}) - d_*| &< 1/300, & |\delta_r^Y(E_d, F_{d,\sigma}) - d_*| &< 1/300 \quad (r \geq 1), \\ |\delta_2^h(E_d, F_{d,\sigma}) - 1/8| &< 1/300. \end{aligned} \quad (23.5)$$

In particular private and visible error exceed $9/20$, while hidden two-selector error is less than $13/100$. More generally the errors in these three comparisons are at most $2\Phi(-1/(4\sigma))$, which tends to zero as $\sigma \downarrow 0$. For $0 < d \leq 2/5$ all three deficiencies are zero. The interacting and noninteracting regimes use the same preparation and detector. The conclusions are unchanged by simultaneously scaling positions, diameter and observation times by any positive factor, with velocities and detector noise unchanged.

Proof. Couple the ideal and physical target by the same (B, W) and the same feedback random table. Lemma 23.2 bounds each potential bit error by $\Phi(-1/(4\sigma))$. With probability at least $1 - 2\Phi(-1/(4\sigma))$ both potential reports agree with their ideal bits; the gates and all emitted transcripts then agree as well. This gives a marked feedback variation bound uniform over all gates. The channels in both directions are the identity on the binary report alphabet and have one retained state. The blind source is already exactly E_* . Stateless comparison therefore changes each fixed-budget deficiency by at most the stated error, without adding memory or hiding an exposed selector.

For completeness, the elementary Gaussian tail bound $\Phi(-x) \leq e^{-x^2/2}/(x\sqrt{2\pi})$ at $x = 3$ gives $\Phi(-3) < 1/600$: use $e^{9/2} > 80$ and $\sqrt{2\pi} > 5/2$. Hence the total error is below $1/300$. The inequalities $d_* - 1/300 > 9/20$ and $1/8 + 1/300 < 13/100$ follow by squaring positive rational bounds.

When $d \leq 2/5$, both potential reports are independent fair noise bits and are independent of W . A private width-one simulator draws those two bits and applies the current gate, exactly reproducing every feedback transcript. Hidden and visible versions can ignore their selectors. Finally elastic scattering is invariant under $(q, d, t) \mapsto (sq, sd, st)$ with v unchanged. This proves the scaling statement; it is geometric similarity, not a particle-number or Boltzmann–Grad limit. $\square$

The theorem combines three effects. Collisions create the sign of the informative transverse velocity. A gate chosen after the first report changes future information. A retained hidden bit, but not a visible one, permits temporal correlation without charging the width-one register. The actual mark remains partially inaccessible in every signature. The conserved longitudinal momentum labels that mark; it does not generate the observed transverse signal.

23.3. Uniformly effective finite presentations. The binary physical interface is already finite, but its probabilities are not generally rational. We make their effective approximation explicit rather than treating individual computability as a uniform algorithm.

Proposition 23.4 (Collision-chart rational presentations). *For computable $d \in [9/10, 11/10]$ and $\sigma > 0$, one algorithm produces rational marked causal instruments and certified two-sided stateless errors tending to zero for $(E_d, F_{d,\sigma})$. On a compact computable parameter interval with $\sigma \geq \sigma_0 > 0$, the errors and the construction are uniform. Thus Proposition 21.6 applies at every fixed finite state and selector signature, not only at $(1, 1)$.*

Proof. On every preparation box the entry time and outgoing velocity are

$$t_c = \frac{-r \cdot g - \sqrt{(r \cdot g)^2 - |g|^2(|r|^2 - d^2)}}{|g|^2}, \quad n = (r + t_c g)/d, \quad v'_1 = V + g/2 - (g \cdot n)n.$$

The positive square root is separated from zero: it equals $-d(g \cdot n) > (9/10)(1443/1000)$. All denominators are likewise bounded away from zero. These explicit bounds and the compact boxes allow interval differentiation of the displayed formulas, producing a rational common Lipschitz upper bound C_{geo} for $v'_{1,y}$ in the packet variables and d . This computation terminates because only addition, multiplication, division by separated positive quantities and a separated square root occur.

Conditional on the microscopic preparation the two potential all-on bits are independent Bernoulli with probability $p = \Phi(h(v'_{1,y})/\sigma)$. Their four probabilities are p^2 , $p(1-p)$, $p(1-p)$ and $(1-p)^2$. The normal distribution function is uniformly computable on bounded intervals, for example by integrating its convergent exponential series with a remainder bound. Its Lipschitz constant is less than one. Together with the 1-Lipschitz detector and the geometric bound, this yields a computable Lipschitz constant at most $2C_{\text{geo}}/\sigma_0$ for these four integrands on fixed-noise compact parameter sets. Equal-box Riemann sums of mesh h therefore have explicit error at most a computable constant times h , plus the chosen rational evaluation error. Parameter uncertainty is included by its interval enclosure.

Integrate the four all-on probabilities separately on each packet, with the exact rational weights (23.1). For each W , round its conditional probability vector to a rational simplex point on a sufficiently fine grid; the rounding error is bounded by the sum of the coordinate errors and the grid mesh. Keep $\mathbb{P}(W = 0) = \mathbb{P}(W = 1) = 1/2$ exact. Sample this finite joint law once at initialization, emit its first bit, and gate its second bit. This is a finite latent realization of the target instrument, not a simulator with extra free state. It makes every causal prefix consistent and never divides by a possibly small prefix probability. Every gated law is a pushforward of the all-on law, so the same marked variation bound holds uniformly under feedback. Identity report channels give both stateless directions. Choosing summable evaluation and mesh errors produces the asserted uniform sequence. $\square$

24. UNIFORM STABILITY OF OPTIONAL PROJECTIONS

Marked comparison controls more than terminal bounded risk when the compared laws share their latent variable. We prove a whole-process estimate for the resulting posterior martingales. The distinction between a probabilistic coupling and an executable causal channel is important in this section.

Let Z take values in a standard Borel space and let Y take values in a standard Borel canonical path space on $[0, T]$, with a right-continuous observation filtration. Consider equivalent joint probability measures P, Q on (Z, Y) , with the same Z -marginal ν . Use the usual completions, which agree for equivalent output laws. For a bounded measurable latent variable $f(Z)$ write

$$M_t^P = \mathbb{E}_P[f(Z) \mid \mathcal{F}_t^Y], \quad M_t^Q = \mathbb{E}_Q[f(Z) \mid \mathcal{F}_t^Y].$$

Take their right-continuous martingale versions. All suprema below include T and can be evaluated on a countable dense time set.

Theorem 24.1 (Marked variation controls the whole posterior process). *Let $\varepsilon = \|P - Q\|_{\text{TV}}$ and $|f| \leq 1$. When both posterior functions are evaluated on the same canonical output path,*

$$\mathbb{E}_Q \sup_{t \leq T} |M_t^P(Y) - M_t^Q(Y)|^2 \leq 80\varepsilon. \quad (24.1)$$

There is a coupling of (Z, Y^P) and (Z, Y^Q) which retains exactly the same latent variable and satisfies

$$\mathbb{P}(Y^P \neq Y^Q) \leq \varepsilon, \quad \mathbb{E} \sup_{t \leq T} |M_t^P(Y^P) - M_t^Q(Y^Q)|^2 \leq 84\varepsilon. \quad (24.2)$$

The coupling is not asserted to be causal or to have finite state. The variation bound itself, not this coupling, is used for resource transport.

Proof. Let $D = dP/dQ$, $D_t = \mathbb{E}_Q[D \mid \mathcal{F}_t^Y]$, and set

$$A_t = \mathbb{E}_Q[(D - 1)f \mid \mathcal{F}_t^Y], \quad B_t = D_t - 1.$$

Bayes' formula gives

$$M_t^P - M_t^Q = \frac{A_t - B_t M_t^Q}{1 + B_t}.$$

Since $\mathbb{E}_Q|D - 1| = 2\varepsilon$, both martingales A, B have terminal absolute expectation at most 2ε . Doob's weak L^1 inequality gives, for $0 < u \leq 2$,

$$\begin{aligned} Q\{\sup_t |M_t^P - M_t^Q| > u\} &\leq Q\{\sup_t |B_t| > 1/2\} + Q\{\sup_t |A_t| > u/4\} + Q\{\sup_t |B_t| > u/4\} \\ &\leq 4\varepsilon + 16\varepsilon/u. \end{aligned}$$

Integrating $2u$ times this bound on $(0, 2)$ gives $16\varepsilon + 64\varepsilon = 80\varepsilon$, proving (24.1). No tightness conclusion is being inferred from finite-dimensional convergence.

Disintegrate both joint laws over their common latent marginal. Measurable maximal coupling of the two output kernels, conditional on Z , has disagreement probability

$$\int \|P(dy \mid z) - Q(dy \mid z)\|_{\text{TV}} d\nu(z) = \|P - Q\|_{\text{TV}}.$$

Such a coupling can be constructed by using the minimum of the two kernel densities with respect to their sum, then their residual kernels. It is measurable on standard Borel spaces. On the event of identical full output paths use (24.1) with the Q -marginal; on its complement the squared difference is at most four. This proves (24.2). The maximal coupling may depend on the full output path; it has no asserted online implementation. $\square$

Corollary 24.2 (Changing-filtration convergence from marked comparison). *Let P_n and P be equivalent joint laws with the same latent marginal ν , and $\|P_n - P\|_{\text{TV}} \rightarrow 0$. They admit a common probability-space realization $(Z, Y, Y^1, Y^2, \dots)$ such that, for every $f \in L^2(\nu)$,*

$$\mathbb{E} \sup_{t \leq T} \left| \mathbb{E}[f(Z) \mid \mathcal{F}_t^{Y^n}] - \mathbb{E}[f(Z) \mid \mathcal{F}_t^Y] \right|^2 \rightarrow 0.$$

The same realization gives convergence uniformly in time in probability for $f \in L^1(\nu)$. It also permits $f_n \rightarrow f$ in $L^2(\nu)$, respectively in $L^1(\nu)$, in the first conditional expectation.

Proof. Glue the conditional maximal couplings of (Z, Y^n) with the fixed reference (Z, Y) by taking their conditional kernels independently on a countable product. The coupling does not depend on f . For the truncation $f^{(K)} = (-K) \vee (f \wedge K)$, Theorem 24.1 bounds the middle difference in the L^2 norm of the supremum by $K\sqrt{84\varepsilon_n}$. Doob's L^2 inequality bounds each truncation tail by $2\|f - f^{(K)}\|_{L^2(\nu)}$. The triangle inequality therefore gives

$$\left\| \sup_t |M_t^{P_n, f} - M_t^{P, f}| \right\|_2 \leq 4\|f - f^{(K)}\|_2 + K\sqrt{84\varepsilon_n}.$$

First let $n \rightarrow \infty$, then $K \rightarrow \infty$. For L^1 use instead the weak maximal inequality for each tail and the bounded-case estimate for the middle term. An additional difference $f_n - f$ is handled by the same maximal inequalities under P_n , whose latent marginal remains ν . $\square$

24.1. A microscopic sufficient condition. Let $(\Phi_t)_{t \in \mathbb{R}}$ be an almost-everywhere regular hard-sphere flow on a labeled phase space, with invariant probability μ . Fix a prepared law ν with $d\nu/d\mu \leq C$, and bounded real observables h, h_n . These may be collision-sensitive tagged observables. Put $\eta_n = \|h_n - h\|_{L^2(\mu)}$. With a common latent initial state $Z \sim \nu$, consider the continuous observation laws

$$\begin{aligned} Y_t &= \int_0^t h(\Phi_s Z) ds + \sigma B_t, \\ Y_t^n &= \int_0^t h_n(\Phi_s Z) ds + \sigma B_t^n, \quad \sigma > 0, \end{aligned}$$

where the Brownian noise is independent of Z in each model. The notation does not prescribe a synchronous coupling.

Theorem 24.3 (Fixed-flow observation-approximation limit). *The joint latent-output laws are equivalent and satisfy*

$$\|P_n - P\|_{\text{TV}} \leq \frac{\sqrt{CT}}{2\sigma} \eta_n. \quad (24.3)$$

For every $|f(Z)| \leq 1$ they admit a latent-preserving coupling with

$$\mathbb{E} \sup_{t \leq T} |M_t^{P_n, f} - M_t^{P, f}|^2 \leq \frac{42\sqrt{CT}}{\sigma} \eta_n. \quad (24.4)$$

In particular $\eta_n \rightarrow 0$ supplies whole-process optional-projection convergence, not merely convergence at finitely many times. This applies when h_n are orbit-mollified graph-core approximants to any bounded $h \in L^2(\mu)$.

Proof. Conditional on Z , the signals are deterministic bounded drifts. The exponential change-of-measure identity for Brownian motion gives

$$D_{\text{KL}}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |(h_n - h)(\Phi_s Z)|^2 ds \leq \frac{CT}{2\sigma^2} \eta_n^2.$$

The last inequality uses $\nu \leq C\mu$ and invariance of μ ; it does not require mixing, differentiability across a collision, or a claim that a tagged velocity belongs to the generator domain. Bounded drifts ensure equivalence and justify the exponential identity. Pinsker's inequality gives (24.3). Applying Theorem 24.1 proves (24.4).

For the last assertion choose a nonnegative smooth approximate identity ρ_n on the time line and set $h_n = \int \rho_n(s) h \circ \Phi_s ds$. Strong continuity of the Koopman group in $L^2(\mu)$ gives $\eta_n \rightarrow 0$ and $\|h_n\|_\infty \leq \|h\|_\infty$. Repeated integration by parts puts h_n in every power domain of the generator, with derivatives obtained from $\rho_n^{(k)}$. This is the orbit-core construction already used in the microscopic domain theory; the new conclusion is the uniform posterior estimate. $\square$

Sampling these observation paths at finitely many checkpoints and applying a common causal output postprocessing are contractions in marked variation. Thus the identity report comparison transfers every bounded finite-register risk at its original state and selector budgets with error bounded by (24.3). That is the resource-comparison consequence. The latent-preserving coupling in (24.4) has a different role: it realizes convergence of the changing observation filtrations and is not used as a finite-register simulator.

The theorem discharges a sufficient conditional-law hypothesis for optional projection in a genuine microscopic observation class. The constants depend on the fixed preparation, time

and noise level. No particle-uniform kinetic estimate, changing microscopic path limit, limiting stochastic-exponential identification or nonlinear BSDE limit is implied without its additional hypotheses. The whole-process estimate is what replaces an unsupported appeal to Doob's inequality as a tightness argument.

25. CHANGING TRAJECTORIES, LIKELIHOODS, AND CONDITIONAL CONSUMERS

25.1. A common preparation, not a common flow. Let $(\mathcal{Z}, \nu)$ be a standard Borel probability space, and let $H, H_n : [0, T] \times \mathcal{Z} \rightarrow \mathbb{R}$ be jointly measurable, with $|H|, |H_n| \leq M$. They may be observations of different flows on different phase spaces, pulled back by specified preparation maps from $\mathcal{Z}$. No common microscopic generator or invariant measure is assumed. Set

$$\eta_n^2 = \int_{\mathcal{Z}} \int_0^T |H_n(t, z) - H(t, z)|^2 dt d\nu(z).$$

On $\mathcal{Z} \times C([0, T])$ let R be the law of independent $Z \sim \nu$ and $Y = \sigma B$, where $\sigma > 0$. Define

$$L_t^n = \exp \left\{ \frac{1}{\sigma} \int_0^t H_n(s, Z) dB_s - \frac{1}{2\sigma^2} \int_0^t H_n(s, Z)^2 ds \right\}, \quad (25.1)$$

$$dP_n = L_T^n dR. \quad (25.2)$$

The reference model P, L uses H . Conditional on Z , these are Gaussian shifts, so under P_n the observed path has law $Y_t = \int_0^t H_n(s, Z) ds + \sigma B_t^n$. Boundedness gives true strictly positive density martingales. Write $\mathcal{F}_t$ for the usual canonical observation filtration and $\mathcal{H}_t = \sigma(Z, B_s : s \leq t)$ for the larger reference filtration. Equivalent laws have the same completed observation stopping times.

Theorem 25.1 (Likelihood and posterior transport for changing paths). *The measures P_n, P, R are equivalent and have common Z -marginal ν . With $A = M^2 T / \sigma^2$ and $\varepsilon_n = \min\{1, \eta_n / (2\sigma)\}$, one has*

$$\|P_n - P\|_{\text{TV}} \leq \varepsilon_n, \quad (25.3)$$

$$\mathbb{E}_R \sup_{t \leq T} |L_t^n - L_t|^2 \leq \frac{8}{\sigma^2} e^{3A} \eta_n^2. \quad (25.4)$$

For $|f(Z)| \leq K$ put

$$N_t^{n,f} = \mathbb{E}_R[L_T^n f(Z) \mid \mathcal{F}_t], \quad \ell_t^n = N_t^{n,1}.$$

Then

$$\mathbb{E}_R \sup_t |N_t^{n,f} - N_t^f|^2 \leq \frac{8K^2}{\sigma^2} e^{3A} \eta_n^2. \quad (25.5)$$

In particular this applies to the projected likelihood ℓ^n . For every $|f| \leq 1$, the posterior functions satisfy the same-path bound $80\varepsilon_n$ of Theorem 24.1, and there is one latent-preserving coupling, independent of f , on which

$$\mathbb{E} \sup_t |\mathbb{E}_{P_n}[f(Z) \mid \mathcal{F}_t^{Y^n}] - \mathbb{E}_P[f(Z) \mid \mathcal{F}_t^Y]|^2 \leq 84\varepsilon_n. \quad (25.6)$$

If $\eta_n \rightarrow 0$ and deterministic microscopic paths $X_n(Z) \rightarrow X(Z)$ in a specified Polish path metric in ν -probability, then on this coupling the microscopic paths, the full and projected likelihood processes evaluated on their own observation paths, and each bounded posterior process converge jointly in probability in that metric and uniformly in time for the last three coordinates.

Proof. Conditional Gaussian change of measure gives

$$\text{KL}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |H_n - H|^2 dt = \frac{\eta_n^2}{2\sigma^2}.$$

The same calculation proves conditional equivalence, and the conditional density has mean one, so the preparation marginal is unchanged. Integrating and applying Pinsker proves (25.3).

For the quantitative density estimate, condition first on $Z = z$ and write $D_t = L_t^n - L_t$. The stochastic differential equation is

$$dD_t = \sigma^{-1} \{D_t H_n(t, z) + L_t(H_n - H)(t, z)\} dB_t.$$

The conditional second moment of L_t is at most e^A . Itô's isometry and $(u + v)^2 \leq 2u^2 + 2v^2$ therefore imply

$$\mathbb{E}_R[D_t^2 | z] \leq \frac{2M^2}{\sigma^2} \int_0^t \mathbb{E}_R[D_s^2 | z] ds + \frac{2e^A}{\sigma^2} \int_0^t |H_n - H|^2(s, z) ds.$$

Gronwall, followed by integration in z , gives $\mathbb{E}_R D_T^2 \leq (2/\sigma^2) e^{3A} \eta_n^2$. The difference is a square-integrable $\mathcal{H}_t$ -martingale. Doob's L^2 inequality proves (25.4). Applying the same maximal inequality to the $\mathcal{F}_t$ -martingale $\mathbb{E}_R[(L_T^n - L_T)f | \mathcal{F}_t]$ proves (25.5). These are direct process estimates, not a claim that finite-dimensional convergence alone implies tightness.

The posterior assertions follow from Theorem 24.1 and its countable conditional-coupling construction. For the last claim, on the event $Y^n = Y$ the two likelihood functions are evaluated on the same (Z, Y) . Their uniform difference tends to zero in R -probability by the estimates just proved, and therefore in P -probability by absolute continuity of the fixed P with respect to R . The complementary event has probability at most ε_n . The same argument applies to ℓ^n , and (25.6) handles the posterior. The microscopic convergence holds on any coupling retaining the same Z . A finite union bound proves joint convergence. Stochastic integral versions are chosen measurably on the canonical product space; equivalence makes these common versions valid under every law in the countable sequence. $\square$

For $f \in L^2(\nu)$ the posterior conclusion follows by truncation exactly as in Corollary 24.2; it also permits $f_n \rightarrow f$ in $L^2(\nu)$. Boundedness of H_n is being used for the uniform likelihood second moments. It is not replaced by an unverified uniform-integrability assertion in a singular or zero-noise limit.

25.2. The projected stochastic exponential. Let $\hat{H}_t^n = \mathbb{E}_{P_n}[H_n(t, Z) | \mathcal{F}_t]$, using a predictable version for Lebesgue-almost every t . Put $\theta_t^n = \hat{H}_t^n / \sigma$.

Proposition 25.2 (Identification before passage to the limit). *The projected likelihood and the innovation satisfy*

$$\ell_t^n = \exp \left\{ \int_0^t \theta_s^n dB_s - \frac{1}{2} \int_0^t (\theta_s^n)^2 ds \right\}, \quad (25.7)$$

$$I_t^n = B_t - \int_0^t \theta_s^n ds. \quad (25.8)$$

Here I^n is a P_n -Brownian motion in $\mathcal{F}_t$. In particular the limit of the projected likelihoods in (25.5) is the projected stochastic exponential of the limiting model. Change of measure commutes with this limit on bounded measurable path tests.

Proof. Project $L_t^n = 1 + \int_0^t L_s^n H_n(s, Z) dB_s / \sigma$ onto the Brownian observation filtration. The integrands are square integrable; testing against bounded elementary predictable stochastic integrals and using Itô's isometry gives

$$d\ell_t^n = \sigma^{-1} \mathbb{E}_R[L_t^n H_n(t, Z) | \mathcal{F}_t] dB_t = \ell_t^n \theta_t^n dB_t.$$

The last equality is Bayes' formula. Since $|\theta_t^n| \leq M/\sigma$, the solution is the positive stochastic exponential (25.7); Girsanov gives the stated innovation. The same argument with H identifies the limiting exponential. Finally $|\mathbb{E}_{P_n} F - \mathbb{E}_P F| \leq 2\|F\|_\infty \varepsilon_n$ for bounded signed F . There is no interchange based only on a formal limit of logarithms. $\square$

25.3. A bounded entropic backward-equation consumer. Fix $\gamma > 0$ and a latent payoff f with $|f| \leq 1$. Define

$$U_t^n = \frac{1}{\gamma} \log \mathbb{E}_{P_n}[e^{\gamma f(Z)} \mid \mathcal{F}_t] = \frac{1}{\gamma} \log \frac{N_t^{n,e^{\gamma f}}}{\ell_t^n}. \quad (25.9)$$

The terminal condition here is the observable conditional certainty equivalent U_T^n , not the generally unobserved $f(Z)$ itself. This specification avoids imposing a latent terminal value on the smaller observation filtration.

Theorem 25.3 (Entropic values and localized backward integrands). *There is a unique bounded entropic value process with terminal value U_T^n ; for a predictable integrand Z^n it satisfies*

$$U_t^n = U_T^n + \int_t^T \frac{\gamma}{2} |Z_s^n|^2 ds - \int_t^T Z_s^n dI_s^n. \quad (25.10)$$

On the coupling of Theorem 25.1,

$$\mathbb{E} \sup_t |U_t^n(Y^n) - U_t(Y)|^2 \leq 84 \left(\frac{e^{2\gamma} - 1}{\gamma} \right)^2 \varepsilon_n. \quad (25.11)$$

For each $r > e^\gamma$ let τ_r^n be the first exit of any of $\ell^n, \ell, N^{n,e^{\gamma f}}, N^{e^{\gamma f}}$ from $(1/r, r)$, capped at T . If $\eta_n \rightarrow 0$, then

$$\mathbb{E}_R \int_0^{\tau_r^n} |Z_s^n - Z_s|^2 ds \rightarrow 0, \quad \mathbb{E}_R \int_0^{\tau_r^n} |\theta_s^n - \theta_s|^2 ds \rightarrow 0. \quad (25.12)$$

Furthermore $\lim_{r \rightarrow \infty} \sup_n R(\tau_r^n < T) = 0$. Thus the localizations exhaust the observation experiment in probability.

Proof. For a positive bounded latent g , write $N_t^{n,g} = \mathbb{E}_R[L_T^n g \mid \mathcal{F}_t]$ and $dN_t^{n,g} = a_t^{n,g} dB_t$. Projection of the full integral gives

$$a_t^{n,g} = \sigma^{-1} \mathbb{E}_R[L_t^n g H_n(t, Z) \mid \mathcal{F}_t], \quad |a_t^{n,g}| \leq (M/\sigma) N_t^{n,g}.$$

The same representation with $g = 1$ gives $d\ell_t^n = b_t^n dB_t$ and $b_t^n/\ell_t^n = \theta_t^n$. Put $N^n = N^{n,e^{\gamma f}}$ and $a^n = a^{n,e^{\gamma f}}$. Itô's formula in (25.9) gives the martingale coefficient

$$Z_t^n = \gamma^{-1} (a_t^n / N_t^n - b_t^n / \ell_t^n).$$

Replacing dB_t by $dI_t^n + \theta_t^n dt$ reduces its drift to $-\gamma |Z_t^n|^2/2$, which proves (25.10). The ratios are bounded by M/σ , so the integrals are well defined. Conversely, exponentiating any bounded solution of (25.10) produces a bounded P_n -martingale with terminal value $e^{\gamma U_T^n}$. Its conditional-expectation representation is unique, and martingale representation then identifies its integrand. This proves uniqueness in the bounded class.

Apply Theorem 24.1 to $g_0 = (e^{\gamma f} - e^{-\gamma})/(e^\gamma - e^{-\gamma}) \in [0, 1]$. The map taking its conditional expectation to $\gamma^{-1} \log \mathbb{E}[e^{\gamma f} \mid \mathcal{F}_t]$ has Lipschitz constant at most $(e^{2\gamma} - 1)/\gamma$. This proves (25.11).

The initial values of N^n and N agree, since Z has the same marginal. Itô's isometry and the terminal version of (25.5) imply

$$\mathbb{E}_R \int_0^T |a_s^n - a_s|^2 ds \rightarrow 0, \quad \mathbb{E}_R \int_0^T |b_s^n - b_s|^2 ds \rightarrow 0.$$

Before τ_r^n , all denominators lie in $[1/r, r]$, and the limiting numerators a, b are bounded in absolute value by $(M/\sigma)r$. For example,

$$\left| \frac{a^n}{N^n} - \frac{a}{N} \right| \leq r |a^n - a| + (M/\sigma) r^3 |N^n - N|.$$

The supremum estimates (25.5), followed by integration in time, prove (25.12). The same inequality applies to b^n/ℓ^n .

Finally $e^{-\gamma\ell^n} \leq N^n \leq e^{\gamma\ell^n}$. Each ℓ^n is a stochastic exponential with uniformly bounded integrand θ^n . Its positive and negative moments of the supremum are bounded uniformly in n : write any power as a stochastic exponential times its bounded compensator and apply Doob's inequality to a higher positive power. The same calculation applies to its inverse. Markov's inequality, the preceding comparison of N^n with ℓ^n , and a union bound prove the exhaustion assertion. This is a localized integrand theorem; it does not assert an unspecified global topology for backward equations with unbounded terminal data or different nonlinear drivers. $\square$

Corollary 25.4 (A changing-filtration stopping consumer). *Let $\psi(t, x) \in [0, 1]$ be measurable and L_ψ -Lipschitz in x , uniformly in t . For a bounded latent f with $|f| \leq 1$, put $M_t^n = \mathbb{E}_{P_n}[f(Z) \mid \mathcal{F}_t]$. Then*

$$\left| \sup_{\tau \leq T} \mathbb{E}_{P_n} \psi(\tau, M_\tau^n) - \sup_{\tau \leq T} \mathbb{E}_P \psi(\tau, M_\tau) \right| \leq \varepsilon_n + L_\psi \sqrt{80\varepsilon_n}.$$

The same statement for the entropic value U has the additional Lipschitz factor $(e^{2\gamma} - 1)/\gamma$ on the square-root term. The suprema range over the common canonical stopping times; attainment is not needed for the estimate.

Proof. For a fixed stopping time, first change the measure while keeping its $[0, 1]$ reward function, at cost at most ε_n . Then compare the two reward functions on the same canonical path under P , using Theorem 24.1, Cauchy-Schwarz and the Lipschitz bound. The estimate is uniform in the stopping time, so it survives both suprema. Equivalence of the observation laws is what allows the same stopping-time class. The entropic assertion follows from the Lipschitz calculation in the preceding theorem. $\square$

25.4. Resource transport is not the posterior coupling. Apply any common finite causal acquisition map to Y and retain a fixed finite mark $w(Z)$. Data processing makes its marked row variations at most ε_n , uniformly over a declared finite feedback family. If the source and target have respective bounds α_n, β_n , Theorem 7.2 gives same-budget deficiency perturbation at most $\alpha_n + \beta_n$, and Proposition 18.4 transports a fixed reset audit. A subsequent consumer of width S composed with a simulator of width K has width KS , with the stated selector price. None of these executable claims uses the latent-preserving maximal coupling. That coupling belongs to the proof of conditional-process convergence, not to the finite-state machine.

These results provide a direct changing-trajectory, likelihood, Girsanov and bounded entropic-value input for a conditional-law theory. They do not assert the strict-dual, rigidity, covariant-form or full kinetic conclusions of a larger program without their separate analytic proofs.

Example 25.5 (Why an amplitude bound is not a tightness argument). Let ξ_1, ξ_2 be independent symmetric signs, and, for $n \geq 5$, let $\mathcal{G}_t^n$ reveal ξ_1 at $1/2 - 2/n$ and ξ_2 at $1/2 - 1/n$. On $[0, 1]$ the bounded martingale

$$M_t^n = \mathbb{E}[\xi_1 + \xi_2 \mid \mathcal{G}_t^n]$$

converges at every finite collection of fixed times to the martingale $M_t = \mathbf{1}_{[1/2, 1]}(t)(\xi_1 + \xi_2)$. Nevertheless its laws are not tight in the Skorokhod J_1 topology. On the event $\xi_2 = -\xi_1$, of probability $1/2$, its path has an excursion of height one on an interval of length $1/n$ and is zero outside that interval. The three-time oscillation modulus of such a path is one at every scale greater than $2/n$. This violates the necessary J_1 tightness modulus criterion. All maximal moments are uniformly bounded by 2^p .

This example concerns the inference from finite-dimensional convergence and a maximal amplitude estimate alone. It does not claim that every possible stronger conditional-kernel hypothesis fails. The uniform estimates in Theorem 25.1 remove the need for that inference in the observation class treated here.

26. OBSERVATION DENSITIES AND COUPLED PROCESS VERSIONS

We record three measure-theoretic facts used in the global transport argument. Their purpose is to make explicit the passage from a tilted physical law to an observation law and from two marginal processes to one joint realization.

Lemma 26.1 (Observation-level return from a bounded tilt). *Let $dP^g/dP = g(Z)/c$, where $c = \mathbb{E}_P g$ and $e^{-\gamma} \leq g \leq e^\gamma$. For any observation sub- σ -field $\mathcal{F}$,*

$$\frac{d(P|_{\mathcal{F}})}{d(P^g|_{\mathcal{F}})} = \mathbb{E}_{P^g} \left[\frac{c}{g} \middle| \mathcal{F} \right] = \frac{c}{\mathbb{E}_P[g | \mathcal{F}]} \quad (26.1)$$

Both this density and its reciprocal lie in $[e^{-2\gamma}, e^{2\gamma}]$. In particular, for any nonnegative $\mathcal{F}$ -measurable V , $\mathbb{E}_P V \leq e^{2\gamma} \mathbb{E}_{P^g} V$.

Proof. For $A \in \mathcal{F}$, integrate c/g over A under P^g to obtain $P(A)$. Conditioning proves the first identity. In the other direction, $d(P^g|_{\mathcal{F}})/d(P|_{\mathcal{F}}) = \mathbb{E}_P[g | \mathcal{F}]/c$; equivalence and inversion prove the second. The bounds follow from the bounds on g , its conditional mean, and c . $\square$

Lemma 26.2 (Domination of the common-path measure). *Let (Y', Y) be a coupling on a standard Borel path space, with marginal laws P', P , and let $E = \{Y' = Y\}$. The subprobability measure $\kappa(A) = \Pr(E, Y \in A)$ is dominated by both P and P' . Consequently, for any nonnegative measurable path functional V ,*

$$\mathbb{E}[\mathbf{1}_E V(Y)] \leq \mathbb{E}_P V, \quad \mathbb{E}[\mathbf{1}_E V(Y')] \leq \mathbb{E}_{P'} V.$$

The assertion also holds on a common latent–observation space when the latent coordinate is shared by the coupling.

Proof. The event defining $\kappa(A)$ is contained in $\{Y \in A\}$ and, on E , in $\{Y' \in A\}$. This proves domination on sets. Integration of simple functions followed by monotone convergence proves the rest. Apply the same argument to the product coordinate for the last assertion. $\square$

Lemma 26.3 (Marginal stochastic integrals on an arbitrary coupling). *On a canonical standard Borel observation path space, suppose that under each of two marginal laws there are a continuous adapted process U , a predictable square-integrable process Z , and an innovation I such that, in that marginal's own filtration,*

$$\int_0^t Z_s dI_s = U_t - U_0 + \frac{\gamma}{2} \int_0^t |Z_s|^2 ds, \quad 0 \leq t \leq T. \quad (26.2)$$

Assume Z is bounded, as in the entropic model here. There are measurable versions of the two continuous integral processes which can be pulled back to any coupling of the observation laws. Each pulled-back version has the correct marginal law and satisfies its own identity (26.2). Neither integral thereby acquires a martingale or Itô-integral interpretation in the enlarged joint filtration.

Proof. A continuous process is determined by its values at rational times. Completed-measurable random variables on the canonical space have Borel versions. Choose such versions at the countably many rational times and extend by continuity on the measurable set where a continuous extension exists, using the zero path elsewhere. This gives a Borel $C([0, T])$ -valued version of U . Likewise take a jointly measurable version of Z modulo $dt \otimes P$; its bounded time integral has a continuous path, and its rational-time values give a measurable continuous version by the same construction. The right side of (26.2) is consequently a measurable continuous path functional. Marginal equality at all rational times, and continuity of the stochastic integral, make it indistinguishable from that integral. Repeat for the other marginal. Pullback along the two coordinate maps of any coupling preserves their respective almost-sure identities and laws. No enlargement theorem for innovations is used. $\square$

In Theorem 27.2, Lemma 26.1 is applied with $\mathcal{F} = \mathcal{F}_T$ to the integrated squared coefficient comparison. Lemma 26.2 justifies restricting a same-path estimate to the equality event of the latent-preserving coupling. Lemma 26.3 fixes the meaning of the coupled backward martingale integrals before their difference is estimated. Thus the coupled notation retains the marginal observation filtrations throughout.

27. GLOBAL TRANSPORT OF THE BACKWARD EQUATION

We retain exactly the common-preparation, bounded-signal and positive-noise hypotheses of Section 25. In particular the canonical joint laws P_n, P are equivalent and have the same latent marginal ν . All comparisons on a single observation path use common measurable versions of the conditional expectations. The following estimate avoids division by a small observation density in the final bound.

Lemma 27.1 (An integrated posterior comparison). *Let P, Q be equivalent laws on a standard Borel product carrying a latent variable Z and an observation Y , and let $\|P - Q\|_{\text{TV}} \leq \epsilon$. If $|u(Z)| \leq M$ and $\mathcal{F}$ is an observation sub- σ -field, then, under either $S = P$ or $S = Q$,*

$$\mathbb{E}_S |\mathbb{E}_Q[u \mid \mathcal{F}] - \mathbb{E}_P[u \mid \mathcal{F}]| \leq 4M\epsilon, \quad (27.1)$$

$$\mathbb{E}_S |\mathbb{E}_Q[u \mid \mathcal{F}] - \mathbb{E}_P[u \mid \mathcal{F}]|^2 \leq 8M^2\epsilon. \quad (27.2)$$

A common latent marginal is not needed for these two inequalities.

Proof. Use a dominating finite measure on $\mathcal{F}$. Write p, q for the densities of the restricted probability laws and a, b for the densities of the signed measures $\mathbb{E}_P[u \mathbf{1}_A], \mathbb{E}_Q[u \mathbf{1}_A]$ on $\mathcal{F}$. Their conditional means are $m_P = a/p$ and $m_Q = b/q$. Outside null sets, $|m_P|, |m_Q| \leq M$, and

$$p(m_Q - m_P) = m_Q(p - q) + (b - a).$$

The integral of $|p - q|$ is at most 2ϵ , by contraction of total variation. The integral of $|b - a|$ is at most $2M\epsilon$, since multiplication by u followed by restriction cannot increase signed variation by more than M . Integrating the displayed identity proves (27.1) under P . Interchanging the two laws proves it under Q . The squared bound follows from $|m_Q - m_P| \leq 2M$. $\square$

Let $|f| \leq 1$, $\gamma > 0$, and use the notation θ^n, U^n, Z^n, I^n of Proposition 25.2 and Theorem 25.3. Here Z^n denotes the scalar backward integrand, whereas the unadorned latent argument of H_n is the preparation variable. Define the deterministic bounds

$$A_n = \sigma^{-2} \{2\eta_n^2 + 16M^2T\epsilon_n\}, \quad (27.3)$$

$$C_n = \frac{2(1 + e^{4\gamma})}{\gamma^2} A_n, \quad (27.4)$$

where $\epsilon_n = \min\{1, \eta_n/(2\sigma)\}$ as before.

Theorem 27.2 (Global innovation and entropic-integrand transport). *Under the hypotheses of Theorem 25.1, under either physical law $S = P$ or $S = P_n$ on the same canonical path,*

$$\mathbb{E}_S \int_0^T |\theta_t^n - \theta_t|^2 dt \leq A_n, \quad \mathbb{E}_S \int_0^T |Z_t^n - Z_t|^2 dt \leq C_n. \quad (27.5)$$

No stopping at a density level is required. If $\eta_n \rightarrow 0$, both integrals also converge to zero in expectation under the reference law R .

On the single latent-preserving coupling of Theorem 25.1, evaluate each process on its own observation path, denoting these evaluations by a bar. Then

$$\mathbb{E} \int_0^T |\bar{\theta}_t^n - \bar{\theta}_t|^2 dt \leq A_n + 4M^2 T \varepsilon_n / \sigma^2, \quad (27.6)$$

$$\mathbb{E} \int_0^T |\bar{Z}_t^n - \bar{Z}_t|^2 dt \leq C_n + 16M^2 T \varepsilon_n / (\gamma^2 \sigma^2). \quad (27.7)$$

If $\eta_n \rightarrow 0$, then $\bar{I}^n \rightarrow \bar{I}$ uniformly in probability. Writing $\bar{J}_t^n = \int_0^t \bar{Z}_s^n d\bar{I}_s^n$ for the stochastic integral in its own observation filtration, and $\bar{J}$ for its limit, one has

$$\mathbb{E} \sup_{t \leq T} |\bar{J}_t^n - \bar{J}_t|^2 \leq 2 \mathbb{E} \sup_{t \leq T} |\bar{U}_t^n - \bar{U}_t|^2 + \frac{8M^2 T}{\sigma^2} \mathbb{E} \int_0^T |\bar{Z}_t^n - \bar{Z}_t|^2 dt. \quad (27.8)$$

The first term on the right has the explicit bound in (25.11). Thus the full bounded entropic backward representation, including its martingale integral, is transported globally. The coupling need not make either innovation Brownian in the joined filtration; each integral is defined under its own marginal law.

Proof. For each deterministic t , under P split

$$\sigma(\theta_t^n - \theta_t) = \mathbb{E}_{P_n}[H_n(t, Z) | \mathcal{F}_t] - \mathbb{E}_P[H_n(t, Z) | \mathcal{F}_t] + \mathbb{E}_P[H_n(t, Z) - H(t, Z) | \mathcal{F}_t].$$

Lemma 27.1, conditional Jensen and $(x + y)^2 \leq 2x^2 + 2y^2$ bound its expected square by $16M^2 \varepsilon_n + 2\mathbb{E}_\nu |H_n(t, Z) - H(t, Z)|^2$. Integration gives the first bound in (27.5) under P . Under P_n use the same decomposition with the signal-difference term conditioned under P_n and the law-change term applied to H . The common latent marginal gives the identical bound. Jointly measurable conditional versions, and predictable versions equal for dt -almost every time, give the same integrated statements.

Put $g = e^{\gamma f}$ and $c = \mathbb{E}_\nu g$. Tilt the two full joint laws by $g(Z)/c$, writing P^g, P_n^g . They still have the same latent marginal $\nu_g = g\nu/c$, and their observation signals remain H, H_n . Since $e^{-\gamma} \leq g, c \leq e^\gamma$,

$$\|P_n^g - P^g\|_{\text{TV}} \leq e^{2\gamma} \varepsilon_n, \quad \eta_{n,g}^2 \leq e^{2\gamma} \eta_n^2.$$

Moreover, the reference martingale coefficient $a^{n,g}$ from the proof of Theorem 25.3 satisfies

$$\alpha_t^n := \frac{a_t^{n,g}}{N_t^{n,g}} = \sigma^{-1} \mathbb{E}_{P_n^g}[H_n(t, Z) | \mathcal{F}_t].$$

Apply the preceding innovation argument to these tilted laws. It bounds the squared integral of $\alpha^n - \alpha$ by $e^{2\gamma} A_n$ under either tilted physical law. For an observation-measurable nonnegative random variable, passage back to the corresponding original physical law costs at most $e^{2\gamma}$: the full density c/g is bounded by this constant, and so is its observation projection. Consequently

$$\mathbb{E}_S \int_0^T |\alpha_t^n - \alpha_t|^2 dt \leq e^{4\gamma} A_n.$$

The identity $Z^n = (\alpha^n - \theta^n)/\gamma$ proves the second bound in (27.5).

The pointwise bounds, valid for all models, are

$$|\theta^n|, |\alpha^n| \leq M/\sigma, \quad |Z^n| \leq 2M/(\gamma\sigma). \quad (27.9)$$

The squared time integrals are thus uniformly bounded random variables. Convergence in P -probability, followed by absolute continuity of the fixed R with respect to P , gives convergence in R -probability. Boundedness then gives convergence of their R -expectations. This argument establishes global reference convergence without claiming a new explicit reference rate.

For the coupling, let $E_n = \{Y^n = Y\}$, so that $\mathbb{P}(E_n^c) \leq \varepsilon_n$. On E_n both processes are functions of the same observation path. The restricted expectation of their squared difference is at most the

corresponding unrestricted expectation under the limiting marginal P . On E_n^c apply (27.9) and integrate for time T . This proves (27.6)–(27.7).

On E_n , the Brownian reference coordinates $Y^n/\sigma, Y/\sigma$ agree, so

$$\sup_{t \leq T} |\bar{I}_t^n - \bar{I}_t|^2 \leq T \int_0^T |\bar{\theta}_t^n - \bar{\theta}_t|^2 dt.$$

Markov's inequality and $\mathbb{P}(E_n^c) \rightarrow 0$ prove uniform convergence in probability. This assertion does not require any moment estimate on the observation difference on the exceptional event.

Finally, the backward equation already proved gives, under each marginal,

$$\bar{J}_t^n = \bar{U}_t^n - U_0 + \frac{\gamma}{2} \int_0^t |\bar{Z}_s^n|^2 ds.$$

The common initial value is $U_0 = \gamma^{-1} \log \mathbb{E}_\nu g$. By (27.9),

$$||\bar{Z}_s^n|^2 - |\bar{Z}_s|^2| \leq \frac{4M}{\gamma\sigma} |\bar{Z}_s^n - \bar{Z}_s|.$$

Subtract the two integral identities, use Cauchy–Schwarz in time and $(x + y)^2 \leq 2x^2 + 2y^2$, and take expectations. This gives (27.8). The integral identity also defines compatible measurable versions on the coupling, without asserting a martingale property for its joint filtration. $\square$

The theorem strengthens the localized conclusion of Theorem 25.3 without changing its terminal condition or its driver. The terminal condition remains the observation-measurable conditional certainty equivalent, not the unobserved latent payoff. Positive fixed noise, bounded signals, and a common preparation are used in the proof. A singular zero-noise limit, an unbounded terminal condition, or a different nonlinear driver would require additional arguments.

28. A UNIFORM CERTIFICATE THROUGH A MOVING COLLISION

We now use all three parts of the argument on one family. The finite marked reference law and the uniform scattering geometry are those of Theorem 23.1 and Lemma 23.2. They are proved above, including the no-collision control. We use a continuous observation of the same detector so that finite reports and conditional processes are generated by one experiment.

Let z denote a preparation in the boxes of (23.3), and let ν be that prepared law. The sign of the impact orientation is $2B - 1$, and the conserved center-of-mass longitudinal sign is the actual mark W . Their correlation is prepared: $\mathbb{P}(B = b, W = w) = 5/16$ for $b = w$ and $3/16$ otherwise. The contact diameter d ranges over $I = [9/10, 11/10]$. Let $X_d(t, z)$ be the two-particle phase path, with outgoing velocities used at the collision, and define

$$H_d(t, z) = h(v_{1,y}^d(t, z)), \quad Y_t^d = \int_0^t H_d(s, z) ds + \sigma B_t^d, \quad 0 < \sigma \leq 1/24.$$

The Brownian noise is independent of the preparation. The source experiment has the same physical preparation and mark but constant reports. The target reports at times 2 and 3 are

$$Z_1^d = \mathbf{1}\{2(Y_2^d - Y_{3/2}^d) > 0\}, \quad Z_2^d = A\mathbf{1}\{2(Y_3^d - Y_{5/2}^d) > 0\}. \quad (28.1)$$

The acquisition gate $A \in \{0, 1\}$ is chosen after Z_1^d ; it does not act on the mechanical trajectory. The continuous path is the auxiliary observation experiment for conditional consumers; the finite simulator is not given this path or an uncharged buffer for computing its increments. The target sensor implements the declared window acquisition.

Lemma 28.1 (Quantitative change of the microscopic path). *For all preparations in the specified boxes, the collision time $t_c(d)$ and outgoing tagged velocities obey*

$$|t_c(d') - t_c(d)| \leq |d' - d|, \quad |v_i^+(d') - v_i^+(d)| \leq 30|d' - d| \quad (i = 1, 2).$$

Writing $\Delta = |d' - d|$, one has

$$\sup_z \int_0^3 |H_{d'}(t, z) - H_d(t, z)|^2 dt \leq \Delta + 2700\Delta^2. \quad (28.2)$$

The phase paths converge uniformly over the preparation in the Skorokhod J_1 metric as $d' \rightarrow d$. More precisely, for the product metric taking the maximum of the per-particle Euclidean position and velocity distances, a time change aligning the collision times gives J_1 distance at most 100Δ .

Proof. Write r and g for the incoming relative position and velocity, and $n(d) = (r + t_c(d)g)/d$ for the contact normal. The collision root satisfies $|r + t_c g| = d$. The nongrazing estimate from Lemma 23.2 is $-g \cdot n > 1443/1000$. Differentiating the root with respect to d gives

$$t'_c(d) = \frac{1}{g \cdot n(d)}, \quad |t'_c(d)| < 1.$$

The incoming bound is $|g| < 3$, so

$$|n'(d)| \leq \frac{|t'_c(d)| |g| + 1}{d} < 5.$$

Since $v_1^+ = V + g/2 - (g \cdot n)n$ and $v_2^+ = V - g/2 + (g \cdot n)n$, differentiation bounds either velocity derivative by $2|g| |n'| < 30$.

Before collision the detector is zero, and after collision it is a constant of absolute value at most one. Its Lipschitz constant is one. The two signals therefore differ on the interval between the two collision times by at most one and afterwards by at most 30Δ . The intervening interval has length at most Δ , and the whole observation lasts three units. This proves (28.2).

Choose the increasing piecewise affine time change fixing 0, 3 and sending $t_c(d)$ to $t_c(d')$. Its uniform displacement is at most Δ . Aligned velocities differ by at most 30Δ , with the same outgoing convention at the jump. Each particle's speed is below four and its velocity jump has norm below three. Without time change, the position difference is bounded by $3\Delta + 3(30\Delta) = 93\Delta$, by integrating the velocity difference before and after the two jump times. Aligning time adds at most 4Δ . The asserted bound follows, with a harmless rounding to 100Δ . $\square$

Theorem 28.2 (Microscopic-to-audit and conditional-law composition). *For the single family (28.1), all of the following hold.*

(a) *At the same width-one private register,*

$$\delta^p > 9/20, \quad \delta_r^v > 9/20 \quad (r \geq 1), \quad \delta_2^h < 13/100,$$

uniformly for $d \in I$ and $0 < \sigma \leq 1/24$. When $d \leq 2/5$ the detector is zero on $[0, 3]$, and all three finite-experiment deficiencies are zero.

(b) *There is one explicit rational reset-audit rule with one all-on training row, eight marked outcomes and $n = 2800$ training runs, whose expected validation score exceeds $2/5$ against every private width-one simulator throughout that same parameter region. In detail, for counts C choose an event $A_C \subseteq \{0, 1\}^3$ maximizing*

$$Q_*(A_C) - C(A_C)/2800, \quad (28.3)$$

with lexicographic tie breaking. Here Q_ is the rational marked reference law of (B, B, W) . The independent validation uses the actual physical target, not Q_* . This finite rule has at most $\binom{2807}{7}$ training-count states at any scheduled epoch; a conservative complete auditor bound is $8(256 + 1)\binom{2807}{7}$. The state cost belongs to the auditor. The certificate is an expected-score inequality, not a confidence claim about one realized score.*

(c) Fix $\sigma > 0$ and let $d_n, d \in I$ with $d_n \rightarrow d$. Set

$$\epsilon_n = \min \left\{ 1, \frac{\sqrt{|d_n - d| + 2700|d_n - d|^2}}{2\sigma} \right\}. \quad (28.4)$$

The joint preparation–continuous-observation laws differ by at most ϵ_n . Every finite resource deficiency above changes by at most ϵ_n , and a fixed audit changes by at most ϵ_n because its source behavior is unchanged. On a latent-preserving coupling, $X_{d_n} \rightarrow X_d$ in J_1 and each bounded posterior converges in supremum L^2 with bound $84\epsilon_n$. The full and projected likelihoods converge in reference supremum L^2 , and jointly in probability with the microscopic paths on that coupling. The projected stochastic exponentials, bounded entropic backward values, localized integrands and stopping consumers have the conclusions of Proposition 25.2, Theorem 25.3, and Corollary 25.4.

(d) On compact parameter sets with $\sigma \geq \sigma_0 > 0$, the finite marked probabilities have uniform rational enclosures. The observable chart, strict same-budget polynomial verification and rational upper tables are therefore effective in the exact sense of Propositions 22.1 and 21.6.

Proof. Every collision occurs before time $3/2$. Conditional on preparation, the two normalized window increments in (28.1) are independent Gaussians with mean $h(v_{1,y}^+)$ and standard deviation $\sqrt{2}\sigma \leq 1/12$. The outgoing detector has sign $2B - 1$ and absolute value greater than $1/4$. Coupling each noisy report to B gives total marked variation less than $1/300$ from the exact finite marked experiment, uniformly for every acquisition gate. The finite calculation of Theorem 23.1 gives

$$d_* = \sqrt{5/2} - 9/8, \quad \delta_*^p = \delta_*^v = d_*, \quad \delta_{*,2}^h = 1/8.$$

Thus private and visible values exceed $d_* - 1/300 > 9/20$, and the hidden value is less than $1/8 + 1/300 < 13/100$. The two-valued hidden witness stores its selector, not an observed mark. In the no-collision regime both window reports before gating are independent fair noise, independent of W . Fresh private coins reproduce them exactly, including feedback gates. This proves (a).

For (b), use only the all-on feedback row. Any private width-one simulator in this row emits independent bits with probabilities p, q , independently of the actual W . The calculation in Theorem 23.1 already gives minimum variation d_* for this row. Its alphabet (z_1, z_2, W) has eight elements, hence $D = 7$. The explicit empirical audit in Theorem 18.1, trained using Q_* , has expected ideal score at least

$$d_* - \sqrt{7/2800} = d_* - 1/20.$$

The candidate training and validation laws do not change when the target is replaced by the physical target. Conditional on any selected event, its target probability changes by less than $1/300$. Therefore the actual score is strictly greater than

$$d_* - \frac{1}{20} - \frac{1}{300} > \frac{2}{5}. \quad (28.5)$$

For completeness, the last strict inequality is the positive-root comparison $\sqrt{5/2} > 947/600$; its squared margin is $5/2 - (947/600)^2 = 3191/360000 > 0$. All probabilities in Q_* are 0, $3/16$ or $5/16$, so (28.3) is a finite rational rule independent of d, σ, p, q . At training epoch k the number of eight-category count vectors is $\binom{k+7}{7} \leq \binom{2807}{7}$. A buffer of eight states records the current two reports and terminal mark; 256 event masks and an idle symbol suffice during event selection and validation. The conservative product bound follows. The clock and the source-reset interface are the declared resources. In particular the proof does not enumerate or claim to store a small complete coefficient table.

For (c), Lemma 28.1 gives the signal-distance hypothesis of Theorem 25.1, with $T = 3$ and $M = 1$. It also supplies its microscopic J_1 hypothesis. That theorem gives (28.4) and all the joint process assertions. The finite acquisition is a common measurable causal map of the continuous

observation, so marked variation contracts for every gate. The source is blind and independent of d . Consequently its executable candidate behaviors are identical across d , and Proposition 18.4 has $\alpha = 0$ and $\beta = \epsilon_n$. The same same-budget comparison applies in each visible fiber and in each hidden mixture. The exponential and backward claims now follow from the identified projected density and the explicitly bounded entropic consumer. This is where the finite certificate and the conditional-law theorem meet the same microscopic approximation.

Finally, the collision root is uniformly separated from grazing, its formula is algebraic on the rational preparation boxes, and the normalized window noise has variance $2\sigma^2$ bounded away from zero on the stated compact sets. The box subdivision and Gaussian-tail enclosure argument in Proposition 23.4 therefore applies with that variance. It returns rational enclosures with a computable uniform error. The rational quotient algorithm and same-budget strict certificate procedure then prove (d). Uniformity as $\sigma \downarrow 0$ is not asserted. $\square$

The all-on audit in part (b) is a single witness for the lower bound in the full feedback problem: a maximum over feedback trees includes this row. The visible lower bound is instead the exact fiber theorem; no claim is made that an auditor with a seed-independent state bound can store an arbitrarily large visible-seed transcript. This separates the two resource budgets rather than silently moving cost from one to the other.

The finite certificate is also preserved under an additional rational presentation error ρ : its lower score decreases by at most ρ when only the target is changed, and by the reset-cost estimate when the source is changed. Thus the certificate is an effective consumer of the physical presentation theorem, not an isolated exact finite example.

Theorem 28.2 and the preceding results prove Theorem C.3. The C2-type consequence is a joint changing-trajectory/likelihood/posterior theorem with an identified stochastic exponential and a fully specified bounded entropic backward consumer. Nothing in that conclusion identifies geometric similarity with kinetic scaling, or supplies particle-uniform recollision estimates by renaming this fixed-particle model.

29. SHARP DECISION MEMORY FOR THE COLLISION EXPERIMENT

We now apply the continuation spectrum to the same marked experiment that supplies the microscopic certificate. Both its impossibility statement and its construction concern the same score, the same private candidate family, and the same training-validation checkpoint.

29.1. The marked family and its first two spectral values. In the all-on row of Section 28, a private width-one candidate has two independent report bits with probabilities $p, q \in [0, 1]$, independently of the fair actual mark. Denote its law by $b(p, q)$ and put

$$a = (1 - p)(1 - q), \quad c = pq.$$

Let $B = \{b(p, q) : p, q \in [0, 1]\}$. The ideal target Q_* is supported on 000, 001, 110, 111, with masses $5/16, 3/16, 3/16, 5/16$, respectively. On these four atoms the candidate masses are $a/2, a/2, c/2, c/2$. The elementary continuation constraint implies

$$\sqrt{a} + \sqrt{c} \leq 1, \quad \min(a, c) \leq \frac{1}{4}. \quad (29.1)$$

Indeed the first inequality is Cauchy-Schwarz applied to $(\sqrt{1-p}, \sqrt{p})$ and $(\sqrt{1-q}, \sqrt{q})$; the second follows from the first. If $a \geq c$, then $c \leq 1/4$ and

$$a + 3c \leq (1 - \sqrt{c})^2 + 3c = 1 - 2\sqrt{c} + 4c \leq 1. \quad (29.2)$$

The symmetric inequality holds when $c \geq a$.

Theorem 29.1 (A sharp two-state obstruction). *For the ideal marked collision family,*

$$\mathfrak{V}_1(B, Q_*) = \frac{1}{8}, \quad \mathfrak{V}_2(B, Q_*) = \frac{3}{8}. \quad (29.3)$$

If $\|Q - Q_\|_{\text{TV}} \leq \beta$, every reset auditor retaining at most two states at the training-validation cut has worst-case expected score at most $3/8 + \beta$, regardless of its finite training length and its upstream or downstream memory. The assertion allows mark-dependent validation events and stochastic state transitions.*

Proof. For one state, use the event $\{000, 111\}$. Its score is $5/8 - (a + c)/2 \geq 1/8$. Conversely, average the two candidates with $(a, c) = (1, 0)$ and $(0, 1)$. Their average is uniform on the four supported atoms and is at total variation distance $1/8$ from Q_* . Every fixed response has average score at most $1/8$, so its minimum over the two candidates is at most $1/8$. This proves the first equality; the average is a bound on scores, not an allowed private candidate.

For the upper bound with two states, consider the three candidates $b_0 = b(0, 0)$, $b_1 = b(1, 1)$ and $b_* = b(1/2, 1/2)$. A response may be set to zero outside the support of Q_* without decreasing its payoff at any candidate. Write its four remaining coordinates as (x, y, z, t) . Its three payoffs are

$$\begin{aligned} f_0 &= (-3x - 5y + 3z + 5t)/16, \\ f_1 &= (5x + 3y - 5z - 3t)/16, \\ f_* &= (3x + y + z + 3t)/16. \end{aligned}$$

Since all four coordinates are in $[0, 1]$,

$$f_0 + f_1 \leq \frac{1}{4}, \quad f_0 + f_* \leq \frac{3}{4}, \quad f_1 + f_* \leq \frac{3}{4}. \quad (29.4)$$

Suppose two responses covered all three candidates with payoff strictly greater than $3/8$. The first inequality forces different responses to cover b_0 and b_1 . The other two inequalities then force both responses to have payoff less than $3/8$ at b_* . This is a contradiction. Hence $\mathfrak{V}_2 \leq 3/8$.

For attainment take the responses, in the same four-atom order,

$$h_0 = (2/3, 0, 1, 1), \quad h_1 = (1, 1, 0, 2/3),$$

with zero response elsewhere. Both have target mean $17/24$. Their payoffs are $17/24 - a/3 - c$ and $17/24 - a - c/3$. If $a \geq c$, the first is larger and (29.2) makes it at least $17/24 - 1/3 = 3/8$. The other case is symmetric. This proves equality. For the perturbed target, apply (9.4) and the operational upper bound of Theorem 9.2. $\square$

The obstruction has an immediate continuation interpretation. Each post-training state supplies one validation response. The three witness candidates in the proof cannot be covered above $3/8$ by two such responses. No increase of sample size can compensate for losing the third continuation at that cut. This argument does not use the unrelated Bernoulli majority calibration.

Lemma 29.2 (Feedback completion of the physical lower bound). *The upper bound $3/8 + \beta$ in Theorem 29.1 remains valid when the auditor may train on any of the original feedback rows and may select a row-event pair after the decision cut. Here every target row is the declared feedback pushforward of Q , with $\|Q - Q_*\|_{\text{TV}} \leq \beta$. In particular, two states cannot evade the bound by using a different sensor gate during training or validation.*

Proof. It suffices to restrict the adversary to an admissible subfamily. For $p, q \in [0, 1]$, let its first emission be $X_1 \sim \text{Ber}(p)$, and, after gate A , let its second emission be AX_2 , with an independent $X_2 \sim \text{Ber}(q)$ and the independent fair actual mark. These are genuine width-one private rows: the gate is a current input, and the second emission uses no retained copy of X_1 .

For any deterministic gate policy $g : \{0, 1\} \rightarrow \{0, 1\}$, the marked row is the pushforward of $b(p, q)$ under $T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w)$. The physical target row is the pushforward of Q by the same map, since acquisition does not act on the path. Including the gate in the transcript

changes nothing: it is also a deterministic function of the displayed coordinates. For a selected row event, the response pulls back to $\mathbf{1}_A \circ T_g$ on the all-on alphabet. A randomized row-event continuation pulls back to a function $h_m \in [0, 1]^\Omega$. Hence, conditional on cut state m , its expected score on this subfamily is exactly $(Q - b(p, q))h_m$.

Whatever training rows were used, their effect is solely the distribution of the retained state. The two-state score is thus a convex combination of two such responses and is bounded above by their envelope. Apply the three-candidate argument of Theorem 29.1, and then the uniform target perturbation bound. Since this is a subfamily of the full private candidate class, it is a valid lower-bound witness for that full class. Randomized feedback is covered by conditioning on its independent random table and then averaging the pulled-back response. $\square$

29.2. A three-response construction. Define three deterministic events, all ignoring the mark,

$$A_D = \{00, 11\} \times \{0, 1\}, \quad A_0 = \{00\} \times \{0, 1\}, \quad A_1 = \{11\} \times \{0, 1\}.$$

Their ideal scores are $1 - a - c$, $1/2 - a$ and $1/2 - c$.

Lemma 29.3 (The three-response envelope). *For the original private candidate image,*

$$\min_{p, q} \max\{1 - a - c, \tfrac{1}{2} - a, \tfrac{1}{2} - c\} = \sqrt{2} - 1. \quad (29.5)$$

In particular $\mathfrak{V}_3(B, Q_) \geq \sqrt{2} - 1$.*

Proof. By symmetry suppose $a \geq c$. Then $c \leq 1/4$ and the maximum is $1 - a - c$ for $a \leq 1/2$, and $1/2 - c$ for $a \geq 1/2$. If $a \leq 1/4$, the first expression is at least $1/2$. If $1/4 \leq a \leq 1/2$, (29.1) gives

$$a + c \leq 1 - 2\sqrt{a} + 2a \leq 2 - \sqrt{2},$$

since the last function is increasing on $[1/4, 1/2]$. If $a \geq 1/2$, then $c \leq (1 - \sqrt{a})^2 \leq (1 - 1/\sqrt{2})^2 = 3/2 - \sqrt{2}$. Both cases give the asserted lower bound. It is attained at $p = q = 1 - 1/\sqrt{2}$, where $a = 1/2$ and $c = 3/2 - \sqrt{2}$. The symmetric point gives the same value. $\square$

Lemma 29.4 (Odd-sample majority selection with a uniform regret bound). *Let $n = 2h - 1$ be odd. In n independent candidate trials, let C_0, C_1 count report pairs 00, 11. Choose A_1 if $C_0 \geq h$, choose A_0 if $C_1 \geq h$, and otherwise choose A_D . The first two cases are mutually exclusive. Its expected ideal score satisfies*

$$g_n(p, q) \geq \sqrt{2} - 1 - \frac{1}{4\sqrt{en}} - \frac{1}{4}e^{-n/8}. \quad (29.6)$$

Its event label has three possible values at the decision cut.

Proof. Because $a + c \leq 1$,

$$\max\{1 - a - c, \tfrac{1}{2} - a, \tfrac{1}{2} - c\} = 1 - a - c + (a - 1/2)_+ + (c - 1/2)_+.$$

The selected payoff is $1 - a - c + (a - 1/2)\mathbf{1}_{\{C_0 \geq h\}} + (c - 1/2)\mathbf{1}_{\{C_1 \geq h\}}$. Its expected loss from this envelope is exactly the sum, for $u = a, c$, of $|u - 1/2|$ times the probability that the binomial majority selects the wrong side of $u = 1/2$. At $u = 1/2$ the contribution is zero. No independence between C_0 and C_1 is used.

For $0 < u < 1/2$ and $k \geq h$, the ratio of the binomial mass at u to that at $1/2$ is

$$[4u(1 - u)]^{n/2} \left(\frac{u}{1 - u} \right)^{k - n/2} \leq [4u(1 - u)]^{n/2}.$$

The fair binomial upper tail is exactly $1/2$ because n is odd. Reflecting k treats $u > 1/2$. The endpoint cases follow directly or by continuity. Thus the wrong-side probability is at most

$$\tfrac{1}{2}[1 - 4(u - 1/2)^2]^{n/2} \leq \tfrac{1}{2}e^{-2n(u - 1/2)^2}.$$

For $x \geq 0$, the maximum of $xe^{-2nx^2}/2$ is $1/(4\sqrt{en})$, reached at $x = 1/(2\sqrt{n})$. At least one of a, c is at most $1/4$ by (29.1); that coordinate has $|u - 1/2| \in [1/4, 1/2]$ and contributes at most $e^{-n/8}/4$. Bound the other coordinate by the unrestricted maximum and use Lemma 29.3. $\square$

29.3. Simultaneous exact widths and the full acquisition schedule. The preceding construction admits more compression than storing its two raw counts. Write 0, 1, * for the three report-pair categories 00, 11 and the two off-diagonal pairs. For $n = 2h - 1$, let F_n be the terminal event-label function of this word. At a trial cut t , let $r = n - t$ and let (u, v) be the two prefix counts. Associate the residual key

$$\mathcal{R}_t(u, v) = \begin{cases} \text{L}, & u \geq h, \\ \text{R}, & v \geq h, \\ (\min(h - u, r + 1), \min(h - v, r + 1)), & u, v < h. \end{cases} \quad (29.7)$$

The symbols L, R mean the constant decisions A_1, A_0 . In a pair key, a coordinate $r + 1$ means that this majority is no longer reachable. The pair $(r + 1, r + 1)$ represents the constant decision A_D .

Theorem 29.5 (Exact profile of the physical audit rule). *The keys in (29.7) are exactly the distinct continuation residuals of F_n . Their number at trial cut t is*

$$r_t(F_n) = \begin{cases} \binom{t+2}{2}, & 0 \leq t \leq h - 1, \\ \binom{n-t+3}{2}, & h \leq t \leq n. \end{cases} \quad (29.8)$$

This is the minimum simultaneous width profile for exact computation of this rule, for deterministic and stochastic machines alike. Its maximum is $\binom{h+2}{2}$ at $t = h$.

For $n = 399$, the maximum trial-cut width is 20301, and the terminal training cut has exactly three states. Acquiring each report pair one bit at a time, discarding its training mark, and then executing independent candidate and target validations gives a deterministic machine with at most $40602 < 2^{16}$ states at every scheduled epoch. One additional ignored training trial may be inserted to give the same 400 training slots and 402 total trials as the earlier certificate. It changes neither the payoff nor these width bounds.

Proof. If a majority has already been reached, its label cannot change. Otherwise, the first coordinate of the pair key is exactly the number of additional 0 symbols needed, with the unreachable threshold clipped to $r + 1$; the second coordinate has the analogous meaning. These data determine the suffix function. Two different reachable pair keys can be distinguished by a suffix containing only one of the two symbols and enough * symbols to fill the remaining positions: use a count between the two different thresholds. The other majority cannot be reached by this suffix. The constant-majority keys are distinct from every pair key because an all-* suffix returns A_D for the latter. Thus the keys are exactly the residuals.

For $t < h$, no majority has been reached and every threshold is at most $r + 1$. The distinct pairs (u, v) with $u + v \leq t$ give $\binom{t+2}{2}$ keys. For $t \geq h$, the two constant-majority keys are reachable. The remaining keys are precisely

$$(i, j) \in \{1, \dots, r + 1\}^2, \quad i + j \geq r + 1.$$

To see surjectivity, use $u = h - i, v = h - j$; these counts are nonnegative, less than h , and sum to at most t because $2h - t = r + 1$. To see necessity before clipping, use the same inequality on $h - u, h - v$; if either coordinate is clipped, the inequality remains automatic. There are $(r + 1)^2 - (r - 1)r/2$ such pairs, with the expression valid also for $r = 0, 1$. Adding the two constant keys gives $(r + 2)(r + 3)/2$. This proves (29.8). The finite-valued realization theorem, Theorem 16.1, proves both the simultaneous realization and stochastic minimality for the specified rule. The two polynomial pieces of the profile give its maximum as stated.

For an explicit transition definition, order the reachable keys lexicographically, putting L, R first. For each key choose the lexicographically first prefix-count pair representing it. After the next category, increment the appropriate count and apply $\mathcal{R}_{t+1}$. Equality of residual functions implies equality of the shifted residuals, so this definition is independent of the chosen representative. The representative is part of the time-indexed transition table, not another stored register. The clock supplies t but no sample history. All these maps are deterministic and rational.

At a bit-level epoch, the state consists of a residual key and, only between the two report bits, the first bit. It therefore has at most $2 \max_t r_t$ values. On reading the second bit, determine the category, update the key and discard both bits. The mark is read and discarded; all-on controls and resets are prescribed by the free schedule. After training, keep just one of the three event labels. During validation, keep that label, the completed candidate-event indicator, and a first-bit buffer when needed. At most $3 \cdot 2 \cdot 2 = 12$ states suffice there; a completed target pair emits its event indicator minus the stored candidate indicator. The final score has three possible values. These phase-specific alphabets may be embedded in a single alphabet of size 40602 because the schedule clock identifies the phase.

The same trial-cut residuals remain distinct for the complete signed-score function, not merely for an externally emitted event name. Any two of A_0, A_1, A_D have different indicator functions. Take an off-diagonal candidate-validation pair, which belongs to none, and a target-validation pair distinguishing the two events. The resulting scores differ. Thus passing to the validation suffix does not merge the residuals already counted. $\square$

Theorem 29.6 (Matched physical decision complexity and stable certification). *For the two-sphere collision family of Theorem 28.2, let $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/24$, and let $Q_{d,\sigma}$ be its all-on marked target. At the charged training-validation cut of the reset interface,*

$$\kappa_{2/5}(B, Q_{d,\sigma}) = 3. \quad (29.9)$$

More concretely, no auditor retaining at most two states at this cut can have expected score greater than $2/5$ uniformly against all private width-one candidates, at any finite training length, even with feedback-row selection as in Lemma 29.2. One rational auditor with three states at this cut does so with 399 informative training trials and two validation trials. With a redundant ignored training trial, it uses exactly 402 total trials, at most 40602 states at every scheduled epoch, and the exact trial-cut profile (29.8) on its informative training segment.

For fixed positive σ and $d_k \rightarrow d$, the same machine changes its expected score by at most

$$\epsilon_k = \min \left\{ 1, \frac{\sqrt{|d_k - d| + 2700|d_k - d|^2}}{2\sigma} \right\}. \quad (29.10)$$

The private, hidden and visible deficiencies, the no-collision control, and the joint microscopic, likelihood, posterior, innovation and global bounded entropic backward convergence retain all conclusions of Theorems 28.2 and 27.2.

Proof. Theorem 28.2 discharges the physical hypotheses and proves $\|Q_{d,\sigma} - Q_*\|_{\text{TV}} < 1/300$, uniformly on the stated parameters. Theorem 29.1 therefore gives the universal cut-width-two upper bound

$$\mathfrak{V}_2(B, Q_{d,\sigma}) < \frac{3}{8} + \frac{1}{300} < \frac{2}{5}.$$

Lemma 29.2 makes this a bound for the full original feedback-row interface, not only all-on training and not only the constructed rule. The upper construction uses all-on trials and works for every private candidate because only its two all-on parameters enter.

Use the deterministic selector of Lemma 29.4 with $n = 399$, but validate against the actual physical target. For each selected event, changing only the target from Q_* loses less than $1/300$.

Its physical expected score is consequently greater than

$$L_{399} := \sqrt{2} - 1 - \frac{1}{4\sqrt{399e}} - \frac{1}{4}e^{-399/8} - \frac{1}{300} > \frac{2}{5}. \quad (29.11)$$

Here is a rational verification of the last strict inequality, independent of decimal evaluation. The bounds $e > 2$, $\sqrt{798} > 28$, $\sqrt{2} > 707/500$, and $399/8 > 49$ imply

$$L_{399} > \frac{207}{500} - \frac{1}{112} - \frac{1}{300} - 2^{-51} > \frac{2}{5},$$

since $207/500 - 1/112 - 1/300 - 2/5 = 73/42000 > 2^{-51}$. The number L_{399} is approximately 0.4032891; the rational inequalities, not this approximation, establish the certificate. Its state realization was proved in Theorem 29.5. The general operational classification now yields (29.9) with both inequalities proved for this same physical task.

Changing d leaves the blind candidate source and all its training laws unchanged. The moving-collision estimate and the common-preparation Gaussian theorem bound the target variation by ϵ_k . For any fixed response its target mean changes by at most this amount; averaging over the unchanged training rule proves (29.10). There is no multiplier equal to the number of training trials. The same physical estimates verify the hypotheses of Theorem 27.2; the observation-level density and coupling versions used there are given in Section 26. This transports the same certificate and the bounded conditional-law consumer along the changing microscopic paths. The no-collision regime $d \leq 2/5$ has the vanishing deficiencies already proved in Theorem 28.2; it is not included in the positive-gap parameter interval. $\square$

29.4. What the two exact complexity statements measure. Equation (29.9) minimizes over every auditor at one specified physical decision cut. Equation (29.8) minimizes all trial-cut widths simultaneously for exact implementation of the stated 402-trial rule. These are different optimizations. The first is a task-level optimum, not an artifact of that rule. The second explains the whole online realization of the same rule, not a claim that every successful auditor must have 20301 states somewhere. The sixteen-bit bound includes bit acquisition and validation but does not bound the description length of time-dependent transition tables. It is an implementation upper bound for peak memory. Independent persistent randomization is charged; fresh randomness from the reset experiments is not a hidden cross-cut memory channel.

The former two-count 400-sample construction and its nineteen-bit bound remain in Section 34. They offer a different approximation to the full marked discrepancy. The new three-response rule does not reproduce that rule or attain its oracle envelope. It instead solves the specified physical certification task with an exactly minimal decision-cut alphabet and an explicit full continuation profile. The score is an expectation, not a confidence guarantee from a single realized validation pair.

30. FULL-PROFILE CERTIFICATION OF THE COLLISION FAMILY

The short audit of Theorem 29.6 is retained with its exact continuation profile. We now give a second audit of the same physical class whose entire acquisition-and-validation register is bounded independently of its accuracy parameters. Its cost is a much longer, explicitly counted, preparation schedule.

30.1. The row-indexed decision invariant. For the original row-event interface define

$$\kappa_c^{\text{rows}}(B, Q) = \min\{K : \mathfrak{R}_K^{\text{rows}}(B, Q) > c\}. \quad (30.1)$$

The minimum is $+\infty$ when the set is empty. This is not an implicit change of the one-row symbol κ_c . By Corollary 9.4, it is the exact minimum decision-cut alphabet for a strict score, allowing a finite training schedule and a post-cut distribution on row-event pairs.

Let W_c^{phys} be the minimum peak width among all finite auditors for the original physical row interface, now charging each report-bit, mark, decision and validation epoch. Its clock is externally scheduled, as in the earlier physical implementation. All persistent random selectors are charged.

Theorem 30.1 (A twelve-state physical audit). *In the physical parameter range of Theorem 29.6, put $\beta = 1/300$. For $0 < \theta < 1/4$ and $0 < \delta < 1$, let L, R be at least the respective quantities in (12.1). There is a deterministic audit with $N = 4LR$ candidate training trials and two validation trials whose expected score is strictly greater than*

$$\sqrt{2} - 1 - \beta - \theta - 2\delta. \quad (30.2)$$

It has at most five states at completed training-trial cuts, ten at intermediate training report-bit cuts, three at the decision cut, and twelve throughout validation. In particular

$$\kappa_{2/5}^{\text{rows}} = 3, \quad 3 \leq W_{2/5}^{\text{phys}} \leq 12. \quad (30.3)$$

One entirely integer schedule takes $\theta = 1/200$, $\delta = 1/1000$, $L = 400$ and $R = 8 \cdot 2^{400}$, so $N = 12800 \cdot 2^{400}$. Its lower bound exceeds $2/5$. The upper bound twelve is not asserted to be the exact minimum integer peak width.

Proof. Use the report-pair probabilities $a = (1-p)(1-q)$ and $c = pq$ from Section 29. In the first $2LR$ training trials run Lemma 12.1 on the indicator of the pair 00. If it returns positive, choose A_1 and ignore the remaining training trials. Otherwise, in the next $2LR$ trials run the same comparison on the indicator of 11, choosing A_0 if positive and A_D if negative. Both phases have fixed lengths, even after an absorbing decision.

A comparison whose parameter is outside its indifference band is correct with probability at least $1 - \delta$. For purposes of analysis, one may define the second comparison on its independent observations even when the implementation ignores them. A union bound gives probability at least $1 - 2\delta$ that neither comparison errs outside its band. Since $\min(a, c) \leq 1/4$ and $\theta < 1/4$, at most one of a, c lies in $[1/2 - \theta, 1/2 + \theta]$.

On this good event the chosen payoff is within θ of

$$E(a, c) = \max\{1 - a - c, 1/2 - a, 1/2 - c\}.$$

Indeed, if one coordinate exceeds $1/2 + \theta$, its associated event is selected and is optimal. If both are below $1/2 - \theta$, A_D is selected and is optimal. In the remaining case the low coordinate's comparison is negative, and choosing either of the two relevant events costs at most the distance of the near coordinate from $1/2$. The range of the three payoffs at any fixed (a, c) is at most one. The expected ideal score is therefore at least $E(a, c) - \theta - 2\delta$. Lemma 29.3 gives $E(a, c) \geq \sqrt{2} - 1$. Changing the target to the physical target loses strictly less than β , proving (30.2).

The first comparator has four states. The second phase has these four states and one extra absorbing label for a previously chosen A_1 , hence five. Retaining the first report bit doubles that count to ten. On the second bit, determine the relevant indicator, update the block state, and discard both bits. The training mark is read and discarded. At the phase boundary the fixed schedule maps the first decision to either the A_1 label or the initial state of the second comparator. At the end it maps the remaining decisions to A_0 or A_D . During validation keep the chosen event, the candidate-event indicator, and a first-bit buffer when needed. This requires at most $3 \cdot 2 \cdot 2 = 12$ states; all events ignore the mark. These are actual phase-compatible transitions, not a bound only on terminal labels.

For the integer schedule, the error estimate in the proof of Lemma 12.1 is $2e^{-8} < 1/1000$. To check it rationally, $e > 8/3$ and $(8/3)^8 > 2000$. Further $\sqrt{2} > 707/500$, and hence

$$\sqrt{2} - 1 - \frac{1}{300} - \frac{1}{200} - \frac{1}{500} - \frac{2}{5} > \frac{11}{3000} > 0.$$

This proves the strict score claim without numerical optimization. The training count is exactly $4LR$; early absorption does not delete trials from that count.

For the lower bounds in (30.3), every full-profile auditor has its decision cut among its charged cuts. The feedback pushforward subfamily of Lemma 29.2 bounds any width-two decision continuation by $3/8 + \beta < 2/5$, for every finite adaptive training schedule. The all-on construction just given is a member of the original row interface and has three decision labels. This proves the row-indexed equality and both peak inequalities. $\square$

Corollary 30.2 (Two physical resource regimes). *The same physical score threshold $2/5$ admits the two schedules*

<i>training trials</i>	<i>decision width</i>	<i>peak width</i>
399	3	40602
$12800 \cdot 2^{400}$	3	12

followed by two validations. The first schedule may be padded to 400 training trials as before. For either fixed machine, changing d to d_k at fixed positive noise level changes its expected score by at most ϵ_k in (29.10). If the candidate training kernels also change, their additional cost is the training term in Corollary 12.3.

Proof. The first line is Theorem 29.6; the second is Theorem 30.1. In the moving-collision comparison the candidate training source is identical. Conditional on its chosen event, only one target validation mean changes. The common-preparation Gaussian bound therefore contributes ϵ_k , independently of the length of the unchanged training. The general perturbation formula applies when this equality of sources is absent. $\square$

The long schedule is an existence bound on the resource frontier, not a practical substitute for the 399-trial rule. In an autonomous model the clock must also be stored; Proposition 11.3 gives its explicit additional factor. Both schedules use deterministic transitions, so no precision-dependent randomizer is hidden in the physical upper bound. The one-collision construction remains the verified physical realization. No many-particle or zero-noise theorem is inferred from the larger class of finite experiments.

31. A THIRD RESOURCE REGIME FOR THE SAME PHYSICAL TASK

The three-response obstruction, the short exact selector, and the constant-width block audit concern the same positive-noise marked collision family. A truncated absorbing comparison supplies a further resource pair without changing that family or its threshold.

Theorem 31.1 (Polynomial-horizon physical realization). *Use the original row-event interface and the parameter range of Theorem 30.1. Let $\beta = 1/300$, $0 < \theta < 1/4$, and $m, n \geq 1$. There is a deterministic audit with $2n$ candidate training trials and two validation trials whose expected signed score is strictly greater than*

$$\sqrt{2} - 1 - \beta - \theta - 2e_{m,n}(\theta). \quad (31.1)$$

It uses at most $2m + 2$ states at completed training-trial cuts, $4m + 4$ at intermediate report-bit cuts, three at the decision cut, and twelve throughout validation. In particular, $\theta = 1/200$, $m = 400$ and $n = 160000$ give score greater than $2/5$, with 320000 training trials and peak width at most 1604.

Proof. Write $a = (1-p)(1-q)$ and $c = pq$ for the actual blind product candidate report probabilities, as in Theorem 30.1. On the first n reports apply Lemma 13.1 to the indicator of 00. If the answer is positive retain A_1 and ignore the second training phase. Otherwise use the next n reports to compare the indicator of 11, retaining A_0 for a positive answer and A_D for a negative answer. Conceptually generate the second phase even when it is ignored. Independence of the

reset preparations gives an error probability at most $2e_{m,n}(\theta)$ for either comparison to be wrong outside its indifference band.

On the complementary event the selected ideal payoff is within θ of $\max(1-a-c, 1/2-a, 1/2-c)$. Indeed $\min(a, c) \leq 1/4$, so at most one coordinate is within θ of $1/2$. If neither is, the two comparisons select the maximizing event; if one is, switching the relevant events changes payoff by at most θ . The three payoffs have range at most one at each (a, c) . Their envelope is at least $\sqrt{2} - 1$ by Lemma 29.3. Replacing the ideal target by the physical target loses strictly less than β and does not change the candidate training law. This proves (31.1).

The walk has $2m + 1$ labels. The second phase adds the previously latched A_1 label, giving $2m + 2$. Retaining the first report bit doubles this number; after the second bit, determine the relevant indicator and discard the pair. The mark is read and discarded. A deterministic clock supplies the phase and the within-phase time. At its end map the remaining walk states to A_0 or A_D . The validation implementation from Theorem 30.1 uses at most twelve states, including the event, first bit, and candidate-event value. No real-valued randomization is needed by this construction.

For the stated integers, $4\theta m = 2n\theta^2 = 8$, so $e_{m,n}(\theta) = 2e^{-8} < 1/1000$. The rational estimates $e > 8/3$, $(8/3)^8 > 2000$, and $\sqrt{2} > 707/500$ give

$$\sqrt{2} - 1 - \frac{1}{300} - \frac{1}{200} - \frac{1}{500} - \frac{2}{5} > \frac{11}{3000} > 0.$$

This proves the strict physical margin, with the stated fixed training count and peak width. $\square$

Corollary 31.2 (Three compatible resource bounds). *The original physical task admits all three rows below, each followed by two validations and having decision width three:*

<i>training trials</i>	<i>peak width</i>	<i>training mechanism</i>
399	40602	<i>exact empirical selector</i>
320000	1604	<i>truncated absorbing comparisons</i>
$12800 \cdot 2^{400}$	12	<i>four-state block comparisons.</i>

The all-auditor row-indexed decision optimum remains $\kappa_{2/5}^{\text{rows}} = 3$ and the unrestricted finite-horizon physical peak still satisfies $3 \leq W_{2/5}^{\text{phys}} \leq 12$. The table gives attainable points, not an optimal physical fixed-sample frontier. In particular the 40602 bound is not an all-auditor lower bound for 399 samples.

Proof. Combine Theorems 29.6, 30.1 and 31.1. The feedback-uniform two-state obstruction in Theorem 30.1 continues to apply to every audit in this enlarged set of constructions. $\square$

For a perturbed blind training source with per-active-trial defect α , the two-phase construction also has the bound

$$2 \min\{1, 2\alpha \min(n, m^2)\} + \beta_c + \beta_t$$

on score change, by the proof of Corollary 15.2. The moving-collision parameter alone leaves that source unchanged, so its training contribution is zero; the previous target transport and bounded Gaussian/entropic consumer are retained without change. If one makes the machine autonomous, an explicit bound is $(6n + 12) \max(4m + 4, 12)$ states, since the $2n + 2$ trials each acquire two bits and a mark, with the phase/decision boundaries included. The clock multiplier is an implementation upper bound, not an optimized autonomous-memory claim.

32. CONFIDENCE-LEVEL CERTIFICATION

An expected signed score is an operational separation, not a confidence statement about one validation pair. We now distinguish these two tasks. The null hypothesis below is equality of the candidate and target validation laws; it need not belong to the private alternative family. All

preparations in the theorem are independent resets with one fixed candidate law. The test cannot be given a new adversarial candidate after each reset.

Theorem 32.1 (Confidence lift with a charged accumulator). *Suppose a base reset audit uses n training trials, one candidate and one target validation, and peak width W , with score $D \in \{-1, 0, 1\}$. Assume its expected score is at least $\gamma > 0$ for every alternative. Run J independent copies and reject equality when*

$$\sum_{j=1}^J D_j \geq J\gamma/2. \quad (32.1)$$

Then both the type-I error under equality and the worst-case type-II error are at most $e^{-J\gamma^2/8}$. The whole test has a realization with at most $W(2J+1)$ states and $J(n+2)$ trials. In particular $J \geq 8\gamma^{-2} \log(1/\alpha)$ suffices for both errors to be at most α .

Proof. Under equality, conditional on the training and selected event, the two independent validations have equal means; hence $\mathbb{E}D = 0$. Under each fixed alternative its mean is at least γ . The copies are independent and $D_j \in [-1, 1]$. The exponential-moment estimate for a variable in an interval of length two is $\mathbb{E}e^{s(D_j - \mathbb{E}D_j)} \leq e^{s^2/2}$, by the same tilted-variance argument used in Theorem 9.2. Markov's inequality and minimization in s give $\Pr(J^{-1} \sum (D_j - \mathbb{E}D_j) \geq a) \leq e^{-Ja^2/2}$, and the same lower-tail bound. Set $a = \gamma/2$.

Keep the running integer sum, with at most $2J+1$ values, beside the base machine's register. At each completion add its emitted score and reset only the base register. The sum is not stored in an external history. The clock supplies the repetition index and the fixed terminal threshold; all $J(n+2)$ trials are counted. $\square$

The 399-trial physical audit has mean strictly above $2/5$, so one may use the rational choice $\gamma = 2/5$ in this theorem. This gives the fully explicit sufficient count $J \geq 50 \log(1/\alpha)$ and $401J$ total trials (or $402J$ with the inherited ignored slot). The confidence guarantee uses repetition and its charged sum, not one pair of validation observations.

Theorem 32.2 (One training phase and repeated validation). *For the physical audit of Theorem 30.1, put $\gamma_0 = \sqrt{2} - 1 - 1/300 - \theta > 0$. Retain its chosen event and use m independent candidate-target validation pairs. Reject equality when their score sum is at least $m\gamma_0/2$. The type-I error is at most $e^{-m\gamma_0^2/8}$ and the worst-case type-II error is at most*

$$2\delta + e^{-m\gamma_0^2/8}.$$

The audit uses $4LR + 2m$ trials and at most $12(2m+1)$ states throughout training and validation. Thus $\delta \leq \alpha/4$ and $m \geq 8\gamma_0^{-2} \log(2/\alpha)$ suffice for error at most α .

Proof. The proof of Theorem 30.1 gives an event of training probability at least $1 - 2\delta$ on which the conditional mean score of the selected event is strictly greater than γ_0 . Conditional on that event and its selected label, the validation pairs are independent. Apply the tail bound in Theorem 32.1 conditionally and add the training failure probability. Under equality every selected event has zero mean, regardless of training, so no training failure term is needed for type-I error. Keep the three-valued event label, the within-pair indicator and first-bit buffer, and a sum with $2m+1$ values. Training itself needs no more than ten states, so the displayed bound covers every phase. $\square$

The tests in this section have a different output task from maximizing an expected witness score. The exact three-state decision-cut theorem is not asserted to be an optimal memory theorem for these binary tests. For perturbations of training and validation laws, the coupling argument of Theorem 11.2 bounds changes of each rejection probability by the sum of the per-call total variation errors. In a source-preserving physical transport, alternatively lower the conditional

mean margin by the single-row target variation and reapply the conditional tail estimate. This latter argument transports the confidence guarantee without pretending that an entire m -sample validation law has the same total variation distance as one validation observation.

33. CONFIDENCE TESTS WITH PRICED FINITE-STATE AMPLIFICATION

The signed-sum construction in Section 32 converts expectation into confidence. There is also a direct sample–memory choice for the amplifier itself. Its formulation keeps the binary testing objective distinct from the expected-score task.

Theorem 33.1 (Two finite-state confidence amplifiers). *Suppose a reset audit uses b training trials, two validations, peak width $W \geq 3$, and reports $D \in \{-1, 0, 1\}$. Under the equality null $\mathbb{E}D = 0$, and under every alternative in a fixed class $\mathbb{E}D \geq \gamma > 0$. Fix a known dyadic $0 < g \leq \gamma$, and put $\theta = g/8$.*

- (i) *For integers $m, n \geq 1$, there is a binary test with both errors at most $e_{m,n}(\theta)$, using $n(b+2)$ preparations and peak width at most $(2m+1)\max(W, 9)$. Thus both errors are at most α with the choices (13.5), with $\delta = \alpha$.*
- (ii) *With L, R as in (12.1), using $\delta = \alpha$ and the same θ , there is another test with both errors at most α , $2LR(b+2)$ preparations, and peak width at most $4\max(W, 9)$. Its very long external schedule is charged separately. The confidence error, not the input parameter, determines these read-only horizon constants.*

Proof. Run independent copies of the base audit with fresh preparations and fresh private bits. Conditional on each output D , generate an additional coin Z with probability

$$\mathbb{P}(Z = 1 \mid D) = \frac{1}{2} + \frac{D - g/2}{4}. \quad (33.1)$$

This is between zero and one, since $0 < g \leq \gamma \leq 1$ and $-1 \leq D \leq 1$. If g has denominator 2^s , the probability has denominator dividing 2^{s+3} and is sampled with $s+3$ fair bits. The coins are independent under each fixed parameter. Their mean is $1/2 - g/8$ under the null and at least $1/2 + g/8$ under the alternatives. Apply the walk of Lemma 13.1, rejecting the null for a positive result, to prove (i). Apply the block comparison of Lemma 12.1 to prove (ii).

At every base-audit epoch retain the amplifier state alongside its W labels. After obtaining D , retain that value and at most three Bernoulli-sampler labels while drawing (33.1). This requires at most nine labels per amplifier state. Update the amplifier and reset the base audit before the next repetition. The terminal rejection decision is a function of the amplifier state and requires no additional persistent random seed. This proves the stated full-cut peak bounds. Even after the amplifier is absorbed, padded base-audit repetitions remain charged to the preparation count. The clock supplying the repetition, base-audit epoch and fair-bit depth is external; storing it multiplies the displayed bound by at most one plus the total number of such epochs. $\square$

For the physical expected-score theorem one may take $g = 3/8$, since $3/8 < 2/5$. The additional coin then needs at most six fair bits. Thus confidence can be purchased either with a growing absorbing counter and polynomial repetition count, or with a constant register and a much longer clocked schedule. Neither statement identifies the minimum state count of the binary testing problem with the three conditional validation responses needed by the expected-score problem. The signed-accumulator and train-once confidence theorems remain available as different resource regimes.

The output-law transfer in Theorem 15.1 applies to a completed binary test. In particular, if perturbations change its output law by at most ε , its type-I and type-II bounds each increase by at most ε . Transporting all repeated raw data instead counts every affected preparation, as in Section 32. Independence of the base repetitions and uniformity over the fixed alternative class

are assumptions of the amplification theorem; adaptation of the alternative between repetitions is not being asserted.

34. A FINITE-REGISTER CERTIFICATE TRANSPORTED THROUGH THE MICROSCOPIC LIMIT

The general empirical audit in Theorem 18.1 treats every coordinate of a marked row as unknown. The private continuation constraint in the collision experiment forces a smaller family. Exploiting this constraint yields a much smaller executable auditor, without granting it knowledge of the candidate law.

In the all-on row of Section 28, every private width-one candidate emits two independent bits with parameters p, q , independently of the fair actual mark W . Put

$$a = (1 - p)(1 - q), \quad c = pq.$$

The ideal target Q_* is supported on 000, 001, 110, 111, with respective masses $5/16, 3/16, 3/16, 5/16$. The candidate masses on these atoms are $a/2, a/2, c/2, c/2$. Every other atom has target mass zero and cannot improve the maximizing event for $Q_* - b$.

Lemma 34.1 (Two-count audit and its exact register implementation). *In $n \geq 1$ independent all-on candidate trials let C_0, C_1 count report pairs 00, 11, respectively, ignoring the training mark. On the four supported atoms use the empirical masses $C_0/(2n), C_0/(2n), C_1/(2n), C_1/(2n)$, and select exactly those atoms whose target mass is strictly greater than this empirical mass. Call the resulting event A_C . For every $p, q \in [0, 1]$,*

$$\mathbb{E}\{Q_*(A_C) - b(A_C)\} \geq \|Q_* - b\|_{TV} - n^{-1/2}. \quad (34.1)$$

For $n = 400$, the rule is implemented by the two truncated counts $\min(C_0, 250), \min(C_1, 250)$, with comparisons to 150 and 250. Including execution of complete transcripts and the two independent validation trials, it has a deterministic streaming realization with at most 504008 retained states at any scheduled epoch. Thus a nineteen-bit register suffices. The schedule clock is free; no other history store is free. The table description and transition evaluation cost are not asserted to be bounded by nineteen bits.

Proof. On any supported atom let $x = Q_*(w) - b(w)$ and let $\hat{x}$ be the empirical discrepancy. If the sign rule selects a different event from an exact maximizing event, the lost contribution is at most $|x - \hat{x}|$. On unsupported atoms both rules can omit the atom. Summing gives regret at most

$$|C_0/n - a| + |C_1/n - c|.$$

The two counts need not be independent. Their marginal binomial variances and Cauchy–Schwarz give expected regret at most

$$\frac{\sqrt{a(1-a)} + \sqrt{c(1-c)}}{\sqrt{n}} \leq \frac{1}{\sqrt{n}}.$$

This proves (34.1).

For the 00 pair, select both marked atoms when $C_0 < 150$, only the atom of target mass $5/16$ when $150 \leq C_0 < 250$, and neither when $C_0 \geq 250$. The same rule applies to 11, with the other mark being the atom of larger target mass. At equality the zero empirical discrepancy is excluded. Increasing a count beyond 250 cannot change the final choice. Two registers with 251 possibilities each therefore implement the rule, with at most nine possible final event masks.

A buffer with eight possibilities is sufficient to execute and record a completed two-bit-plus-mark transcript. During training the pair of truncated counts and this buffer use at most $8 \cdot 251^2 = 504008$ states. After training, discard the counts and retain the event mask. During validation at most nine masks, one target-event indicator with two values, and the same eight-state transcript buffer are needed, a total of 144 states. The final signed score takes three values. Since the epoch is

known, labels can be reused between these phases; their state counts need not be multiplied. The update maps and threshold comparisons are parameter-independent deterministic maps. They are thus stochastic rows with entries zero or one, or equivalently a compatible continuation cover by Theorem 8.2. Fresh sampling randomness belongs to the specified reset experiments, not to an uncharged auditor register. Finally $504008 < 2^{19}$. $\square$

Theorem 34.2 (Finite-register microscopic certification and global transport). *Consider the marked hard-sphere family of Theorem 28.2, with $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/24$. There is one rational streaming audit, independent of d, σ, p, q , using 400 candidate training resets and one independent candidate and target validation trial, whose expected signed score is strictly greater than $2/5$ against every private width-one candidate. It uses at most 504008 retained auditor states. The candidate remains private width one; the auditor's register is a distinct, charged resource. The physical deficiencies still satisfy*

$$\delta^p > 9/20, \quad \delta_r^v > 9/20 \ (r \geq 1), \quad \delta_2^h < 13/100.$$

For $d \leq 2/5$ all three finite-experiment deficiencies vanish.

Fix a positive σ and a sequence $d_k \rightarrow d$ within the collision interval. Put

$$\rho_k = |d_k - d| + 2700|d_k - d|^2, \quad \epsilon_k = \min\{1, \sqrt{\rho_k}/(2\sigma)\}.$$

The same 400-reset rule changes its expected score by at most ϵ_k under this perturbation. At the same time the microscopic paths, full and projected likelihoods, posteriors, and bounded entropic values have the joint convergence of Theorem 28.2. Their innovations, backward integrands and backward martingale integrals additionally have all the global conclusions of Theorem 27.2, with $T = 3$, $M = 1$, $\eta_k^2 \leq \rho_k$, and $\varepsilon_k \leq \epsilon_k$. On compact parameter sets bounded away from zero noise, the same audit has the effective probability-enclosure and rational-verification properties of Theorem 28.2(d).

Proof. The private width-one continuation restriction gives precisely the candidate family used in Lemma 34.1; the actual mark is not available to its transition rows. The exact finite separation in Theorem 23.1 gives

$$\min_{p,q} \|Q_* - b(p, q)\|_{\text{TV}} = d_* := \sqrt{5/2} - 9/8.$$

The scattering and observation estimates proved in Theorem 28.2 put the physical marked target within $1/300$ of Q_* , uniformly in all stated parameters. Train the two-count rule against the fixed rational Q_* , but use the actual physical target for validation. Each selected event's target probability changes by less than $1/300$. Hence its actual expected score is greater than

$$d_* - \frac{1}{20} - \frac{1}{300} > \frac{2}{5}. \quad (34.2)$$

The last strict inequality is exact: it is equivalent to $\sqrt{5/2} > 947/600$, whose squared margin is $3191/360000$. The rule's executable memory bound was proved in Lemma 34.1. Its total trial count is 402, not 400 once validation is included. The private, hidden and visible deficiency bounds and the no-collision control are inherited from the same physical theorem, rather than inferred from an auditor whose signature was changed.

For changing d_k , Lemma 28.1 supplies the signal modulus ρ_k and the pathwise J_1 bound $100|d_k - d|$. Theorem 25.1 gives the joint marked variation bound ϵ_k and the likelihood/posterior convergence. The candidate source is blind and its behavior does not depend on d . Thus training laws and candidate validation laws are unchanged, and Proposition 18.4 has $\alpha = 0$ and $\beta = \epsilon_k$. There is no factor of 400 in this transport cost. The same preparation and signal bounds verify every hypothesis of Theorem 27.2, proving the global backward assertions. Finally the audit uses rational thresholds and finitely many deterministic row maps. The positive-noise enclosure

theorem therefore applies directly; an additional target-presentation error ζ reduces its score by at most ζ . $\square$

The two-count construction is specific to the private behavior family. A hidden mixture can destroy the independence of its two reports, and its mark relation must still be checked. The bound is therefore not asserted against the hidden class. The visible lower bound has its separate fiberwise proof; an arbitrarily large visible alphabet is not silently stored in the nineteen-bit auditor. The certificate concerns expected score, not a confidence statement about a single realized validation pair. Repeating validation for statistical confidence would incur an additional sample and storage budget.

The theorem uses a single mechanism: the continuation restriction fixes the feasible candidate laws and permits the compressed audit; scattering supplies its marked separation; common-preparation variation transports the same executable test and the global conditional-law consumer through a change in the microscopic trajectory. The collision transduces a prepared impact variable into a signal. It does not create the prepared correlation between that variable and the mark, and it is not a particle-number limit.

35. EXPLICIT PHASE TRANSITIONS FOR THE TWO-COUNT CERTIFICATE

For comparison with the new residual quotient, we give the complete phase-state specification of Lemma 34.1. It realizes the original 400-training-trial rule exactly, not the majority rule of Section 29. Let

$$C = (C_0, C_1) \in \{0, \dots, 250\}^2, \quad \tau(x) = \min(x, 250).$$

The free clock records the training index, the within-trial observation epoch, the validation phase, and deterministic processing epochs. It contains no data-dependent information. At the start, $C = (0, 0)$.

For a completed transcript (x_1, x_2, w) define the deterministic update

$$T(C, x_1, x_2, w) = (\tau(C_0 + \mathbf{1}_{\{(x_1, x_2) = (0, 0)\}}), \tau(C_1 + \mathbf{1}_{\{(x_1, x_2) = (1, 1)\}})).$$

The value of w does not enter this training update. The phase transitions are the following; x denotes the current input, not an earlier unretained report.

Phase	Retained data	Next transition
Training start	C	Read x_1 and keep (C, x_1) .
Second report	(C, x_1)	Read x_2 and keep (C, x_1, x_2) .
Terminal mark	(C, x_1, x_2)	Read w and keep (C, x_1, x_2, w) .
Count update	(C, x_1, x_2, w)	Replace by $T(C, x_1, x_2, w)$; discard the transcript.
Event formation	C	Replace by the event mask $A(C)$ defined below.
Candidate validation	A , partial transcript	Complete the transcript; replace it by $i = \mathbf{1}_A(x_1, x_2, w)$.
Target validation	(A, i) , partial transcript	Complete the transcript and emit $\mathbf{1}_A(y_1, y_2, v) - i$.
Score	A value in $\{-1, 0, 1\}$	Terminate.

The event formation map, including both threshold ties, is

$$\begin{aligned} 000 \in A(C) &\iff C_0 < 250, & 001 \in A(C) &\iff C_0 < 150, \\ 110 \in A(C) &\iff C_1 < 150, & 111 \in A(C) &\iff C_1 < 250. \end{aligned}$$

No other atom is included. For each report pair there are three choices: both marked atoms, only its larger-target-mass atom, or neither. Thus there are at most nine masks. Clipping at 250

preserves every comparison, including equality; no flag for a count exceeding the clipping threshold is required.

At a training epoch the possible states are bounded by $8 \cdot 251^2 = 504008$: the factor eight dominates every partial or complete transcript buffer. At validation epochs the largest alphabet has at most $9 \cdot 2 \cdot 8 = 144$ values, including the already computed candidate-event indicator. Event formation and final emission need at most nine and three states, respectively. Embed each phase alphabet into $[504008]$ by a fixed injective encoding. Because the phase is supplied by the free clock, these encodings need not have disjoint images. The next row decodes the current phase’s encoding and applies the displayed transition. This gives formal deterministic rows with entries zero or one at every epoch. The buffers, event mask, and validation indicator are all charged. No unpriced data store is introduced by label reuse.

APPENDIX A. THE PROOF DEPENDENCY CHAIN

An arrow below means an actual use of a theorem in the displayed proof, not an assertion of priority or an artificial dependency between papers. The preserved exact physical lower and upper bounds meet at the decision cut. The new full-profile theorem supplies an explicit training realization of general decision covers; its proof does not invoke the historical aggregate kinetic claims.

Input	Consequence using that input
Marked private rows and normalized continuations, Theorems 7.2, 8.2	The product behavior image and its validation continuation responses.
Decision continuation spectrum, Theorem 9.2	Universal two-state physical obstruction, Theorem 29.1; the strict-cut classification (29.9).
Product constraint (29.1) and three-response envelope, Lemma 29.3	Uniform finite-sample score, Lemma 29.4.
Finite-valued residual realization, Theorem 16.1	Exact full training profile and bit-level construction, Theorem 29.5.
Marked scattering and moving-collision estimate, Theorem 28.2	Both physical cut bounds and transport of the very same auditor, Theorem 29.6.
Common-preparation Gaussian comparison, Theorem 25.1; the three version lemmas in Section 26	The global conditional-law consumer, Theorem 27.2, on that moving-collision family.

The prior binary audit theorem remains a separate calibration, not a premise of the physical memory lower bound. The original two-count certificate remains a separate full-discrepancy approximation, not a claimed optimal implementation. The source-preserving build, rational checks and finite enumerations accompanying this article test the stated implementation and identities; they are not independent analytic proof certification.

A.1. The full-profile resource chain. The new central arrow is

strict response cover $\longrightarrow$ finite block tournament $\longrightarrow$ rational whole-schedule auditor.

Theorem 12.2 uses the decision spectrum, the finite block lemma and the dyadic sampler. It does not use a training oracle. Theorems 11.1 and 11.2 track approximation, calibration and precision on complete causal schedules. Theorem 17.1 uses one shared capacity budget for all cuts, and Theorem 32.1 retains the score accumulator. Together these establish a finite controlled-experiment instance of the program’s resource–precision–statistical-error objective. A universal singular-modulus rate or an infinite-horizon kinetic realization is not an input or a consequence of this finite theorem.

The current C2-scoped consumer remains the bounded, common-preparation, positive-noise Gaussian/entropic theorem. The original B4 normalized nonlinear resolvent formula cannot be inherited: its displayed linear difference identity fails on constant inputs. For a separately justified graph H with single-valued $R_\lambda = (I - H/\lambda)^{-1}$, the correctly typed identity is instead

$$R_\lambda h = R_\mu [(\lambda/\mu)h + (1 - \lambda/\mu)R_\lambda h].$$

Indeed both sides solve the same μ -resolvent inclusion. This conditional identity does not establish the graph's range or its kinetic corrector. Those historical targets remain in the program rather than being assigned proof credit from a finite resource theorem.

APPENDIX B. A NAMESPACE MAP OF THE ELEVEN-COMPONENT PROGRAM

The namespaces in this map separate historical paper targets from GTF statements proved here. Every historical component is retained. An interface means that the indicated consumer can be applied after its model-specific hypotheses are proved; it is not an automatic proof of the historical component’s aggregate theorem.

Paper	Retained historical role	GTF relation and precise proof credit
A1	Protocol and finite task reduction	Marked comparison, priced composition and decision-cut certification; no automatic all-order physical response conclusion.
A2	Independent primary geometric chain	Its protocol adapter is historical; the primary geometric chain is not a premise or consequence of a theorem in this article.
A3	Dynamical/potential-space targets	Conditional comparison interface; no new assertion of the historical dynamical closure.
A4	Conditional kernels and forgetting	Historical finite-model consumer; common-preparation posterior estimates feed the current C2-scoped consumer when their hypotheses hold.
B1	Microscopic realization targets	Declared marked experiment interface; the present scattering construction is fixed-particle.
B2	Dynamic action and kinetic targets	Historical action estimates are not used to prove the present decision-cut theorem.
B3	Balance, response and corrector targets	No current proof depends on the historical aggregate balance/kinetic claims.
B4	Nonlinear kinetic semigroup	The historical normalized-resolvent identity and its aggregate downstream claims are not premises here. Their independent repair remains a program obligation.
C1	Statistical acquisition/inference	Original behavior image, finite-reset experiments, and the present response-envelope and microscopic certification consumers.
C2	Duals, response, memory and projections	The current bounded Gaussian/entropic transport replaces the optional-projection step only in its explicitly proved class; it does not certify the historical strict-dual, form-response or rigidity aggregate.
D1	Phase and compact statistical targets	Historical finite-model consumer retained; no new complete phase-theory closure is inferred.

The machine-readable companion records the exact predecessor graph and source identities, including the original B4 and C2 statements inspected at their frozen commit. Names such as `gtf20:decision-cut` and `historical:B4:r17-main` denote different proof obligations. The organizing theorem is a general comparison-and-certification result with an essential physical consumer. It does not replace the entire historical pipeline by a declaration that every aggregate theorem follows from it. The broader program remains intact as a mathematical program.

APPENDIX C. PRESERVED ORGANIZING STATEMENTS

The following two statements retain the earlier organizing labels and claims for cross-reference. Their full proofs are included in the canonical modules above. The unchanged predecessor article and its complete historical development are appended in the separate companion, so that reorganization does not discard earlier derivations.

Theorem C.1 (Decision continuation complexity and physical certification). *The following assertions hold with all persistent data charged to the appropriate machine.*

(i) *The supremum $v_{n,K}$ of worst-case expected scores with n training samples and K states at the decision cut satisfies*

$$0 \leq \mathfrak{V}_K - v_{n,K} \leq \min \left\{ \sqrt{(|\Omega| - 1)/n}, \sqrt{2 \log(2K)/n} \right\}.$$

The least K permitting a strict score greater than c at some finite training length is the least size of a cover of B by sets $\{b : (Q - b)h > c\}$. This spectrum is Lipschitz under uniform total variation perturbations of candidate and target laws.

(ii) *For the ideal marked private product family B of the two-sphere experiment, $\mathfrak{V}_1 = 1/8$ and $\mathfrak{V}_2 = 3/8$. Throughout the collision interval $9/10 \leq d \leq 11/10$, $0 < \sigma \leq 1/24$, the minimum decision-cut width for uniform physical score greater than $2/5$ is exactly three. The lower bound applies to every finite-reset auditor with the specified interface, not merely a particular counter implementation; feedback-row selection does not evade it.*

(iii) *One deterministic rational rule has physical score greater than*

$$\sqrt{2} - 1 - \frac{1}{4\sqrt{399e}} - \frac{1}{4}e^{-399/8} - \frac{1}{300} > \frac{2}{5}.$$

It uses 399 informative training trials, an optional ignored training trial and two independent validations, hence fits the 402-trial schedule. It has exactly three decision-cut states. On its informative training segment, the minimum simultaneous widths for exact stochastic implementation are

$$r_t = \begin{cases} \binom{t+2}{2}, & 0 \leq t \leq 199, \\ \binom{402-t}{2}, & 200 \leq t \leq 399. \end{cases}$$

Their maximum is 20301. Including individual report-bit acquisition and the validations, at most $40602 < 2^{16}$ retained states suffice.

(iv) *At an arbitrary cut of a finite-valued decision, a weighted graph of distinguishing suffixes gives a distribution-sensitive error bound through its minimum monochromatic weight under K labels. Applied to Bayes margins, it yields a quantitative constrained-audit regret bound. At zero error it recovers the exact simultaneous residual profile, even for stochastic implementations.*

(v) *For fixed positive σ and changing d , the same physical rule has the target-variation modulus $\min\{1, \sqrt{|d' - d| + 2700|d' - d|^2/(2\sigma)}\}$, with no training-count factor. The microscopic paths, likelihoods, posterior processes, innovations, and bounded entropic backward values, integrands and martingale integrals have the joint transport conclusions proved below. The private and visible deficiencies exceed $9/20$, whereas a hidden two-valued selector has error less than $13/100$. In the no-collision regime $d \leq 2/5$ all three finite-experiment deficiencies vanish.*

Proof. Part (i) is Theorem 9.2. The exact ideal spectral values are Theorem 29.1; Lemmas 29.3 and 29.4 and Theorem 29.6 prove the matching physical construction. Theorem 29.5 proves (iii), including all acquisition and validation phases. Theorem 16.1 and Corollary 16.2 prove (iv). The microscopic estimates in Theorem 28.2, the explicit coupling-version lemmas in Section 26, and Theorem 27.2 prove (v). $\square$

Theorem C.2 (Continuation complexity and stable causal certification). *With these information conventions the following statements hold.*

(i) *A finite causal kernel is realizable at a width profile if and only if it admits a compatible normalized continuation cover of that profile. For a Boolean terminal test, the exact minimal profile is the number of distinct continuation responses at each cut, even when the implementing*

machine is stochastic. If a cut has $r \geq 2$ distinct responses and only K retained states, its total wordwise error is at least $(r - K)_+/(r - 1)$.

(ii) For $0 < h < 1/2$, consider $n = 2m + 1$ independent binary samples from one of $\text{Ber}(1/2 - h)$ and $\text{Ber}(1/2 + h)$, and the signed event-discrepancy audit against $\text{Ber}(1/2)$. Its unrestricted value is $h\|P_+ - P_-\|_{\text{TV}}$. The exact training widths needed to attain it are

$$r_t = \min\{t + 1, n - t + 2\}, \quad 0 \leq t < n.$$

A smaller width at any cut has the strictly positive value deficit (19.4). Thus the continuation invariant controls both an implementation and a statistical obstruction.

(iii) In the prepared two-sphere experiment of Theorem 34.2, one rational audit using 400 candidate training resets, two validation trials, and at most 504008 retained auditor states has expected discrepancy greater than $2/5$ against every private width-one candidate. The same audit works for every contact diameter $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$. The private and visible deficiencies exceed $9/20$, whereas a hidden two-valued selector has error less than $13/100$. Removing the collision, by taking $d \leq 2/5$, makes all three finite-experiment deficiencies zero.

(iv) With positive noise fixed, changing d changes both the collision time and the microscopic path. If $\rho = |d' - d| + 2700|d' - d|^2$, the same audit's expected score changes by at most $\min\{1, \sqrt{\rho}/(2\sigma)\}$. Along $d' \rightarrow d$, the microscopic paths, full and projected likelihoods, and bounded posterior processes converge jointly in the senses specified below. For the associated bounded entropic backward equation, values converge in supremum L^2 and integrands converge globally in time-integrated L^2 , including on the common-latent coupling. Innovations converge uniformly in probability, and backward martingale integrals converge in supremum L^2 .

C.1. Inherited baseline. The following summary is retained from the predecessor. The present article strengthens its memory and backward-integrand conclusions in Theorem C.2; the earlier 2800-trial audit remains valid.

Theorem C.3 (Finite-resource certification and microscopic transport). *The following assertions hold with the conventions just described.*

(i) *Finite causal kernels have a register realization with a prescribed width profile exactly when their normalized continuation laws admit compatible finitely generated polytopes of those widths. At an erasure cut carrying a latent letter of law π , the optimal error with K retained states is*

$$1 - \sum_{i=1}^{\min(K, N)} \pi_{(i)}, \quad \pi_{(1)} \geq \cdots \geq \pi_{(N)}.$$

Independent hidden or visible selectors do not improve this cut value.

(ii) *The reset-audit value is attained, has a finite-dimensional moment minimax representation on the original B , and satisfies*

$$0 \leq \delta(B, Q) - v_n^{\text{aud}}(B, Q) \leq \sqrt{D/n}, \quad D = \sum_j (|\Omega_j| - 1).$$

The exponent $1/2$ cannot be improved uniformly over compact candidate classes. This assertion is about reset samples, not about the minimal degree of unrestricted polynomial tests.

(iii) *Suppose two prepared microscopic observation models have a common preparation variable, bounded signals H, H' on $[0, T]$, and the same positive Gaussian noise level σ . Set*

$$\eta^2 = \mathbb{E} \int_0^T |H - H'|^2 dt, \quad \varepsilon = \min\{1, \eta/(2\sigma)\}.$$

Their joint marked observation laws differ by at most ε . This estimate transports resource spectra at their stated budgets and transports each fixed finite audit with its explicitly charged reset cost. It

also gives whole-process posterior convergence. The unprojected and projected likelihoods converge in the supremum L^2 norm under the common Wiener reference when $\eta \rightarrow 0$. For bounded latent payoffs, the associated entropic backward equations converge in value, and their integrands converge after the stated denominator localization.

(iv) In the two-sphere model of Theorem 28.2, one rationally specified audit with 2,800 training runs has expected discrepancy strictly greater than $2/5$ against every private width-one simulator, uniformly for $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$. A hidden two-valued selector has error less than $13/100$; every visible-selector width-one error exceeds $9/20$. For $d_n \rightarrow d$ the microscopic paths, likelihoods and posterior processes converge in the precise senses of that theorem, and the same audit remains valid. For $d \leq 2/5$ all three deficiencies vanish.

APPENDIX D. UNRESTRICTED-HORIZON MULTICUT REALIZATION

The following organizing statement retains the unrestricted finite-horizon comparison and its complete proof dependencies. The interfaces are those fixed in the introduction.

Theorem D.1 (Multicut realization and certification). *The following statements hold under the interfaces and resource conventions just described.*

(i) For $d = |\Omega|$, let W_c be the minimum peak width over all finite training schedules, including every random-bit and validation cut, and let W_c^{rat} restrict to finite fair-bit implementations. Then

$$\kappa_c(B, Q) \leq W_c(B, Q) \leq W_c^{\text{rat}}(B, Q) \leq 12d \kappa_c(B, Q).$$

For a K -response cover, a finite-horizon tournament realizes this bound with $2LR(K-1)$ training observations, where $L = \lceil \log(2/\delta)/(4\theta) \rceil$ and $R = \lceil 2^L \log(2/\delta) \rceil$. Its uniform score loss is at most $2^{1-r} + 2\rho + 4(K-1)\theta + 2(K-1)\delta$, with dyadic precision r and response-mean calibration error ρ .

(ii) Approximate normalized continuation lists with compatible one-step updates and local causal errors η_t realize their entire width profile with total error at most $\sum_t \eta_t$. Full-profile risk bounds also charge the sum of the per-call calibration errors and the row-rounding errors. Composition multiplies the simultaneous simulator and auditor profiles. A shared word-capacity constraint turns weighted residual obstructions at several cuts into a single Bayesian regret lower bound. A family of point-source experiments has exact decision spectra and matching linear peak-width growth; overlapping independent retention tasks have simultaneous sharp exact profiles and quantitative approximate lower bounds.

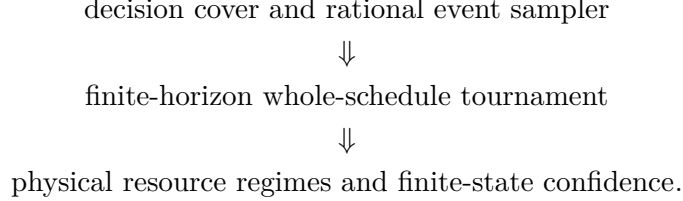
(iii) In the prepared two-sphere collision family, for $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$, the row-indexed decision complexity for score greater than $2/5$ is exactly three. A deterministic physical audit has peak width at most twelve throughout report-bit acquisition and validation, with an explicit finite preparation count. The earlier 399-training-trial audit is also retained, with peak width 40602 and its exact simultaneous residual profile. For the complete physical task these give $3 \leq W_{2/5}^{\text{phys}} \leq 12$; the upper endpoint is not claimed to be an exact integer optimum.

(iv) Independent repetition turns either positive expected-score certificate into a test with both errors at most α , charging the score accumulator and every preparation. The short physical audit needs $J \geq 50 \log(1/\alpha)$ repetitions. The physical certificate and the bounded common-preparation Gaussian/entropic conditional-law consumer are stable under the moving-collision perturbations specified below.

Proof. Part (i) is Theorem 12.2. Part (ii) follows from Theorems 11.1, 11.2, 17.1, 17.2, 17.3 and Proposition 11.3. Part (iii) is Theorem 30.1 together with the preserved Theorems 29.5 and 29.6. Part (iv) follows from Theorems 32.1 and 32.2, Corollary 30.2, and the common-preparation transport results in Theorems 28.2 and 27.2. $\square$

APPENDIX E. DEPENDENCIES OF THE STATISTICAL RESOURCE FRONTIER

The general preparation bound uses the decision spectrum only to supply a finite response list and its strict margin. Dyadic event realization supplies its executable randomizers. The absorbing-walk lemma then supplies both a finite comparison error and the expected number of active updates. Thus



The occupation-weighted transfer theorem acts on the induced retained kernels of each implemented machine. In the independent reset setting it consumes the walk's active-time estimate. Its proof does not invoke any claim about convergence of changing filtrations.

The exact revelation theorem is an independent, fully solved finite statistical model inside the same profile-constrained decision layer. Its necessary direction applies to every stochastic machine; its sufficient direction constructs compatible nested labels. It is not used as a physical collision lower bound. The lower bound three in that family still comes from the original feedback-row obstruction. Likewise, the complete finite-word residual profile of the short physical selector is retained as an exact implementation theorem and not relabeled as a statistical optimum.

These distinctions give an actual resource–precision–error consumer for the foundation blueprint: preparation and state budgets come from Theorem 13.2, calibration and rationalization from its explicit coin construction, perturbation from Theorem 15.1, and a confidence-level readout from Theorem 33.1. The exact frontier provides a sharp case of the same optimization with nonconstant alphabet and arbitrary state profiles. None of these arrows presupposes the historical B4 kinetic range theorem, the broader C2 rigidity aggregate, or an unproved dependency of the independent A2 chain.

APPENDIX F. THE CONSTRUCTIVE RESOURCE THEOREM

The constructive statement and its proof are retained here as a single reference point for the supporting development.

Theorem F.1 (Continuation profiles and statistical resources). *The following statements hold in the declared clocked finite-state model.*

(i) *Every strict K -response cover of envelope v has, for integers $m, n, r \geq 1$ and dyadic calibration error ρ , a rational whole-schedule realization with*

$$N = n(K - 1), \quad W \leq 3dK(2m + 1),$$

and worst-case score at least

$$v - 2^{1-r} - 2\rho - 4(K - 1)\theta - 2(K - 1)(e^{-4\theta m} + e^{-2n\theta^2}).$$

Here $K \geq 2$; one response needs no training. The previously available constant-width realization gives, in the same model, $\kappa_c \leq W_c \leq W_c^{\text{rat}} \leq 12d\kappa_c$.

(ii) *Suppose a parameter in $\{1, \dots, M\}$ has known ordered prior π , and report t independently reveals that parameter with probability q_t and otherwise reports $*$. Among all stochastic machines having at most K_t labels after each report, the exact Bayesian success probability is*

$$a\pi_1 + \sum_{t=1}^n w_t \sum_{j=1}^{k_t} \pi_j, \quad k_t = \min(M, K_t, \dots, K_n),$$

where $a = \prod_t (1 - q_t)$ and $w_t = q_t \prod_{s>t} (1 - q_s)$. One compatible deterministic machine attains all the cut bounds simultaneously. The formula yields exact sample–peak tradeoffs and exact losses under checkpoint insertion.

(iii) If P_t, Q_t are the induced retained-state transition kernels of a fixed execution with common initial law, its terminal-law change is bounded by

$$\min \left(1, \sum_t \mathbb{E}_P \|P_t(X_{t-1}, \cdot) - Q_t(X_{t-1}, \cdot)\|_{\text{TV}} \right).$$

For absorbed reset comparisons this prices active updates even when training laws differ. It does not remove ignored preparations from the schedule or from the full raw-data law.

(iv) In the marked positive-noise collision experiment, a 320000-training-trial audit with peak width at most 1604 has score greater than $2/5$. The 399-trial, 40602-state audit and the 12-state long-clock audit are also retained. The exact row-indexed decision width remains three. Separate finite-state amplifiers yield specified type-I and type-II errors, with a polynomial-horizon growing counter or a constant register and a longer clocked schedule.

Proof. Part (i) is Theorem 13.2 together with Theorem 12.2. Part (ii) is Theorem 14.1 and its corollaries. Part (iii) is Theorem 15.1 and Corollary 15.2. Part (iv) follows from Theorem 31.1, Corollary 31.2, and Theorem 33.1. $\square$

REFERENCES

- [1] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272. doi:10.1214/aoms/1177729032.
- [2] L. Le Cam, Sufficiency and approximate sufficiency, *Ann. Math. Statist.* **35** (1964), 1419–1455. doi:10.1214/aoms/1177700372.
- [3] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28. doi:10.1524/strm.2002.20.14.1.
- [4] A. Nayyar, A. Mahajan and D. Teneketzis, Decentralized stochastic control with partial history sharing: a common information approach, *IEEE Trans. Automat. Control* **58** (2013), 1644–1658. doi:10.1109/TAC.2013.2239000. The theorem numbers cited here refer to the author preprint, arXiv:1209.1695.
- [5] H. S. Witsenhausen, On the structure of real-time source coders, *Bell System Tech. J.* **58** (1979), 1437–1451. doi:10.1002/j.1538-7305.1979.tb02263.x.
- [6] I. Gallagher, L. Saint-Raymond and B. Texier, *From Newton to Boltzmann: Hard Spheres and Short-Range Potentials*, Zurich Lectures in Advanced Mathematics, European Mathematical Society, 2014. The flow and boundary discussion used here is in Sections 4.1–4.3 of arXiv:1208.5753v2.
- [7] C. Widder, J. Zimmer and T. Schilling, On the generalized Langevin equation and the Mori projection operator technique, preprint (2025), revised 2026. arXiv:2503.20457v6.
- [8] C. Widder and T. Schilling, Generalised Langevin dynamics: significance and limitations of the projection operator formalism, preprint (2026). arXiv:2604.20453v2.
- [9] H. F. Trotter, Approximation of semi-groups of operators, *Pacific J. Math.* **8** (1958), 887–919. doi:10.2140/pjm.1958.8.887. Theorem 5.2, p. 902, is the approximation theorem cited here.
- [10] H. Weisshaupt, LeCam’s randomization criterion in the setting of locally convex spaces without lattice structure, EURANDOM report 2002–017, 22 April 2002. Lemma 1 and Theorem 5, pp. 14–16. Author report.
- [11] S. Basu, Algorithms in real algebraic geometry: a survey, author preprint, arXiv:1409.1534 (2014). Published in *Panoramas et Synthèses* **51** (2017), 107–153. Theorem 2.1 and Section 2.1 in the author preprint are used here.
- [12] M. C. Paull and S. H. Unger, Minimizing the number of states in incompletely specified sequential switching functions, *IRE Trans. Electron. Comput.* **EC-8** (1959), 356–367. Publisher record.
- [13] E. Torgersen, *Comparison of Statistical Experiments*, Encyclopedia of Mathematics and its Applications 36, Cambridge University Press, Cambridge, 1991.
- [14] N. Saldi, A topology for team policies and existence of optimal team policies in stochastic team theory, preprint (2017), arXiv:1711.00639.
- [15] S. Yüksel and N. Saldi, Convex analysis in decentralized stochastic control, strategic measures and optimal solutions, author preprint arXiv:1609.07685v2.
- [16] A. Nerode, Linear automaton transformations, *Proc. Amer. Math. Soc.* **9** (1958), 541–544. doi:10.1090/S0002-9939-1958-0135681-9.

- [17] S. Kiefer, Notes on equivalence and minimization of weighted automata, preprint (2020), arXiv:2009.01217v1.
- [18] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176. doi:10.2140/pjm.1958.8.171.
- [19] H. Weisshaupt, A generalization of Le Cam’s randomization criterion, *Studia Sci. Math. Hungar.* **43** (2006), 443–486. doi:10.1556/SScMath.43.2006.4.3. Theorem numbers refer to the author’s 39-page preprint, Randomization.pdf.
- [20] E. de Klerk and M. Laurent, Error bounds for some semidefinite programming approaches to polynomial minimization on the hypercube, *SIAM J. Optim.* **20** (2010), 3104–3120. doi:10.1137/100790835.
- [21] M. Laurent and L. Slot, An effective version of Schmüdgen’s Positivstellensatz for the hypercube, arXiv:2109.09528v2, 2023.
- [22] E. de Klerk and J. C. Vera, The link between 1-norm approximation and effective Positivstellensätze for the hypercube, *Numer. Algebra Control Optim.* **16** (2026), 84–104. doi:10.3934/naco.2024060. Theorem numbers refer to arXiv:2404.04190v2.
- [23] M. Fliess, Séries rationnelles positives et processus stochastiques, *Ann. Inst. H. Poincaré Sect. B* **11** (1975), 1–21. Original article. Theorems 3.1 and 5.4 and Corollary 2, pp. 8–9 and 18–19.
- [24] T. M. Cover, Hypothesis testing with finite statistics, *Ann. Math. Statist.* **40** (1969), 828–835. doi:10.1214/aoms/1177697590. Theorem 2, pp. 833–834, and Section 4, pp. 834–835.
- [25] W. Hoeffding, Probability inequalities for sums of bounded random variables, *J. Amer. Statist. Assoc.* **58** (1963), 13–30. doi:10.1080/01621459.1963.10500830.
- [26] M. E. Hellman and T. M. Cover, Learning with finite memory, *Ann. Math. Statist.* **41** (1970), 765–782. doi:10.1214/aoms/1177696958.
- [27] J. N. Tsitsiklis, Extremal properties of likelihood-ratio quantizers, *IEEE Trans. Commun.* **41** (1993), 550–558. doi:10.1109/26.223779. The original article is available from the author’s MIT publication page.
- [28] J. Müller and G. Montúfar, The geometry of memoryless stochastic policy optimization in infinite-horizon POMDPs, *International Conference on Learning Representations*, 2022. arXiv:2110.07409. Theorem numbers refer to the 45-page author version.
- [29] M. Putinar, Positive polynomials on compact semi-algebraic sets, *Indiana Univ. Math. J.* **42** (1993), 969–984. doi:10.1512/iumj.1993.42.42045.
- [30] V. Powers, Rational certificates of positivity on compact semialgebraic sets, preprint (2009), arXiv:0911.1331. Theorem 6 is the explicit-ball rational positivity result used here.